PORTFOLIO

Finding a space for leapfrog development plan in the Indian planning system

A DESSERTATION

Submitted to

The department of architecture and planning

of Maulana Azad National Institute of Technology, Bhopal

in partial fulfilment of the requirements

for the degree of Bachelors of Planning

By

Niteshkumar S K
211109041

Maulana Azad National Institute of Technology, Bhopal

2024

Guided by: Professor Dr. Vinay Mohan Das

Copyright@2024 by Niteshkumar S K. All rights reserved.

Abstract

In this dissertation report, the understanding and prior knowledge of leapfrog development theories and practices around the world is studied along with the current planning system practised in India (2024). Through this study the pre-requisites such as approaches, methodologies, environment & conditions that are considered and required were formulated for the following thesis paper that would be made in the aim of emphasising the need for leapfrog development plan for developing but lagging India & also prepare its brief guidelines with my intended definition and cause clearly.

Review of various literatures and case study of South Korea, China and India states out the importance of leapfrog strategies in “latecomer” nations development. And me figuring the research gap to further explore in terms of Indian context and its future development.

Keywords: Leapfrogging, development plan, Indian planning system

Contents

1. Introduction……..……5
2. Literature review ……6
   2.1. Introduction
   2.2. Indian planning system
   2.3. Lagging and latecomers
   2.4. Catchup theories and latecomer's advantages
   2.5. What is Leapfrogging: origin
   2.6. Leapfrog development
   2.7. Demand & technological window of opportunities
   2.8. Endogenous cause of leapfrog cycle
   2.9. Role of government & policies
   2.10. Challenges & risk involved in leapfrogging
3. Desk appraisals……18
   3.1. Case study on South Korea: R & D programme and industrial development
   3.2. Case study on India: Success of UPI- Digital transaction services
   3.3. Case study on China: Transfer technologies and localization
4. Result & learnings……26
5. Research gap…….27
6. Following Thesis- background…….28
   6.1. Aim
   6.2. Objectives
7. Conclusion……...…….29
8. References……..……..30

Appendix

1. Introduction

A nation's development is affected by various factors and its effects. From its history, geography, demographics & culture to its current affairs, global trends, local economy & the government initiatives such as schemes, programs and development strategies or plans. The result of those development plans is viewed from the convergence outcome of physical, social & economic benefits to the public and the country. The path for the development and growth that is chosen is the backbone which supports the entire framework of factors and body of implementation to do so. By the history of ancient and modern development we can easily confront that preparing a strategic planned path is better than foolishly expecting for the natural growth & development by its own.

Every change that occurs in the world that affects the countries development both positively & negatively whether it is any new technological revolution, geo-political affairs, natural calamities, etc. are not just due to natural effects but majorly through the plans and prediction of human knowledge involved. The success of Korean industrial economy is an effect of their planned policy focus such as national R & D programme. While the failure of disaster (flood) management of Chennai city, India is caused by illegal real-estate activities as the effect of absence of master plan/land-use plan for 8 years.

Due to social, political & economic – history of nations, their development is varied drastically from developed nations to developing nations to under-developed nations. So, the path that is taken to achieve the required goals are also varied. Not every nation should follow the same path or always should follow the path of developed nations cause not all countries are same or their goals and another important development factor “Time” plays a major role in the path choosing and its resultant development. The path used by the developed nation could be no more effective due to change of time and conditions. And international economy or sectoral economy is based on market competitiveness among the nations.

“Lagging” is a negative development phenomenon occur due to gap formed by the front lead of the developed nation in the competing sector/field while the latecomers lag behind and can’t able to equally compete or enjoy the economic benefits like the leading developed nation. That’s where the catch-up theories and strategies comes into notion as an enabler of cheat skill to catch up with the developed nations. Theoretically various approaches, models and classification are published & implemented all over the world across the time. From which we can broadly classify catch-up approaches into 3 pathways patterns- following pathway, stage-skipping pathway & creating new pathway (Lee and Lim 2001). In that stage-skipping and path-creating pattern is what we call “leapfrogging”, where we skip various non-required stages, risks, methodologies & technologies to catch up and also aims to get ahead of leading nations.

So, through the following sections we would discuss my understanding and criticism of various leapfrog theories & models along with their challenges in implementation. Case studies of various nation’s success stories in using leapfrog strategies both directly and indirectly. And also, the Indian perspective through showcase of current Indian planning system and try to find a space to accommodate a new leapfrog development plan on the system.

2. Literature review

Aim:

The aim of this literature review is to understand the prior theoretical knowledge and take away the research gap in relation to my study on "Need for integrated leapfrog development plan for developing but lagging India".

2.1 Introduction:

India is a vast nation with wide range of history and culture. Unlike any other country India's development and growth morphology is unique from being down bottom in the early decades of post-independence to economic crisis and failure of socialistic approach leads in opening the country's economy only in 1990s and become one of the latecomers in the economic race along with other nations. Through various strategies & plans and boom in IT industries, India now being one of fastest growing nation in the world.

But the growth gap existing between the developed nations and latecomers' countries like India is huge. Catch up theories (Lall 2000; Kim 1980; Westphal, Kim, and Dahlman 1985; Hobday 1995) has shown that the latecomer tends to achieve economic development by integrating and adapting the developed nations technologies and practices. This is consistent with product life cycle theory (Vernon 1966). Just through follow or catch up we can't achieve greater growth in this rapidly changing environment. So, using the time, capital, knowledge & technologies we should promote a strategy that would help us to skip stages, upgrade, avoid familiar risks & constantly adapt to enable catch-up and ahead growth in the world economy. That is through planned leapfrogging.

2.2 Indian planning system

The Indian planning system can be categorised broadly into two major planning type:

> **Economic (socio-economic) planning system** - *Fifteen Year Vision, Seven Year Strategy and Three-Year Action Agenda* documents (before- Five year plans)
>
> **Urban and Regional planning system**- Core area planning, special investment planning and Rural area development planning (GPDP).

India began with centralized economic planning, heavily influenced by socialist ideologies with *five-year plans* launched in 1951 to promote a rapid rise in the standard of living of the people by efficient exploitation of the resources of the country, increasing production and offering opportunities to all for employment in the service of the community. Subsequent plans addressed industrialization, education, health and poverty alleviation. Economic liberalization in the 1990s marked a significant shift, reducing the role of centralized planning and promoting market-driven policies.

In India, Town Planning is a state function and is not included in the concurrent list of the Central Government. Ministry of Urban Development (MoUD), Government of India (GoI) provides national level policy guidelines and legislations for urban planning and development in the country.

2.2.1 History and evolution

The foundation for systematic urban and regional planning was laid during the colonial period (pre-independence) with Bombay town planning act of 1915 followed by some theoretical efforts such as setting up of National Planning Committee by Indian National Congress in 1938, The Bombay Plan & Gandhian Plan in 1944, Peoples Plan in 1945 (by post war reconstruction Committee of Indian Trade Union), Sarvodaya Plan in 1950 by Jaiprakash Narayan were steps in this direction. (MOSPI-GoI, 2022)

Post independence, the process began with setting up of Planning Commission in March 1950 and in 1955 Government of India formed Town Planning Organization (TPO) to prepare the master plan of Delhi. In 1957, Central Urban and Regional Planning Organization (CRUPO) was created. The two organizations (TPO and CRUPO) were merged in 1962 to form Town and Country Planning Organization (TCPO). Since then, TCPO is working as the technical advisory agency to the Ministry of Urban Development on various issues related to urban and regional planning in the country. The TCPO is an apex body in the field of urban and regional planning in India and it works as a subordinate office to the central Ministry of Urban Development. The Model Town Planning and Regional Development Law was prepared by the TCPO in 1970s. This law has formed the basis for several state-level Town and Country Planning Acts in the country. (D. Mahadevia, R. Joshi, R. Sharma 2009)

Town planning at state level is looked after by state level Town and Country Planning offices. The state level town planning agencies in India follow different Acts, thus Town Planning guidelines across the country may vary from state to state. So do the nomenclature & structures of the plans.

2.2.2 Morden changes

On 1st January 2015, the National Institution for Transforming India or NITI Aayog came into existence as the government's premier think tank with the objective to evolve a shared vision of national development priorities, sectors and strategies with the active involvement of States by promoting competitive federalism and encouraging innovation.

And to prepare *Fifteen Year Vision, Seven Year Strategy and Three-Year Action Agenda* documents which represents a departure from the Five-Year Plan process, followed with a handful of discontinuities until the fiscal year 2016-17. The 12th Five Year Plan was the last of these plans. It has been felt that with an increasingly open and liberalized economy, we needed to rethink the tools and approaches to conceptualizing the development process. (NITI Aayog)

The Ministry of urban development formulated a comprehensive framework called urban and regional development planning, formulation and implementation (URDPFI) guidelines in 2014 to address the evolving needs for urban planning as an upgrade to UDPFI guidelines 1996. UDPFI guidelines suggested urban development planning system consists of a set of four inter-related plans with then existing national five year plans (Inter-relationship with FYP & various below mentioned plan is given in Appendix 1) as follows:

a. Perspective Plan
b. Development/ Master Plan
c. Annual Plan and
d. Plans of Projects/Schemes

Later, URDPFI guidelines (2014) suggested the planning system framework on the basis of hierarchy of plans, spatial extent, scale of planning, details provided in the plan and the function & their speciality as given in the below table

Planning system	Scope and purpose of the plan	Time frame*	Various plans; indicative list							
Core area of planning										
Perspective Plan	To develop vision and provide a policy framework for urban & regional development and further detailing	20-30 years	Long Term Perspective Vision document	Concept plan	Mission statement	--	--	--	--	--
Regional Plan	To identify the region and regional resources for development within which settlement (urban and rural) plan to be prepared and regulated by DPC	20years	Regional Plan (Mobility 1)	Sub-regional plan	--	--	--	--	--	--
Development Plan	To prepare a comprehensive Development Plan for urban areas, Peri-urban areas under control of Development authority/ Metropolitan Planning Committee.	20-30 years (Review every 5 years)	District Development Plan (Mobility 1)	City/ Metropolitan Development Plan (Mobility 2)	Master Plan City Utility (30 years)	Revised Development Plan	--	--	--	--
Local Area Plan	To detail the sub-city landuse plan and integration with urban infrastructure, mobility and services.	5-20 year (Review every 5 yrs)	Town Planning Schemes	Zonal Plan / Sub-city plan	Ward Committee Plan	Coastal Zone Mgmt Plan	Urban Redevelopment Plan	--	--	--
Specific and investment planning					1.4.1.		1.4.2.		1.4.3.	
Special Purpose Plan	To identify the needs of the special areas which require special plan within the framework of the development plan.	5-20 year (within city utilities 30 year plan)	City Development Plan (as per JnNURM)	Comprehensive Mobility Plan (as per JnNURM)	City Sanitation Plan (as per JnNURM)	Disaster Management Plan (as per NDMA)	Slum Redevelopment Plan (as per RAY)	Tourism Master Plan	Environmental Conservation Plan	Heritage Conservation Plan
Annual plan	To translate Development Plan in the context of annual physical & fiscal resource requirement. To monitor plan implementation with performance milestones.	1 year	Investment plan	Audit and monitoring plan	--	--	--	--	--	--
Project/ Research	To focus on project related investments, costing and returns & for the studies required prior to or post plan formulation. This should be a continuous process to support planning and implementation at all stages and promotes innovation in practice.	5-20 year	Pre-feasibility & feasibility study	Detailed Project Report	Schemes & Sub-projects	Surveys & Studies	Project such as: Riverfront development projects	--	--	--

Note: Consultative meetings and the regional workshops during URDFI formulation have pointed out the need to review the plan period of 20 years and extend the same to 30 years. However general consensus was in the favour of 20 years. Intermittent review in 5 year gap would be required.

Table1. Hierarchy & scope of Indian planning system, URDPFI guidelines 2014.

A) Core area of planning

a. Perspective Plan
b. Regional Plan
c. Development Plan (previously known as Master Plan)
d. Local Area Plan

B) Specific and investment planning

a. Special Purpose Plan
b. Annual Plan
c. Project/ Research

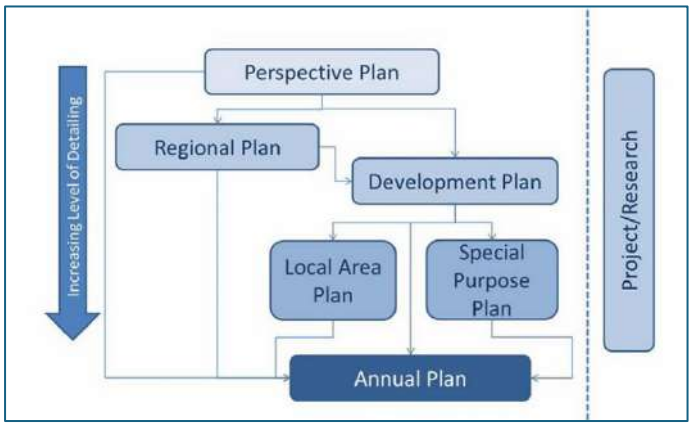


Fig.1 inter-relation between plans, URDPFI guidelines 2014.

Gram Panchayats have been mandated for the preparation of Gram Panchayat Development Plan (GPDP) for economic development and social justice utilizing the resources available to them under "Sabki Yojana Sabka Vikas" from 2018. And RADPFI 2021 recommended changes in GPDP planning formulation along with integration of rural spatial development plans.

2.3 Lagging and latecomers

Latecomers are enterprises/countries who are “forced” into an industry late due to historical reasons and they are lagging back & at a competitive disadvantage due to the initial lack of technology and market accesses (Hobday, 1995; Mathews, 2002). The acquisition of initial resources and capabilities by latecomer is of crucial importance. In this context, firms from emerging economies are often defined as “resource-poor late entrants” (Mathews, 2002). The aspect of “resource-poor” implies that one of the most fundamental differences between firms in the advanced and in developing economies is that the former have diverse resources available internally, from other firms or from the context, whereas the latter do not have all these critical resources (Franco Malerba, Keun Lee 2021).

The debate relating to latecomers is framed in discussions about the countries that arrive late to mastering industrialization in achieving economic prosperity through the use of knowledge (B. O.Oyeyinka, Padmashree G.Sampath 2010). In other words, a new divide has emerged among the latecomers themselves, and with it, greater conceptual complexity in the ways of our understanding of the divergent ways of economic development. We have thus separated "fast followers" and new "late comers".

The importance of identification and acceptance of the place a country has in the economy is crucial as described by Gerschenkron (1962)- the policies toward the backward countries are unlikely to be successful if they ignore the basic peculiarities of economic backwardness. Only by frankly recognizing their existence and strength, and by attempting to develop fully rather than to stifle what Keynes once called the "possibilities of things".

The differentiation in growth rate among the countries is explained with convergence club models (Azariadis and Drazen 1990), the world economy constitutes three convergence clubs: an advanced club, an underdeveloped club and a developing club. (Yixiao Sun 2001). Different convergence clubs exhibit different convergence behaviour in terms of their speeds of convergence and their long run growth rates. It is envisaged from these differences that catching up, forging ahead, and falling behind happen simultaneously as individual economies interact with each other.

Some countries are trapped not only because of backward technology but also because of detrimental social characteristics. To a large extent, the social characteristics depend on the distributions of social and economic variables rather than the levels of these variables. (Yixiao Sun 2001)

2.4 Catch-up theories & Latecomer’s advantages

Sectoral innovation system is a dynamic system, which will change over time. These changes include both progressive continuous changes and breakthrough discontinuous changes (Anderson and Tushman, 1990). In the literature economic catch-up is defined as a narrowing of a latecomer firm’s or country’s gap vis-à-vis a leading country or firm (Fagerberg and Godinho, 2005) or catch-up as a learning and capability-building process, which may take a long time and may require the mastery of a wide range of capabilities (Bell, 1984; Bell and Pavitt, 1993).

Catching-up is considered as a question of relative speed in a race along a fixed track, and technology is understood as a cumulative unidirectional process (Perez, 1988). In order to catch-up with incumbents, latecomers must overcome competition barriers set by incumbents. They can take advantage of some disadvantageous factors for incumbents, namely, the trap of incumbents or take advantage of latecomer advantages (Lieberman and Montgomery, 1998;; Klingebiel and Joseph, 2016; Kim and Park, 2019) that incumbents do not have, to catch up and surpass incumbents.

In recent contributions by Malerba and Nelson (2011), Lee (2013), and Lee and Malerba (2017) propose that catch-up by latecomers does not mean just the act of cloning, but often the creation of divergent trajectories compared to the ones of leading firms or countries. In different view (Lee and Lim 2001, Lee 2019) is that latecomers do not simply follow the advanced countries' path of technological development; rather, they sometimes do something new, skip certain stages or even create their own path that is different from those of the forerunners.

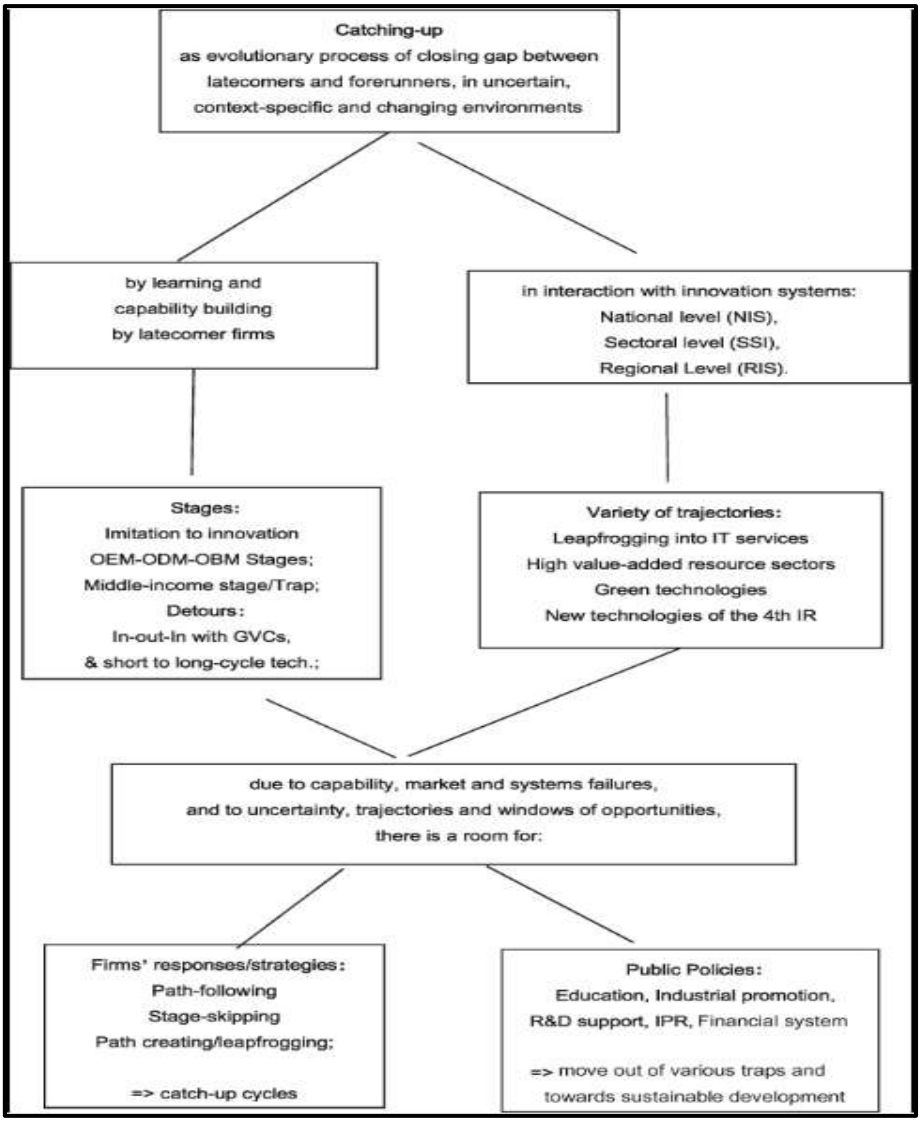


Fig 2. Evolutionary perspectives on economic catch-up
(source: Franco Malerba, Keun Lee 2021)

The evolutionary perspective of catch-up by latecomers (Franco Malerba, Keun Lee 2021) given in Figure 2 illustrates the main theoretical blocks of the perspective that views catch-up as an evolutionary process that cannot be planned in any detail, and has lots of failures and false starts, which can and need to be understood and evaluated based on actual outcome.

2.5 What is Leapfrogging: origin

The origins of the leapfrogging thesis may be regarded as going back to the idea of the so-called latecomers' advantage by Gerschenkron (1962; 1963) that these countries may adopt and use the technology only after it becomes matured enough to have the standardized capital goods suitable for mass production.

However, more recently, it has frequently been observed that in the catching-up process, the late comer does not simply follow the path of technological development of the advanced countries. They perhaps skip some stages or even create their own individual path, which is different from the forerunners. This observation is consistent with the emerging literature on leapfrogging**.** For example, Perez (1988) observes that every country is a beginner in terms of the newly emerging techno-economic paradigm, which implies the possibility of leapfrog ging by latecomers like NIEs.

The idea of leapfrogging is that some latecomers may be able to leap-frog older vintages of technology, bypass heavy investments in previous technology system, and catch-up with advanced countries Hobday, 1995. The increasing tendency toward globalization and development of information technology makes the leapfrogging argument ever more plausible.

Perez and Soete (1988) discussed the latecomers' advantages for leapfrogging in terms of the following three aspects: entry barrier, accessibility of knowledge, and the possibility of lock-in by the incumbents

2.6 Leapfrog development

Lee and Lim (2001) classified the catch-up strategy of later-comers into three categories: path-following strategy, path-skipping strategy and path-creating strategy. Path-following strategy means that the late mover follows the same path as the early mover. But latecomers have been down this road for a shorter time than early movers. Path-skipping strategy means that latecomers follow this path to a certain extent, but skip somestages, thus shortening the technological development time, and then catch up with the first mover. Path-creating strategy refers to the exploration of technological development routes by latecomers, which uses new breakthrough innovations or new generation technologies to explore their own technological development. This method enables latecomers to create their own path, which is different from the path taken by the pioneers (Lee and Malerba, 2017).

The discontinuity of capability disruption means that the cumulativeness of technology is destroyed and the established industry structure is broken, which increases the possibility of rapid rise of late comers, such as the mobile phone industry from the analog era to the digital era (Giachetti and Marchi, 2017).

1) **Compared with the Path of the Incumbent (Lee and Lim 2001)**
a) Stage-skipping
b) Path-creating
cf) Path-following catch-up
2) **Two variations of Path-Creating Leapfrogging (Lee and Ki 2017)**
a) Follow-on Innovation-Based Leapfrogging
b) Radical Innovation-Based Leapfrogging
3) **Inter- vs. Intra- Sectoral Leapfrogging (Lee 2019)**
a) Intra-sector Leapfrogging
b) Inter-sector Leapfrogging

Table 2. Variations of technological leapfrogging (source: Keun Lee 2021)

Latecomers do not simply follow the advanced countries' path of technological development; rather, they sometimes do something new (Malerba and Lee, 2021). The path- creating strategy (Lee and Lim 2001) refers to the case of a latecomer exploring its own path of technological development by utilizing a new generation of technology. Figure 3 shows that, in this strategy, the late entrant chooses the emerging or third- generation technology ahead of the incumbent. This strategy is consistent with the idea of leapfrogging discussed by Perez and Soete (1988). An apparent advantage of this path- creating or leapfrogging is that this strategy is focused on technologies with high long- term potential or productivity as shown in Figure 3. However, a risk is that the emerging or new technology is neither stable nor reliable, and it has low productivity or high costs at its early stage as shown in Figure 3. Despite the high potential of this emerging technology, a firm that adopts the technology has to endure high costs. Thus, losses during the initial stage in the market might be incurred.

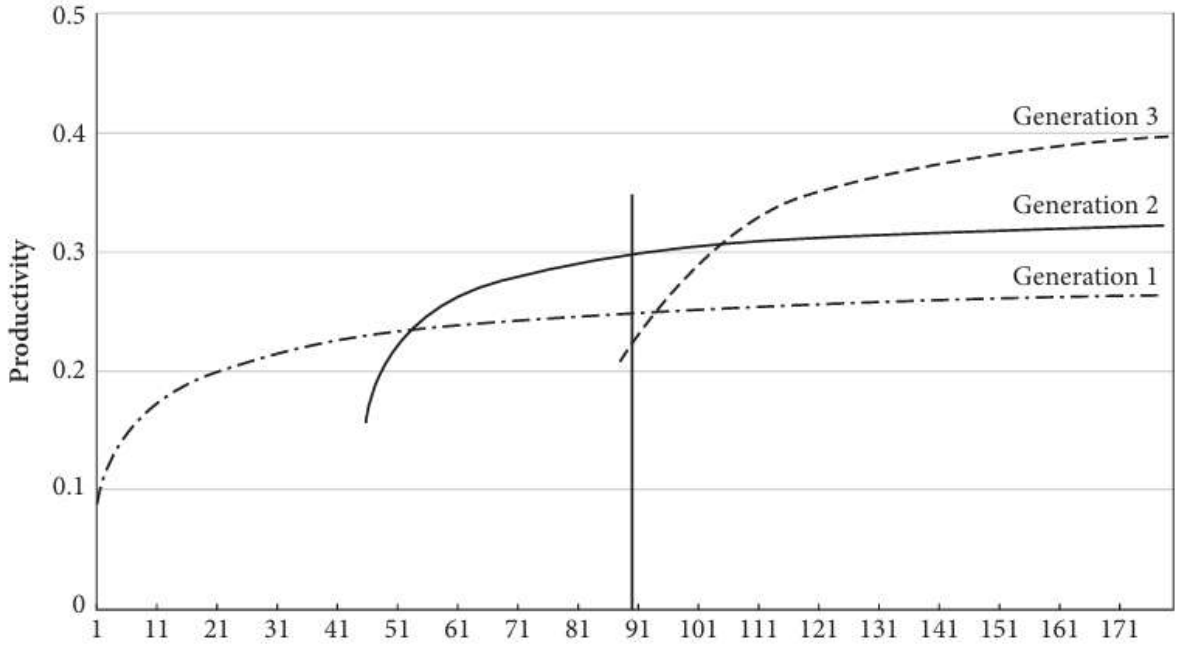


Fig 3. Leapfrogging and Path- following Strategies of Latecomers
Notes: Path- following strategy = To adopt the oldest (generation 1) technology.
Source: Lee (2019: Figure 5- 1) which is from Lee and Ki (2017), adapted from Lee et al. (2016)

Another dimension of leapfrogging can be conceived in terms of inter- sectoral and intra-sectoral leapfrogging (lee 2019) depending upon whether it is happening with the same sector or across different sectors. The inter- sectoral leapfrogging is, to a certain extent, similar to "long jump" in Hidalgo et al. (2007) which argues that latecomer economies must shift to core product spaces that are located far away from their current or periphery position. By contrast, intra- sectoral leapfrogging involves jumping across generations of technologies within the same sector. Intra- sectoral leapfrogging is easier or less risky than the inter- sectoral long jump, as long as latecomers have already built certain absorptive capabilities, such as manufacturing experiences, in the given sectors.

The catch-up cycle theory has explained industrial leadership change in relation to windows of opportunity, catch-up cycles and strategic choices (Lee and Ki, 2017; Lee and Malerba, 2017; Figueiredo and Cohen, 2019; Ahn, 2020)

Lee and Lim (2001) adjust sectoral innovation system model to make it more suitable for the study latecomer enterprises' catch up. In their framework, technological, market opportunity and strategies of the participants jointly determine R&D efforts, while R&D efforts and the availability of knowledge further determine R&D results and ultimate market performance of firms, see Figure 4.

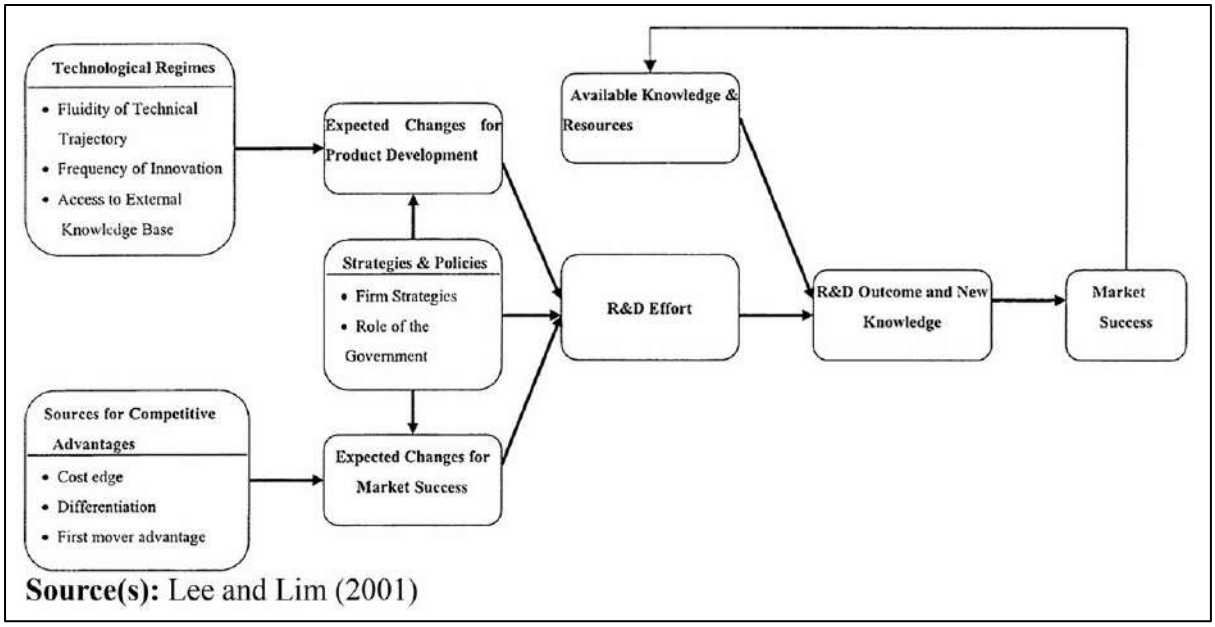


Fig 4. Lee and Lim (2001) Model of technological and market catch-up

In his most recent book, Lee (2019) proposed a comprehensive theory of eco nomic catch- up, consisting of "late entry→ three detours→ leapfrogging," and positioned leapfrogging as the final stage of catching- up after the detour stage of building technological capabilities. The three detours, in terms of innovation, include the following: the first detour is to promote minor innovations via petit patents rather than the high level of innovation via regular patents, the second detour is to increase the share of domestic value- added in exports rather than keep relying on GVCs, and the third or final detour is to specialize in short- cycle technologies rather than long- cycle technologies

2.7 Demand & Technological window of opportunities in leapfrogging

The concept of "windows of opportunity" was first used by Perez and Soete (1988) to refer to the role of the rise of new techno-economic paradigms in the leapfrogging of latecomers who take advantage of a new paradigm and overtake incumbents.

By seizing windows of opportunity in the process of sectoral innovation system change, latecomers have a greater chance to catch up with or even surpass incumbents and become new industry leaders through leapfrogging. However, having a window of opportunity does not necessarily mean that latecomers will be able to catch up, they need to adopt appropriate catch-up strategies (Chen et al., 2023; Min et al., 2023)

The discontinuities in the dynamics of a sectoral system have been labelled "windows of opportunity" by Lee and Malerba (2017). Lee and Malerba (2017) focus on three of them. One is a "technological window" related to major changes in technologies. A second is a "demand window" related to a new type of demand, a major increase in local demand, or a business cycle (in particular, the arrival of a downswing in business cycle which creates difficulties for incumbents, whereas latecomers may have lower entry cost than in normal periods, as discussed by Mathews, 2005). A third window of opportunity is an "institutional/public policy window" related to a public intervention in the industry or a drastic change in institutional conditions. It must be noted that some of these windows are not exogenous, but emerge out of the evolution of the sector.

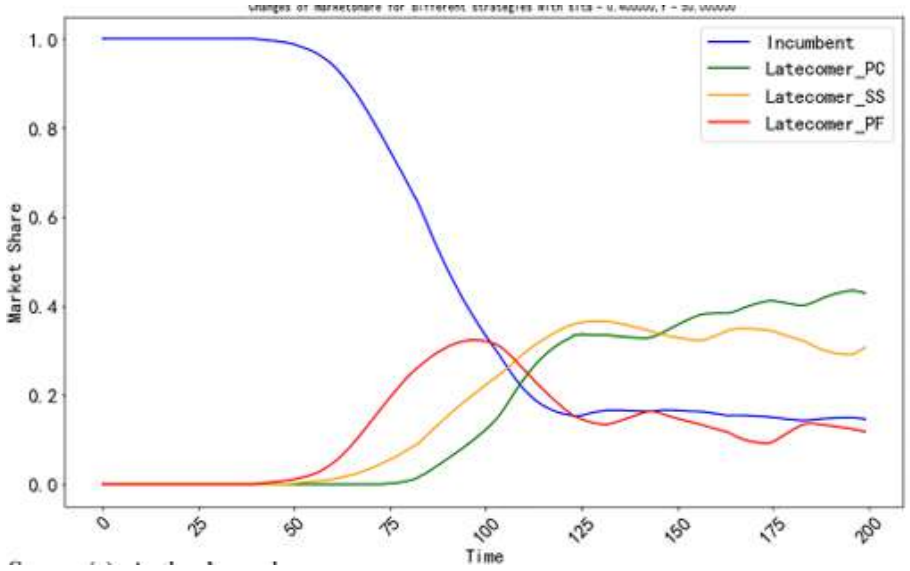


Fig 5. Changes in market share of latecomers with different catch-up strategies
(Source: Yicun Li, Yuanyang Teng, Dong Wu and Xiaobo 2023)

In recent research study findings (Yicun Li, Yuanyang Teng, Dong Wu and Xiaobo 2023), latecomers can seize the technological opportunity window through path-creating strategy to catchup with technology, but without the demand opportunity window, the market catch-up with path creation strategy cannot be guaranteed. Under different size demand opportunity windows, latecomers need to adopt different catch-up strategies to catch up with the market. However, when only demand from lagging behind to going beyond opportunity windows exist,

latecomers cannot catch up with technology and the scale effect of incumbent firms is enhanced. There is a combination effect between the demand window of opportunity and the technology window of opportunity, the demand window affects the “degree” of change in survival rates, while the technology window affects the “speed” of change. The demand window provides security, the technology window provides the possibility of a breakthrough for technology ability.

Lee and Lim (2001) explain that the level of required R&D efforts depends on the probability of success of the R&D effort. Here, the success should be considered in terms of probability of the actual development of target products, as well as the expected marketability competitiveness of to-be-developed products. In other words, we have to separate the physical development of products from their success in markets, like the distinction between invention and innovation. Such separation is needed because the market success of the products is not guaranteed even if target product is developed. In general, firms will devote more R&D resource when they are sure of the linkage between more R&D input and more R&D outputs. The technological regimes come in as determinants of the expected chance for product development, whereas such factors as cost edge, product differentiation, and first-mover advantages come in as determinants of the expected competitive ness of the to-be-developed products.

2.8 Endogenous cause of leapfrog cycle

Technological leapfrogging is triggered by major exogenous changes in technology (Brezis, Krugman, and Tsiddon 1993). In contrast, the cause of perpetual cycles in technological leapfrogging has scarcely been studied. While we may regard the perpetual cycles of leapfrogging as responses to the perpetual exogenous changes in technology, this explanation for leapfrogging cycles is essentially exogenous based on macro shocks in technology.

Endogenous theory that explains the perpetual cycle in technological leadership as a market-driven equilibrium phenomenon (Furukawa, Yuichi 2012). As long as the profitability of innovation is sufficiently high, leapfrogging can take place endogenously and perpetually if the international knowledge spillovers are reasonably efficient. Even if the spillovers are not completely efficient, leapfrogging is still possible to occur in equilibrium. The lagging country’s knowledge accumulates not only through its own innovations but also through the ow of spillovers from the leading country s innovations.

In this sense, the growth engine of knowledge in the lagging country is dual: innovating by itself and learning from abroad. Although the leading country innovates faster than the lagging country, knowledge growth in the leading country is driven only by domestic innovations. This creates the possibility of leapfrogging.

2.9 Role of government & policies

In low- income and middle- income economies in particular, public support for the connections with the world economy can become self- defeating due to non- existent capabilities. The issue is how to generate such capabilities.

According to "Using large Scale Programs to Help Develop Technological Capabilities Cases in China" by Xudong Gao, the biggest challenge of latecomer disadvantage shows up when the initiator of a large- scale program is not a government agency but a firm (either a user or a producer). In fact, when the government is the initiator, technology transfer from MNEs is relatively easy. This is because the government has strong bargaining power, especially when it chooses to leverage the huge domestic market. This can also help explain why South Korea is very successful in catching and leapfrogging in CoPs (Park 2012).

The below table 3 is summarization of Kuznetsov, Y. and Sabel, C. (2011, 2017), New Open Economy Industrial Policy for latecomers to implement leapfrogging.

	Vertical Industrial Policy—*Backward Linkages*	Horizontal Industrial Policy—*Market Failures*	New Open Economy Industrial Policy—*Missing Connections*
Motivation for private agents	Rents (infant industry protection)	Subsidies (when private returns are believed to be lower than social returns)	Quasi-rents—rent opportunities which are contingent on one's effort or performance
Focus	*Micro-level* and sectoral ("picking winners")	*National level*: institutional infrastructure ("backing winners")	*Mezzo-level*: connections between agents ("matching winners")
Main conceptual axis	*Government capabilities* enable and monitor firm-level learning	*Background conditions* to mitigate market failures and distortions. Assuring balance of macro aggregations and eliminating the many micro impediments to growth	*Search network*—to identify successive constraints and then people or institutions that help mitigate (in part) the difficulties associated with these constraints
Main problem	State capture: development of capabilities gets subverted by entrenched interests	Absence of a link between macro changes and increase in micro potential	Gap between micro innovations and improvements in macro conditions; deep constraints remain binding; micro changes do not necessarily achieve critical mass
Examples of programs and policies	Infant industry protection	Reduction of regulatory burden; creation of venture capital funds	Supplier development, early stage venture capital and consortia programs

Source: Based on Kuznetsov and Sabel (2011) and Kuznetsov and Sabel (2017).

Table 3. Three generations of industrial and innovation policy: vertical, horizontal and new open economy- compiled by Xudong Gao 2021

As we discussed about the two windows of opportunities in leapfrogging demand & technology in above section. There is third window of opportunity that can be opened by the government. This opportunity usually generates an asymmetric environment for incumbents and entrants through a range of regulations and supportive actions for entrants. Latecomers can utilize such asymmetries to offset initial cost differences associated with late entry. Governments may intervene through the establishment of R&D programs that affect the learning process and the accumulation of capabilities of domestic firms or through the provision of subsidies, tax reduction, export support, regulations, and public standards. he importance of active government policies in the catching up of countries (e.g., Korea and China) in several sectors is discussed by Malerba and Nelson (2012).

Other examples are provided by public policy in the Chinese telecommunications equipment industry (Mu and Lee, 2005), as well as in Taiwanese industries (Mathews, 2002). At the institutional level, Guennif and Ramani (2012) showed how changes in the regulatory system opened up opportunities for Indian firms in the pharmaceutical industry.

2.10 Challenges & risk involved in leapfrogging

Large technology systems usually involve huge investment and high risks, and cannot be developed without social commitments, usually including legal (especially in the form of contracts), organizational (involving producers, users, and financiers), and political (involving various actors associated with the state commitments). Huge investment and high risks also imply that much more attention needs to be given to the maintenance of reversibility and adaptability in infrastructural development (Fu 2017; Walker 2000).

One early paper (Lee et al. 2015) identified the following two risks involved in leap frogging. The first kind of risk is that of choosing right technologies out of several possible emerging standards, and the second risk is how to create the initial market after the choice of technology to produce new goods. In the early stage of technological paradigm, there tend to be available alternative technologies, among which one dominant or successful technology shall emerge eventually in the later stages. Therefore, if the catching- up country invests in wrong technologies, the country will fail in gaining returns from investments.

Next, even after the catching- up country becomes successful in choosing the right technology, it still needs to be successful in competition with other competitors from the advanced country. In mitigating the second risk (Keun Lee, 2021) of the existence of initial markets or not, we may first emphasize the importance of standard which is a critical factor in the market of the new innovations, in particular digital technology. When the involved products are in the area of information or other emerging technology, an isolated development that does not pay attention to the issue of standards might lead to a failure of the whole project.

The next important thing is to manage the risks involved in leapfrogging. The primary risk mitigation with leapfrogging has to do with how to make a right choice among the several alternative technologies. In this regard, cooperation with public R&D organs, universities, and other entities is critical as such R&D consortium can reduce the involved risk by pooling knowledge together. These collaborating entities may con tribute critically by conducting “technology watch” to interpret and monitor the state- of- the art trend of R&D activities in foreign countries (Lee et al. 2015). Despite the possibility of mitigating the risk of making right or wrong choice among emerging technologies, the issue of whether sensible targeting is possible has always been a controversial issue. But, in such debate against targeting, design failure is often mixed up with targeting failure (Lee 2017).

In standard setting, collaboration and getting partnership with rivals or suppliers of complementary products are important. Also important is who creates and gets to the market first, as the size of the market determines the success or failure of one standard against other. Again, in this competition for standard setting and market creation, the role of the government can be noted as it can play the role of facilitating the adoption of specific standards and thereby influencing the formation of markets at the right times.

3. Desk Appraisals:

3.1 Case 1

South Korea: R & D programme and industrial development

In the evolution of the industries in Korea, including the D-RAM, automobile, mobile phone, consumer electronics, personal computer, and machine tool industries. There three different patterns of catching-up, which are defined in comparison with the paths taken by forerunning firms in the advanced countries. The patterns include path-creating catching-up (CDMA mobile phone), path-skipping catching-up (D-RAM and automobile), and path-following catching-up (consumer electronics, personal computers and machine tools). The first two cases of catching-up may be considered as leapfrogging (K. Lee, C. Lim 2001)

The structural changes to the growing South Korean economy were accompanied by increased levels of sophistication that could no longer be met by imported technologies. The Seoul government responded to these changes by shifting the policy focus from learning to development, and in 1982 launched a national R&D programme. This was the first programme specifically designed for industrial technology development, and it was accompanied by tax and fiscal incentives to promote industrial R&D. These actions were further reinforced in the 1990s by large scale mission-oriented programmes aimed at developing strategic technologies. Private industries joined in by switching their strategy from imitating and assimilating mature foreign technologies to in-house R&D, drawing upon emerging new foreign technologies. (Sung Chulchung 2015)

1980's	• Launching of the national R&D program (1982) • Promoting private firm's research: financial, tax and other incentives to stimulate R&D investments
1990's	• Promotion of university research: SRC, ERC, etc. • Introduction of new types of nat'l R&D programs - Highly Advanced Nat'l Program, The 21st Century Frontier R&D Program. • Establishment of inter-ministerial coordination body: NSTC • Opening of technology stock market --KOSDAQ
2000's	• Introduction of National Technology Road Map (NTRM) - To suggest TRMs for key technologies to secure products/functions. • Introduction of overall coordination system - NSTC

Table 4. Timeline R & D initiatives by government of South Korea
(source: Sung Chulchung 2008)

During the 1980s and up to 2000, R&D spending in South Korea soared from 0.5% of GDP to more than 3%, with the private sector's share increasing to 80% towards the end of the 1990s. This period witnessed some remarkable achievements, almost all of them resulting from local and international joint R&D efforts drawing on foreign technologies.

However, in contrast to the original argument by Perez and Soete 1988 that leapfrogging or new entry tend to be initiated by research activities of public R&D capacities, it was found that with the exception of automobiles where only private R&D was involved, all the important R&D projects involved both private and public capacities. Also, it was found that the entry was not only driven by endogenous generation of knowledge and skills but by collaboration with foreign companies.

The consumer electronics and personal computers, which once achieved a rapid catching-up by path-following taken by the forerunner firms, are found to be experiencing setbacks primarily due to the continuing substantial gap in technological capabilities. Which shows that the persisting gap in terms of the technological regimes of the industries, determines the degree of R&D effort by the latecomer firms.

World and Korean patent grants in the US: annual increase rates between 1986–1993

	Annual increase rate of US patents registered (a)	Annual increase rate of US Patents by Korea (b)	Speed of catching up $= b - a$
Whole industry	4.8	50.3	45.5
Machine tool	−0.9	38.9	39.8
Automobile	3.0	19.6	16.6
Personal computer	29.4	65.5	36.1
Dram	52.5	105.8	53.4
Consumer electronics	16.9	31.6	14.7
Sound and communication	8.2	103.4	95.2

Table 5. World and Korean patent grants in the US: annual increase rates between 1986–1993 (source: K. Lee, C. Lim 2001)

In the case of the forerunner firms, they start with a new product concept and then develop parts, and finally, assemble them. How ever, in the case of the latecomer firms, typically in the case of “reverse engineering” by Korean firms, they started with the assembly production of imported parts, then developed low- to high-tech parts, and learned to design the existing products with some modification, and finally, reached the stage of the new product concept creation

The Korean government first designated the CDMA system development as a national R&D project as early as 1989. In 1991, the contract to introduce the core technology from and to develop the system together with the US- based Qualcomm was forged. In 1993, the Ministry declared CDMA to be the national standard in telecommunication. Given the high frequency of innovation and the high fluidity of trajectory, the telecommunication industry does not provide the latecomers any incentives for R&D effort. Expected profits and other related gains from first- mover advantages served as a strong attraction, and the high risks were shared by the government- led R&D consortium and knowledge alliance with Qualcomm. The ETRI also contributed to reducing technological uncertainty by providing accurate and up- to- date information on technology trends and by identifying the correct R&D targets that were more promising than alternatives (Lee et al. 2005, Keun Lee 2021)

A long list of success with the public–private R&D consortium, from digital telephone switches to memory chips (D- RAM), wireless phones (CDMA), and finally digital TV in Korea,

confirms the positive role of the government and the GRIs in technological catch- up by the latecomer firms. The private firms that participated in the public–private consortium all acknowledged the important function of the government in providing legitimacy to the large projects that are often difficult for private firms to support. The consortium also served as a field to pool together the domestic resources from various sources, especially resources in the universities. The contribution of public research laboratories is also critical in conducting the role of "technology watch" to interpret and monitor the state- of- the art trend of R&D activities in foreign countries.

The Korean experiences suggest that a path-following or skipping catching-up is more likely to happen largely by private initiatives in industries where innovations are less frequent or cumulative and the innovation path is more predictable, and thus, the catching-up target is more easily identified, whereas a path-creating catching-up is more likely to happen by public–private collaboration where the involved technology is more fluid and the risk is high with bigger capital requirements.

3.2 Case 2

India's UPI: A global front-runner in digital payment systems

If there is one Indian innovation that has grabbed global headlines in recent years, it is undoubtedly the UPI (Unified Payments Interface) payments system. Today, more than 40% of all payments done in India are digital, with UPI having a lion's share, used by over 30 crore individuals and over 5 crore merchants.

UPI is used at all levels from street vendors to large shopping malls. Today, among all countries in the world, India is the country with the highest digital transaction, accounting for nearly 46% share, as per the 2022 data (much more 2024). India is followed by Brazil, China, Thailand and South Korea. From just one million transactions in 2016, UPI has since crossed the landmark 10 billion (1,000 crores) transactions.

*this data excludes the transactions having debit/credit to the same account for the month of August 2018 onwards

Month	No. of Banks live on UPI	Volume (in Mn)	Value (in Cr.)
Oct-24	632	16,584.97	23,49,821.45
Sep-24	622	15,041.75	20,63,994.71
Aug-24	608	14,963.05	20,60,735.57
Jul-24	605	14,435.55	20,64,292.40
Jun-24	602	13,885.14	20,07,081.16
May-24	598	14,035.84	20,44,937.05
Apr-24	581	13,303.99	19,64,464.52
Mar-24	572	13,440.00	19,78,353.23
Feb-24	560	12,102.67	18,27,869.35
Jan-24	550	12,203.02	18,41,083.97
Dec-23	522	12,020.23	18,22,949.42
Nov-23	516	11,235.29	17,39,740.61
Oct-23	505	11,408.79	17,15,768.34

Table.6 UPI product statistics (source: NPCI 2024)

The Indian government and RBI (Reserve Bank of India) played significant roles in driving the success of UPI. With RBI offering incentives for banks to join the NPCI, and in the promotions of UPI.

The research work of Rohit Bansal, Kanti Mohan Saini, Amrit Kumar Jha 2024 (Observer Research Foundation) shows that a *1 percent* increase in UPI transaction volume strongly correlates (R^2=0.75) with a *0.03 percent increase in GDP growth*, highlighting UPI's role in economic expansion.

Two key factors have contributed to the trust users place in UPI. Firstly, UPI's endorsement by the Indian government as part of the **Digital India initiative** has lent it credibility and legitimacy. Wide adoption of Government-backed initiatives like the Pradhan Mantri Jan Dhan Yojana (PMJDY) are evidence of trust, integrating UPI and financial inclusion programmes. Secondly, UPI's integration with Aadhaar provides a robust security framework, ensuring that

transactions are secure and identities are reliably authenticated. Additionally, UPI offers real-time transaction updates and easy dispute resolution, fostering a transparent environment where users feel in control.

Microloan disbursements increased by 100 percent from 2018 to 2023, with a correlation between UPI transaction volume and microloan disbursements (Bansal, Rohit, et al 2023-2024)

Continuous advancement – such as integrating AI based service system in government portal. National Payment Corporation of India (NPCI) has launched an artificial intelligence (AI) based Chabot, PAi, to create awareness around its products like FasTag, RuPay, UPI, AePS on a real time basis.

This is yet another NPCI's initiative to improve digital financial inclusion in India. The AI powered virtual assistant, PAi is available round the clock helping users with access to accurate information on NPCI products. Customers can ask their queries in English and Hindi, via text or voice on the websites of NPCI, RuPay, and UPI Chalega. Through PAi, users get verified automated responses to their queries on all NPCI's products. PAi would also be accessible to global RuPay Cardholders as well.

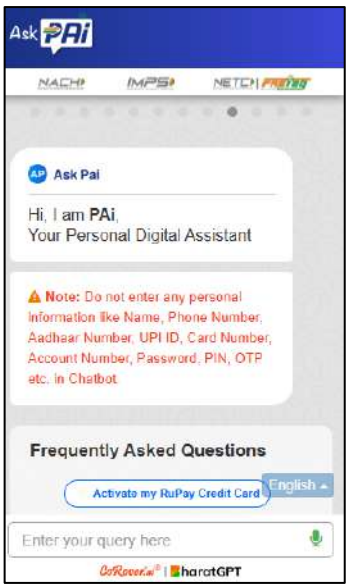


Fig. 6 PAi - The AI powered virtual assistant

India's ambition to export UPI to other countries represents a strategic move to enhance its positioning as a trusted development partner, particularly in digital finance. The success of UPI domestically has sparked interest from several countries, say, Japan, UAE, France and now Maldives, as part of their pursuit for financial modernization. By promoting UPI globally, India is engaging in technological diplomacy. Successfully implementing UPI in other markets could establish India as a leader in providing scalable and secure digital payment solutions.

Industry participants in India are now eyeing the opportunity for UPI to facilitate international money transfers. By 2024, prominent stakeholders foresee UPI-enabling transactions with countries that have large Indian populations like Singapore, UAE, and others. This would provide Indian laborers and students overseas with convenient access to funds and open up new streams of remittances back home. Experts suggest that this could significantly benefit the Indian economy as well.

3.3 case 3

China: Transfer technologies and localization

One of the strategies that proved to be of special importance in China's rapid progress in technological capability development is the use of large- scale programs to help with technological capability development. In sectors such as the telecommunications industry, the high- speed rail industry, the power generation equipment industry, the power transmission industry, the subway equipment industry, etc., China was lagging behind the technological innovation frontier before the related large- scale programs but is now among the world leaders. The high- speed rail industry is illustrative. Although China did not invent the high- speed rail technology, China has made an impressive achievement in building up its high- speed rail system.

In 2004 the former Ministry of Railway (MOR) decided to massively transfer technology from multinational enterprises (MNEs) to help with China's high- speed rail development. On 26 October 2010, CRH380A, a highly localized high- speed train went into operation. On 3 December 2010, CRH380A reached an operation speed of 486.1 km/h, the highest in the world. Locally developed and manufactured new generation high- speed trains went into operation on 26 June 2017, indicating that local firms had developed strong techno logical capabilities in not only manufacturing but also designing and engineering. By the end of 2017, China had more than 25,000 km of high- speed rail under operation, accounting for about 60 percent of high- speed rail in the world. In 2017, more than 7 billion people travelled by high- speed trains in China.

The impact of government policies and social environment is complicated (Carlson 1991; Hughes 1987; Walker 2000). If we first look at the impact of government policies. In contrast to the argument that the Chinese government is always very supportive of local firms' technological capability development (Kshetri et al. 2011), the role of government policies in China is complicated. The following are three examples.

The first example is the subway equipment localization. In this case, the key initiator was the Shenzhen Metro, but the central government's policy was supportive in the sense that it created a favorable environment. Specifically, because local governments are the owners and investors of the subway companies, they usually faced a big dilemma. On the one hand, they saw huge benefits from successful localization of subway equipment because the financial burden for them would be lower after localization. On the other hand, they did not want to expose their subway systems to a higher rate of operation risks, especially serious accidents, by using locally made equipment, because they did not believe that locally made equipment could be as reliable as that of MNEs'. In this situation, the policy of the central government proved to be very important and supportive. In February 1999, the State Council issued a regulation, requiring that the localization rate of all subway vehicles and other equipment reach 70 percent in order to get approval for a city to build up its subway systems. NDRC also provided financial resources to support locally developed equipment. Leveraging the policy of the central government, the Shenzhen Metro was able to gain support from local government to promote subway equipment localization.

The second example is the Yun 10 program, which could indicate the changing and ineffective nature of government policies. This program was started in 1972 and stopped in 1986. This is the first attempt by China to develop large civil aircraft. Although Yun 10 did its first fly test in 1980, it became more and more difficult to get support from the central government. In 1982, even after the Shanghai Municipal Government promised to provide half of the budget to continue the program, the central government refused to provide more financial support and the program was stopped in 1986. Although there are many debates about the success and failure of Yun 10, policy change seems to be the key factor. Specifically, from the early 1980s, China began to explore the possibility to co- develop large civil aircraft with MNEs. In fact, in April 1985 China signed a contract with McDonnell- Douglas Corporation to assemble MD-82. It turned out that the collaboration with MNEs to develop large civil aircraft failed (Wang 2012; Zhao 2018).

The third example is the TD- SCDMA program. Why was it so difficult to pro mote TD-SCDMA in China, even when it became clear that technology transfer from MNEs was becoming more and more difficult and indigenous innovation was becoming more and more important? One reason is the transitional nature of government policy (Gao 2014). On the one hand, a key policy of opening the Chinese economy to the outside was to transfer technology from advanced economies. In fact, to some extent, this policy proved to be very successful (Zhang 2000).

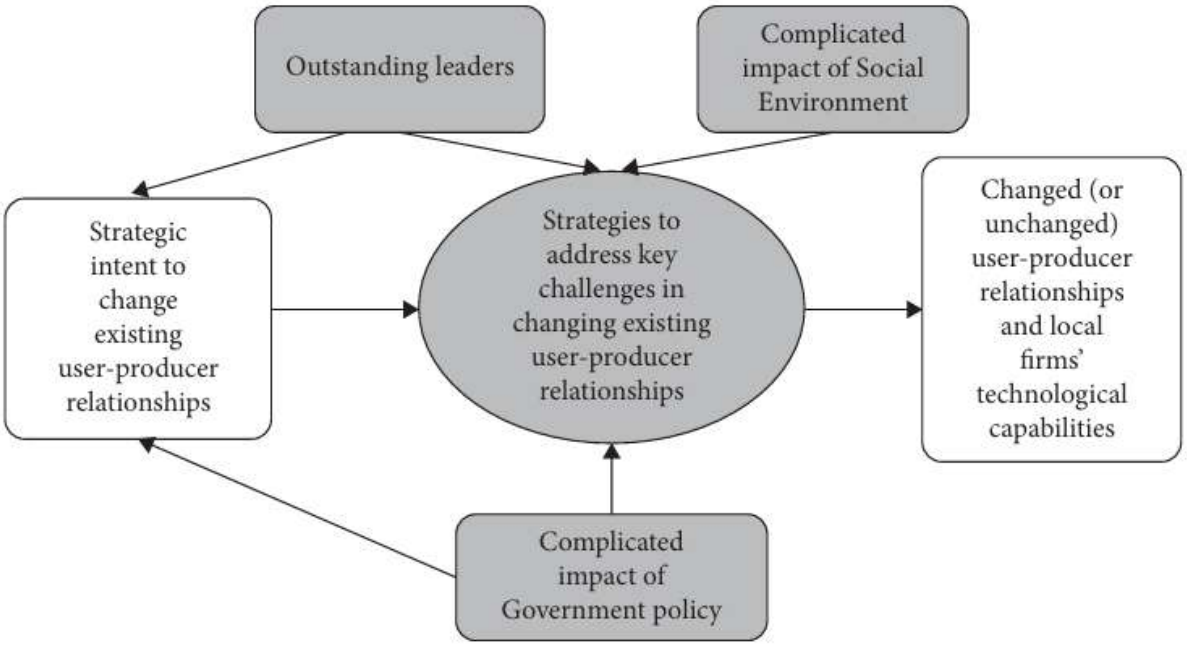


Fig.7 Key factors affecting Chinese programs

On the other hand, the policy of encouraging indigenous technological innovation reflected an inevitable trend. MNEs were becoming more and more reluctant to transfer technology to local firms. This meant that local firms have had to rely more and more on internal technology development. Without indigenous technological innovation, it'll be very hard for local firms to survive. For a specific firm, it might be easier to balance technology transfer and indigenous

technological innovation. However, for government agencies, the internal conflict between the two policies is obvious and big. Specifically, when some stakeholders argue that choosing WCDMA or CDMA 2000, which are mainly based on MNEs' technologies, rather than TD-SCDMA, is following the policy of opening to the outside, it'll be very hard for the government not to support these stakeholders. However, this means that the support for TD- SCDMA has been weakened.

A key policy of opening the Chinese economy to the outside is to transfer technology from advanced economies, and this policy proved to be successful for many firms. The impact of this is obvious: Chinese society had a strong faith in the benefits of technology transfer. In this situation, it proved to be very difficult to believe in indigenous technological innovation, let alone to provide strong support. However, society is not homogenous. It's not hard to find stakeholders who have different opinions about the technology transfer policy and strongly believe in the benefits of indigenous technological innovation. It proved that these stakeholders could play very important roles, even though they are in the minority. As mentioned earlier, Datang developed an informal social network to help promote its TD- SCDMA technology. Although most people and organizations were not active in supporting or even opposing TD-SCDMA, some highly influential stakeholders could change the balance between those supporting TD- SCDMA and those opposing it.

For example, although the industrialization trials showed that TD- SCDMA was suitable for building large- scale nationwide telecom networks in July 2005, service providers were not interested in conducting commercialization trials. To deal with this challenge, Datang invited three noted scientists (the president of the Chinese Academy of Sciences, the president of the Chinese Academy of Engineering, and the Chairman of the Chinese Association of Science and Technology) to provide support. These three scientists wrote a letter to the top leaders of the central government, requesting that the government support TD- SCDMA. The result was that some top leaders asked related government agencies to support TD- SCDMA. This proved to be of crucial importance, and the development of TD- SCDMA entered a new stage.

Technological innovation at state- owned enterprises (SOEs). In China, SOEs are important players in the economy. For example, in 2018, the sales revenue for SOEs was RMB 58.75 trillion Yuan. In fact, 120 firms from mainland China made the list of 2018 Fortune 500, and eighty- three of them are SOEs. Specifically, in the past ten years we have participated in a series of studies about technological innovation at large SOEs sponsored by different divisions of the State-Owned Assets Supervision and Administration Commission of the State Council (SASAC) and the National Development and Reform Commission (NDRC) (Gao 2019; Ralston et al. 2006; Shen and Feng 2010; Shi 1998).

The biggest challenge of latecomer disadvantage shows up when the initiator of a large- scale program is not a government agency but a firm (either a user or a producer). In fact, when the government is the initiator, as discussed earlier, technology transfer from MNEs is relatively easy. This is because the government has strong bargaining power, especially when it chooses to leverage the huge domestic market in China. This can also help explain why South Korea is very successful in catching and leapfrogging in CoPs (Park 2012).

4. Result & Learnings

Recognized the India's conscious effort in adapting and updating its development strategies and planning system. Also, its lacking in some parts.

Understood how the demand window affects the degree of change in survival rates and provides market security. the technological window's impact on the speed of change, enabling breakthroughs in technology capabilities. And the need for a combination of both demand and technology windows to effectively implement catch-up strategies.

Should be the producer not just consumer of technology. Just by following the path of developed nation and dependency towards foreign technologies and FDIs would add more gap in the competitive development.

Not just catch-up but get ahead in the economic race by choosing the appropriate path of development as combination endogenous and exogenous changes.

Realized the importance of aligning R&D efforts with both market success and the physical development of products, emphasizing the distinction between invention and innovation.

Explored the knowledge on endogenous theories explaining perpetual cycles of leapfrogging, emphasizing market-driven equilibrium and international knowledge spillovers. Recognized that the dual growth engine for lagging countries involves both self-innovation and learning from international spillovers.

Highlighted the government's critical role in facilitating leapfrogging through R&D support, regulations, subsidies, tax reductions, and export assistance. From the case studies showing how government-initiated programs (e.g., in South Korea) have enabled successful technological catch-up.

Comprehensive alignment of various schemes, programs & initiatives to build a converged uni-directional development towards the target, which we saw from the case of UPI's success by integrating UPI and financial inclusion programmes on the frame of Digital India initiative.

Identified risks like choosing the wrong technologies among emerging standards and creating initial markets post-technology selection.

Emphasized the importance of setting standards, fostering partnerships, and timing market entry for successful leapfrogging. And the necessity of maintaining reversibility and adaptability in large-scale technological investments.

The need for tailored strategies for latecomer nations with sector specific model. Due to evident failure of development in some sectors and nations.

Not forcing ourselves - to advance everything - like if we try to make everything non-man based solutions - we may affect the livelihood of lower income pop. - so, technology should be supporting public not replacing them.

5. Research gap

Economic benefits of leapfrogging in India

To explore the variation of economic benefits and impacts that can be induced by leapfrogging in various sectors & fields, scale & size, location and at international economic stage.

Formulation of R & D and S & T based development strategies for India

To explore the benefits and effects of Research & Development and science & technology strategies by increasing its GDP share, private industrial supporting and initiating policies, private involvement in R & D, Indian institutional integration and etc.

Evaluating India's existing leapfrog initiatives

To study the existing leapfrog initiatives in India such as Smart city scheme, digital India, make in India, etc. In understanding their effect on rapid development and advancement.

Holistic development & advancement approach for India

To promote a new holistic development and advancement approach around the principle of leapfrogging by incorporating the existing schemes and programs as implementing vehicle.

The approach is – introduction of new level of plan to Indian planning system – *Leapfrog development plan*
(not the actual name of the plan but the concept)

Hybrid plan – It would work as a hybrid plan that consists of both visionary(target) plan and specific sector/industry/project development plan

National and state visionary & project proposal plan- the plan would be prepared for both national development goals and state development goals as decentralised plan on focus of single aligned vision

Sectional integration- recommended to add a section of leapfrog development proposal or else incorporate the leapfrog development vision & goals of both national & state in the core area & other level planning

Space/position suited for leapfrog development plan in Indian planning system

This leapfrog development plan has a potential to be formulated along the side of

15-year vision document at – National level- As National leapfrog development plan

Perspective plan at – State level- As State leapfrog development plan

6. Following Thesis -background

Thesis: Need for integrated Leap frog development plan for developing but lagging India

6.1 Aim of the study:

To understand the lag/gap in India's development and promote personalized & Integrated Leap frog development plan for India

6.2 Objectives/stages:

1. To evaluate the current practice and development strategies of India along with the planning systems implementation success in economic POV

2. To estimate the current gap between the other competing countries in- practices, technology, production, gross & net value, R&D expenditure & emphasis etc.

3. To estimate the required growth rate to i) leap and ii) land ahead - through present value determination - how much we can financially invest today to attain the future goal in the market and profit

4. Logistics transportation corridor plan: Case study on current India's logistics vs new hyperloop tech based on IIT Madras research - low cost than metro

5. To highlight the critical strategies for adopting leapfrogging principles in national planning and finding a space for leapfrog development plan in the Indian planning system

MAIN PART:

6. To define and prepare the contents guidelines with my intended definition & cause clearly and reforms for India's leapfrog development plan – which also includes integration of existing reforms, actions and schemes (such as make in India, smart city mission, etc.)

And made based on the idea of

- ✓ “Should be the producer not just consumer of technology”
- ✓ “Don't forget the necessity” and basic requirements to be achieved both socially and economically
- ✓ “Not forcing - to advance everything”
- ✓ “NOT just CATCH-UP but get AHEAD”

Note: Methodologies of the following thesis study is yet to be finalised but some of the explored candidates are TECHNO-ECONOMIC ASSESSMENT (TEA) which is includes project efficiency & production capacity comparism analysis. Net present value, return on investment

7. conclusion

In conclusion, the exploration of leapfrog strategies, alongside theories of catch-up and latecomer advantages, reveals a multifaceted approach to economic and technological development. Latecomer economies often leverage opportunities created by emerging technological paradigms, bypassing traditional pathways through strategic research and development (R&D) initiatives and well-crafted government policies. These policies are critical in fostering innovation, enabling the effective use of knowledge infrastructure, and supporting industries that have the potential for rapid development.

Moreover, latecomers' success frequently hinges on the ability to capitalize on strategic windows of opportunity, avoiding the inefficiencies that incumbents may face due to legacy technologies. The role of government intervention cannot be overstated, as it helps facilitate technology transfer, provides incentives for R&D, and nurtures institutional frameworks conducive to innovation. However, these strategies are not without challenges and risks, such as market volatility, technological uncertainty, and the potential for intense competition with established leaders. Therefore, choosing the right developmental path requires careful assessment of these risks and a forward-looking vision that balances long-term potential with the resources and capabilities at hand.

The understanding about Indian planning system and the literatures knowledge provides significant contribution to my following thesis about "The need for leapfrog development plan for developing but lagging India"-by Ultimately, designing a leapfrog development plan for India will involve balancing the dual goals of rapid modernization and equitable growth while managing the risks and uncertainties inherent in adopting cutting-edge technologies. With informed policy decisions and an emphasis on collaborative public-private partnerships, India can harness these strategies to bridge development gaps and position itself competitively on the global stage.

References:

1. Ahn, S.J. (2020), “Three characteristics of technology competition by IoT-driven digitization”, Technological Forecasting and Social Change, Vol. 157, 120062.
2. Azariadis, C. and Drazen, A. (1990), Threshold Externalities in Economic Development. Quarterly Journal of Economics, Vol. 105(2): 501-526
3. Banji Oyelaran-Oyeyinka, Padmashree Gehl Sampath. (2010), Latecomer Development: Innovation and knowledge for economic growth. Book series-Routledge Studies in Development Economics.
4. Bansal, Rohit, et al. "Regression Study on Unified Payment Interface (UPI) and Economic Growth: India Evidences." Fellowship Program in Management, Indian School of Business, 2023-2024.
5. Bell M. (1984), ‘Learning and the accumulation of industrial technological capacity in developing countries,’ in Fransman M. and King K. (eds), Technological Capability in the Third World. Palgrave Macmillan: London, pp. 187–209.
6. Bell R. M., Pavitt K. (1993), ‘Technological accumulation and industrial growth: contrasts between developed and developing countries,’ Industrial and Corporate Change, 2(2), 157–210.
7. Fagerberg J., Godinho M. M. (2005), ‘Innovation and catching-up,’ in Mowery D. C., Fagerberg J., and Nelson R. R. (eds.), The Oxford Handbook of Innovation, pp. 514–543. Oxford University Press: New York.
8. Figueiredo, P.N. and Cohen, M. (2019), “Explaining early entry into path-creation technological catch up in the forestry and pulp industry: evidence from Brazil”, Research Policy, Vol. 48 No. 7, pp. 1694-1713.
9. Gerschenkron, A. (1962), Economic Backwardness in Historical Perspective. (Cambridge, Mass: Harvard University Press).
10. Giachetti, C. and Marchi, G. (2017), “Successive changes in leadership in the worldwide mobile phone industry: the role of windows of opportunity and firms’ competitive action”, Research Policy, Vol. 46 No. 2, pp. 352-364.
11. Hobday, M. (1995), Innovation in East Asia: The Challenge to Japan (Cheltenham: Edward Elgar).
12. IDFC First Bank, All you need to know about the UPI forecast for 2024- *https://www.idfcfirstbank.com/finfirst-blogs/finance/growth-of-upi-in-india*
13. Keun Lee. (2021), Economics of Technological Leapfrogging (chapter 5), The Challenges of Technology and Economic Catch-up in Emerging Economies (Book)- © Oxford University Press 2021.
14. Kim, Y.-Z., and K. Lee (2009), ‘Making a Technological Catch-Up in the Capital Goods Industry: Barriers and Opportunities in the Korean Case’, in F. Malerba, ed., Sectoral Systems of Innovation and Production in Developing Countries, (Cheltenham, UK: Edward Elgar Publishing), 259–86.
15. Klingebiel, R. and Joseph, J. (2016), “Entry timing and innovation strategy in feature phones”, Strategic Management Journal, Vol. 37 No. 6, pp. 1002-1020.
16. Kuznetsov, Y. and Sabel, C. (2011), New Open Economy Industrial Policy: Making Choices without Picking Winners, PremNote no. 161, World Bank,

https://openknowledge. worldbank.org/handle/10986/11057/, accessed 30 January 2021.
17. Kuznetsov, Y. and Sabel, C. (2017), 'Managing Self-Discovery: Diagnostic Monitoring of Project Portfolio', in Radosevic et al., eds, Advances in Theory and Practice of Smart Specialization (London: San Diego; Cambridge, MA; Oxford: Academic Press, Elsevier), 52–73.
18. Lall, S. (2000), 'The Technological Structure and Performance of Developing Country Manufactured Exports, 1985–98', Oxford Development Studies, 28/3, 337–69.
19. Lee, K., and Lim, C. (2001), 'Technological Regimes, Catching-up and Leapfrogging: Findings from the Korean Industries', Research Policy, 30/3, 459–83.
20. Lieberman, M.B. and Montgomery, D.B. (1998), "First-mover (dis) advantages: retrospective and link with the resource-based view", Strategic Management Journal, Vol. 19 No. 12, pp. 1111-1125.
21. Mahadevia, Darshini & Joshi, Rutul & Sharma, Rutool. (2009). Integrating the Urban Poor in Planning and Governance Systems, India. 10.13140/RG.2.2.36783.12961.
22. Malerba F., Nelson R. R. (2011), 'Learning and catching-up in different sectoral systems: evidence from six industries,' Industrial and Corporate Change, 20(6), 1645–1675.
23. Mathews, J.A. (2002), "Competitive advantages of the latecomer firm: a resource-based account of industrial catch-up strategies", Asia Pacific Journal of Management, Vol. 19, pp. 467-488.
24. Ministry of Statistics & Programme Implementation, 'Statistical Year Book India 2022, chapter 7: Five Year Plan'-ch7.pdf
25. NPCI press release (19.5.2020), NPCI launches Personal AI (PAI) virtual assistant to create consumer awareness.pdf
26. RADPFI guidelines (Revised), 2021- Ministry of Panchayati Raj Government of India
27. Rohit Bansal, Kanti Mohan Saini, Amrit Kumar Jha (2024), Decoding India's UPI phenomenon: A digital revolution with global implications, (Observer Research Foundation)
28. Sung Chulchung 2008, Korean Strategy for the Promotion of R&D and Innovation, Science and Technology Policy Institute (STEPI), Korea.
29. UDPFI guidelines, 1996.
30. URDPFI guidelines, 2014. Ministry of Urban Development
31. Xudong Gao, (2021) Using Large- Scale Programs to Help Develop Technological Capabilities: Cases in China (chapter 14), The Challenges of Technology and Economic Catch-up in Emerging Economies (Book)- © Oxford University Press 2021.
32. Yixiao Sun. (2002), Catching up, Forging ahead, and Falling behind: A Panel Structure Analysis of Convergence Clubs.

Appendix-I: Inter-relation between various plans given in UDPFI guidelines 1996.

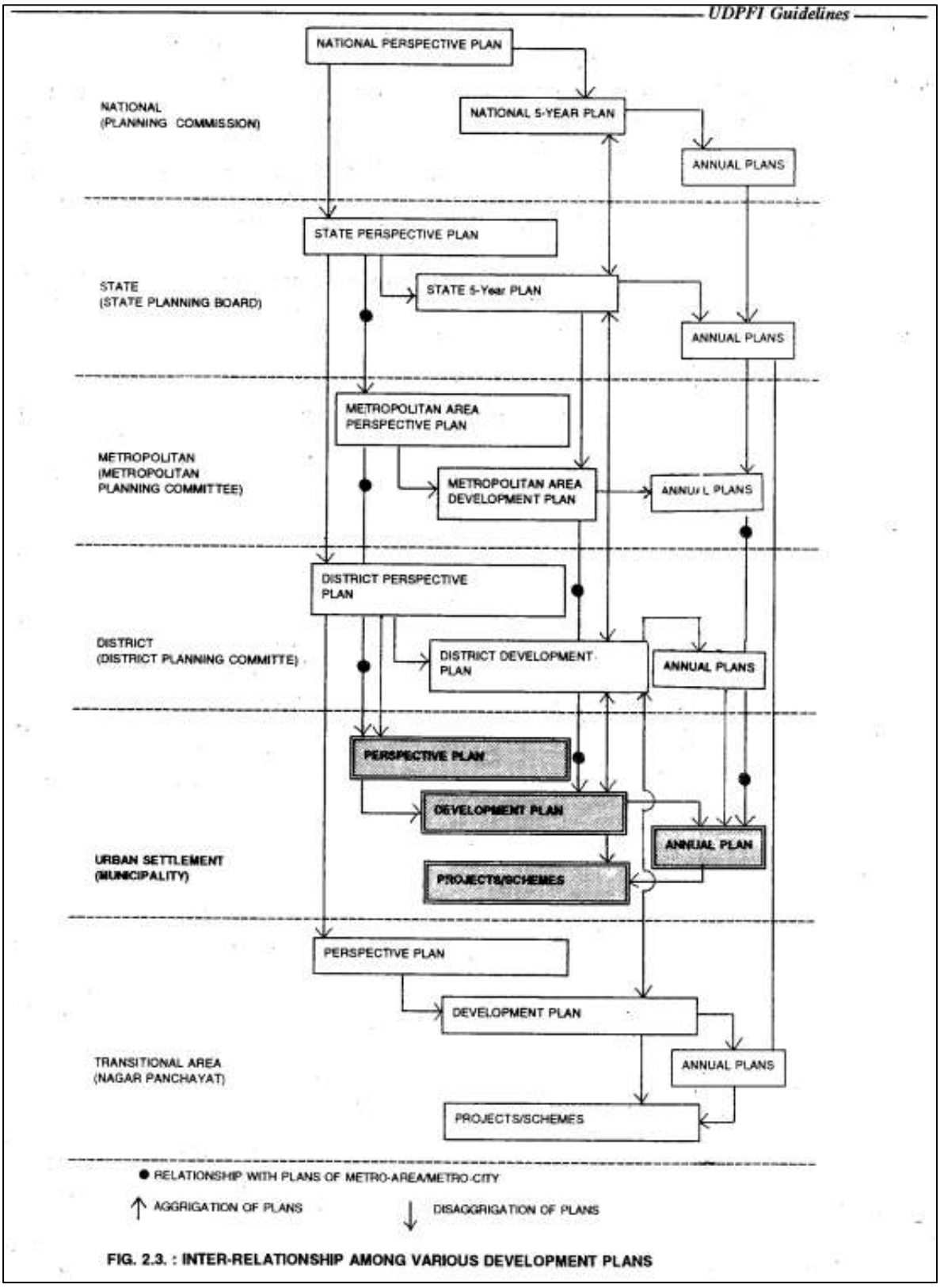


FIG. 2.3. : INTER-RELATIONSHIP AMONG VARIOUS DEVELOPMENT PLANS

PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

Bachelor of Planning

Niteshkumar S K
[Sch. No. 211109041]

Department of Architecture and Planning
Maulana Azad National Institute of Technology, Bhopal
462003

May 2025

PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

A Thesis Project
submitted in partial fulfillment of the requirements for
the award of the degree of

Bachelor of Planning

By
Niteshkumar S K
Scholar No. 211109041

Under the Guidance of
Dr. Vinay Mohan Das

Department of Architecture and Planning
Maulana Azad National Institute of Technology, Bhopal

May 2025

© MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY - 2025

ALL RIGHTS RESERVED

CANDIDATE'S DECLARATION

I **NITESHKUMAR S K** Scholar No. **211109041** hereby declare that the Thesis Project **Planning hyperloop freight corridor between Bengaluru to Chennai** submitted by me in partial fulfilment of the requirement for the award of degree of Bachelor of Planning at the Department of Architecture and Planning, Maulana Azad National Institute of Technology, Bhopal, India, is a record of the work carried out by me. The matter/result embodied in this project has not been submitted to any other University or Institute for the award of any degree or diploma.
If, in future, it is found that the above statement is false, then I have no objection in withdrawal of Bachelor of Planning Degree awarded to me by the Institute.

Signature of the Student

Date

RECOMMENDATION

Based on the above declaration of the student, the undersigned recommends that the work submitted by the candidate may be considered for the end term examination of the subject PLA-424, Planning Thesis.

Thesis Guide

CERTIFICATE OF EXAMINATION

The Bachelor of Planning Thesis Viva-Voce examination of **NITESHKUMAR S K** Scholar No **211109041,** was held on …….

Dr. Nakul Dhagat	Dr. Manmohan Kapshe	Dr. Preeti Onkar
Head of Department	**Thesis Coordinator**	**Thesis Coordinator**

सार:

यह शोध प्रबंध रिपोर्ट बेंगलुरु से चेन्नई के बीच हाइपरलूप मालवाहन गलियारे की योजना को एक "लीपफ्रॉग विकास" परियोजना के रूप में प्रस्तुत करती है। यह अध्ययन क्षेत्र चेन्नई-बेंगलुरु औद्योगिक गलियारे (CBIC) के 91051 वर्ग किलोमीटर के प्रभाव क्षेत्र को शामिल करता है, जो तीन राज्यों, 16 जिलों और 8 प्रमुख औद्योगिक नोड्स में फैला हुआ है।

इस अध्ययन का मुख्य उद्देश्य हाइपरलूप के लिए आवश्यक "राइट ऑफ वे" के आधार पर मार्ग संरेखण का निर्धारण करना और उन नोड्स व टर्मिनलों की पहचान करना है जिन्हें यह गलियारा कवर करेगा, साथ ही यह सुनिश्चित करना है कि यह मौजूदा औद्योगिक और परिवहन बुनियादी ढांचे के साथ स्थानिक संगतता बनाए रखे और CBIC विकास योजना प्रस्तावों के साथ मेल खाए। इस अध्ययन में JICA की CBIC व्यापक क्षेत्रीय रिपोर्ट को अध्ययन क्षेत्र के डेटा के मुख्य स्रोत के रूप में उपयोग किया गया है।

इस प्रकार, यह अध्ययन कई सीमाओं और पूर्वानुमानों के साथ किया गया है, जिससे इस अध्ययन को डेस्क मूल्यांकन अनुसंधान के रूप में पूरा करना संभव हो सका। मुख्यतः हाइपरलूप ट्रांजिट तकनीक की व्यवहार्यता को एक पूर्वमान्यता के रूप में स्वीकार किया गया है।

प्रस्तावित टर्मिनलों और नोड्स के संभावित मार्गों के लाभ और हानियों के स्थानिक और आर्थिक विश्लेषण, मिश्रित संरेखण पद्धति और मात्रात्मक विकल्प तुलना के आधार पर निम्नलिखित अंतिम मार्ग प्रस्तावित किया गया है:

तुमकुरु से बेंगलुरु (रेल मार्ग) - बेंगलुरु से पालयमनेर एक्सचेंज तक (NH 48 का भाग) - पालयमनेर एक्सचेंज से रानीपेट के बनावरम तक (NE 7) - NE 7 से रेल द्वारा अरक्कोनम होते हुए चेन्नई-पोननेरी तक (रेल मार्ग) = 398 (~400 किमी)।

संकेतशब्द: हाइपरलूप, मार्ग नियोजन, माल परिवहन, लीपफ्रॉग विकास

ABSTRACT

This thesis report details the work on planning a hyperloop freight corridor between Bengaluru to Chennai as a leapfrog development project. With CBIC (Chennai Bengaluru industrial corridor) of 91051 sq. km influence area spread out to 3 states, 16 districts and 8 key industrial nodes as study area.

The core objective of determining route alignment based on right of way required for hyperloop and identifying the nodes and terminals the corridor would cover while ensuring the correlation & spatial compatibility with the existing industrial and transport infrastructure and align with the CBIC development plan proposals. In this study JICA'S CBIC comprehensive regional report used as the main source of data about the study area to achieve the above objectives.

Thus, this study deals with an ample number of limitations and pre assumptions that made it possible to achieve the end outcome of this study as desk appraisal research. Majorly the assumption of feasibility of hyperloop transit technology.

Based on the advantages and disadvantages of the potential routes identified by proposed terminals and nodes with spatial and economic analysis, the mixed alignment methodology and quantitative alternatives comparism, the following final route have been proposed:

Tumkuru to Bengaluru (RAIL) - Outside Bengaluru to till Palamaner exchange (NH 48- part) - Palamaner exchange to RAIL at Ranipet: Banavaram (NE 7) - From NE 7 to rail exchange to **Chennai-Ponneri** via Arakkonam (RAIL) =**398(~400 km).**

Keywords: Hyperloop, route planning, freight, leapfrog development

ACKNOWLEDGEMENT

I am profoundly grateful the Maulana Azad National Institute of Technology, Bhopal for providing me with the invaluable opportunity to be the part of this esteemed centre of knowledge and completing my bachelor's in planning discipline. This experience has been a significant learning journey, and I am deeply thankful for the support and guidance I received throughout the course and my research activity. Firstly, I would like to extend my sincere gratitude to Dr. Prof Vinay mohan das sir, my thesis guide for his invaluable mentorship and guidance and His detailed insights and practical knowledge have greatly enhanced the quality of my work in the field.

I would also like to express my heartfelt thanks to all my teachers for equipping me with enough knowledge to execute this research study and I would like to thank my family and friends who have supported me in terms of various needs through both direct and indirect contribution to my work.

Lastly I am thanking myself for enduring the challenges and hardship that came along the way.

Thank you

TABLE OF CONTENTS:

CHAPTER 1-Understanding hyperloop and route planning

1. Introduction 1
2. Hypothesis and Assumptions..... 1
3. Aim.....2
4. Objectives.....2
5. Scope 3
   5.1 Physical Planning Scope
   5.2 Economic growth scope
   5.3 Research and home hyperloop technology development scope
6. Limitation.....4
   6.1 Planning and study Limitations
   6.2 Technological and Structural Limitations
   6.3 Financial Limitations
7. Literature guides.....5
8. Methodology 6
9. Literature review 7
   9.1 Introduction to Hyperloop
   9.2 Technical Design and Principles
   9.3 Hyperloop compared to other HSR, its corridor selection, corridor and terminals, travel time, capacity and efficiency
   9.4 Hyperloop technology route needs
   9.5 Right-of-way planning requirements
   9.6 Economic Feasibility and Cost Analysis
   9.7 Capsule and payload components
   9.8 Safety and Regulatory Challenges
   9.9 Environmental Impact
   9.10 Current Developments and Future Prospects
10. Data collection.....13

CHAPTER 2-CBIC and Planning parameter

11. Study area.....14
    11.1 CBIC
    11.2 Historical Evolution

11.3 Economic Impact
12. Existing characters and conditions…. 21
12.1 Demographics and related parameters
12.2 Physical characteristics of study area
12.3 Existing connectivity and transportation in CBIC
12.4 Economic parameters
CHAPTER 3 -Proposal
13. Route planning proposal…..39
13.1 Proposal methodology and techniques
13.2 Terminal and nodes proposal
13.3 Possible straight line route connectivity
13.4 Route alignment
13.5 Constraints & Advantages
14. Mixed alignment alternatives 49
15. Finalised route alignment plan 50
16. Conclusion….....52
17. References 53

LIST OF TABLES:

Table No 1: Hyperloop compared to other HSR 9

Table No 2: Cost analysis with respect to existing metro transit 11

Table No 3: Capsule and payload components… 12

Table No 4: Districts covered in CBIC Influence area 16

Table No 5: Area of CBIC covered districts 17

Table No 6: CBIC Population Projection (2033) 24

Table No 7: Sectors and districts 34

Table No 8: Possible route connectivity 43

Table No 9: CBIC Road &Rail 44

Table No 10: Route alignment-Along NH 48(old NH4) 45

Table No 11: Route alignment-Along NE 7 46

Table No 12: Route alignment- Along Rail Line 47

Table No 13: Mixed alignment alternatives analysis 49

LIST OF MAPS:

Map No 1: CBIC-NICDC 14
Map No 2: Industrial corridor and dedicated freight corridor in India 15
Map No 3: Overview of Corridor influence zone 17
Map No 4: CBIC State overview 18
Map No 5: CBIC Total Population Distribution 21
Map No 6: CBIC Population Density 22
Map No 7: CBIC Urbanization ratio 23
Map No 8: Land use-land cover map CBIC 25
Map No 9: Topography of CBIC region 26
Map No 10: Protected areas of CBIC region 26
Map No 11: CBIC rail connectivity 27
Map No 12: CBIC Road connectivity 29
Map No 13: CBIC Air connectivity 30
Map No 14: potential nodes (3 selected) identified for development in CBIC 32
Map No 15: Three Regions for Development of Logistic Parks 33
Map No 16: Sectors and districts 34
Map No 17: Potential terminals &nodes 40
Map No 18: Ponneri node 41
Map No 19: Ponneri node Development Plan – under CBIC 41
Map No 20: Tumkuru node 42
Map No 21: Tumkuru node master plan of CBIC 42
Map No 22: Possible route connectivity 43
Map No 23: CBIC Road &Rail 44
Map No 24: Route alignment-Along NH 48(old NH4) 45
Map No 25: Route alignment-Along NE 7 46
Map No 26: Proposal NE 7 by NHAI 46
Map No 27: Route alignment- Along Rail Line 47

Map No 28: Chittoor turn NH 48.....48
Map No 29: Mixed alignment alternatives.....49
Map No 30: Final route plan hyperloop freight corridor.....50
Map No 31: Satellite spatial overview.....51
Map No 32: Arakkonam and Kanchipuram rail accessory node.....51
Map No 33: Route sum-up of Bangalore-Chennai hyperloop freight corridor.....52

LIST OF FIGURES:

Fig No 1: Hyperloop system decomposition…. 8

Fig No 2: 2(a) Total Population-All CBIC;(b) Total Population-All CBIC breakdown 24

Fig No 3: Banglore- ChennaI slope analysis.....26

Fig No 4: Freight capacity potential.....28

Fig No 5: a) Dwell time of cargo import;(b) Comparison of tonnage handled.....31

Fig No 6: Required freight mode for various industries..... 32

Fig No 7: Contribution of top sectors to corridor's manufacturing GVA in 2023-34..... 35

Fig No 8: Corridor GDP in Rs'000 Bn 36

Fig No 9: Export and Import Time dealy 37

Fig No:10 Containers handled at CFS(BIS).....38

Fig No:11 (a)CBIC States generation mix;(b)Fuel source & potential.....38

LIST OF ABBREVIATIONS:

CBIC : Chennai Bengaluru industrial corridor

HSR : High speed rail

JICA : Japan International Cooperation Agency

NH : National Highway

NE : National Expressway

GIS : Geo Information Software

CHAPTER 1-Understanding hyperloop and route planning

1. Introduction

Freight corridors, such as the Dedicated Freight Corridors (DFCs) in India and high-speed rail freight lines in Europe and China, have revolutionized cargo transport by streamlining logistics, reducing travel times, and improving economic connectivity (Singh & Verma, 2017). These projects are designed to handle large volumes of goods efficiently through electrified railways optimized for heavy loads. In this context, Hyperloop technology emerges as a futuristic alternative, promising even faster transit times, reduced environmental impact, and seamless integration with multimodal logistics hubs (Mokhtarian et al., 2020). Faster transfer of goods is the modern quality of economics that made everything and all products and services around as an instant deliverable. The demand for faster shipping in micro, medium and macro industries keep on increasing year by year but through traditional freight we couldn't neither handle the demand due to saturation and traffic nor could reduce the travel time with existing freight infrastructure and its operating speed. So dedicated freight corridor has become necessary, we could see the government initiatives to bring it on operations. But still joining late to DFCs makes us slow comparatively on worlds stage on economic race. Moving on and skipping older/current technology and build and adapt to new way of things is the best choice to catch up and get head.

The Chennai-Bangalore Industrial Corridor (CBIC) is a significant infrastructure initiative by the Government of India, aiming to enhance industrial development and economic integration between Chennai in Tamil Nadu and Bangalore in Karnataka. Spanning approximately 560 kilometres, the corridor is designed to bolster commerce, reduce transit times, and create a robust industrial belt in southern India.

By incorporating hyperloop freight technology in CBIC, we would able to achieve the above advantages in large scale in long term scenario.

2. Hypothesis and Assumption

Every change that occurs in the world that affects the countries development both positively and negatively whether it is any new technological revolution, geo-political affairs, natural calamities, etc. are not just due to natural effects but majorly through the plans and prediction of human knowledge involved. Due to social, political & economic

– history of nations, their development is varied drastically from developed nations to developing nations to under-developed nations. “Lagging” is a negative development phenomenon occur due to gap formed by the front lead of the developed nation in the competing sector/field while the latecomers lag behind and can’t able to equally compete or enjoy the economic benefits like the leading developed nation.

The idea of leapfrogging is that some latecomers may be able to leap-frog older vintages of technology, bypass heavy investments in previous technology system, and catch-up with advanced countries Hobday, 1995. The increasing tendency toward globalization and development of information technology makes the leapfrogging argument ever more plausible.

So, through the following thesis paper I would be emphasising the need for leapfrog development plan for developing India briefly and following that, with the assumption (based on real case outcomes and literature study- covered in dissertation) that path skipping-creating strategy by advancing the development methodologies and technologies would provide better and fast growth, here I would be preparing a planning framework for an hyperloop corridor between Chennai and Bangalore.

3. Aim

To develop a **physical planning framework** for an Hyperloop freight corridor from Chennai to Bangalore along the Chennai-Bangalore Industrial Corridor (CBIC), by **determining route alignment** based on Right of way requirement for hyperloop and **identifying the nodes and terminals** the corridor cover, integrating freight handling facilities and supporting logistics hubs and ensure corelation & spatial compatibility with the existing industrial and transport infrastructure.

4. Objectives

1. Route Planning and Alignment:
    - Identify a feasible alignment for the Hyperloop corridor based on land cover, CBIC industrial clusters/ nodes, and existing freight movement patterns.
    - Optimize alignment to maximize the productivity of CBIC and minimize travel distance

2. Station and Terminal planning:

    - Propose Hyperloop freight terminals at key industrial nodes, ensuring connectivity of core CBIC influence area with existing transport networks (rail, road, and ports).
    - **Land-Use Considerations:** Buffer zones, industrial growth areas, and transit-oriented logistics zones.
    - **Intermediate Cargo Transfer Hubs:** Locations optimized based on industrial freight demands to functionate the route for CBIC

5. Scope

5.1 Physical Planning Scope:

➢ **Corridor Length and Alignment:**

CBIC Influence Area spread across the States of Karnataka, Andhra Pradesh and Tamil Nadu, along the corridor between Chennai-Bengaluru (around 560 km) covering 16 districts.

Total CBIC Area: 91,059 Sq. Km

Approximately 350-400 km alignment following NH-48 and the existing Chennai-Bangalore rail corridor.

Key Industrial Nodes: 8 potential nodes (3 selected) identified for development in CBIC

➢ **Terminal Planning:**

- **Freight Terminals:** Chennai Port, Ponneri node, Tumkur National Industrial Manufacturing Zone and Bangalore Industrial Zone.
- **Intermediate Cargo Transfer Hubs:** Chittoor, Kolar and Kanchipuram.

5.2 Economic growth scope:

Expected contribution of top sectors to CBIC corridor's manufacturing GVA (2033-34)

- Business As Usual (BAU)- Rs.198,524cr (JICA 2014)
- Business Induced Scenario (BIS)- Rs.903,565cr (JICA 2014)

Estimated Port traffic in the CBIC Region (BIS)- Over 685 MT by 2033

5.3 Research and home hyperloop technology development scope:

Indian Institute of Technology Madras(IIT Madras)-incubated Deep Tech Start-up TuTr Hyperloop is collaborating with the Institute on developing Intellectual property (IP) in hyperloop technology.

"*The initial focus will be on cargo movement to enhance the productivity of various stakeholders by solving their pain points. This will be a first step towards the deployment of high-speed mobility corridors for passenger transportation across India and globally*."- Prof. Satyanarayanan Chakravarthy, who is also a faculty in the Department of Aerospace Engineering Department, IIT Madras

Railway Minister Ashwini Vaishnaw reviewed the progress of the test project in Chennai, 15th March, 2025. Minister Ashwini Vaishnaw stated that *the project is currently in the experimental stage, but in the future, it has the potential to redefine transportation.*

6. Limitations

6.1 Planning and study Limitations:

- Work is based on several major assumption – hyperloop's feasibility
- Data lack and limit:
    - ❖ Data are completely through secondary sources – JICA 2014 CBIC summary report
    - ❖ Lack of actual demographics, economic trends data (work is based on estimation and projection on 2011 census and JICA analysis)
    - ❖ Limited pre-processed data from reports
- Methodology of the paper – Desk appraisal (no stakeholders, no visit)
- Neglecting ground reality and work is fully based on GIS model of the case.
- No land suitability analysis, Feasibility analysis, No EIA, No Socio-economic Impact Assessment- purely focused on Physical aspect of route alignment planning.

6.2 Technological and Structural Limitations:

- New unexplored and amateur technology in practice
- Station infrastructure Requirements: Requires advanced unloading and loading infrastructure to support and as well as not hinder the operation of hyperloop.
- Integration with Existing Logistics: Adapting Hyperloop terminals to existing transport networks.
- Hyperloop specific technological limitations – maintenance, regulation, control etc.

6.3 Financial Limitations:

- High Initial Cost: Hyperloop infrastructure require higher capital investment than pre-existing mode of alternatives.

7. Literature guides:

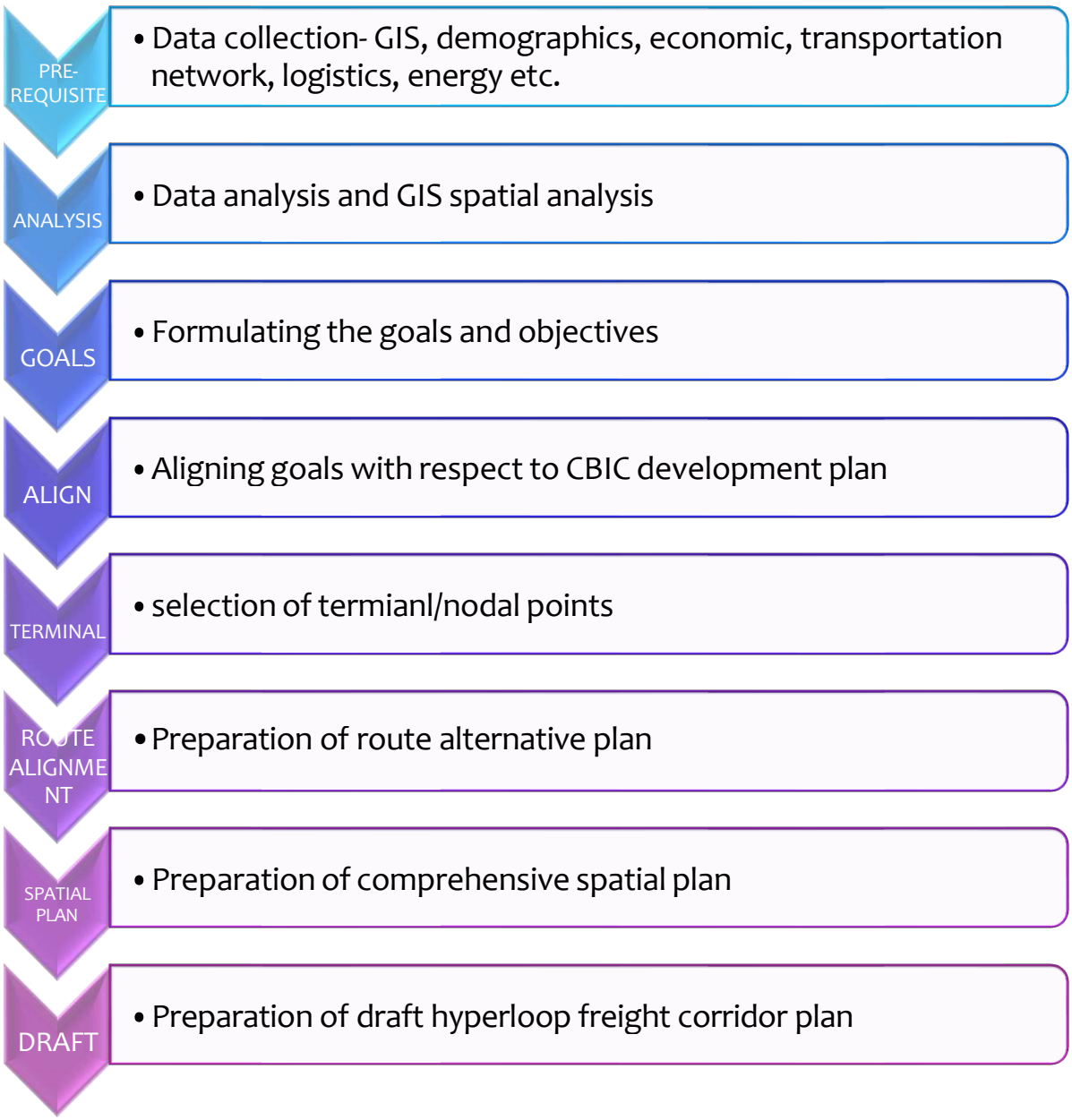
Planning guides:

1) The Midwest Connect Hyperloop Feasibility Study 2020: Mid-Ohio Regional planning commission (MORPC)
2) Joint Feasibility Study for Mumbai-Ahmedabad High Speed Railway Corridor 2015 by JICA & Ministry of Railways, India.

Concept and technological guide:

1) Great Lakes Hyperloop Feasibility Study: HyperloopTT Express Freight – HyperloopTT
2) Animated video showing how the HyperloopTT Express Freight will operate is here https://youtu.be/l-eE9gRXFoI
3) Hyperloop Commercial Feasibility Analysis: High Level Overview by Catherine L. Taylor, David J. Hyde, Lawrence C. Barr (U.S. Department of Transportation; John A Volpe National Transportation Systems Center)

8. Methodology



9. Literature Review:

9.1. Introduction to Hyperloop

Hyperloop is the concept of high-speed ground transit involving pods with cargo or passengers travelling at 1000km/hr through a low pressurised tube, cutting air resistance and friction making it faster than air transportation. It was popularized by Elon musk through his white paper in 2013 (Musk, 2013). This system works on magnetic levitation (maglev) and linear induction motors to achieve near-supersonic speeds while maintaining energy efficiency (Guzmán & García, 2017). Hyperloop aims to achieve long distances travel faster than traditional rail and air mode, with the advantage of being more environmentally sustainable due to its potential reliance on renewable energy sources (Bachman, 2015).

9.2. Technical Design and Principles

The Hyperloop works on the principle of minimizing drag by reducing air density is significantly to nearly vacuum-like conditions through low-pressure tubes, (González & Sanz, 2018). This allows pods to move with minimum aerodynamic resistance. The propulsion in hyperloop is achieved through linear electric motors, providing bursts of acceleration at specific intervals while magnetic levitation keeps the pods floating without direct contact with the ground track, thus reducing friction (Zhang et al., 2020). Various levitation technologies have been explored in researches to be applied in hyperloop, including electrodynamic suspension (EDS) and electromagnetic suspension (EMS), each with its pros and cons concerning cost, stability, and energy consumption (Liu & Wang, 2019). The Alpha Paper postulates average travel speeds of 300 mph/480 kmh (v_u) through urban areas and difficult course sections and 760 mph/1,220 kmh (v_t) through regular course sections.

Hyperloop involves magnetic levitation (or Maglev) conveyance and propulsion vehicles in a tube/conduit guideway that allows for a semi-vacuum travel-way environment to reduce air resistance. It is not vacuum propelled, such as bank teller transport tubes. The tube is typically configured as an elevated structure, although it can also be configured near ground level, in an open cut or in an underground tunnel configuration. The tube needs to be grade-separated or barrier protected from other transportation modes.

Hyperloop portals are passenger/freight access points with station-like features. Portals allow hyperloop passengers and cargo shippers to have the advantage of direct travel to individual destinations, instead of stopping at each community along the corridor, the way air and rail travelers do.

The five physical clusters are:

- Hyperloop as a system: this includes research that encompasses the entire system and that cannot be considered under other disaggregated levels. Examples may include efficiency and energy studies of the system in operation.
- Substructure (including foundations and bridge work): focuses mostly on structural engineering design for the supporting structure.
- Tube: considers aspects related to the tube structure.
- Tube pod interface: focuses on research on the interface between the tube and the pod. Examples may include aerodynamic phenomena as a consequence of the pressure variation.
- Pod: focuses on aspects related to the pod (e.g., levitation, suspension, powertrain, electronics) Hyperloop pods are on-demand vehicles that move to and from passenger portals or freight intermodal facilities with freight portals that link to a mainline via branches. These branches are sometimes referred to as on/off ramps.

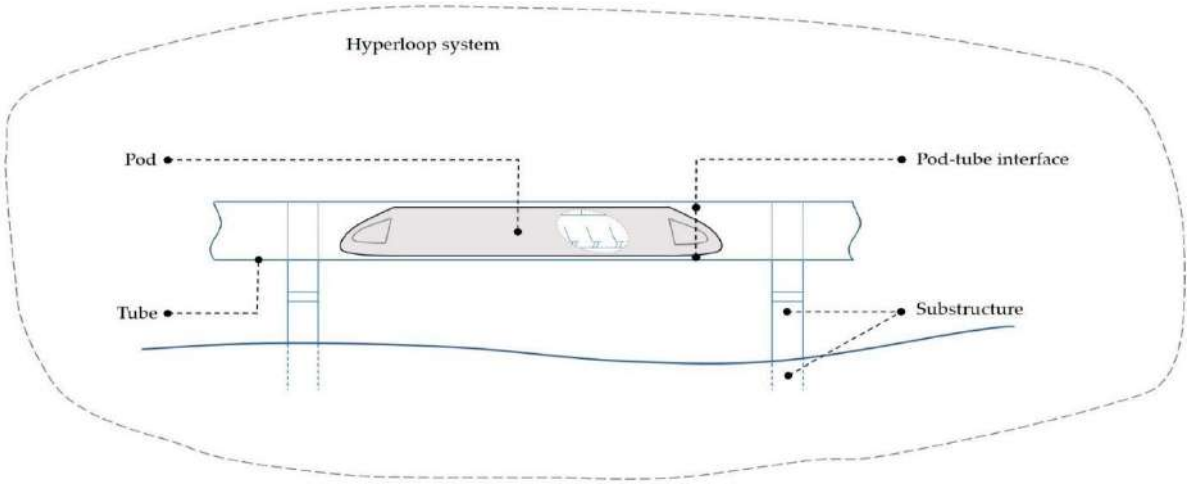


Fig No:1 Hyperloop system decomposition
Source: Gkoumas & Christou, 2020

9.3 Hyperloop compared to other HSR, its corridor selection, corridor and terminals, travel time, capacity and efficiency

	Hyperloop's Commercial Claims	Hyperloop's More Realistic and Expected Implications
Hyperloop compared to other operational high-speed ground transportation (HSR and maglevs).	Hyperloop claims to be superior to other HS ground transportation (operational, economic, environmental and social performance).	Compared to HSR and maglevs, hyperloop will have: - Higher speed; - High reliability regardless of weather conditions; - Higher fares; - Higher energy efficiency; - Lower comfort and safety.
Criteria for the selection of a hyperloop corridor		Hyperloop should connect two important growing cities of the same megaregion (minimum 500–800 km apart) with: - High economic productivity; - Important intercity connection; - Existing low interconnections with inefficient transportation modes; - Political and economic support; - A flat and straight alignment.
Corridor and terminals	Hyperloop is a guided transportation mode with a specific infrastructure along the route it covers.	- Using existing transportation infrastructure minimises total land requirement for new infrastructure, whereas land used for new infrastructure creates opportunity costs. - Hyperloop corridor would be mostly elevated on pillars 30 m apart. - It will be challenging to incorporate too sharp turns on the hyperloop corridor, lowering its flexibility. Rapid changes in the land use patterns will be visible in hyperloop regions such as: - Accelerated urban development; - Reconstruction of the urban hierarchy; - Hyperloop terminals may lie anywhere between the centre of city and its outskirts.
Travel time, travel capacity and efficiency	Hyperloop claims to be: - More energy efficient; - Having a lower station-to-station travel time (due to higher frequency between pods) than flights and transnational rail. - Hyperloop will meet the demand of 840 passengers during off-peak hours and 3360 passengers during peak hours (headway of 30 s).	- Hyperloop travel time will be like air travel, undermining the overall speed advantage and increasing journey time. - Terminals within the city centre will be more accessible due to the central location and existing access options. - Hyperloop will run for 19 h and use 5 h for maintenance. - A single pod can hold only up to 28 passengers. - The headway between pods is unrealistic and questionable due to safety reasons at such high-speed (headway of 80 s seems more realistic hence reduced capacity).

Table No:1 Hyperloop compared to other HSR
source: Premsagar & Kenworthy(2022)

9.4 Hyperloop technology route needs

Hyperloop technology is designed for mainline speeds comparable to commercial airliners. Therefore, the technology necessitates specific route conditions to operate at optimal speeds. In ideal conditions where the horizontal and vertical curvature of the corridor does not constrain vehicle speed, vehicles glide at high cruising speeds for long distances due to ultra-low aerodynamic drag. With both friction and air resistance

greatly reduced, the vehicles are projected to travel up to 670 mph (1,080 km/h). While the mainline is designed to maintain optimal speeds, the local access branch lines are designed for travel at slower speeds and offer more flexibility for vertical and horizontal curvature. Given these technology and system characteristics, the sections below describe how route alternatives were analysed relative to hyperloop technology needs.

Route planning goals:

1. Connect potential portals (stations or freight access points) using a straight-line path to the extent possible
2. Provide a mainline designed for average speeds of 500 MPH, and provide point-to-point service along the entire corridor, with local access branches or "ramps" required to access portals
3. Maximize opportunities for fast travel times and system efficiencies by minimizing elevation changes and alignment curves, particularly along the mainline
4. Minimize impact on neighbourhoods and businesses
5. Minimize impact to potential environmentally sensitive areas
6. Choose alignments that will minimize infrastructure costs, including minimizing the need for tunnelling and complex construction methods. Elevated and near at-grade guideway configurations are generally preferred
7. Identify opportunities for using existing public rights-of-way

9.5 Right-of-way planning requirements

The hyperloop guideway will be configured so that the exterior of the tube is accessible for routine maintenance with the exception of tunnelled segments, where maintenance access is typically provided by vertical shafts that descend to the tunnel

Highway Alignment: either in a median or edge configuration. The guideway is preferred to be configured near ground level on short levelling structures. Roadway barriers would protect hyperloop structures from highway traffic.

Railroad Alignment: parallel to or configured above active railroad. Project partners will need to reach agreements with rail right-of-way owners and operating railroads in order to share right-of-way.

Greenfield Alignment: along acquired right-of-way between urbanized areas.

Tunnel Alignment: which may be required to navigate through hills or under urbanized communities. All right-of-way design will require safety and technical considerations, which will involve input from technical experts for any alignment that shares a right of-way with another mode (such as highways or railroads)

9.6 Economic Feasibility and Cost Analysis

While the Hyperloop promises faster transit times, its economic feasibility remains contentious. Initial cost estimates by Musk suggested around $6 billion for a Los Angeles–San Francisco route; however, independent studies argue this figure is optimistic, with actual costs likely exceeding $20 billion due to land acquisition, safety systems, and infrastructure complexities (Harris, 2017). Additionally, maintenance costs could be high because of the sophisticated vacuum and maglev systems (Kaur & Singh, 2019). On the other hand, proponents argue that long-term operational efficiency, reduced fuel dependency, and high-speed logistics could offset these initial investments (Nakamura et al., 2021).

Metro System	Cost of Construction (USD Million per km in 2011)	Number of Times Less than the Minimum Cost of Constructing Hyperloop (USD 139 Million/km)	Kilometres of Metro Construction Possible with the Total Cost of Proposed Hyperloop of 120 km (Approximately USD 16.680 Billion)	Population of Metropolitan Area [29,67,68]
Delhi metro	Average: USD 26 Underground: USD 74	5.3 times	642 km	16,349,831
Jaipur metro	USD 50	2.8 times	334 km	3,046,163
Mumbai metro	USD 19	7.3 times	878 km	18,394,912

Table No:2 Cost analysis with respect to existing metro transit source: : Premsagar & Kenworthy(2022)

9.7 Capsule and payload components

Capsule component (alpha)	Weight (kg)	Payload component adjusted	Weight (kg)
Capsule structure and doors	3,500	Capsule structure and doors	3,100
Interior and seats	2,700	Propulsion system	800
Propulsion system	800	Suspension and air bearings	1,300
Suspension and air bearings	1,300	Batteries, motor, and coolant	5,500
Batteries, motor, and coolant	5,500	Air compressor	2,500
Air compressor	2,500	Emergency braking	800
Emergency braking	800	Payload	12,000
Passengers and luggage	1,400		
Car and cargo	7,500		
Total	26,000	Total	26,000

Table No:3 Capsule and payload components
source: Van Goeverden, C. D., Milakis, D., Janic, M., & van Wee, B. (2016)

9.8 Safety and Regulatory Challenges

Ensuring passenger safety in a vacuum environment presents unique challenges. The risks include sudden decompression, system malfunctions, and emergency evacuation difficulties (Johnson & Baker, 2018). Unlike conventional rail systems, Hyperloop's reliance on near-vacuum tubes complicates rescue operations in case of accidents. Moreover, regulatory frameworks for such technology are underdeveloped globally, with no established safety standards for high-speed vacuum transit systems (Martínez et al., 2020). These gaps raise concerns about the readiness of Hyperloop for public deployment, necessitating rigorous testing and international collaboration for safety standardization.

9.9 Environmental Impact

Hyperloop advocates highlight its potential for environmental sustainability. Unlike traditional aircraft, Hyperloop can be powered entirely by renewable energy, such as solar panels installed along the tube infrastructure (Davies & Kim, 2019). Additionally, its energy efficiency per passenger-kilometer is projected to be significantly lower than that of cars, trains, or airplanes (Singh & Zhao, 2020). However, concerns remain

regarding the environmental costs of construction, including land disruption, resource-intensive materials, and the ecological impact of extensive linear infrastructure cutting across natural habitats (Li & Green, 2021).

9.10 Current Developments and Future Prospects

Various companies, including Virgin Hyperloop and TransPod, have conducted prototype tests, demonstrating the feasibility of basic Hyperloop systems (Thompson, 2020). As of 2023, Virgin Hyperloop successfully completed human trials, though at reduced speeds compared to the theoretical maximum. Despite these advancements, commercial deployment faces hurdles, including scaling the technology, securing investment, and achieving regulatory approval (Jensen & Li, 2022). Future research focuses on improving energy efficiency, reducing construction costs, and enhancing passenger comfort to make Hyperloop a viable alternative to existing transport modes (Klein et al., 2023).

10. Data collection

Data completely collected from secondary sources; major sources are:

- Comprehensive Integrated Master Plan for Chennai Bengaluru Industrial Corridor Final Report- JICA 2015
- The Chennai-Bengaluru Industrial Corridor -A Preliminary Report
  Author: Siddharth KJ, Independent Researcher
  Published by: Centre for Financial Accountability
- National Industrial Corridor Development Corporation (NICDC) & other portals.

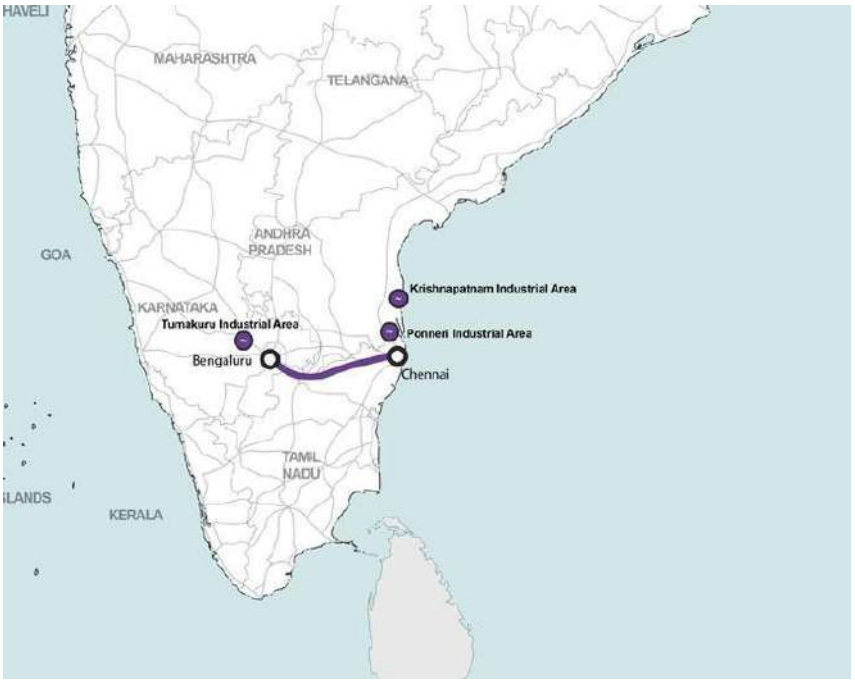
GIS spatial data is used for the physical analysis and planning

CHAPTER 2-CBIC AND Planning parameters

11. Study Area: Chennai-Bangalore Industrial Corridor (CBIC)

11.1. CBIC

The Chennai-Bangalore Industrial Corridor (CBIC) is a significant infrastructure initiative by the Government of India, aiming to enhance industrial development and economic integration between Chennai in Tamil Nadu and Bangalore in Karnataka. Spanning approximately 560 kilometers, the corridor is designed to bolster commerce, reduce transit times, and create a robust industrial belt in southern India.



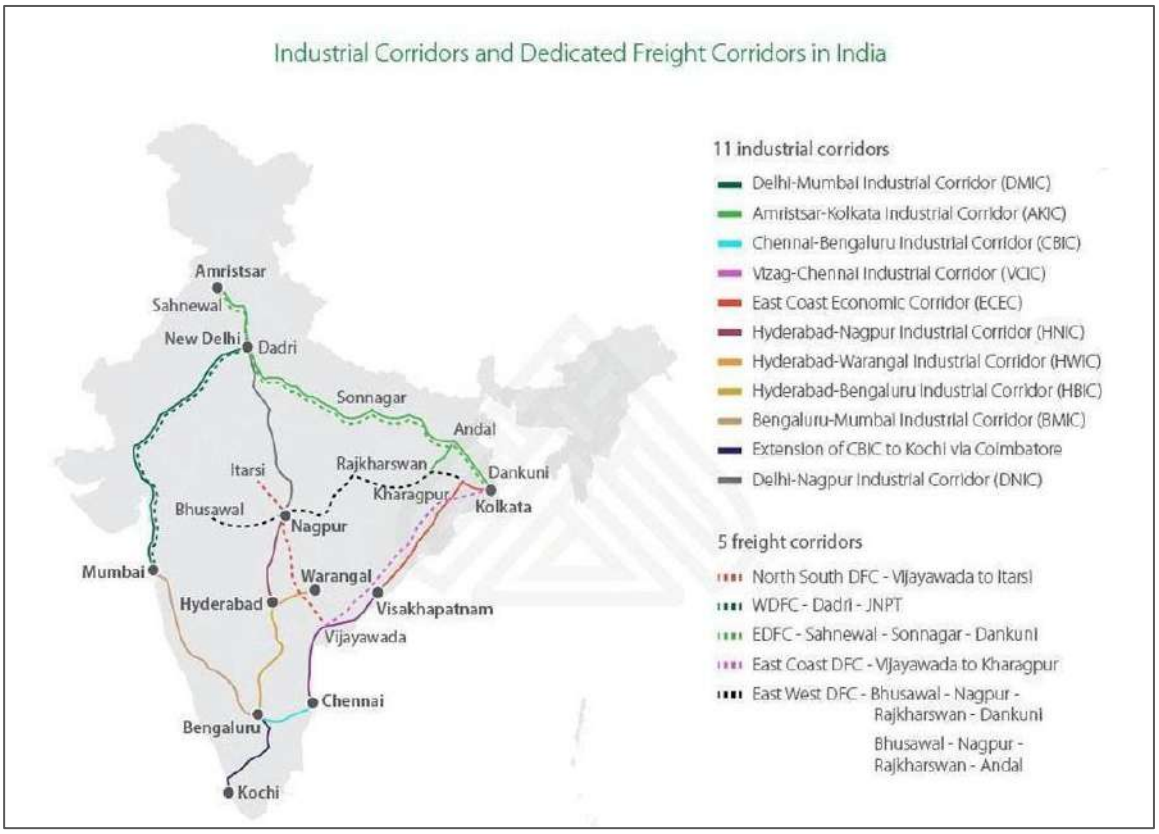
Map No:1 CBIC-NICDC
Source: (NICDC)

11.2. Historical Evolution

11.2.1 Conceptualization and Planning

The National Industrial Corridor Development Programme (NICDP) of Government of India (GoI) has been undertaken with the stated objective of "developing greenfield industrial nodes in India which can compete with the best manufacturing and investment destinations in the world". So far approval has been granted to following 11 industrial corridors in 4 phases spread across 17 states.

The idea of the CBIC emerged in the early 2010s, inspired by the success of the Delhi-Mumbai Industrial Corridor (DMIC). Recognizing the economic potential of, in collaboration with the Japan International Cooperation Agency (JICA), initiated studies to assess the feasibility of the corridor. The project aimed to address infrastructure bottlenecks and promote regional industry agglomeration in Tamil Nadu, Karnataka, and Andhra Pradesh



Map No:2 Industrial corridor and dedicated freight corridor in India

Source: National Industrial Corridor Development Corporation (NICDC)

In the Comprehensive Regional Perspective Plan submitted by the JICA study team (with PricewaterhouseCoopers and Nippon Koei) in October 2015, three nodes were identified between Chennai-Bengaluru (around 560 km) with NH-48 as the backbone for the corridor: Ponneri (Tamil Nadu), Tumkur (Karnataka) and Krishnapatnam (Andhra Pradesh)

Targeted Year: The Master Plan to cover 20years, during 2013-2033.

11.2.2 Development Phases

The CBIC project is managed by the National Industrial Corridor Development Corporation (NICDC), which oversees the planning and implementation of industrial corridors across the country.

The CBIC project was structured into multiple phases:

- **Phase I:** Focused on comprehensive planning and feasibility studies, leading to the preparation of a regional master plan.
- **Phase II:** Involved detailed master planning for identified industrial nodes, including Ponneri in Tamil Nadu, Tumakuru in Karnataka, and Krishnapatnam in Andhra Pradesh.
- **Phase III:** Emphasized infrastructure development, policy implementation, and attracting investments to the corridor.

Throughout these phases, the project received support from both central and state governments, with active participation from international agencies.

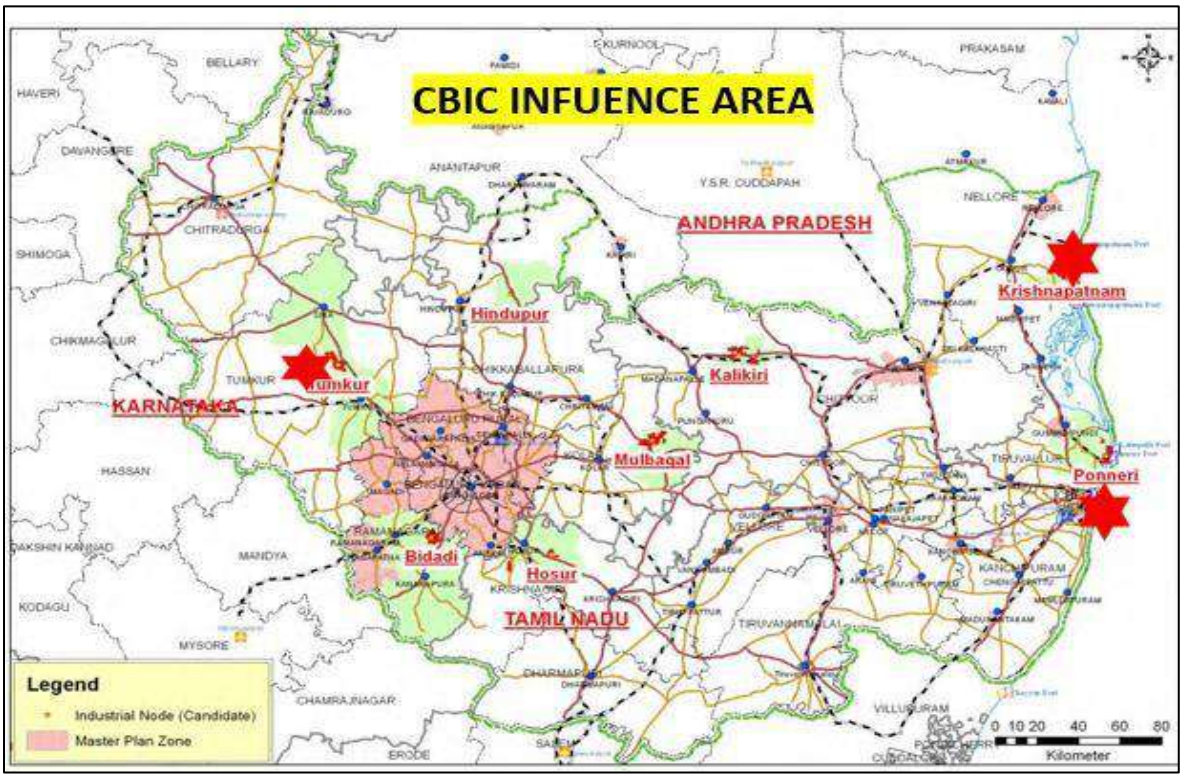
11.2.3 Area covered

CBIC Influence Area spread across the States of Karnataka, Andhra Pradesh and Tamil Nadu, along the corridor between Chennai-Bengaluru (around 560 km) covering 16 districts.

Total CBIC Area: 91,059 Sq. Km

Districts covered		
Tamil Nadu	***Karnataka***	***Andhra Pradesh***
1. Chennai	8. Bengaluru (urban and rural)	14. Chittoor Potential area
2. Tiruvallur	9. Ramnagara	15. South part of Nellore
3. Kancheepuram	10. Kolar	16. South west part of Anantapur
4. Tiruvannamalai	11. Chikkaballapura	
5. Vellore	12. Tumkur	
6. Dharmapuri	13. Chitradurga	
7. Krishnagiri		

Table No:4 Districts covered in CBIC Influence area
source: CBIC-Comprehensive Regional Plan report

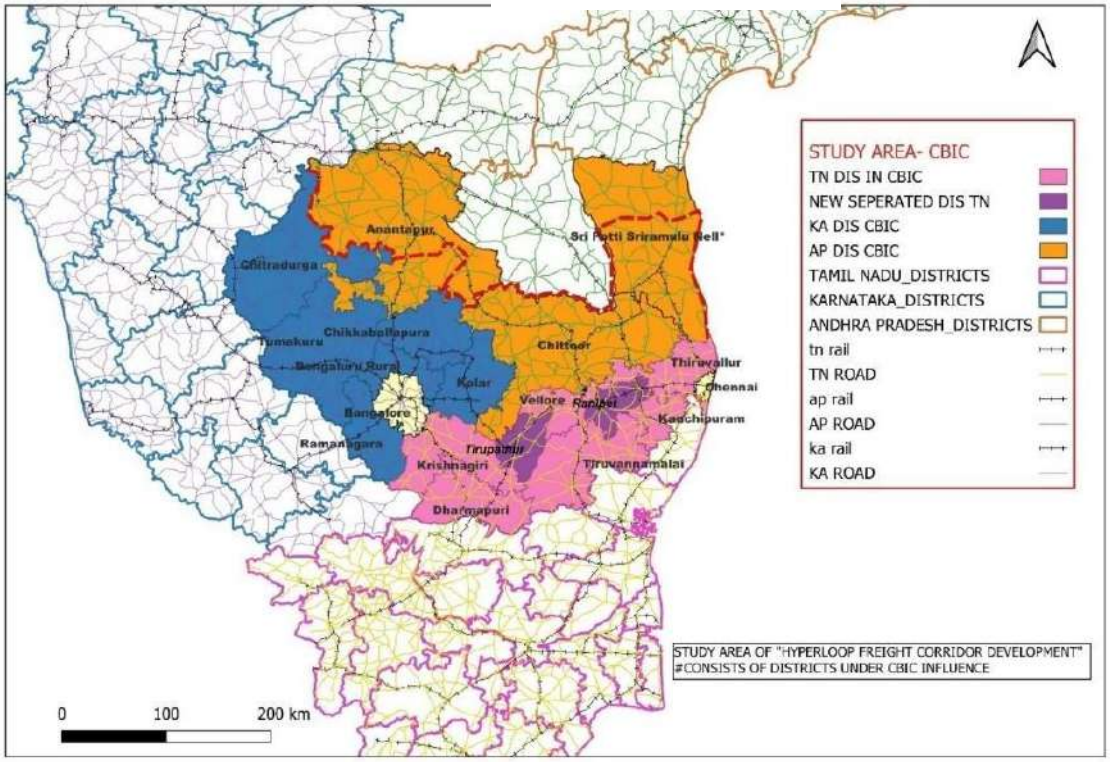


Map No:3 Overview of Corridor influence zone
source: CBIC-Comprehensive Regional Plan report

State	CBIC districts	Area covered (Sq. Km)
Tamil Nadu	Chennai	175
	Thiruvallur	3,394
	Kancheepuram	4,483
	Tiruvannamalai	6,188
	Vellore	6,075
	Dharmapuri	4,497
	Krishnagiri	5,129
Karnataka	Bengaluru	2,196
	Bengaluru rural	2,298
	Ramanagara	3,516
	Kolar	3,979
	Chikkaballapura	4,244
	Tumkur	10,597
	Chitradurga	8,436
Andhra Pradesh	Chittoor	15,152
	Nellore	6,400
	Anantapur	4,300
Total CBIC		**91,059**
All India		**3,166,414**

Table No:5 Area of CBIC covered districts
source: CBIC-Comprehensive Regional Plan report

CBIC State overview



Map No:4 CBIC State overview
source: Author

11.2.4 Industrial Clusters

The corridor is home to a diverse workforce engaged in various sectors:

- **Manufacturing:** Automobiles, electronics, and textiles are prominent industries- Chennai, Kanchipuram and chittoor.
- **Information Technology:** Bangalore, known as the "Silicon Valley of India," hosts numerous IT companies.
- **Logistics and Warehousing:** Strategic location of the corridor facilitates robust logistics operations on selected nodes for development planning - Ponneri in Tamil Nadu, Tumakuru in Karnataka, and Krishnapatnam in Andhra Pradesh.

Industrial clusters in regions like Sriperumbudur, Oragadam, and Hosur have attracted significant investments, leading economic growth.

Industrial clouds in regions such as - Sriperumbudur, Oragadam, and Hosur have attracted significant investments leading economic growth.

11.2.5 Industrial Nodes

The CBIC project proposes the development of high-impact, market-driven nodes at strategic locations within the corridor. These nodes are envisioned as self-sustained regions with world-class infrastructure, ensuring seamless connectivity for freight movement to and from ports and logistics hubs. The primary nodes identified for development include:

- **Ponneri (Tamil Nadu):** Focus on engineering components manufacturing, logistics, and warehousing.
- **Tumakuru (Karnataka):** Emphasis on aerospace, information technology, and biotechnology sectors.
- **Krishnapatnam (Andhra Pradesh):** Centered around port development, petrochemicals, and power sectors.

These nodes aim to create a transparent and investment-friendly environment conducive to business establishment and growth.

11.3. Economic Impact

11.3.1 Investment Inflows

The CBIC has attracted both domestic and international investments:

- **Foreign Direct Investment (FDI):** Several multinational corporations have established operations, leveraging the strategic advantages of the corridor.
- **Public-Private Partnerships (PPP):** Collaborative projects have been initiated to develop infrastructure and industrial facilities.

The corridor's development is expected to further boost investor confidence and economic activities.

11.3.2 Employment Generation

The industrial activities have led to significant job creation:

- **Direct Employment:** Opportunities in manufacturing units, IT parks, and logistics centers.

- **Indirect Employment:** Growth in ancillary industries, service sectors, and support services.

This employment surge has contributed to improved living standards and economic prosperity in the region.

11.3.3 Economic growth

Expected contribution of top sectors to corridor's manufacturing GVA in 2033-34

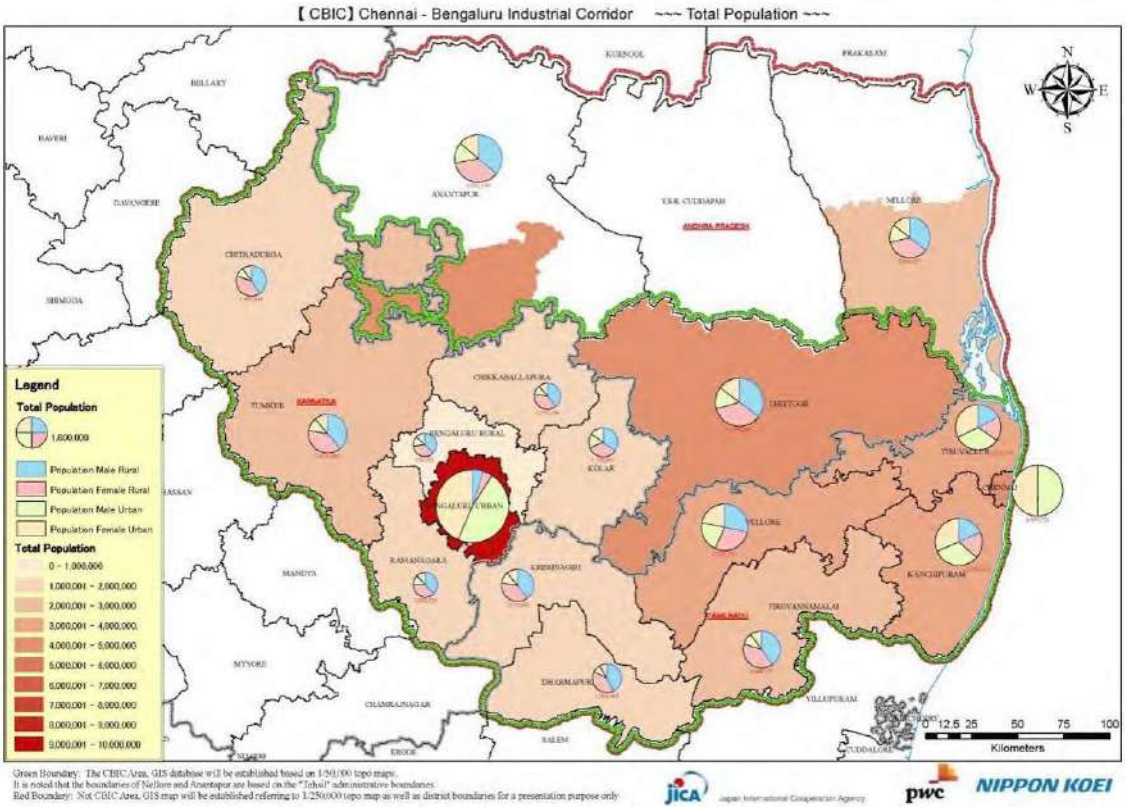
- Business As Usual (BAU)- Rs.198,524cr (JICA 2014)
- Business Induced Scenario (BIS)- Rs.903,565cr (JICA 2014)

12. Existing character

12.1. Demographics and related parameters

12.1.1 Total population:

According to Census 2011, the total population of India is 1,210.57million, while that of the CBIC region is 47.53million. Compared with 2001, the total population of India has increased by 17.69% from 1,028.61 to 1,210.57million while that of the CBIC region was 26.60% from 37.54 to 47.53million. It clearly indicates the population growth speed of CBIC region for the last decade has been greater than that of all India.



Map No:5 CBIC Total Population Distribution
Source: PwC, NK Analysis, data from respective state governments

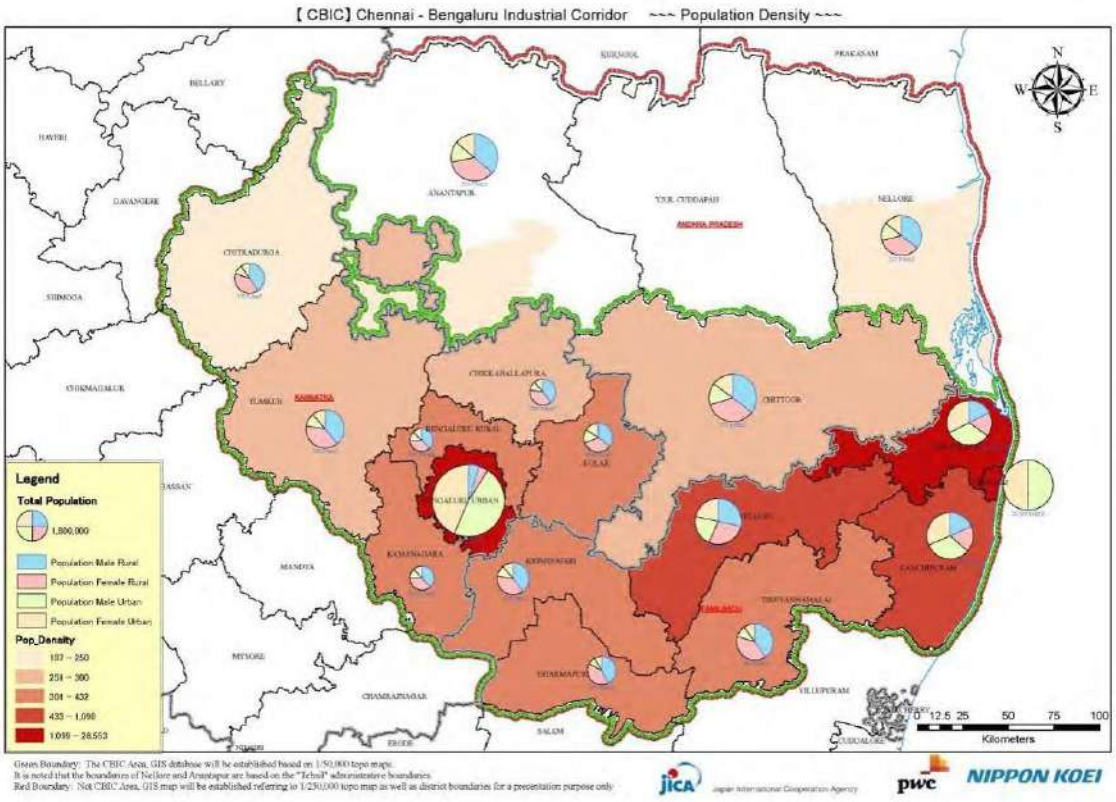
- The most populated district is Bengaluru which has 9.62 million (20.24% of the CBIC region) in 2011.
- Chennai (4,646,732) in Tamil Nadu state and Chittoor (4,174,064) in Andhra Pradesh, the 2nd populated districts in the region, have respectively over 4 million

Also, the 3 eastern districts in Tamil Nadu, Tiruvallur, Kancheepuram, and Vellore, have a population of over 3.5 million respectively despite their relatively smaller areas. It could be suggested that east coast districts closer to the Bay of Bengal are populous area rather than inland area except Bengaluru district.

12.1.2 Density

Average density of the CBIC region is 464 persons per Sq. km as per census 2011. The density map above typically suggests that the density of Chennai in Tamil Nadu state and Bengaluru in Karnataka state are enormously higher dense than the other district.

Also, it can be suggested that in the eastern side of the CBIC region, Tiruvallur, Vellore and Kanchipuram districts in Tamil Nadu states nearer the Bay of Bengal are populous and dense which is higher than the average of the CBIC region



Map No:6 CBIC Population Density

Source: PwC, NK Analysis, data from respective state governments

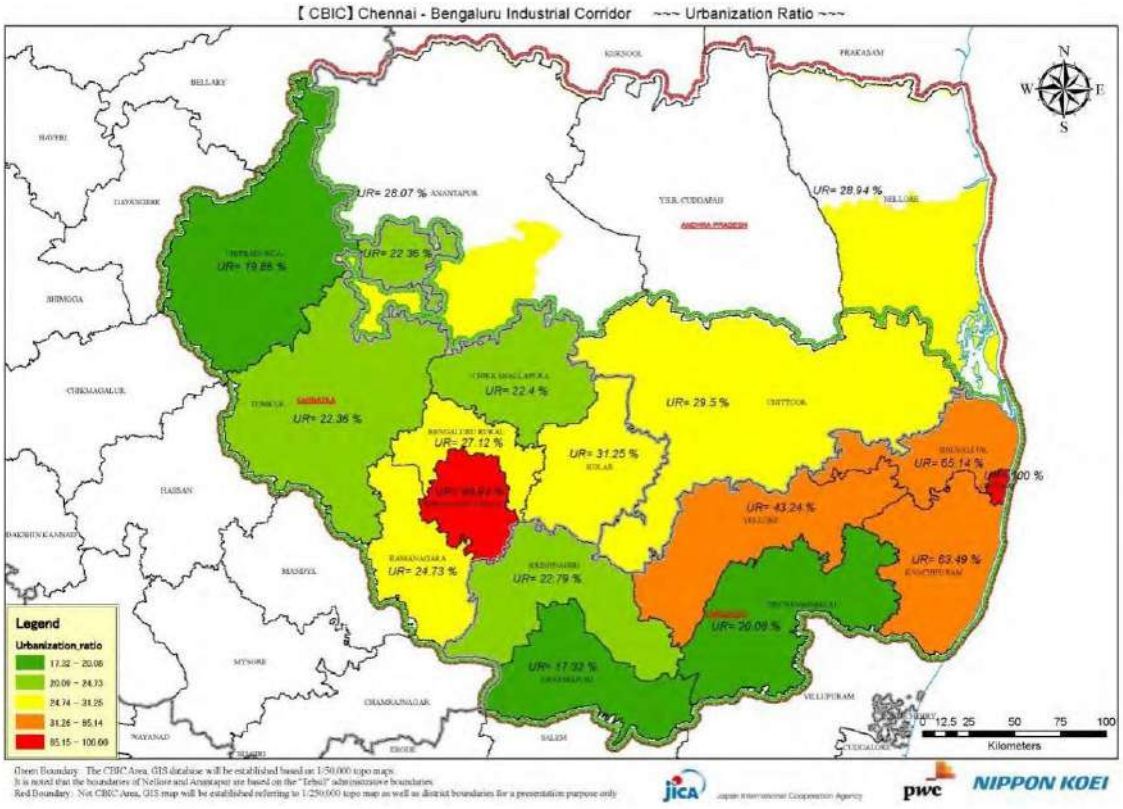
Between Chennai and Bengaluru districts, the ones along NH-4 (one of the alignments in CBIC) tend to be denser than the other districts. Especially the inland or north districts of CBIC region are less dense than NH-4 between Chennai and Bengaluru.

12.1.3 Urbanization ratio

The all-India urbanization ratio is 31.15%, while that of the CBIC region is 51.17%.

Chennai and Bengaluru, which are two most populous districts in Karnataka state and Tamil Nadu state respectively, are the most urbanized districts in the CBIC region.

Additionally, in eastern area, the districts of Tiruvallur and Kanchipuram have an urbanization ratio over 60%. The other inland districts such as Dharmapuri in Tamil Nadu and Chitradurga in Karnataka are under the average urbanization ratio of the CBIC region



Map No:7 CBIC Urbanization ratio

Source: PwC, NK Analysis, data from respective state governments

Population Projection (2033)

Regarding to future population of all CBIC area until 2033 under Business in Induced Scenario (BIS), the population will increase by 47% from 54,464 thousand in 2013 to 79,182 thousand in 2033. The compound average population growth rate from 2013 to 2033 is estimated as 1.89%, which is slightly higher than actual growth rate from 2001 to 2011 of 1.82%.

On the other hand, the average growth rate from 2013 to 2033 under Business-as-Usual scenario (BAU) scenario is estimated as 1.37%. In 2033, population will reach at 70,432 thousand, the level which will be achieved in 2026 under BIS.

(in 000)			Actual	Projected	Projected	Projected	Projected
State	District	Level	2011	2013	2018	2023	2033
TN	Chennai	District	4,647	4,816	4,946	5,144	5,597
	Thiruvallur	District	3,728	4,051	4,680	5,477	7,550
	Kancheepuram	District	3,998	4,367	5,113	6,063	8,583
	Tiruvannamalai	District	2,465	2,536	2,722	2,907	3,335
	Vellore	District	3,936	4,054	4,359	4,664	5,372
	Dharmapuri, Krishnagiri	District	3,387	3,520	3,874	4,243	5,120
KA	Bangalore	District	9,622	10,445	12,808	15,185	17,593
	Bangalore Rural, Ramanagara	District	2,074	2,124	2,253	2,373	2,650
	Kolar, Chikkaballapura	District	2,792	2,859	3,031	3,190	3,558
	Tumkur	District	2,679	2,711	2,788	2,847	2,989
	Chitradurga	District	1,659	1,697	1,793	1,881	2,083
AP	Chittoor	District	4,174	4,284	4,567	4,849	5,472
	Nellore	District	2,964	3,039	3,235	3,429	3,856
	Anantapur	District	4,081	4,194	4,485	4,776	5,422

Table No:6 CBIC Population Projection (2033)
source: PwC, NK Analysis, data from respective state governments

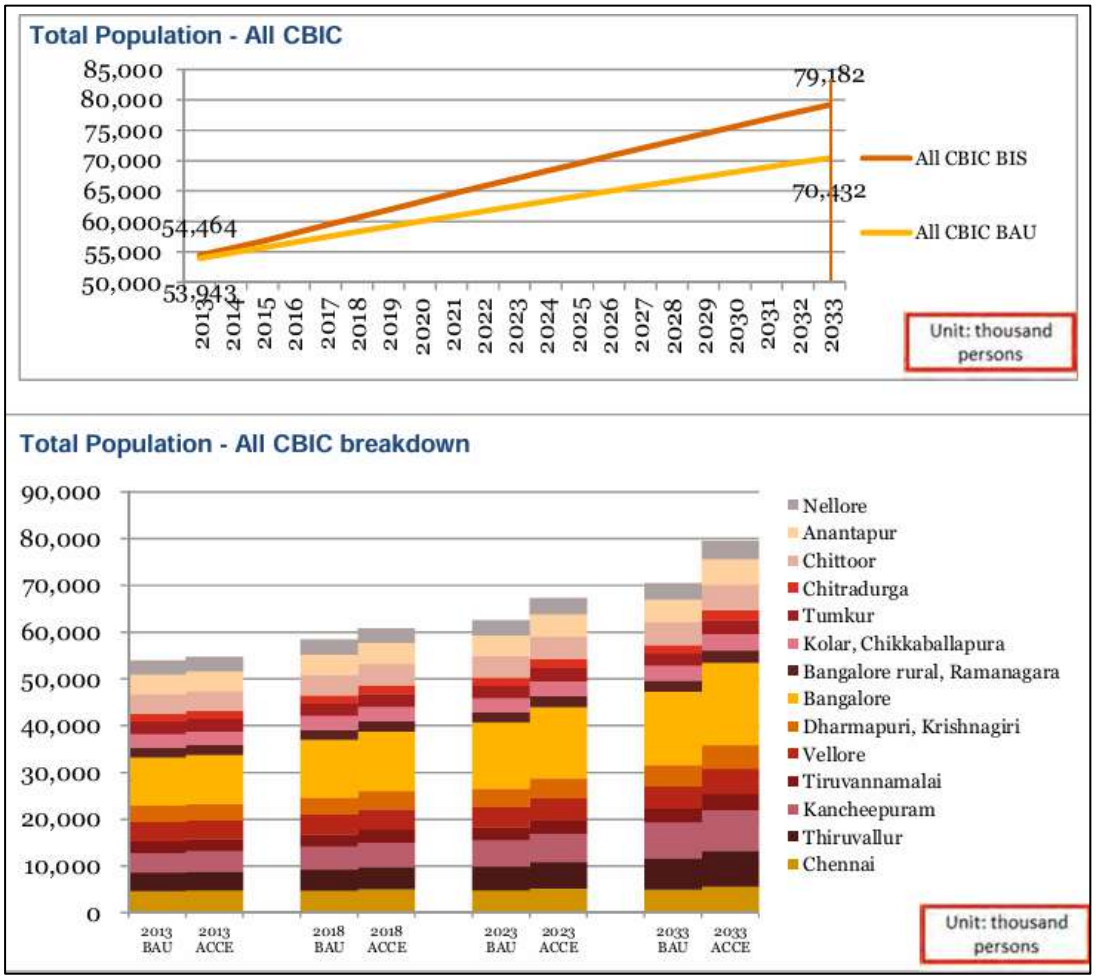
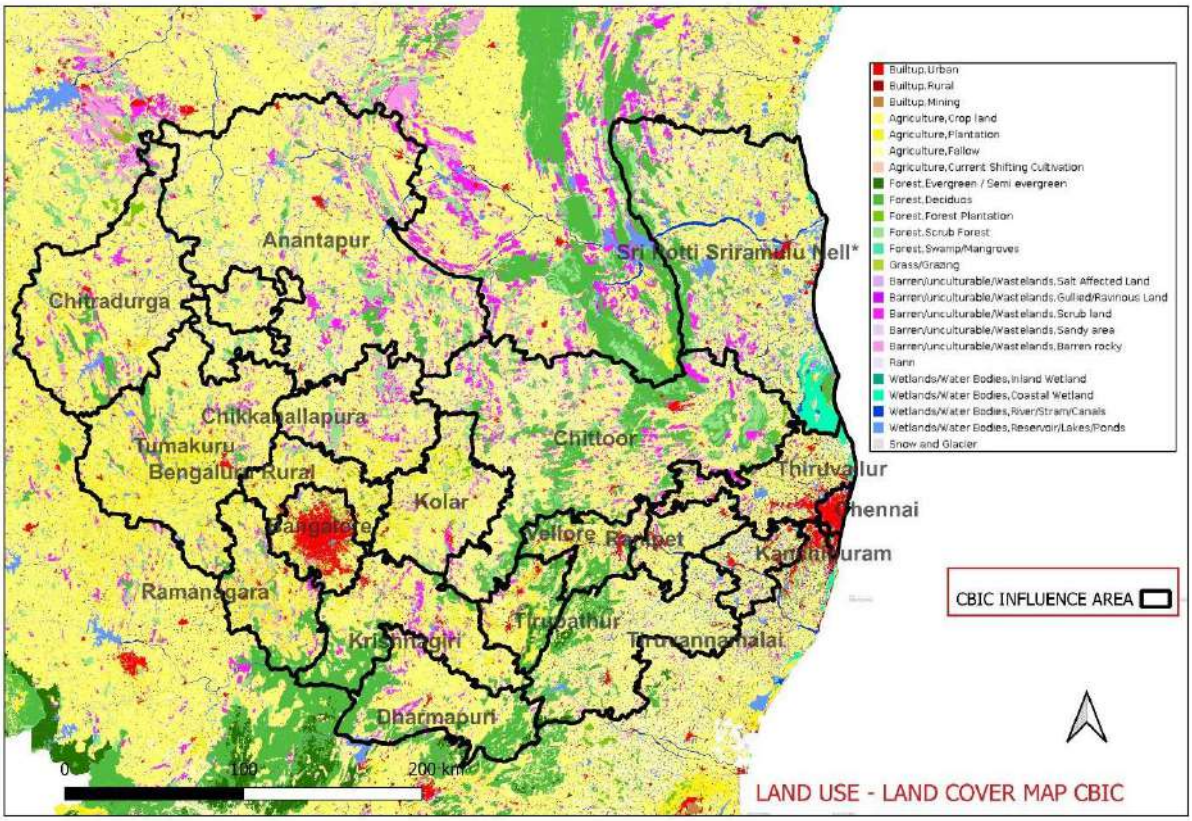


Fig No:2(a) Total Population-All CBIC;(b) Total Population-All CBIC breakdown
Source: PwC, NK Analysis, data from respective state governments

12.2 Physical characteristics of study area

12.2.1 Land cover of CBIC region



Map No:8 Land use-land cover map CBIC
Source: Author

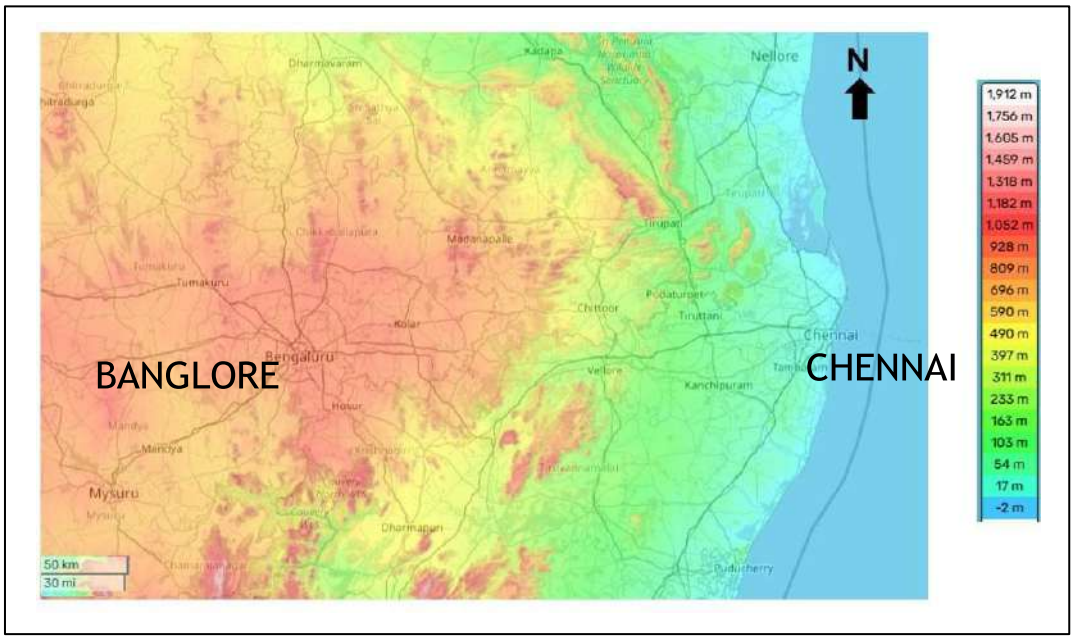
As expected, the megapolitan Bengaluru and Chennai are the mass clusters of built area in CBIC.

Other than them Kolar, Kanchipuram, Chittoor, Ranipet, Tumkuru are other important urban built areas in CBIC

In between the megapolis it is filled with agricultural crop land and forest areas including sanctuary and reserve forest.

12.2.2 Topography and protected areas of CBIC region

Along Bangalore to Chennai (west to east) the land area steeps greatly and uneven spikes in topography is seen in middle of the 2 metropolitan cities due to presence hill forests like Tiruvannamalai and Tirupati.



Map No:9 Topography of CBIC region

Source: Author

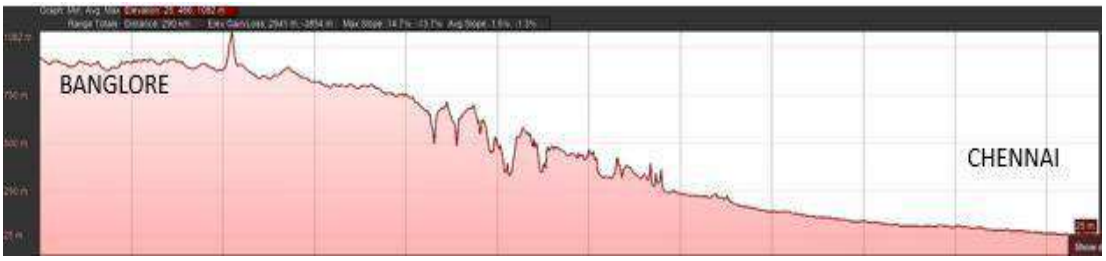
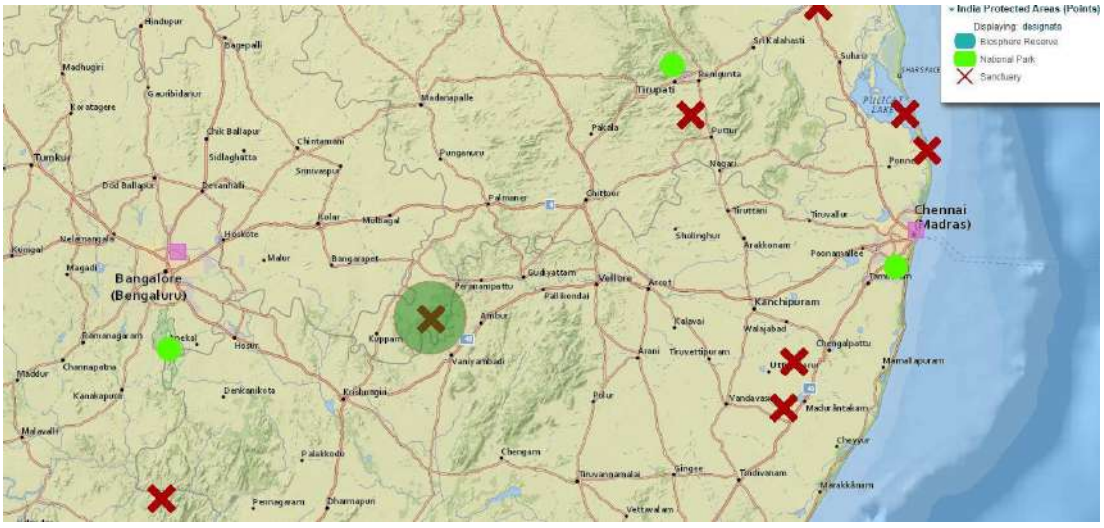


Fig No:3 Banglore- ChennaI slope analysis

Source: Gkoumas & Christou, 2020



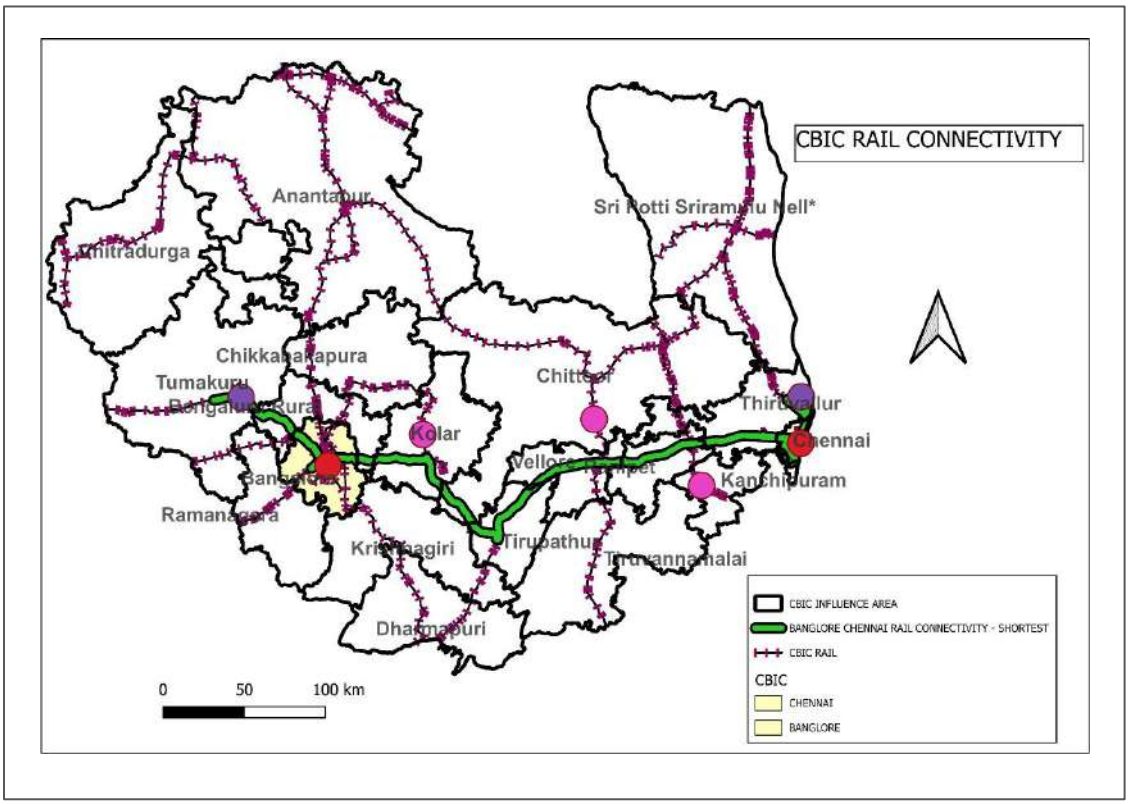
Map No:10 Protected areas of CBIC region

Source: Conservation Biology Institute/ Author

Cauvery north wildlife sanctuary and Bannerghatta national park – along south of Bengaluru and jawadhu hills vellore, koundinya wildlife sanctuary – south of Chittoor, Guindy national park and banavaram reserve forest are important protected environmental zones which may act as physical constraint in route planning.

12.3 Existing connectivity and transportation in CBIC

12.3.1 Rail connectivity and freight:



Map No:11 CBIC rail connectivity

source: Author

The railway network in the CBIC area is 2,806 Km long covering three zones of the Indian Railways– Southern Railway, South Western Railway and South-Central Railway.

The network contains 22 routes and 66-line sections, linking Chennai and Bengaluru and providing through routes to other cities like Mumbai and Hyderabad.

Capacity saturation occurs when the number of trains each way per day **exceeds 90%** of the estimated daily train capacity. Nearly 37% of this network in terms of route km has either already reached or is fast approaching capacity saturation.

The Chennai– Bengaluru route currently carries **10.5m tonnes of through-freight (2013 JICA)**. The freight includes bulk cargo (mainly coal, limes and dolomite), break bulk (foodgrains and fertilizers) and containers. It is forecasted that in the BIS case, by year 2033, the Chennai– Bengaluru route **will carry 65.2m tonnes** of freight in total in both directions.

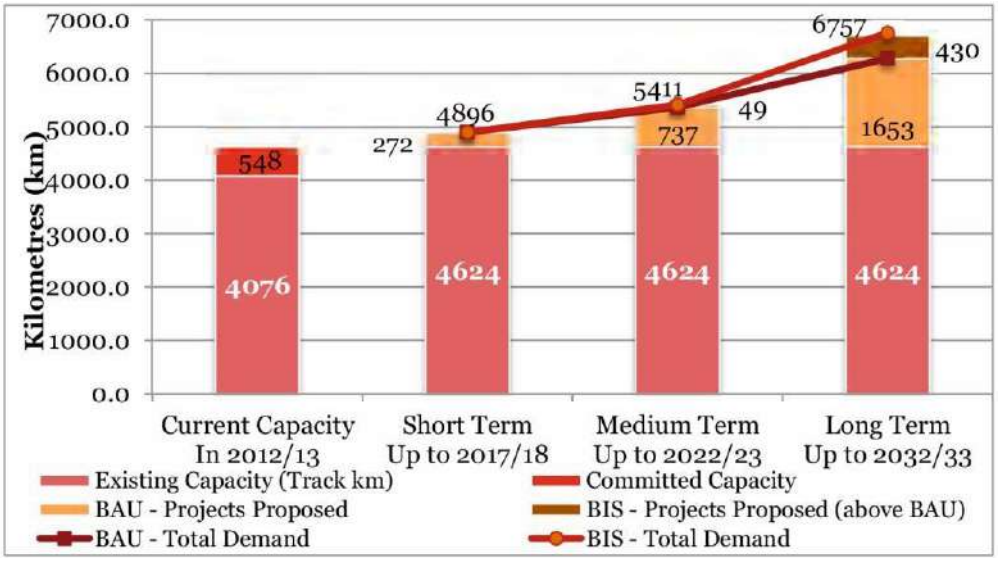


Fig No:4 Freight capacity potential

source: CBIC-Comprehensive Regional Plan report

It is expected that rail will also start carrying cars for export via Ennore. Coal from Ennore and Krishnapatnam is also expected to be a major driver of rail freight growth.

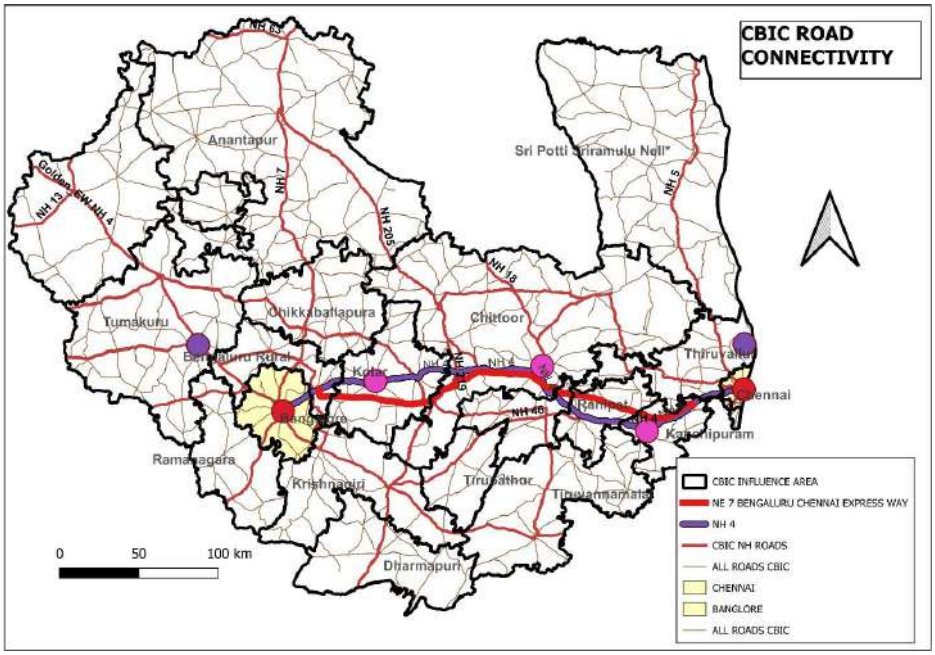
For expansion of railway route capacity, 2,133 km of additional running lines will need to be constructed over the 20-year forecast period (2013-2033) to satisfy freight and passenger demand which would be generated under the BIS case. This will include 686 km of new running lines on the Chennai– Bengaluru route and 1,447 newlines on all other routes.

Transportation via road is significant as compared to rail in the CBIC region. The modal split of cargo, mainly containers, moving for imports and exports from Chennai port is 95%(road) and the **rail is 5%.**

The current rail work is heavily dominated by passenger services and **less orientated towards freight traffic.**

Further, critical portions of the rail network in the CBIC region stands utilized at over **100% causing congestion** and limited bandwidth to increase the frequency of freight trains.

12.3.2 Road connectivity and freight:



Map No:12 CBIC Road connectivity
source: Author

Approx. 2,942 km of national highways and 5,343 km of state highways have been identified in the CBIC area + ongoing/proposed/announced projects (data from 2015)

Vehicle congestion ratio of national highway network will reach almost 50% and 90% in Karnataka state in year 2018 & 2033 respectively, In Tamil Nadu state in year 2023 & >2033, respectively, while Andhra Pradesh state will maintain less than 50% of vehicle congestion ratio even in year 2033

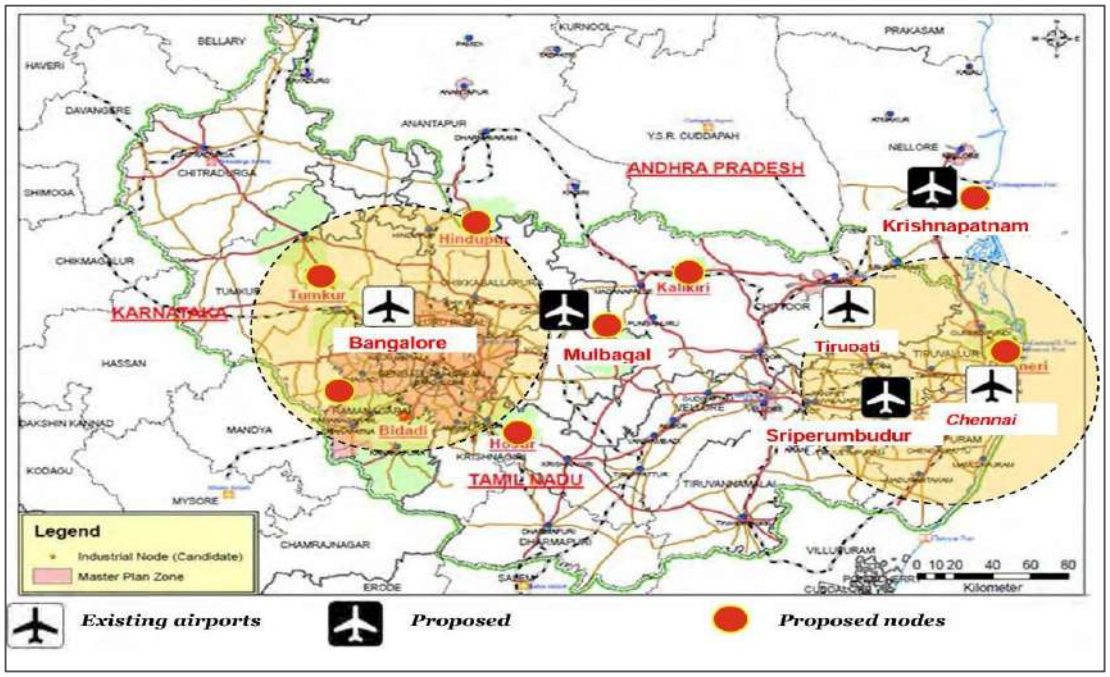
The length of National Highway 4 (NH 4) from Bangalore to Chennai is approximately 345 km. However, NH 4 has been renumbered as part of NH 48 under the new National Highway numbering system in India. This highway is a crucial segment of the Golden Quadrilateral and connects Bengaluru, Kolar, Chittoor, Ranipet, Sriperumbudur, and Chennai. It is a four-lane to six-lane highway

Bangalore-Chennai Expressway (NE-7): This access-controlled, signal-free expressway has been accorded special priority under the CBIC project(260-kilometer). As of December 2024, the 71-kilometer stretch of the expressway in Karnataka is open and toll-free. The entire expressway is expected to be fully operational by August 2025.

Transportation via road is significant as compared to rail in the CBIC region. The modal split of cargo, mainly containers, moving for imports and exports from Chennai port is 95%(road) and the **rail is 5%.**

- Dedicated Freight Corridor: Plans are underway
- HSR: Discussion stage -But this hyperloop is planned to replace the above options.

12.3.3 Air connectivity and freight:



Map No:13CBIC Air connectivity

source: CBIC-Comprehensive Regional Plan repor

Chennai International Airport (MAA):

- January 2025: Handled approximately 28,606.7 metric tons of cargo, down from 32,214.3 metric tons in December 2024.
- Fiscal Year 2024: Handled over 340,000 metric tons of freight, comprising more than 258,000 metric tons of international freight and over 81,000 metric tons of domestic freight.
- In the fiscal year 2020, both Bengaluru and Chennai airports each accounted for approximately 11% of India's total cargo traffic.

Bengaluru's Kempegowda International Airport

- In the financial year 2023-2024 (FY24), (BLR) handled a total cargo volume of 439,524 metric tonnes, marking a 7.1% increase compared to the previous fiscal year.
- This comprised 266,186 metric tonnes of international cargo (a 4% year-on-year growth) and 173,338 metric tonnes of domestic cargo (a robust 13% increase)

source: Global Economic Insights, Statista, ICRA, freight trends global

Dwell time of cargo import and export

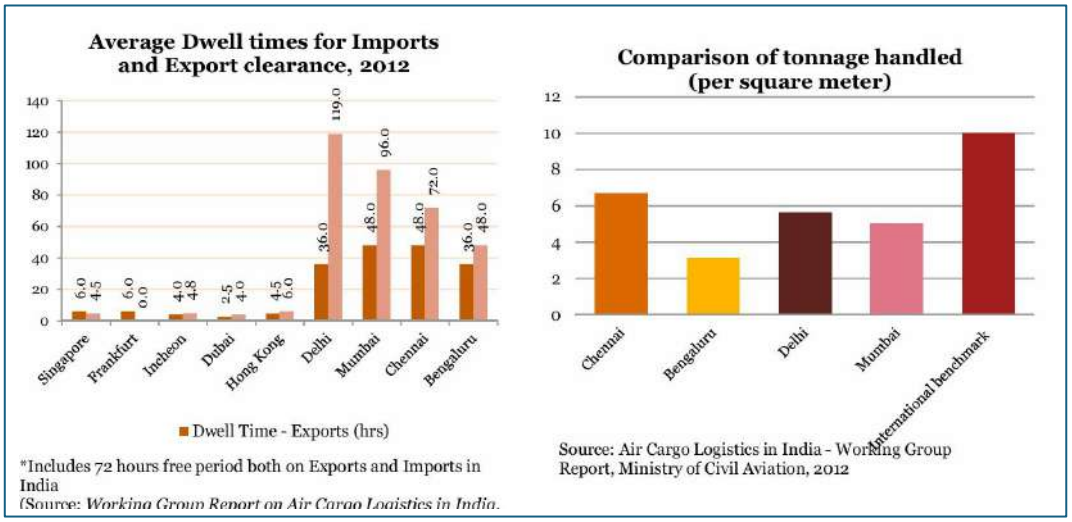


Fig No:5 (a) Dwell time of cargo import;(b) Comparison of tonnage handled
source: CBIC-Comprehensive Regional Plan report

The dwell time of export cargo in Chennai airport is 48 hours compared to around 4.5 to 6 hours in the airports at Hong Kong, Singapore and Frankfurt.

A comparison of the average tonnage handled per square meter shows that Chennai and Bangalore handle about 6.6 and 3.14 tonnes per square meter respectively. This when compared to international benchmark of 10 tonnes per square meter is very low.

12.3.4 Required freight mode for various industries

- ❖ Auto, electrical machinery, metallurgy, computer electronics – require mass freight transport service- rail
- ❖ Chemical, pharma and food processing industries need airways due to its time sensitive quality.
- ❖ Hyperloop is planned to serve these industries that require constant deliverable like electronics and time sensitive goods.

Industries	Water	Power	Road connectivity	Rail connectivity	Ports	Airports
Metallurgy	◔	●	●	●	◐	◔
Medical equipment	◔	●	◔	◔	◔	◔
Food processing	◔	◐	◔	◔	◐	◔
Textiles	●	◔	◔	◔	●	◔
Electrical machinery	◐	◔	◔	◔	◔	◔
Machinery	◐	◔	◔	◔	◔	◔
Chemicals	●	●	●	◐	◔	◔
Pharma	●	●	●	◐	◔	◔
Auto	◔	●	◔	●	◔	◔
Computer, electronics	◔	●	◔	◔	◐	◐

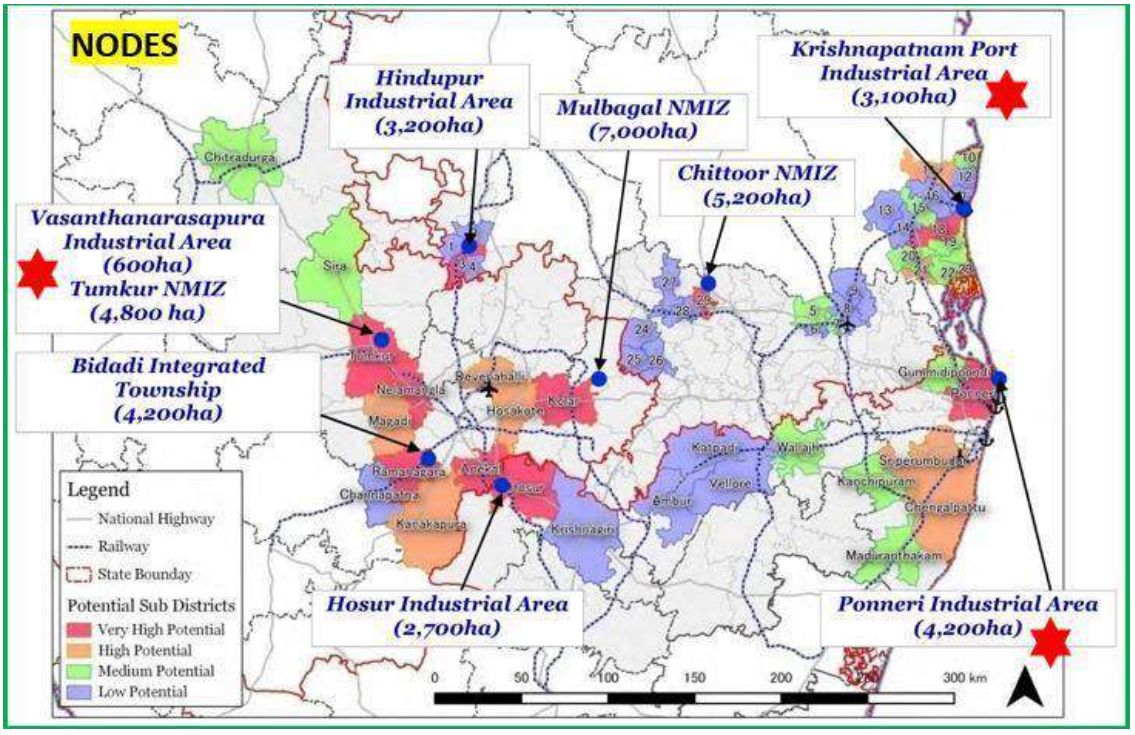
Importance: ● Critical ◔ High ◐ Medium ◔ Low

Fig No:6 Required freight mode for various industries
source: CBIC-Comprehensive Regional Plan report

12.4 Economic parameters

12.4 .1Key Industrial Nodes:

8 potential nodes (3 selected) identified for development in CBIC

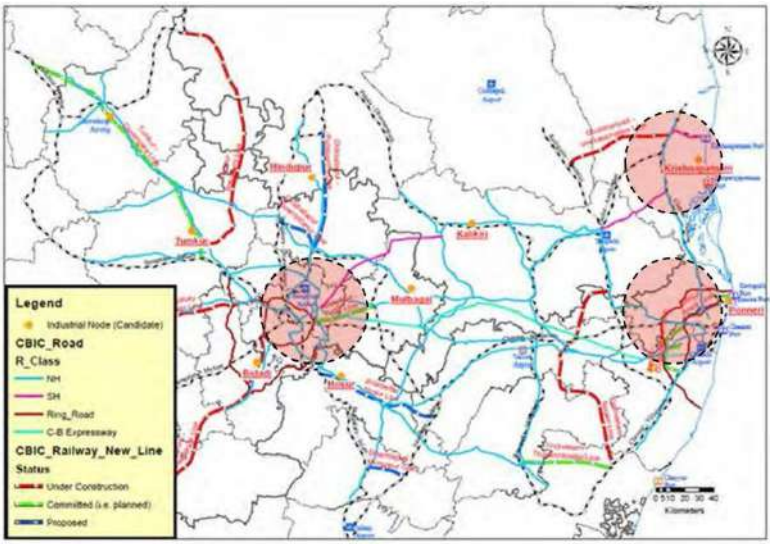


Map No:14 8 potential nodes (3 selected) identified for development in CBIC
source: CBIC-Comprehensive Regional Plan report

- **Ponneri (Tamil Nadu):** Focus on engineering components manufacturing, logistics, and warehousing.

- **Tumkur (Karnataka):** Emphasis on aerospace, information technology, and biotechnology sectors.
- **Krishnapatnam (Andhra Pradesh):** Centred around port development, petrochemicals, and power sectors

12.4.2 Three Regions for Development of Logistic Parks:

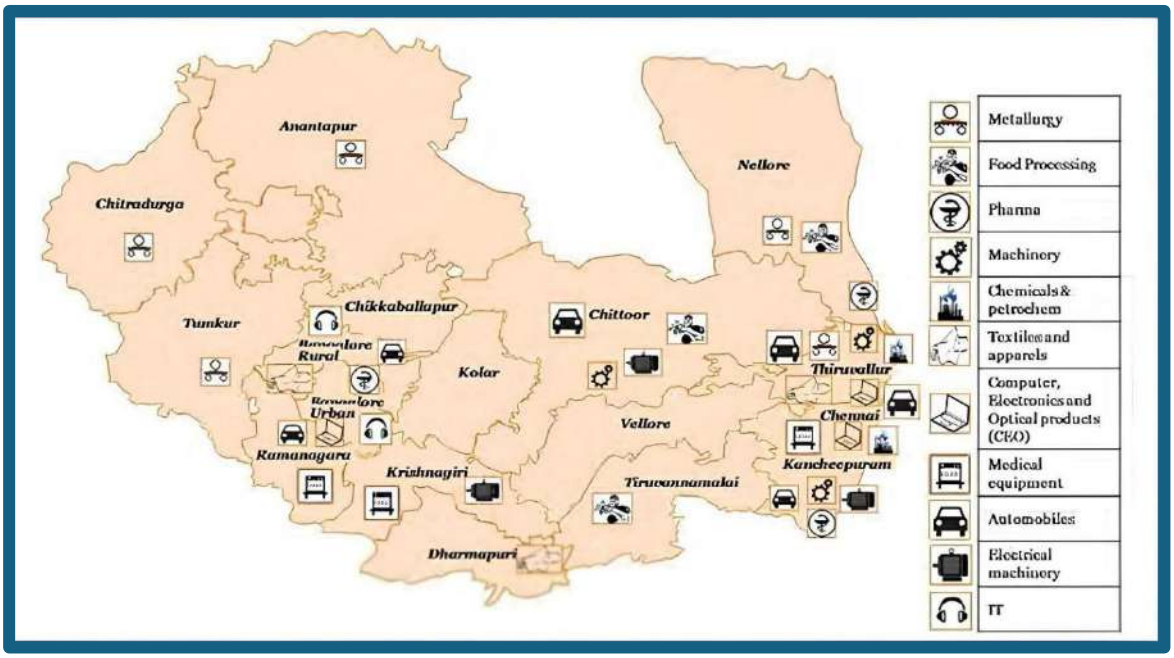


At this stage, three regions have broadly been identified for development of such logistics parks which include the Bengaluru outskirts, Chennai outskirts (near Sriperumbudur) and Krishnapatnam region.

Map No:15 Three Regions for Development of Logistic Parks

source: CBIC-Comprehensive Regional Plan report

12.4.3 Sectors and districts



Map No:16 Sectors and districts

source: CBIC-Comprehensive Regional Plan report

- Cluster advantage: Automobiles- clustered around Chennai; Corridor accounts for about 60% of India's automotive export
- IT/ITeS-about 400 of the Fortune Global 500 companies outsource their IT/ITeS/other services to firms in Bengaluru & Chennai
- Textiles- Kancheepuram hub of silk weaving and handloom industries of India
- Food Processing– strong availability of raw materials in Nellore
- Leather– Vellore accounts for about 37% of the country's export of leather products
- Bengaluru is the 4th largest technological cluster in the world after Silicon Valley, Boston and London. About 50% of MNC, R&D centres in India are based in Bengaluru
- Based on the identified key districts for investment in sectors that may require faster mass transit like hyperloop kanchipuram and chittoor has potential to be intermediate nodes along the corridor.

Sector	Output growth India (CAGR)	Key districts for investment in the corridor
Computer, electronics and optical products	15% (2006-11)	Chennai, Kancheepuram, Bengaluru urban, Bengaluru rural
Metallurgy	14% (2004-12)	Chitradurga, Anantapur, Chittoor, Nellore, Tiruvallur, Thiruvannamala, Kancheepuram, Tumkur, Bengaluru rural
Textiles and Apparel	17% (2006-11)	Tiruvallur, Dharmapuri, Kancheepuram, Bengaluru Rural, Bengaluru Urban, Chitradurga
Food processing	20% (2009-11)	Nellore, Chittoor, Tiruvannamalai, Dharmapuri
Pharmaceuticals	14% (2008-12)	Bengaluru Urban, Bengaluru Rural, Chennai, Kancheepuram, Nellore
Chemical and petrochemicals	11% (2009-13)	Bengaluru Urban, Bengaluru Rural, Ramnagara, Chennai, Tiruvallur
Electrical machinery	23% (2009-11)	Kancheepuram, Chittoor, Chennai, Bengaluru urban, Bengaluru Rural, Krishnagiri, Tiruvallur
Machinery	14% (2009-11)	
IT and financial	8% (2008-12)	Bengaluru Urban, Bengaluru Rural, Chennai

Table No:7 Sectors and districts
source: CBIC-Comprehensive Regional Plan report

12.4.4 Economic growth

By 2033-34, at an accelerated average growth rate of 12-13%, CBIC GDP is expected to growth to more than twice the size it would have grown under business-as-usual case. Manufacturing contribution shall increase from around 17-18% currently to 24-25% in 2033-34.

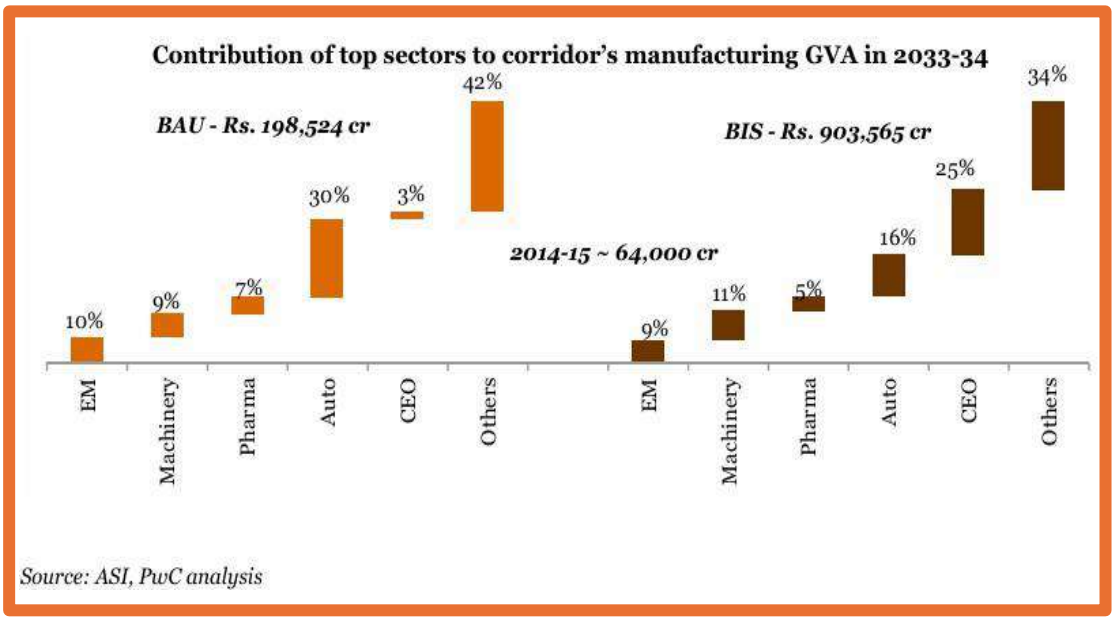


Fig No:7 Contribution of top sectors to corridor's manufacturing GVA in 2023-34
source: ASI, PwC analysis

Expected contribution of top sectors to corridor's manufacturing GVA in 2033-34

- Business As Usual (BAU)- Rs.198,524cr (JICA 2014)
- Business Induced Scenario (BIS)- Rs.903,565cr (JICA 2014)

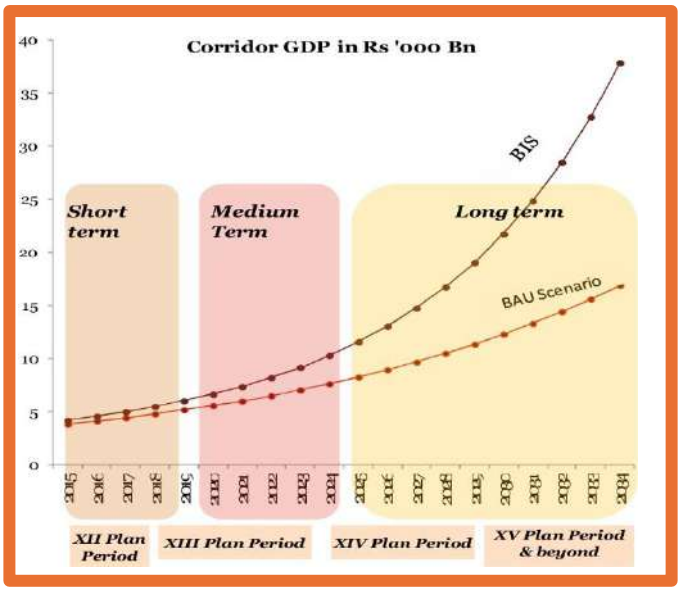


Fig No:8 Corridor GDP in Rs'000
source: CBIC-Comprehensive Regional Plan report

12.4.5 Exports

Automobiles, Textiles & Apparels, Computer, electronic and Optical products (CEO), Pharmaceuticals and Food processing sectors are likely to drive exports in the corridor.

Automobiles and CEO sector has the potential to drive over 50% of the exports from the corridor followed by Textiles & apparels, Pharmaceuticals and Food Processing.

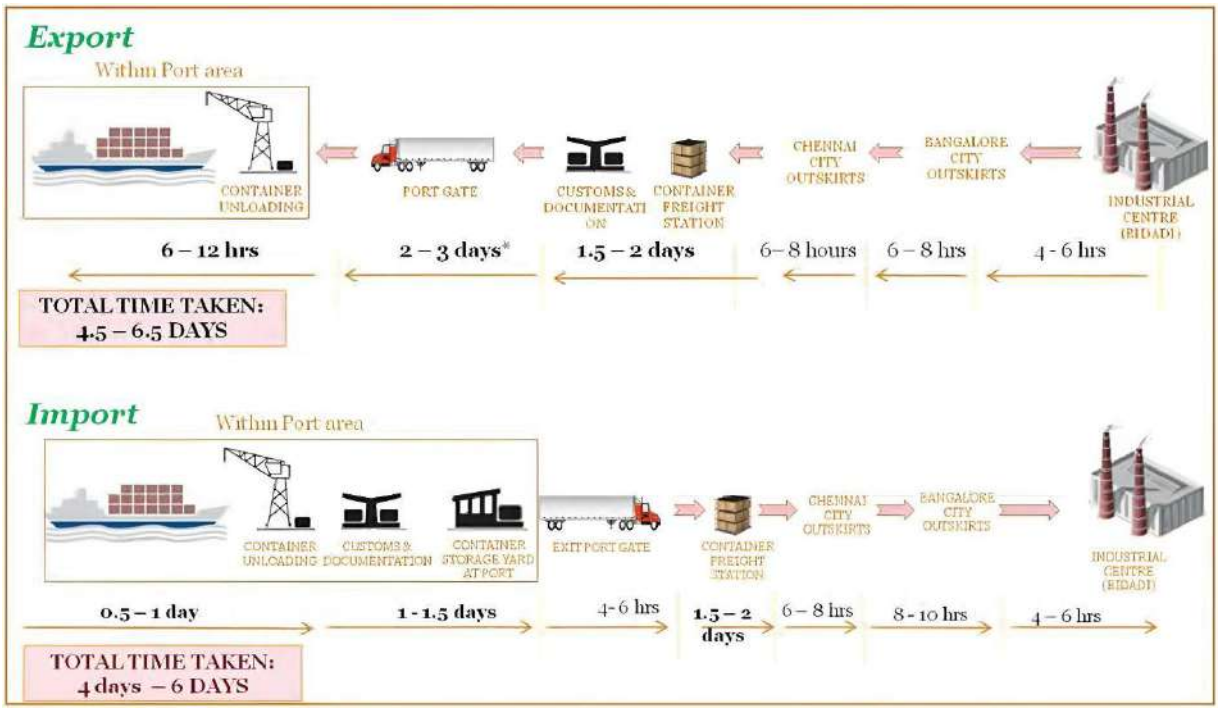


Fig No:9 Export and Import Time dealy
source: CBIC-Comprehensive Regional Plan report

- Spending almost 20 hrs in traveling between a 6-hr travel city in normal speed freight rail
- Need to reduce delay in the way that it optimises the paper time delay.

12.4.6 Ports

Manufacturing led growth expected to drive port traffic in the CBIC region to over 685 MT The ports serving the CBIC region include Chennai Port, Ennore Port, Kattupalli Port and Krishnapatnam port.

These ports jointly handled traffic of 92 MT in the year 2012-13. The manufacturing-led growth in the CBIC region is expected to be the major driver for increased traffic at ports in the CBIC region. The traffic at the ports is expected to reach 423 MT in the BAU scenario and to over 685 MT in the BIS case by the year 2033.

This increased cargo traffic at the ports in the CBIC region are likely be comprised of the major commodity segments such as bulk and containers collectively accounting for over 75% of the total traffic at the ports in 2033 in the BIS case.

12.4.7 Container Freight Stations

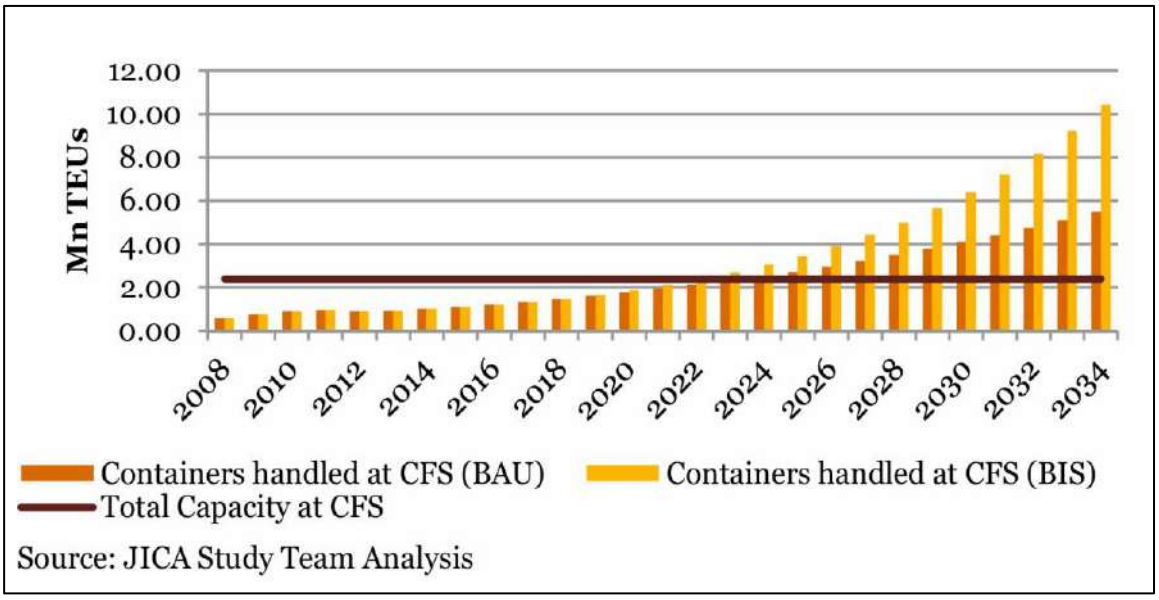


Fig No:10 Containers handled at CFS(BIS)

source: CBIC-Comprehensive Regional Plan report

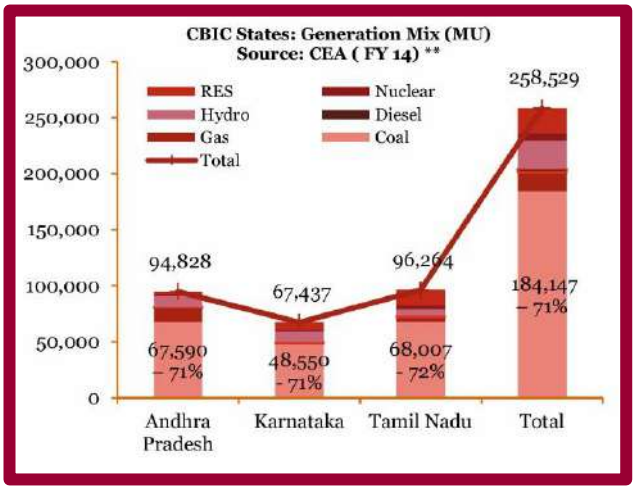
The major logistics infrastructure in the corridor region comprises of and the Inland Container Depots (located in and around Chennai) and one Inland Container Depot at Whitefield, Bengaluru.

The modal split of the container traffic to and from Chennai Port comprises of 95% of container movement by road and only 5% by rail.

At present, most of the 26 Container Freight Stations located in and around Chennai handle only around 36% of the total export container traffic movement by road to Chennai Port while the remaining container traffic comprises of factory stuffed containers.

At this stage, three regions have broadly been identified for development of such logistics parks which include the Bengaluru outskirts, Chennai outskirts (near Sriperumbudur) and Krishnapatnam region.

12.4.8 Energy generation:



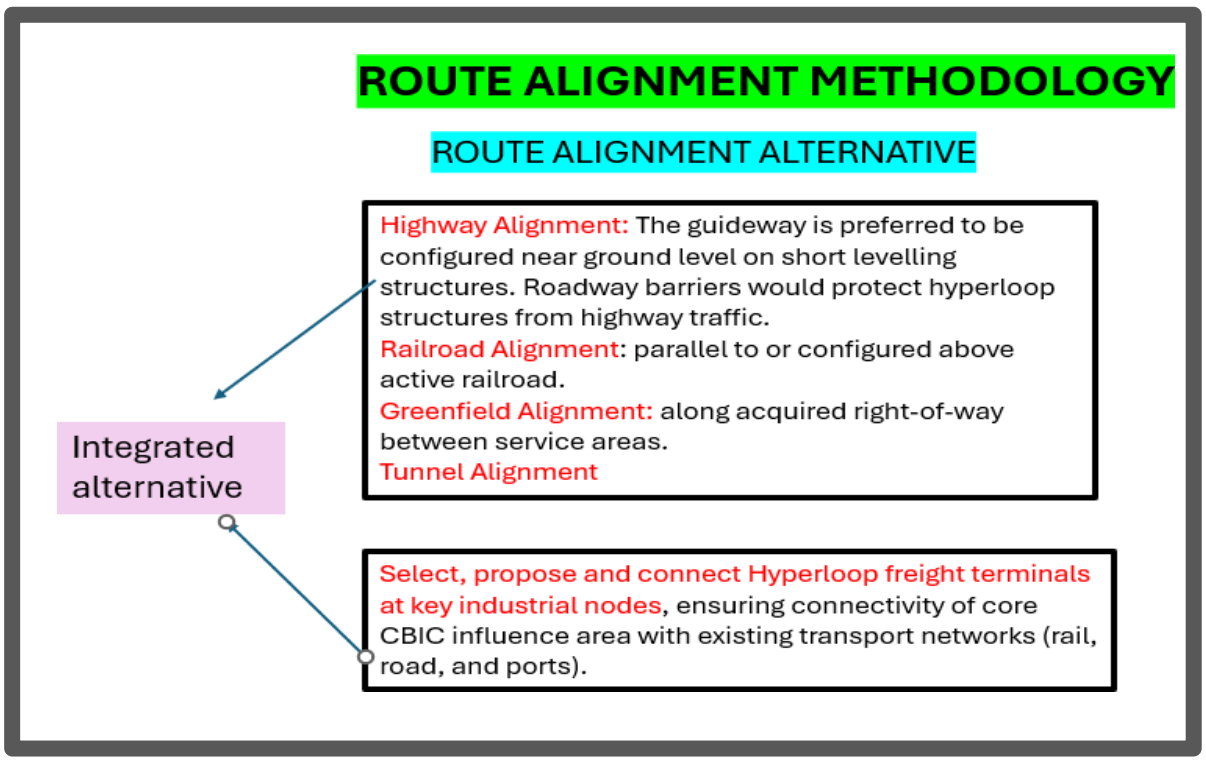
Fuel source (MW)	Potential	Mid-term 2022	Long-term 2032	Agency for capacity addition
Wind	26,029	16,387	20,484	
Solar	118,000	3,300	5,000	
Biomass	3,124	1,771	2,214	
Small Hydro	2,926	1,695	2,119	
Waste to Heat	425	250	350	
RES Total	150,504	23,404	30,167	STATE
% of potential		16%	20%	
Hydro	30,980	14,891	22,336	STATE
% of potential		48%	72%	
Nuclear	20,000	4,055	8,555	CENTRAL
% of potential		13%	28%	
Gas	45,000	7,200	10,000	STATE
% of potential		16%	22%	
Coal - Balance		**6,650**	**73,598**	**MIX**

Fig No:11 (a)CBIC States generation mix;
(b)Fuel source & potential
source: CBIC-Comprehensive Regional Plan report

CHAPTER 3 -PROPOSAL

13. Route planning and alignment proposal

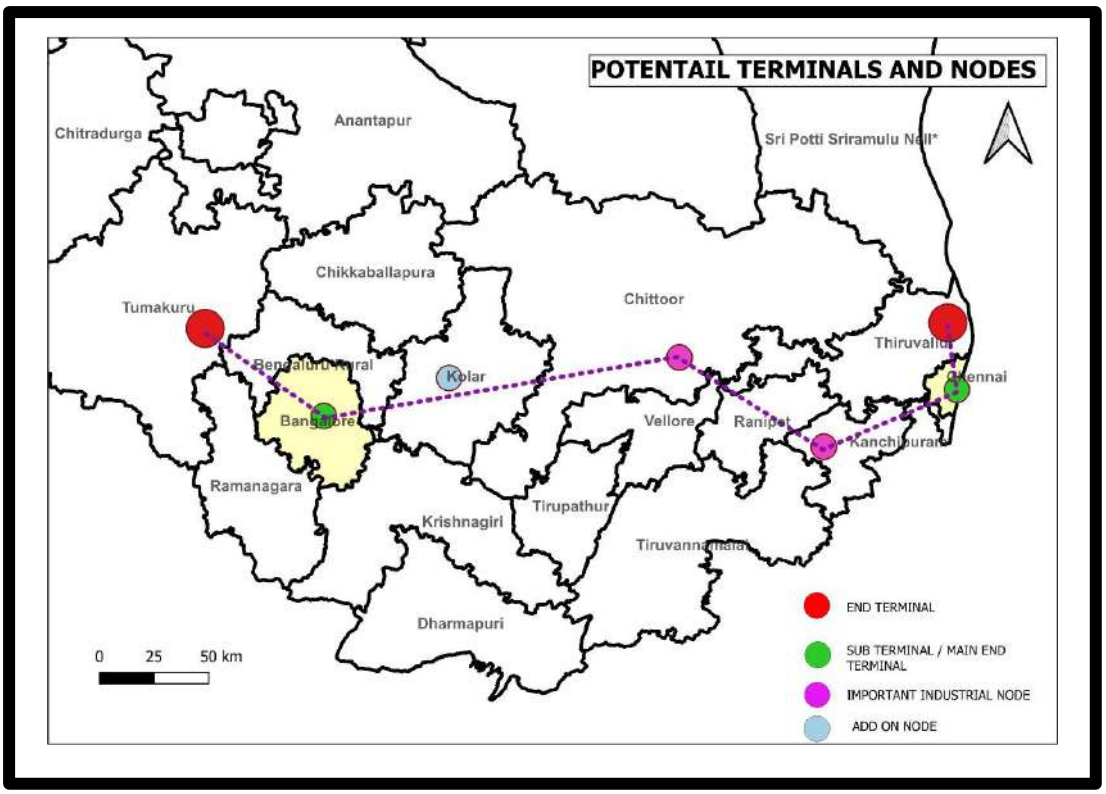
13.1 Proposal methodology and techniques



ROUTE ALIGNMENT GOALS

1. Optimize alignment to maximize the productivity of CBIC
2. Include key industrial zones that require alternate to rail transport, time sensitive goods production nodes, rail/ air travel required industries and Port connectivity required industries
3. Minimize travel distance – BEST/KEY SHORT ROUTE – low cost
4. More toward straight lined route – avoid curves, angles- based on turning radius
5. Align along with existing Road or Rail corridor
6. Avoid more greenfield route
7. Avoid natural/ other constraints
8. Avoid tunnels, urban areas

13.2 Terminal and nodes:



Map No:17 Potential terminals &nodes
source: Author

IMPORTANT NODES TO BE INCLUDED

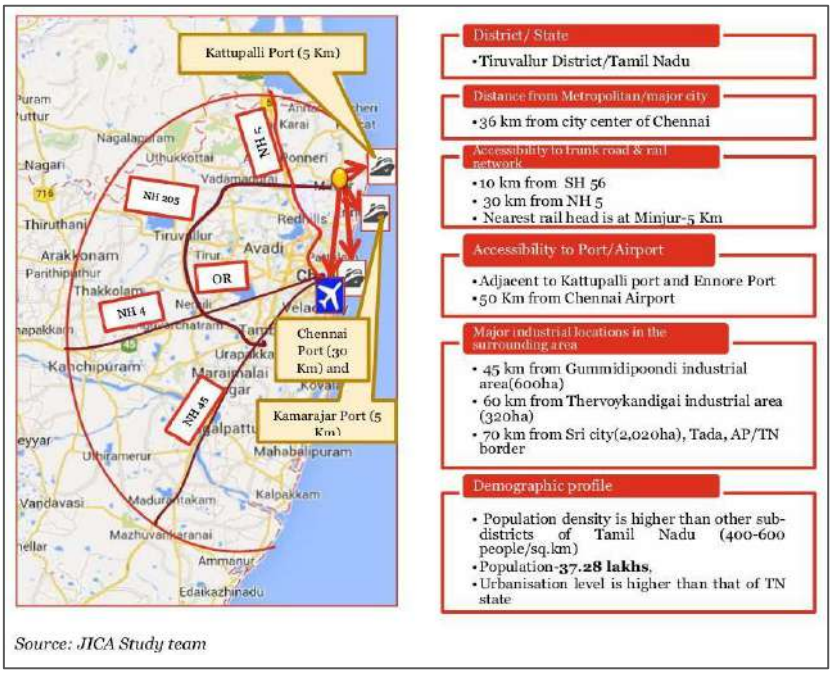
- **Ponneri (Terminal)**
- Chennai (sub terminal)
- Kanchipuram
- Chittoor
- Kolar and Mulbagal
- Bengaluru urban and rural (sub terminal)
- **Tumkuru (Terminal)**

❖ Ponneri and Tumkuru industrial is planned and developed to serve the CBIC area in transportation, as logistics hub and provide other industrial development services such storage and distribution network and etc. So, they have the best potential to function as terminal to the hyperloop freight corridor.

❖ Whereas Chennai and Bengaluru city core function as sub terminal with their local station and loading/unloading infrastructure to directly support megapolises industrial activities.

❖ Kanchipuram (with support of Arakkonam node) and Kolar function as node of cargo transfer. Chittoor would be central intermediate cargo transfer hub as an inclusive of state Andra Pradesh influence area in CBIC

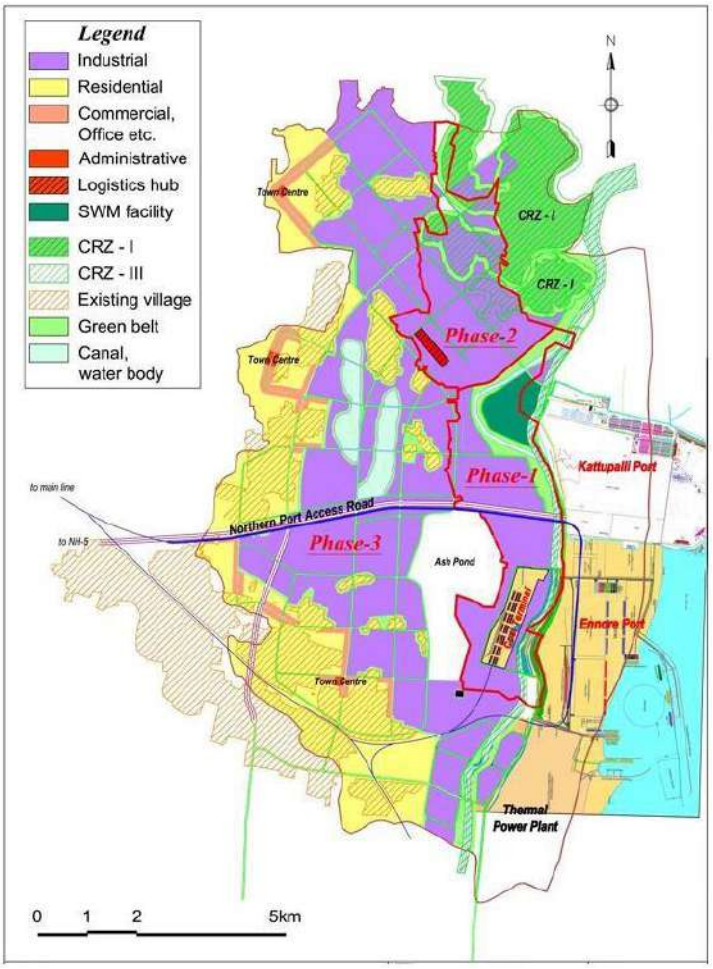
Map No:18 Ponneri node



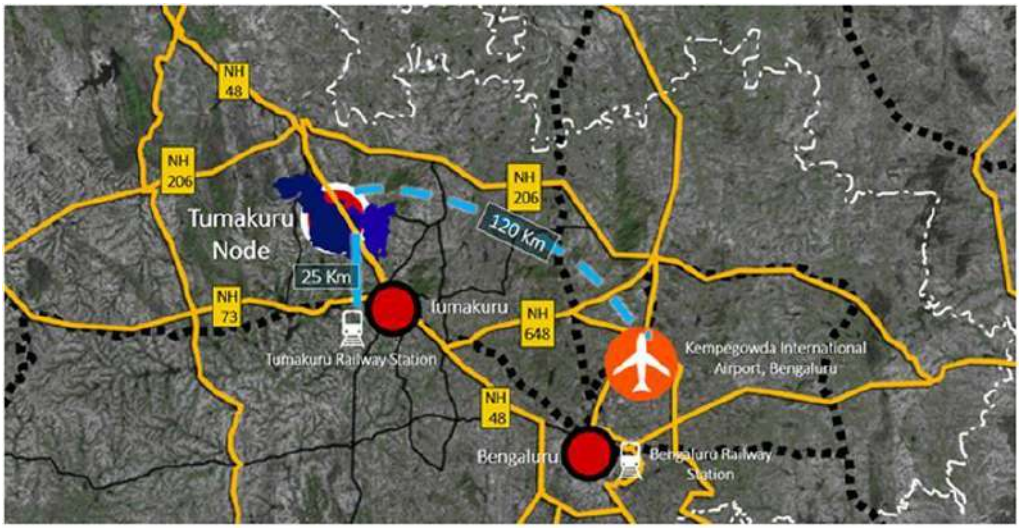
Source: JICA Study team

Map No:19 Ponneri node Development Plan – under CBIC

Source: JICA Ponneri development plan

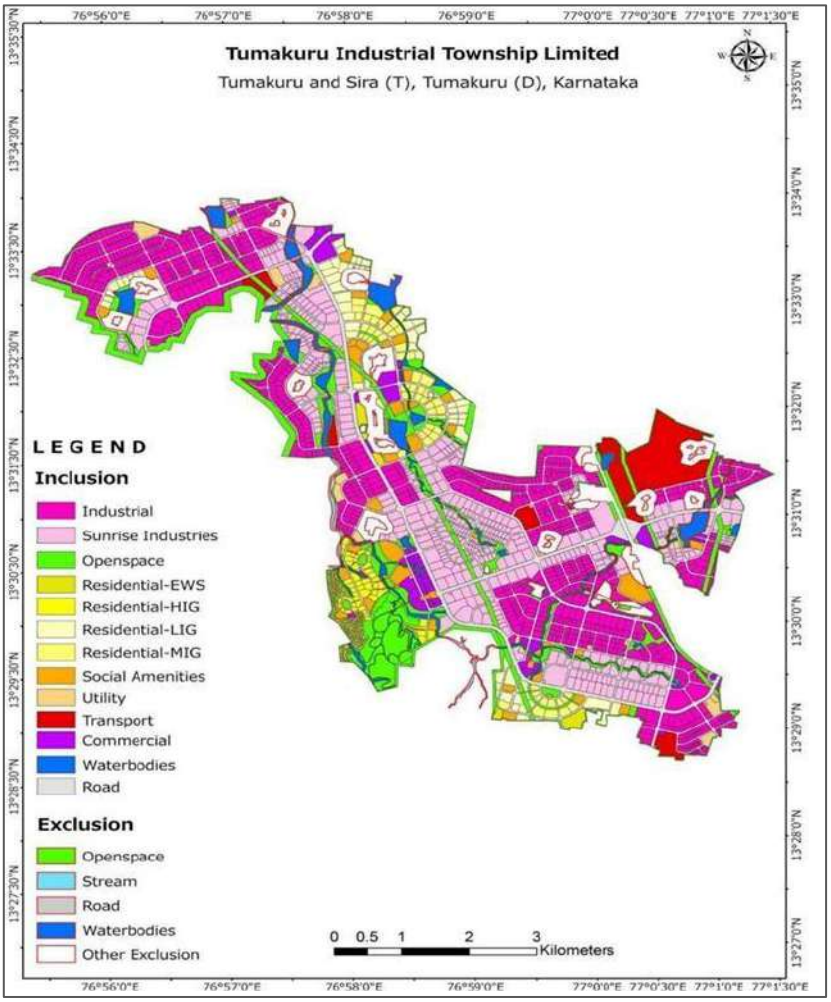


Map No:20 Tumkuru node

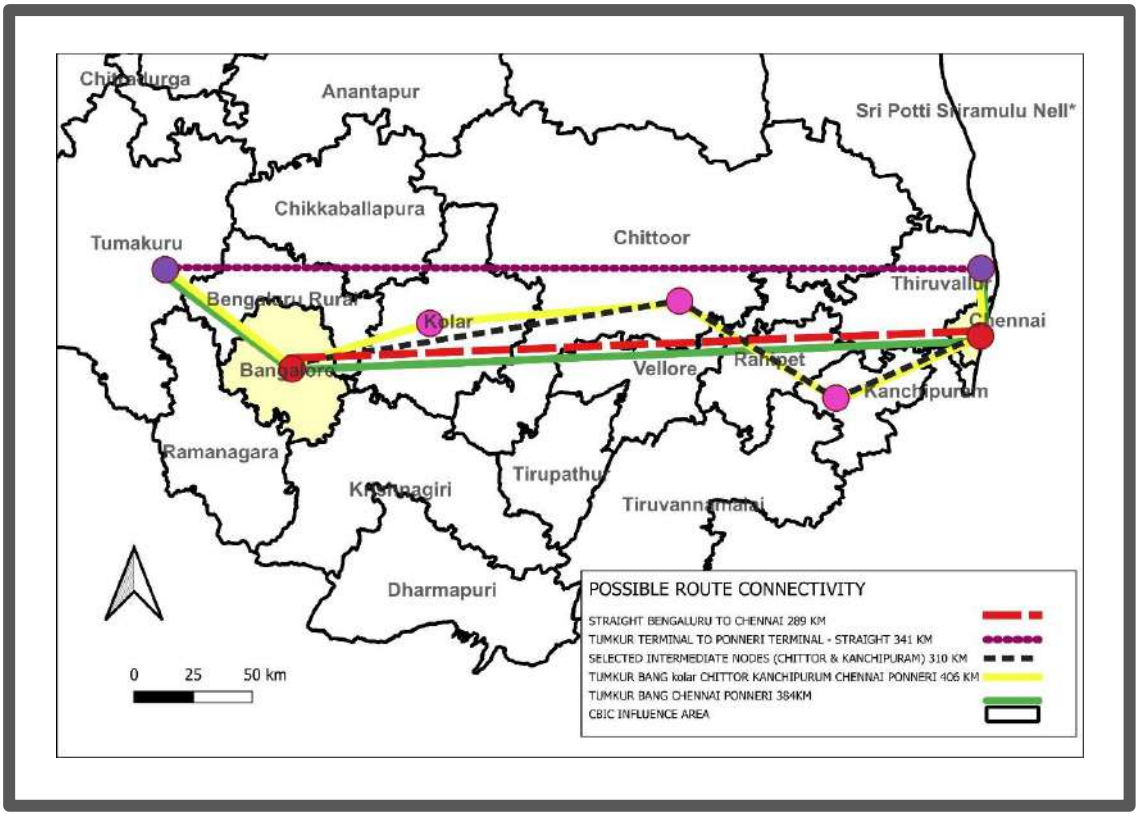


Map No:21 Tumkuru node master plan of CBIC

Source: JICA



13.3 Possible straight line route connectivity:



Map No:22 Possible route connectivity
source: Author

Route straight connectivity	**Distance**
Straight Bengaluru to Chennai	289 km
Straight Tumkuru to Ponneri terminal	341 km
Straight Tumkuru-Bengaluru-Chennai-Ponneri	384 km
Straight Bengaluru-Chittoor-Kanchipuram-Chennai (with selected nodes	310 km
Straight Tumkuru-Bengaluru-Kolar-Chittoor-Kanchipuram-Chennai-Ponneri	406 km

Table No:8 Possible route connectivity
source: Author

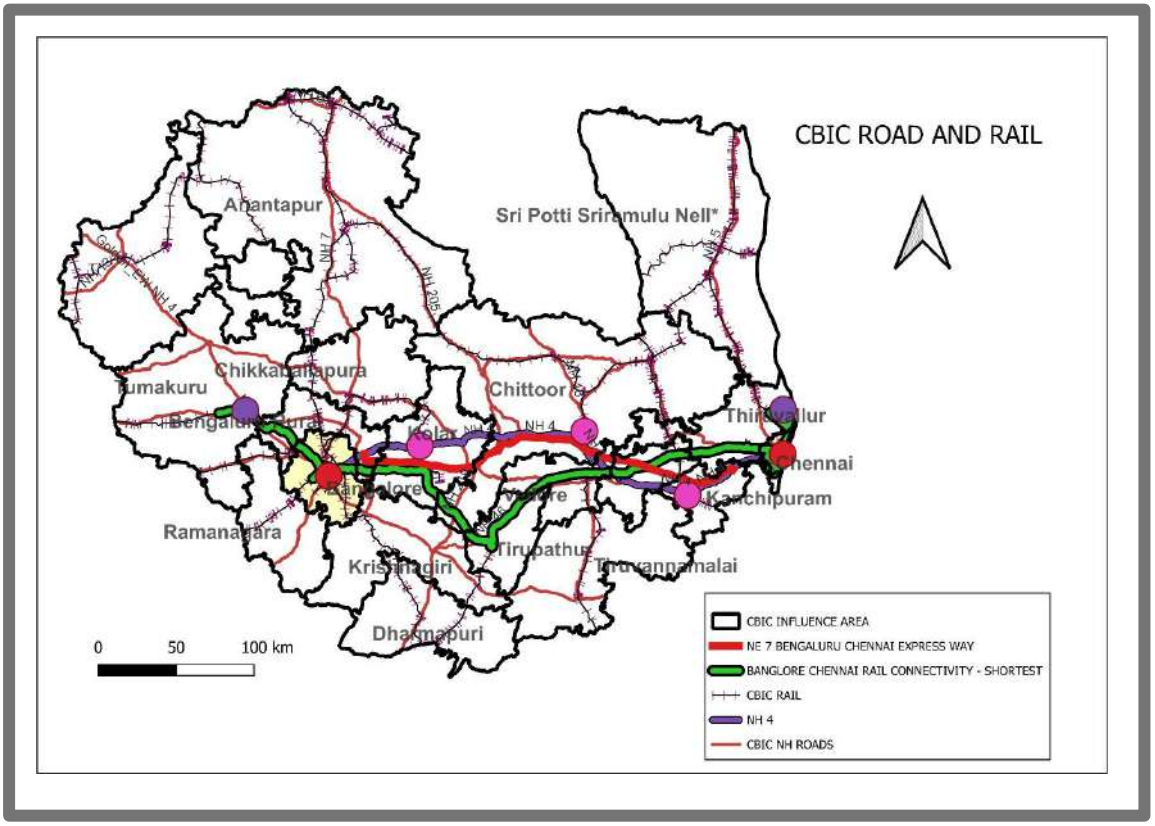
Best potential route connectivity – inclusive of all important terminals and nodes

Straight [Tumkuru-Bengaluru-Kolar-Chittoor-Kanchipuram-Chennai-Ponneri]

~406 km

13.4 Route alignment

13.4.1 Potential Shortest best available route alignment:

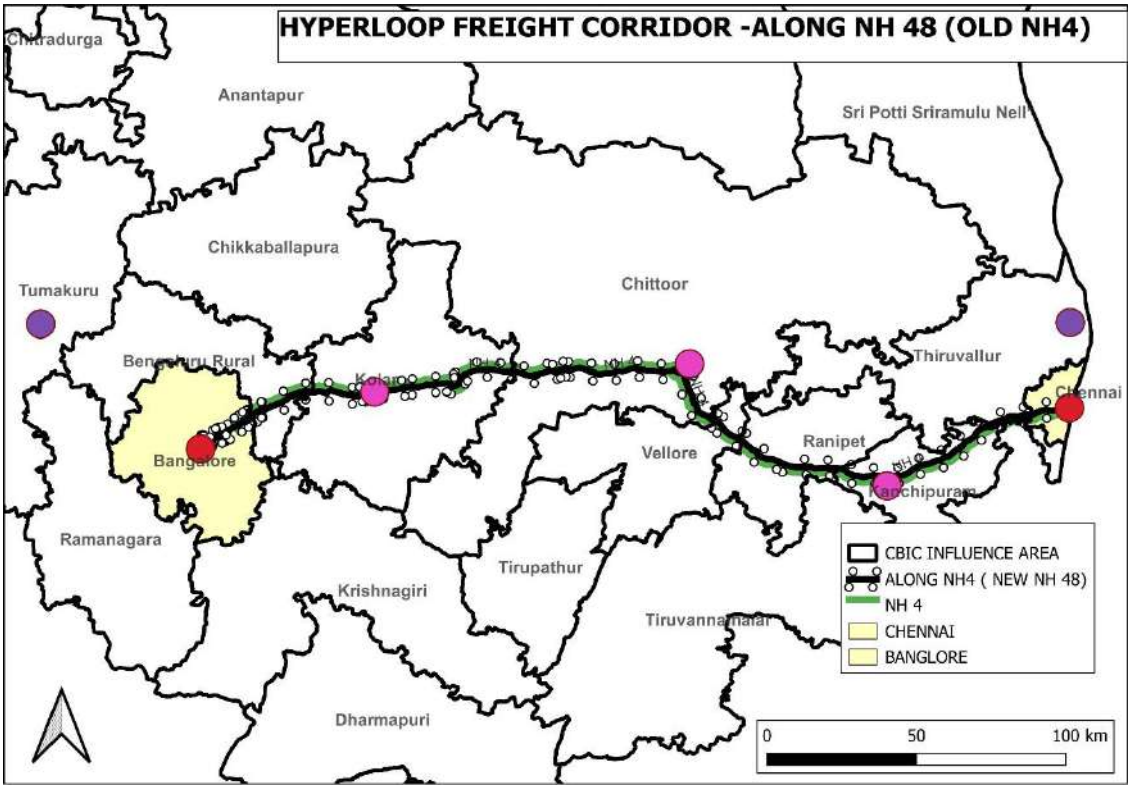


Map No:23 CBIC Road &Rail
source: Author

Route alignment (various mode & type)	**Distance**
Road alignment- Along NH 48	~345 km
Road alignment- Along NE 7 Bengaluru Chennai express way	~260 km (Outside cities)
Rail alignment – Along Bengaluru – Tirupathur – Chennai (include Tumkuru and Ponneri)	~454 km

Table No:9 CBIC Road &Rail
source: Author

13.4.2 Route alignment-Along NH 48(old NH4)

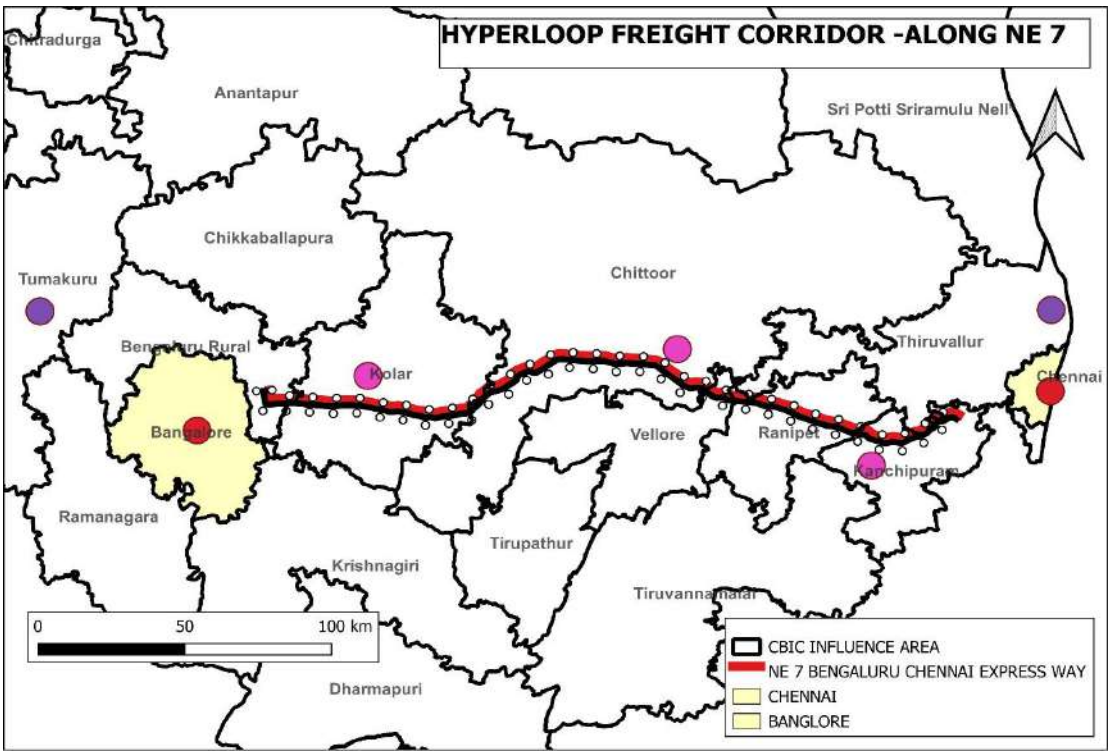


Map No:24 Route alignment-Along NH 48(old NH4)
source: Author

PARAMETERS	QUALITY/ QUANTITY
Align along	**NH 48 (OLD NH 4)**
Route – terminal/node/intermediate	Bengaluru- Kolar-Mulbagal-Chittoor-Kanchipuram-Chennai
Length	**~345 KM**
Route quality	Angular bend on Chittoor Chittoor-Kanchipuram-Chennai curve
Constraints	In urban core – less space availability along NH 48 (may need mix and fly over route)
Need for green field add on	Yes – to straighten it
Need for tunnel	No (not outside city)

Table No:10 Route alignment-Along NH 48(old NH4)
source: Author

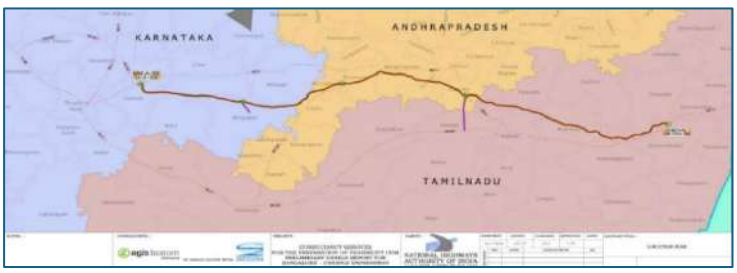
13.4.3 Route alignment-Along NE 7



Map No:25 Route alignment-Along NE 7
source: Author

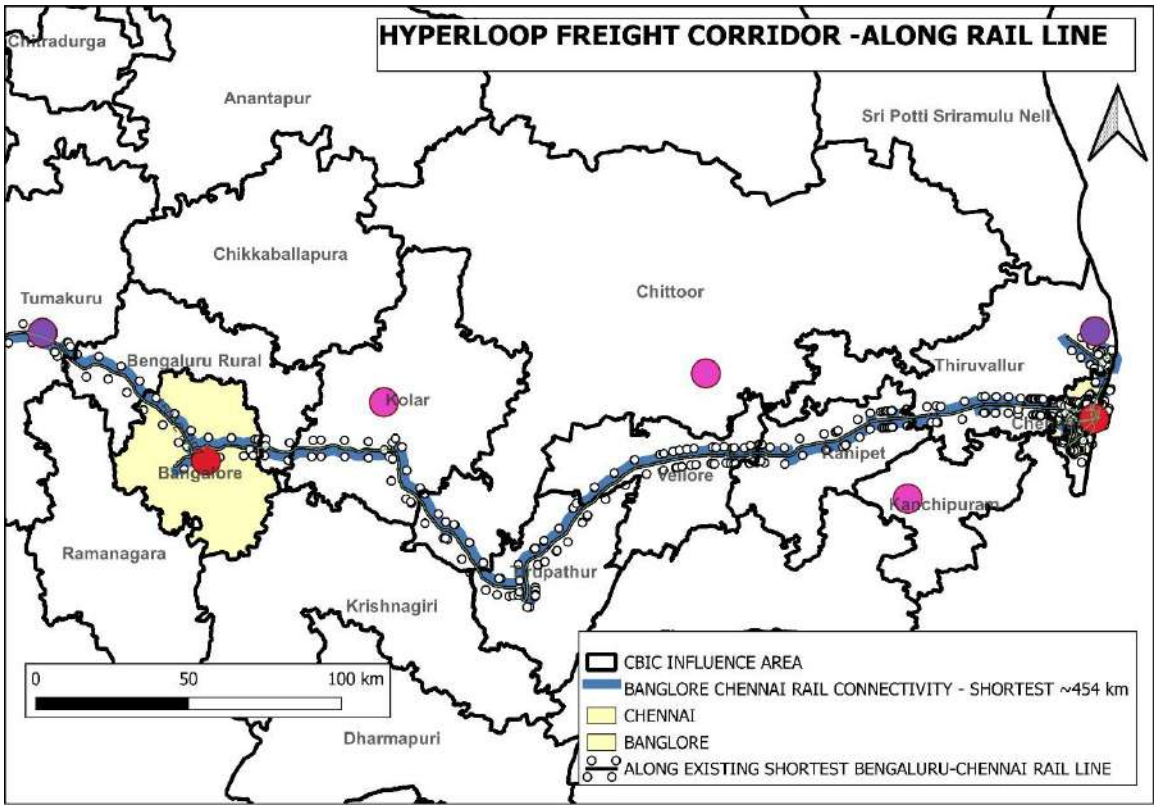
PARAMETERS	**QUALITY/ QUANTITY**
Align along	**NE 7 – BENGALURU CHENNAI EXPRESS WAY**
Route	NH 48(Bengaluru) – Hoskote-Venkatagirikota (Kolar)-Palamaner-Ranipet-Sriperumbadur-NH48(Chennai)
Length	260 KM + (20 + 20 NH48) = **~300 KM**
Route quality	Short distance angular at nodes – especially in main accelerating point - centre
Constraints	**Not straight at accelerating path**
Need for green field route	YES
Need for tunnel	NO(not outside city)

Table No :11 Route alignment-Along NE 7 (*source: Author)*



Map No:26 Proposal NE 7 by NHAI (source: NHAI)

13.4.4 Route alignment-Along Rail Line



Map No:27 Route alignment- Along Rail Line
source: Author

PARAMETERS	**QUALITY/ QUANTITY**
Align along	**BENGALURU CHENNAI RAIL LINE**
Route	**Tumkur**-Bengaluru-Kolar-Tirupathur-Vellore-Ranipet-Chennai-**Ponneri**
Length	~454 km + ~75 tumkur/ponneri = ~530km
Route quality	Not straight, too bendy, terminal connectivity
Constraints	NOT straight
Need for green field route	About 1/4th length of rail line required
Need for tunnel	NO
Advantage	Easy to enter and align in urban area

Table No:12 Route alignment- Along Rail Line
source: Author

13.5 Constraints & Advantages

13.5.1 Lack of straightness

1. In NH 48 at chittor almost right angle turn – probabily at maximum operational acceleration.

At this scenario exchange from NH 48 to NE 7 via GREEN FIELD ROUTE

Map No:28 Chittoor turn NH 48
source: NHAI/ Author

2. Rail line outside cities – zig zag- not feasible for heavy acceleration
3. NE 7 - Short distance angular at nodes – especially in main accelerating point – centre

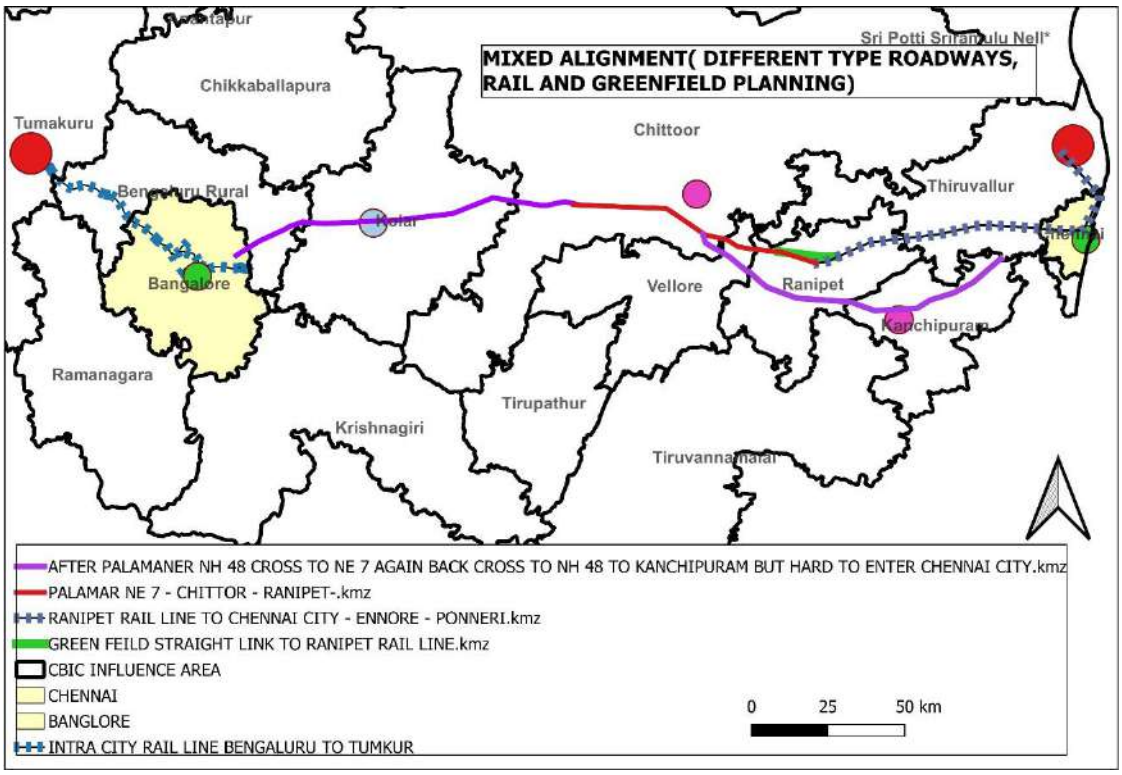
Unease entering urban centre

Along NH 48 and NE 7 need fly over and tunnel in entering heavy built-up area

Advantages:

1. Rail -easy align along existing rail line corridors
2. Rail -direct connectivity to End Terminals with already existing rail route
3. NH 48 and NE 7 complement each other in bringing out the straightest route possible with their mix

14. Mixed alignment alternatives:

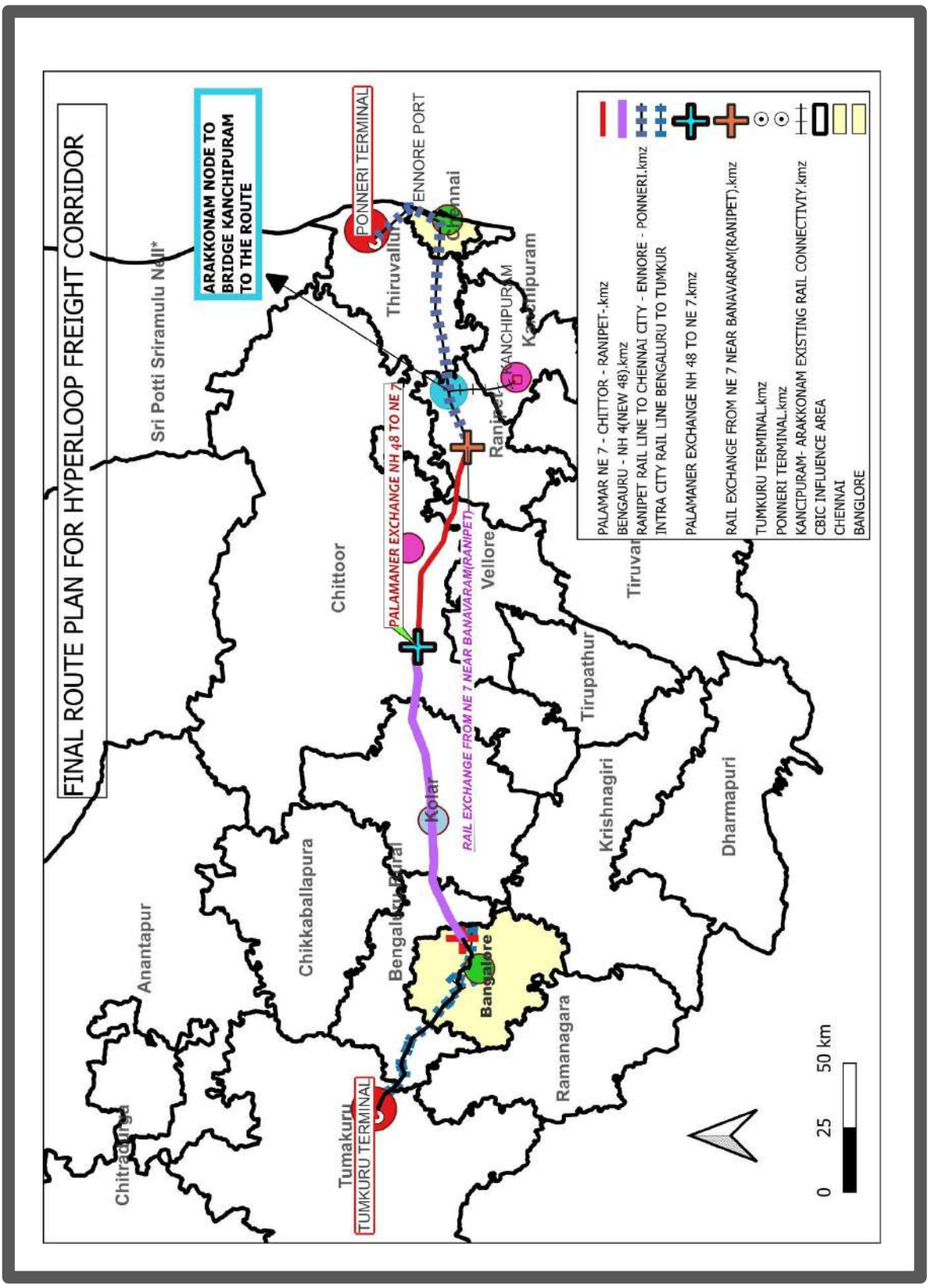


Map No:29 Mixed alignment alternatives
source: Author

Mix alignment components	Length
Tumkuru to Bengaluru (RAIL)	**~75 km**
Alternative [A] Outside Bengaluru to Chennai via Kanchipuram (NH 48-full)	**~345 km**
Alternative [B] Outside Bengaluru to till Palamaner exchange (NH 48- part)	**~119 km**
From [B] Palamaner exchange to RAIL at Ranipet: Banavaram (NE 7)	**~83 km**
From NE 7 to rail exchange to Chennai-Ponneri via Arakkonam	**~121 km**
Alternative [A] total	**420 + Chennai-Ponneri flyover along NH 48 (50km) = ~470**
Alternative [B]	**398(~400 km)**

Table No:13 Mixed alignment alternatives analysis. *source: Author*

15. Finalised route alignment plan

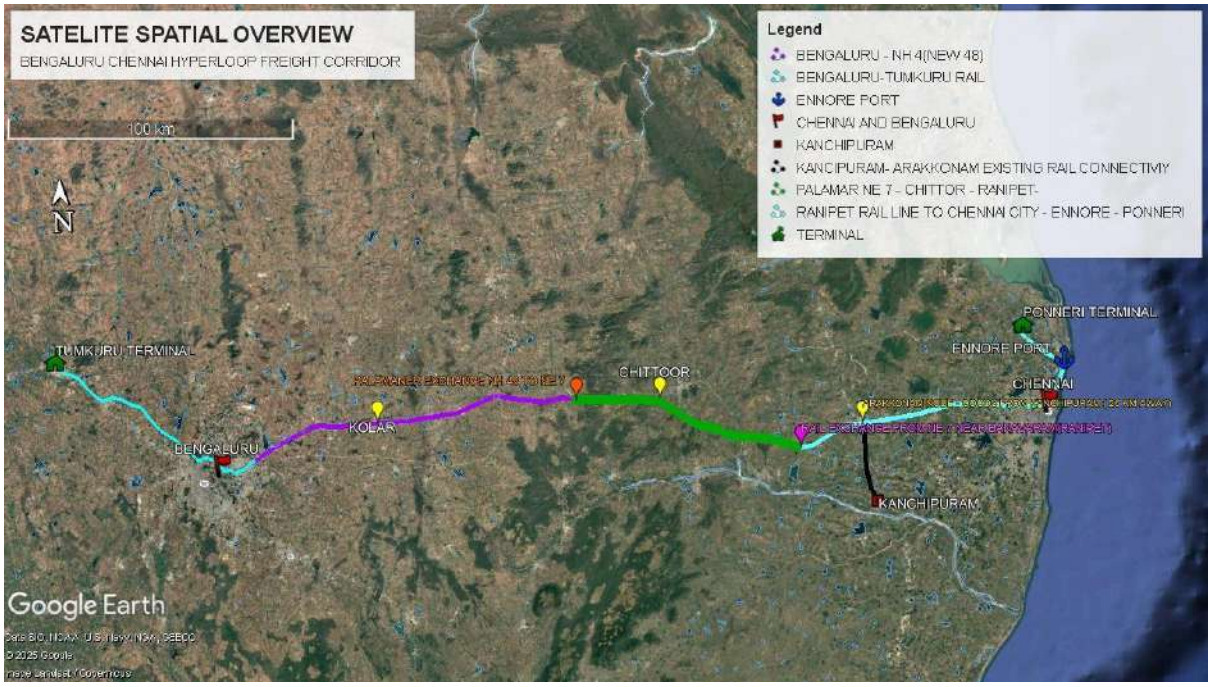


Map No:30 Final route plan hyperloop freight corridor.

15.1 Trip route and distance:

ARAKKONAM TO BENGALURU - ~ 230 KM
KOLAR TO CHENNAI ~ 240 KM
CHITTOR TO CHENNAI ~ 140KM
TUMKUR TO BANG ~ 75 KM
PONNERI TO CHENNAI ~ 30 KM
TOTAL 400 KM
CHENNAI TO BANGLORE ~ 295 KM

15.2 Satellite overview of proposed Route:

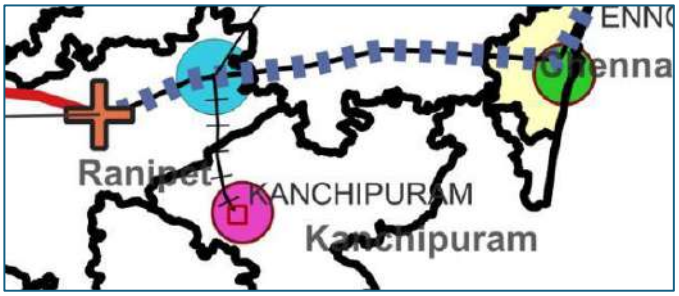


Map No:31 Satellite spatial overview
source: Author

15.3 Arakkonam and Kanchipuram rail accessory node:

KANCHIPURAM TO ARAKKONAM ~ 27 KM (by EXISTING RAIL ROUTE/ROAD)

This would able the corridor to serve Kanchipuram as cargo transfer accessory node.



Map No:32 3Arakkonam and Kanchipuram rail accessory node
source: Author

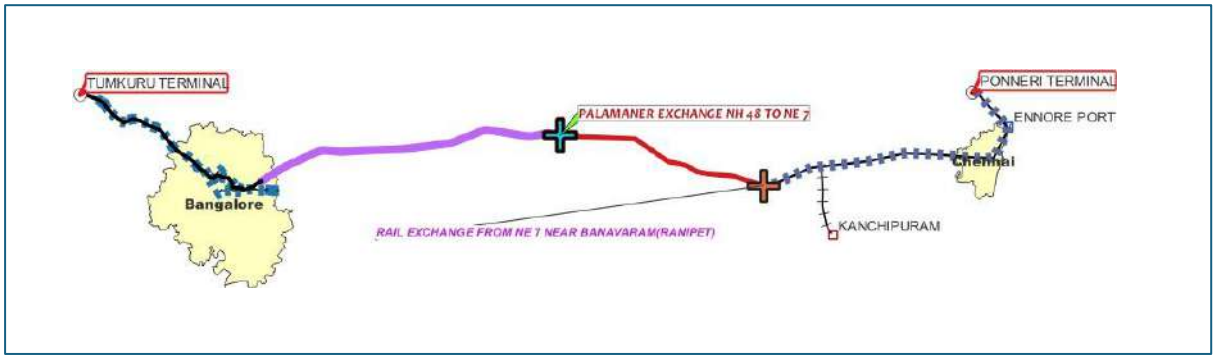
16. Conclusion:

The final route alignment with the identified and proposed terminals and nodes is the outcome of spatial-economic and alternative comparison analysis. Which covers the important industrial nodes of the CBIC region and includes the development plan proposed nodes (tumkur and ponneri) which were intended to act as logistics hubs of the CBIC region would function as end terminals of the hyperloop corridor. While Chennai and Bangalore would function as sub terminals to support the city's industries directly.

The route plan is made out as a desk appraisal which deals with several assumptions and limitations such as feasibility of hyperloop, fully based on secondary sources of data and GIS based spatial analysis excluding the ground reality.

Next and further incremental scope of this study would be enabled using this route plan as a preliminary study. This route alignment can be updated with a buffer zone method analysis based on ground reality, primary data and site study.

Hyperloop is the potential future transit mode that may provide upper hands to India with our development rate intensity, knowledge and research and youth workforce in the world competitive forum among the other developed nations as an underdog raising and shining by incorporating leapfrog development strategies in the national and regional planning formulation.



Map No:33 Route sum-up of Bangalore-Chennai hyperloop freight corridor
source: Author

17. References:

1) Bachman, J. (2015). "The Hyperloop: A High-Speed Future," *Technology Review Journal*, 12(4), pp. 67–74.
2) Comprehensive Integrated Master Plan for Chennai Bengaluru Industrial Corridor Final Report- JICA 2015
3) Davies, J., & Kim, H. (2019). "Sustainability Prospects of Hyperloop Transportation," *Environmental Science and Technology Journal*, 53(8), pp. 4420–4430.
4) González, A., & Sanz, P. (2018). "Hyperloop Transportation System: Engineering Challenges and Prospects," *Journal of Modern Transport*, 26(3), pp. 211–220.
5) Guzmán, J., & García, F. (2017). "A Study of Magnetic Levitation Systems for High-Speed Transportation," *Transportation Research Procedia*, 25, pp. 234–245.
6) Harris, T. (2017). "Economic Viability of Hyperloop Systems: Myth or Reality?" *International Journal of Transportation Economics*, 44(2), pp. 145–160.
7) Jensen, M., & Li, F. (2022). "Commercialization Barriers in Emerging Transport Technologies," *Global Mobility Review*, 15(3), pp. 140–153.
8) Johnson, D., & Baker, S. (2018). "Safety Assessment in Vacuum-Based Transportation Systems," *Journal of Risk Analysis and Management*, 21(2), pp. 134–149.
9) Kaur, P., & Singh, R. (2019). "Cost-Benefit Analysis of High-Speed Transportation Systems," *Economic Review of Transportation*, 33(1), pp. 89–104.
10) Klein, R., Novak, S., & Patel, J. (2023). "The Future of Hyperloop: Technological and Policy Innovations," *Advanced Transportation Studies*, 40(1), pp. 12–25.
11) Li, M., & Green, T. (2021). "Ecological Footprint of High-Speed Infrastructure Projects," *Journal of Environmental Impact Assessment*, 12(4), pp. 250–265
12) Liu, J., & Wang, X. (2019). "Magnetic Levitation Technologies for Hyperloop: A Review," *Advances in Mechanical Engineering*, 11(5), pp. 1–15.
13) Martínez, R., López, G., & Ahmed, S. (2020). "Regulatory Challenges for Emerging Transportation Technologies: The Case of Hyperloop," *Transportation Policy Journal*, 18(1), pp. 56–70.
14) Musk, E. (2013). *Hyperloop Alpha*. SpaceX and Tesla White Paper.

15) Nakamura, K., Patel, A., & Wong, T. (2021). "Hyperloop Economics: A Sustainable Investment?" *Global Transport Review*, 38(4), pp. 201–218.

16) National Industrial Corridor Development Corporation (NICDC) & other portals.

17) Premsagar, Smriti & Kenworthy, Jeffrey. (2022). A Critical Review of Hyperloop (Ultra-High Speed Rail) Technology: Urban and Transport Planning, Technical, Environmental, Economic, and Human Considerations. Frontiers in Sustainable Cities. 4. 10.3389/frsc.2022.8422

18) Siddharth KJ, *The Chennai-Bengaluru Industrial Corridor -A Preliminary Report*- Published by: Centre for Financial Accountability

19) Singh, A., & Zhao, L. (2020). "Energy Efficiency in Next-Generation Transport: A Hyperloop Case Study," *Renewable Energy Reviews*, 29(3), pp. 101–110.

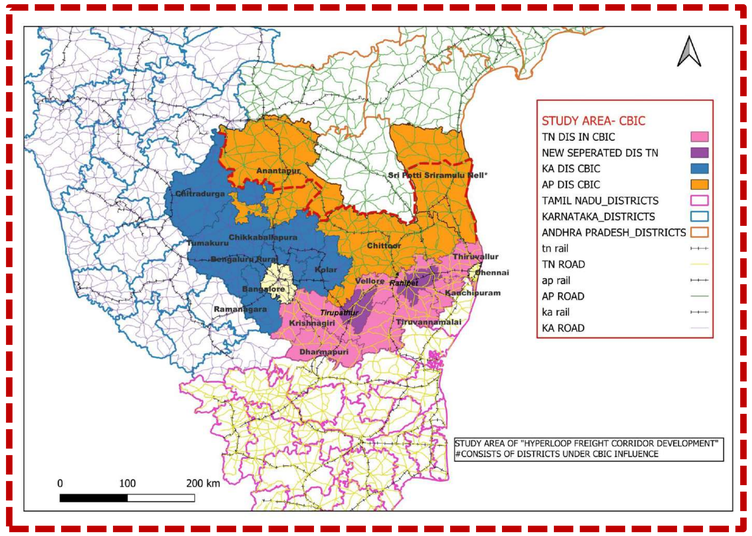
20) Thompson, L. (2020). "Virgin Hyperloop's Test Achievements and Technological Insights," *Innovations in Transport Technology*, 34(2), pp. 89–97.

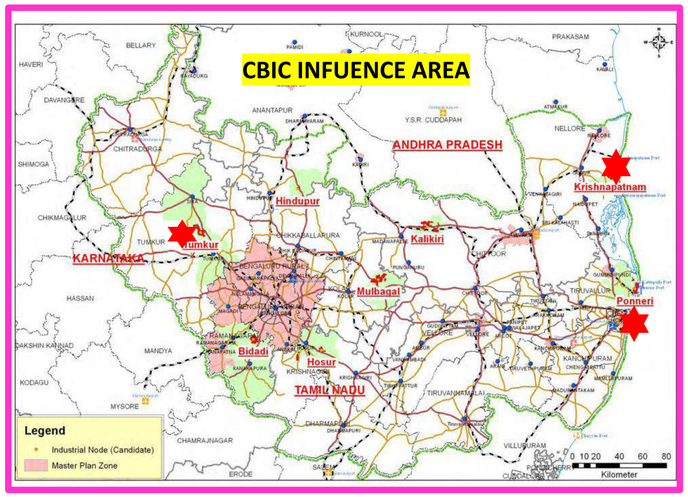
21) Van Goeverden, C. D., Milakis, D., Janic, M., & van Wee, B. (2016). Shared value potential of transporting cargo via Hyperloop. Frontiers in Built Environment, 2, Article 17.

22) Werner, M. (2015). *Hypercargo. Ein deutscher Hyperloop nach Transrapid und Co?: Überlegungen zu einer Güterverkehrsumsetzung für Elon Musks bahnbrechende Technologie auf deutschem Boden*. GRIN Verlag.

23) Zhang, Y., Lee, S., & Chen, H. (2020). "Linear Motors for Hyperloop: A Comparative Analysis," *IEEE Transactions on Industrial Electronics*, 67(9), pp. 7740–7750.

CBIC DEVELOPMENT PROJECT BY NICDC AND JICA





source: CBIC-Comprehensive Regional Plan report

Source: NICDC

Districts covered		
Tamil Nadu	***Karnataka***	***Andhra Pradesh***
1. Chennai	8. Bengaluru (urban and rural)	14. Chittoor
2. Tiruvallur	9. Ramnagara	Potential area
3. Kancheepuram	10. Kolar	15. South part of Nellore
4. Tiruvannamalai	11. Chikkaballapura	16. South west part of Anantapur
5. Vellore	12. Tumkur	
6. Dharmapuri	13. Chitradurga	
7. Krishnagiri		

- Total CBIC Area: 91,059 Sq. Km
- Covering 16 districts in 3 states
- Key Industrial Nodes: 8 potential nodes (3 selected) identified for development in CBIC

State	CBIC districts	Area covered (Sq. Km)
Tamil Nadu	Chennai	175
	Thiruvallur	3,394
	Kancheepuram	4,483
	Tiruvannamalai	6,188
	Vellore	6,075
	Dharmapuri	4,497
	Krishnagiri	5,129
Karnataka	Bengaluru	2,196
	Bengaluru rural	2,298
	Ramanagara	3,516
	Kolar	3,979
	Chikkaballapura	4,244
	Tumkur	10,597
	Chitradurga	8,436
Andhra Pradesh	Chittoor	15,152
	Nellore	6,400
	Anantapur	4,300
Total CBIC		**91,059**
All India		**3,166,414**

source: CBIC-Comprehensive Regional Plan report

Planning hyperloop freight corridor between Bengaluru to Chennai

Aim

To develop a physical planning framework for an Hyperloop freight corridor from Chennai to Bangalore along the Chennai-Bangalore Industrial Corridor (CBIC),
by determining route alignment based on Right of way requirement for hyperloop and identifying the nodes and terminals the corridor cover, develop a model station and terminal infrastructure requirements, integrating freight handling facilities and supporting logistics hubs and ensure correlation & spatial compatibility with the existing industrial and transport infrastructure.

MAIN

Objectives

Route Planning and Alignment

Propose Hyperloop freight terminals/nodes at key industrial nodes

Leap frog and Hyperloop

LEAP FROG DEVELOPMENT STRATEGY

Skipping old methodologies, practices and technologies and landing on new advanced practice

Latecomers advantage

Suitable planning strategy for India

(My dissertation)

Finding a space for leapfrog development plan in the Indian planning system

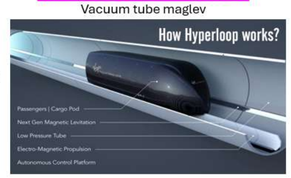
my idea was to explain the significance of

- New generation planning framework with leapfrog development strategy as one the core component in planning
- Using new methodologies and techniques in planning- (NOT JUST CONSUMER TECHNOLOGY BUT PRODUCER)
- Developing and planning new and advanced technology based projects

HYPERLOOP TECHNOLOGY

Vacuum tube maglev

travel up to 670 mph (1,080 km/h)



AS LEAPFROG DEVELOPMENT PROJECT
"Planning hyperloop freight corridor between Bengaluru to Chennai"

WHY?

New potential home technology
Indian Institute of Technology Madras(IIT Madras)-incubated Deep Tech Start-up TuTr Hyperloop is collaborating with the Institute on developing Intellectual property (IP) in hyperloop technology

TRAVEL TIME ~7HRS TO ~30 MIN - Avishkar Hyperloop team

"**The initial focus will be on cargo movement** to enhance the productivity of various stakeholders by solving their pain points. This will be a first step towards the deployment of high-speed mobility corridors for passenger transportation across India and globally."- **Prof. Satyanarayanan Chakravarthy, who is also a faculty in the Department of Aerospace Engineering Department, IIT Madras**

Proceeding with assumption that hyperloop is feasible
Focus on physical planning- mainly route alignment

The rail network in the CBIC region stands utilized at over 100% causing congestion

Lack of dedicated freight corridors in CBIC leads to delays and traffic congestion

So CBIC is a EFFORT to ERADICATE the barriers that restricts faster and efficient development force in industrial sectors

Bangalore-Chennai Expressway(NE-7): This access-controlled, signal-free expressway has been accorded special priority under the CBIC project
Dedicated Freight Corridor: Plans are underway
HSR: Discussion stage

Hyperloop freight corridor to take over the place of above 2 project

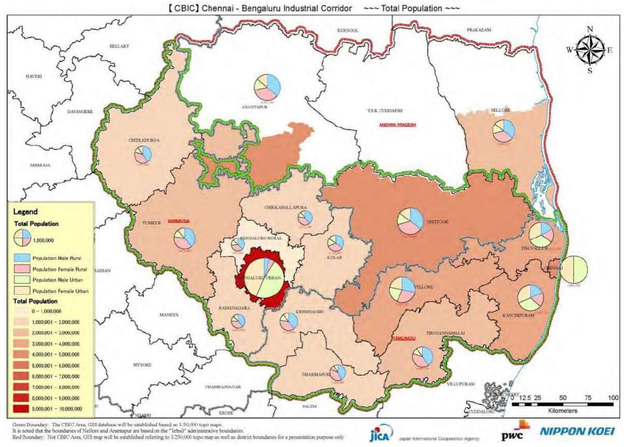
Note : " NOT ground level route plan, ONLY node/terminal connectivity route plan and route alignment"

STUDY AREA – CHENNAI BANGLORE INDUSTRIAL CORRIDOR

PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

NAME: NITESHKUMAR SK
SCHOLAR NO: 211109041
VIII SEM – B. PLAN

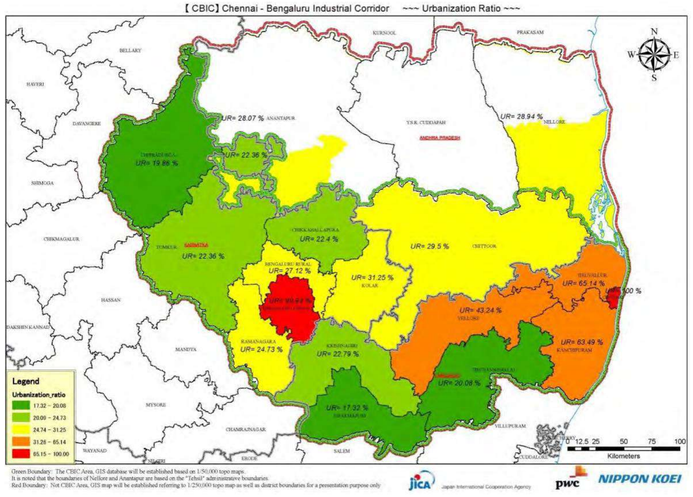
DEMOGRAPHICS OF STUDY AREA



Total population: *source: CBIC-Comprehensive Regional Plan report*

According to Census 2011, the total population of India is 1,210.57million, while that of the **CBIC region is 47.53million.**
Compared with 2001, the total population of the CBIC region was 26.60% from 37.54 to 47.53million**.** ***It clearly indicates the population growth speed of CBIC region for the last decade has been greater than that of all India (17.69%)***

- The most populated district is Bengaluru which has 9.62 million (20.24% of the CBIC region) in 2011.
- Chennai (4,646,732) in Tamil Nadu state and Chittoor (4,174,064) in Andhra Pradesh , the 2nd populated districts in the region, have respectively over 4 million

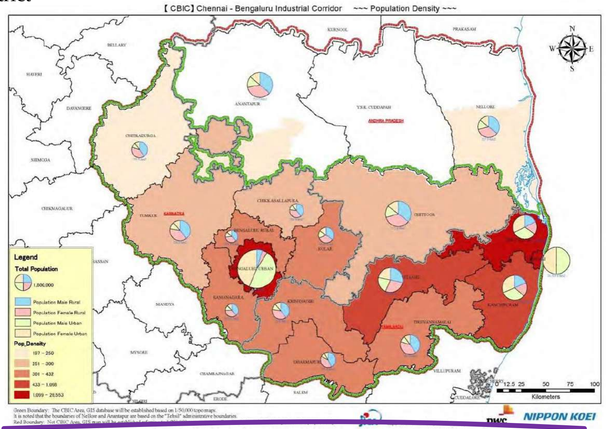


Urbanization ratio *source: CBIC-Comprehensive Regional Plan report*

The all India urbanization ratio is 31.15%, while that of the CBIC region is **51.17%.**

Chennai and Bengaluru, which are two most populous districts in Karnataka state and Tamil Nadu state respectively, are the most urbanized districts in the CBIC region. Additionally, in eastern area, the districts of Tiruvallur and Kanchipuram have an urbanization ratio over 60%.

Density
Average density of the CBIC region **is 464 persons per Sq. km** as per census 2011. The density map above typically suggests that the density of Chennai in Tamil Nadu state and Bengaluru in Karnataka state are enormously higher dense than the other district



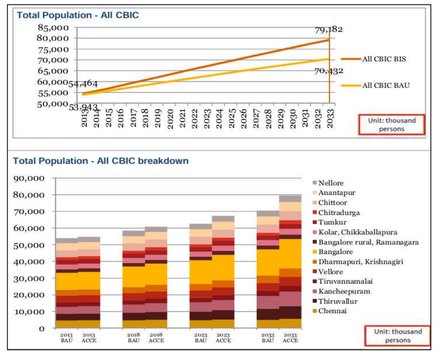
Between Chennai and Bengaluru districts, the ones **along NH-4 (one of the alignments in CBIC)** tend to be denser than the other districts. Especially the inland or north districts of CBIC region are less dense than NH-4 between Chennai and Bengaluru.

Population Projection (2033)

Regarding to future population of all CBIC area until 2033 under Business in Induced Scenario (BIS), the population will increase by 47% from 54,464 thousand in 2013 to 79,182 thousand in 2033. The compound average population growth rate from 2013 to 2033 is estimated as 1.89%, under Business-as-Usual scenario (BAU) scenario is estimated as 1.37%. In 2033, population will reach at 70,432 thousand, the level which will be achieved in 2026 under BIS

(in 000)

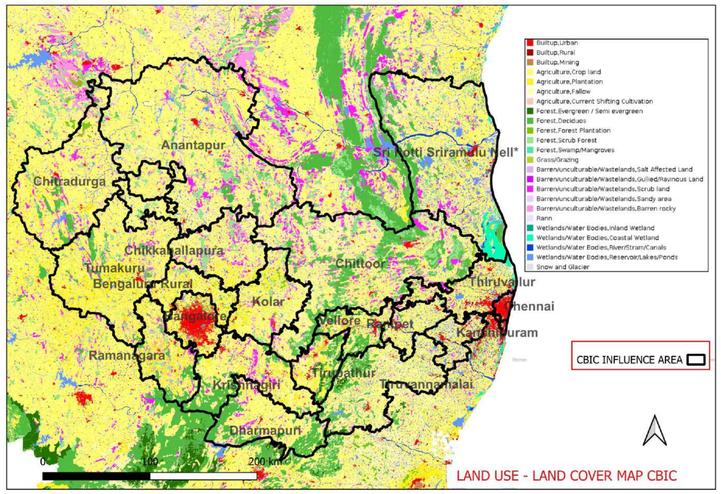
State	District	Level	Actual 2011	Projected 2013	Projected 2018	Projected 2023	Projected 2033
TN	Chennai	District	4,647	4,816	4,946	5,144	5,597
	Thiruvallur	District	3,728	4,051	4,680	5,477	7,550
	Kancheepuram	District	3,998	4,367	5,113	6,063	8,583
	Tiruvannamalai	District	2,465	2,536	2,722	2,907	3,335
	Vellore	District	3,936	4,054	4,359	4,664	5,372
	Dharmapuri, Krishnagiri	District	3,387	3,520	3,874	4,243	5,120
KA	Bangalore	District	9,622	10,445	12,808	15,185	17,593
	Bangalore Rural, Ramanagara	District	2,074	2,124	2,253	2,373	2,650
	Kolar, Chikkaballapura	District	2,792	2,859	3,031	3,190	3,558
	Tumkur	District	2,679	2,711	2,788	2,847	2,989
	Chitradurga	District	1,659	1,697	1,793	1,881	2,083
AP	Chittoor	District	4,174	4,284	4,567	4,849	5,472
	Nellore	District	2,964	3,039	3,235	3,429	3,856
	Anantapur	District	4,081	4,194	4,485	4,776	5,422



source: CBIC-Comprehensive Regional Plan report

PHYSICAL CHARACTERISTICS OF STUDY AREA

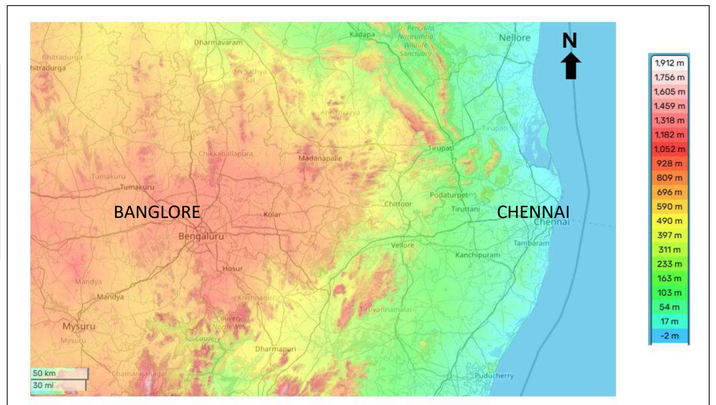
LAND COVER OF CBIC REGION

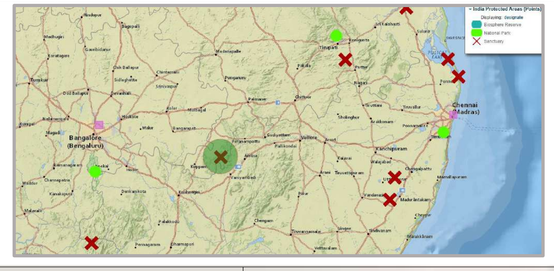


TOPOGRAPHY OF CBIC REGION

source: bhuvan/author

ALONG BANGLORE TO CHENNAI (WEST TO EAST) THE LAND AREA STEEPS GREATLY AND UNEVEN SPIKE IN TOPOGRAPHY IS SEEN IN MIDDLE OF THE 2 METROPOLITAN CITIES





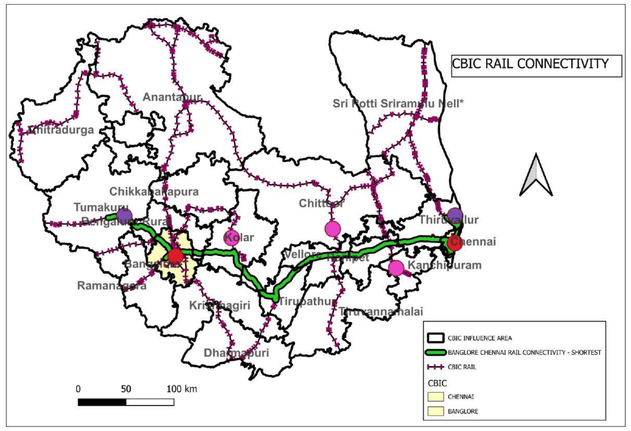
PROTECTED AREA
NATURAL SANCTURY IN BETWEEN STUDY AREA ACT AS PHYSICAL CONSTRAITS TO TRANSPORT PLANNING

DEMOGRAPHICS AND PHYSICAL CHARACTERS – CBIC

PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

NAME: NITESHKUMAR SK
SCHOLAR NO: 211109041
VIII SEM – B. PLAN

source: author

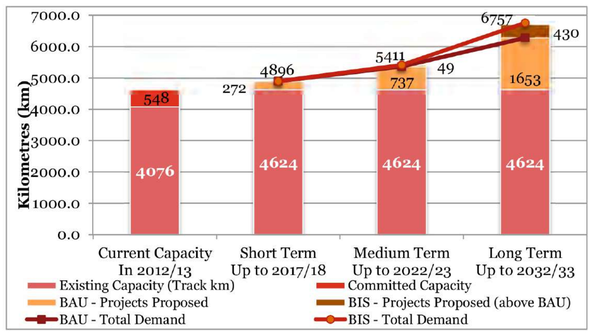


The railway network in the CBIC area is 2,806 Km long covering three zones of the Indian Railways– Southern Railway, South Western Railway and South Central Railway.
The network contains 22 routes and 66 line sections, linking Chennai and Bengaluru and providing through routes to other cities like Mumbai and Hyderabad.

Capacity saturation occurs when the number of trains each way per day exceeds 90% of the estimated daily train capacity. Nearly 37% of this network in terms of route km has either already reached or is fast approaching capacity saturation.

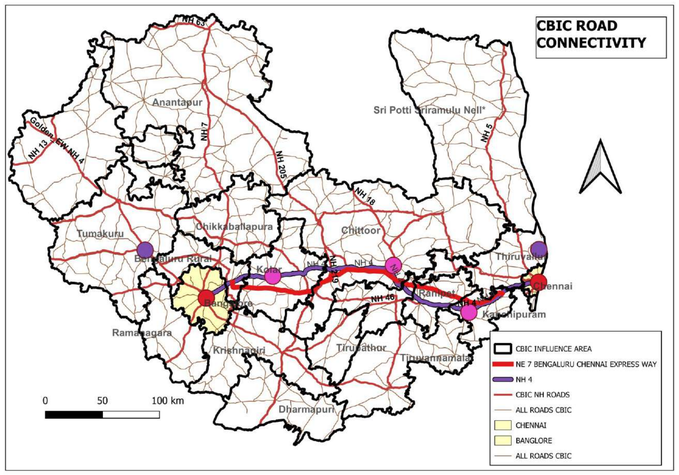
The Chennai– Bengaluru route currently carries **10.5m tonnes** of through-freight. The freight includes bulk cargo (mainly coal, limestone and dolomite), break bulk (foodgrains and fertilizers) and containers. It is forecasted that in the BIS case, by year 2033, the Chennai– Bengaluru route will carry **65.2m tonnes** of freight in total in both directions.
It is expected that rail will also start carrying cars for export via Ennore. Coal from Ennore and Krishnapatnam is also expected to be a major driver of rail freight growth.

For expansion of railway route capacity, 2,133 km of additional running lines will need to be constructed over the 20 year forecast period (2013-2033) to satisfy freight and passenger demand which would be generated under the BIS case. This will include 686 km of new running lines on the Chennai– Bengaluru route and 1,447 newlines on all other routes.



source: CBIC-Comprehensive Regional Plan report

source: author



Approx. 2,942 km of national highways and 5,343 km of state highways have been identified in the CBIC area + ongoing/proposed/announced projects (data from 2015)

- vehicle congestion ratio of national highway network will reach almost 50% and 90% in Karnataka state in year 2018 & 2033 respectively,
- In Tamil Nadu state in year 2023 & >2033, respectively
- while Andhra Pradesh state will maintain less than 50% of vehicle congestion ratio even in year 2033

95% freight transportation is via ROAD

The length of National Highway 4 (NH 4) from **Bangalore to Chennai** is approximately **345 km**. However, NH 4 has been renumbered as part of **NH 48** under the new National Highway numbering system in India. This highway is a crucial segment of the **Golden Quadrilateral** and connects **Bengaluru, Kolar, Chittoor, Ranipet, Sriperumbudur, and Chennai**. It is a **four-lane to six-lane** highway

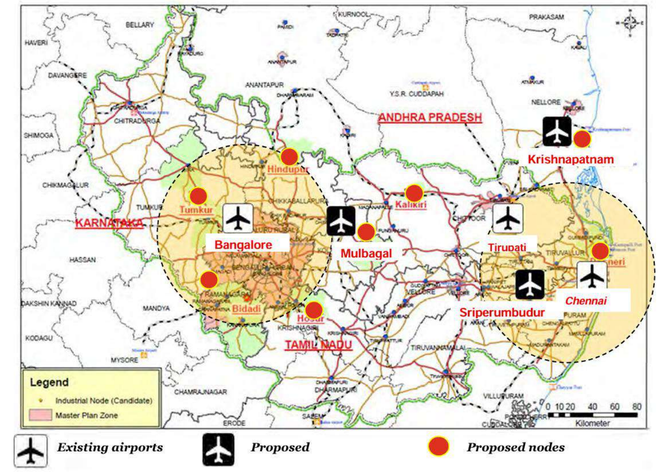
Bangalore-Chennai Expressway(NE-7): This access-controlled, signal-free expressway has been accorded special priority under the CBIC project **(260-kilometer)**
As of December 2024, the **71-kilometer** stretch of the expressway in Karnataka is open and toll-free. The entire expressway is expected to be fully operational by August 2025.

Dedicated Freight Corridor: Plans are underway
HSR: Discussion stage

Required freight mode for various industries

Industries	Water	Power	Road connectivity	Rail connectivity	Ports	Airports
Metallurgy	◔	●	●	●	◐	◔
Medical equipment	◔	●	◔	◔	◔	◔
Food processing	◔	◐	◔	◔	◐	◔
Textiles	●	◔	◔	◔	●	◔
Electrical machinery	◐	◔	◔	◔	◔	◔
Machinery	◐	◔	◔	◔	◔	◔
Chemicals	●	●	●	◐	◔	◔
Pharma	●	●	●	◐	◔	◔
Auto	◔	●	◔	●	◔	◔
Computer, electronics	◔	●	◔	◔	◐	◐

Importance: ● Critical ◕ High ◐ Medium ◔ Low



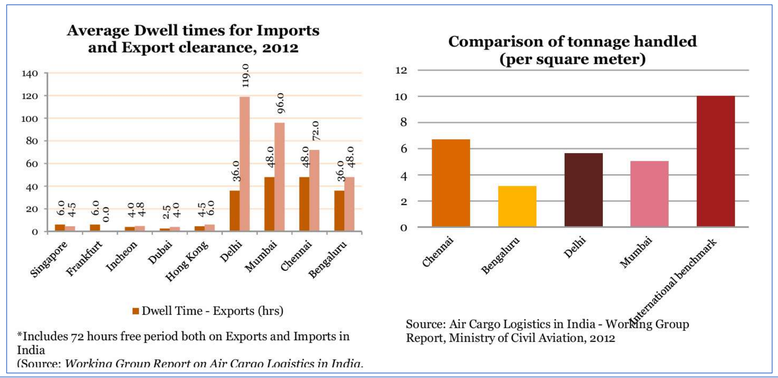
Chennai International Airport (MAA):
January 2025: Handled approximately 28,606.7 metric tons of cargo, down from 32,214.3 metric tons in December 2024.

Fiscal Year 2024: Handled over 340,000 metric tons of freight, comprising more than 258,000 metric tons of international freight and over 81,000 metric tons of domestic freight.

In the fiscal year 2020, both Bengaluru and Chennai airports each accounted for approximately 11% of India's total cargo traffic.

Bengaluru's Kempegowda International Airport
In the financial year 2023-2024 (FY24), (BLR) handled a total cargo volume of 439,524 metric tonnes, marking a 7.1% increase compared to the previous fiscal year.

This comprised 266,186 metric tonnes of international cargo (a 4% year-on-year growth) and 173,338 metric tonnes of domestic cargo (a robust 13% increase)



The dwell time of export cargo in Chennai airport is 48 hours compared to around 4.5 to 6 hours in the airports at Hong Kong, Singapore and Frankfurt.

A comparison of the average tonnage handled per square meter shows that Chennai and Bangalore handle about 6.6 and 3.14 tonnes per square meter respectively. This when compared to international benchmark of 10 tonnes per square meter is very low.

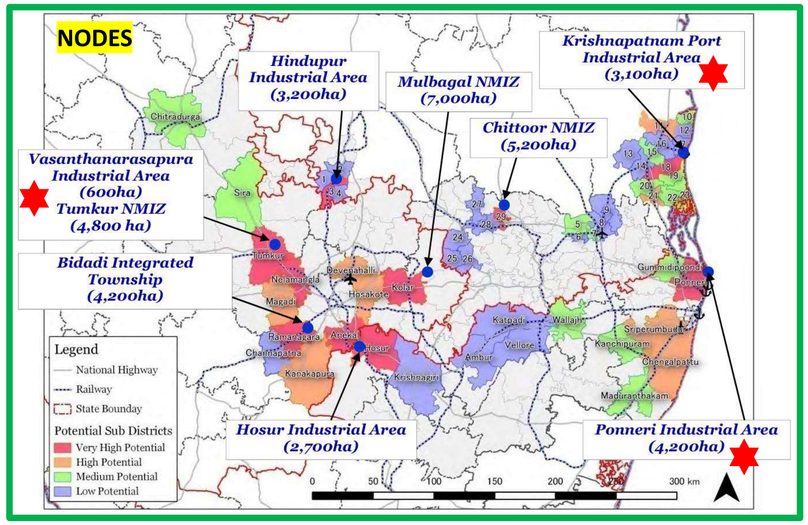
source: Global Economic Insights, Statista, ICRA, freighttrendsglobal

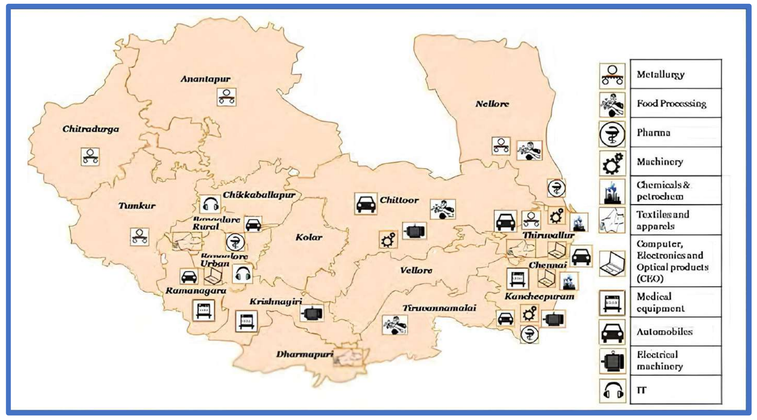
TRANSPORTATION AND FREIGHT - CBIC

PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

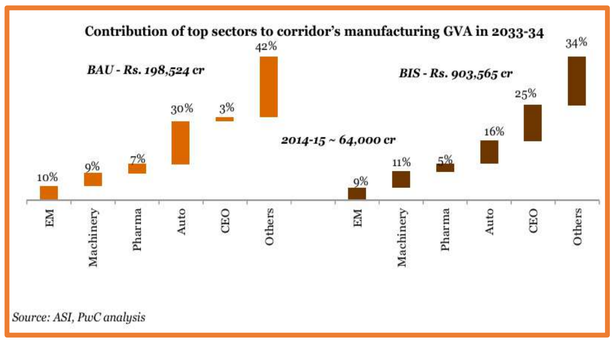
NAME: NITESHKUMAR SK
SCHOLAR NO: 211109041
VIII SEM – B. PLAN

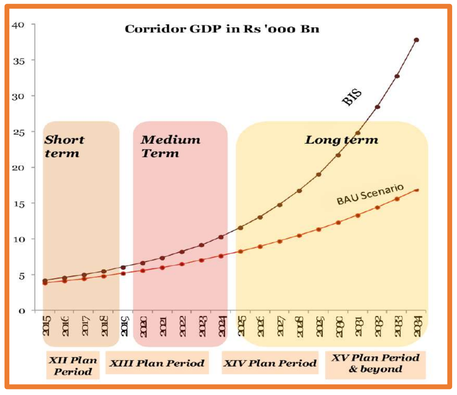
source: CBIC-Comprehensive Regional Plan report

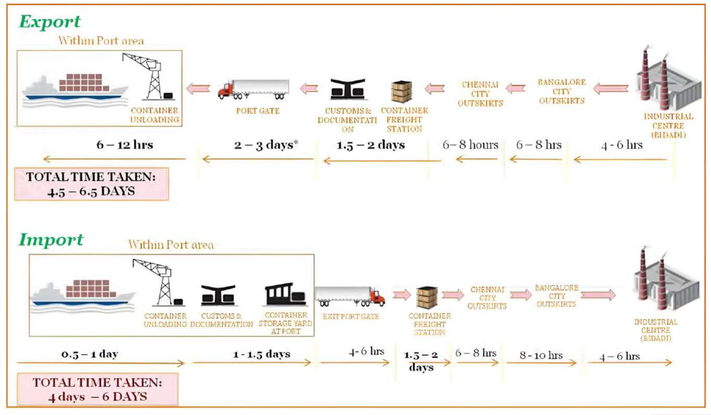


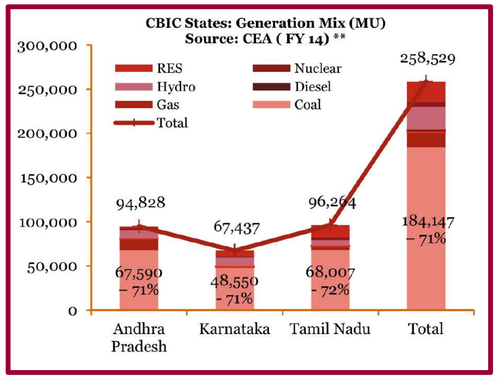


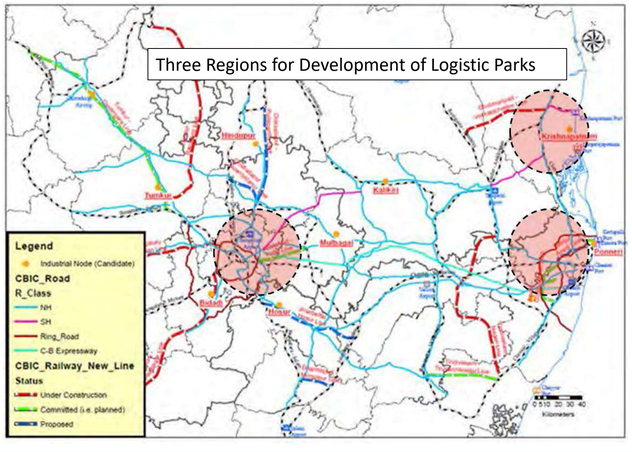
Sector	Output growth India (CAGR)	Key districts for investment in the corridor
Computer, electronics and optical products	15% (2006-11)	Chennai, Kancheepuram, Bengaluru urban, Bengaluru rural
Metallurgy	14% (2004-12)	Chitradurga, Anantapur, Chittoor, Nellore, Tiruvallur, Thiruvannamala, Kancheepuram, Tumkur, Bengaluru rural
Textiles and Apparel	17% (2006-11)	Tiruvallur, Dharmapuri, Kancheepuram, Bengaluru Rural, Bengaluru Urban, Chitradurga
Food processing	20% (2009-11)	Nellore, Chittoor, Tiruvannamalai, Dharmapuri
Pharmaceuticals	14% (2008-12)	Bengaluru Urban, Bengaluru Rural, Chennai, Kancheepuram, Nellore
Chemical and petrochemicals	11% (2009-13)	Bengaluru Urban, Bengaluru Rural, Ramnagara, Chennai , Tiruvallur
Electrical machinery	23% (2009-11)	Kancheepuram, Chittoor, Chennai, Bengaluru urban, Bengaluru Rural, Krishnagiri, Tiruvallur
Machinery	14% (2009-11)	
IT and financial	8% (2008-12)	Bengaluru Urban, Bengaluru Rural, Chennai





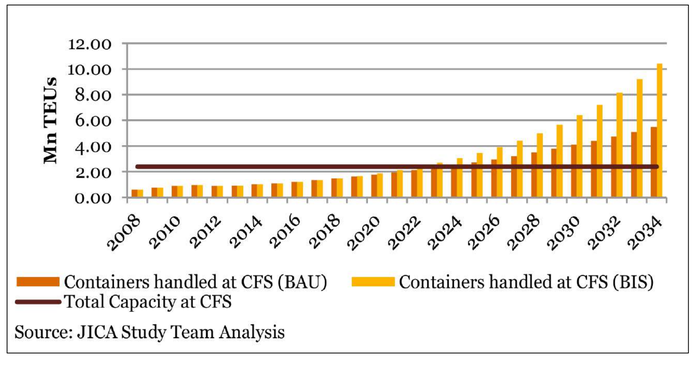






At this stage, three regions have broadly been identified for development of such logistics parks which include the Bengaluru outskirts, Chennai outskirts (near Sriperumbudur) and Krishnapatnam region.

Container Freight Stations



Spending almost 20 hrs in traveling between a 6 hr travel city in normal speed freight rail
Need to reduce delay in the way that it optimise the paper time delay.

The major logistics infrastructure in the corridor region comprises of and the Inland Container Depots (located in and around Chennai) and one Inland Container Depot at Whitefield, Bengaluru.

The modal split of the container traffic to and from Chennai Port comprises of 95% of container movement by road and only 5% by rail.

At present, most of the **26 Container Freight Stations located in and around Chennai** handle only around 36% of the total export container traffic movement by road to Chennai Port while the remaining container traffic comprises of factory stuffed containers.

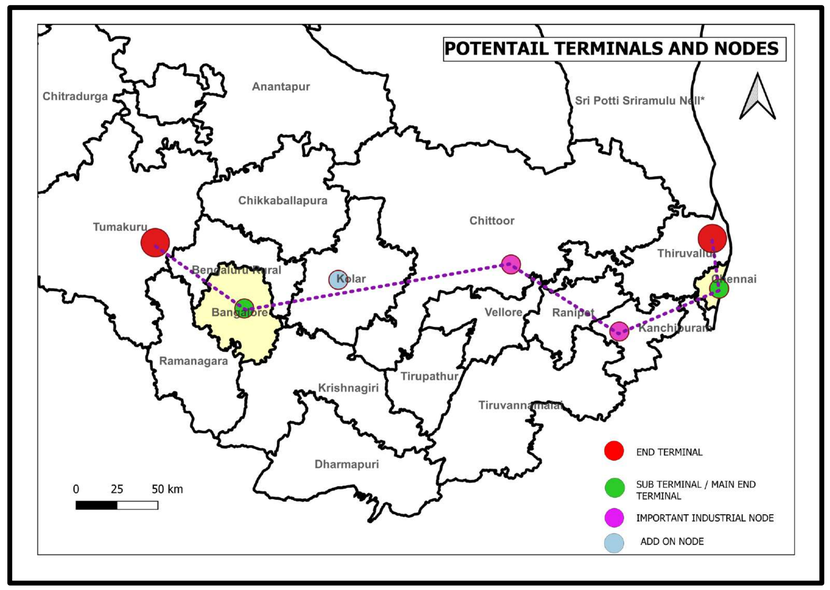
Source wise breakup of meeting demand for CBIC region in long term, FY 2032

Fuel source (MW)	Potential	Mid-term 2022	Long-term 2032	Agency for capacity addition
Wind	26,029	16,387	20,484	
Solar	118,000	3,300	5,000	
Biomass	3,124	1,771	2,214	
Small Hydro	2,926	1,695	2,119	
Waste to Heat	425	250	350	
RES Total	150,504	23,404	30,167	STATE
% of potential		16%	20%	
Hydro	30,980	14,891	22,336	STATE
% of potential		48%	72%	
Nuclear	20,000	4,055	8,555	CENTRAL
% of potential		13%	28%	
Gas	45,000	7,200	10,000	STATE
% of potential		16%	22%	
Coal - Balance		**6,650**	**73,598**	**MIX**

ECONOMY AND LOGISTICS - CBIC

PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

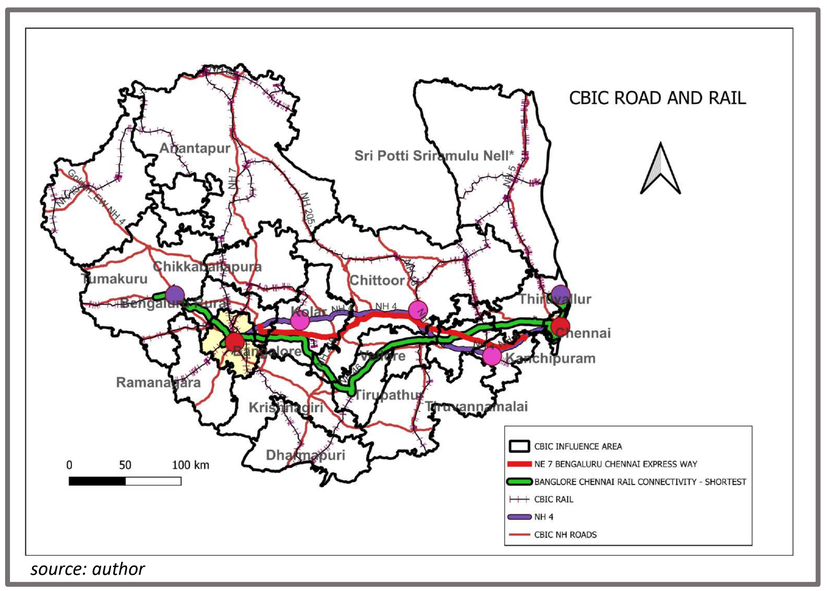
NAME: NITESHKUMAR SK
SCHOLAR NO: 211109041
VIII SEM – B. PLAN



source: author

IMPORTANT NODES TO BE INCLUDED

- **PONNERI (TERMINAL)**
- CHENNAI (SUB TERMINAL)
- KANCHIPURAM
- CHITTOOR
- KOLAR AND MULBAGAL
- BENGALURU URBAN AND RURAL (SUB TERMINAL)
- **TUMKURU (TERMINAL)**



source: author

ROUTE ALIGNMENT GOALS

1. Optimize alignment to maximize the productivity of CBIC
2. Include key industrial zones that require alternate to rail transport, time sensitive goods production nodes, rail/ air travel required industries and Port connectivity required industries
3. Minimize travel distance – BEST/KEY SHORT ROUTE – low cost
4. More toward straight lined route – avoid curves, angles- based on turning radius
5. Align along with existing Road or Rail corridor
6. Avoid more greenfield route
7. Avoid natural/ other constraints
8. Avoid tunnels, urban areas

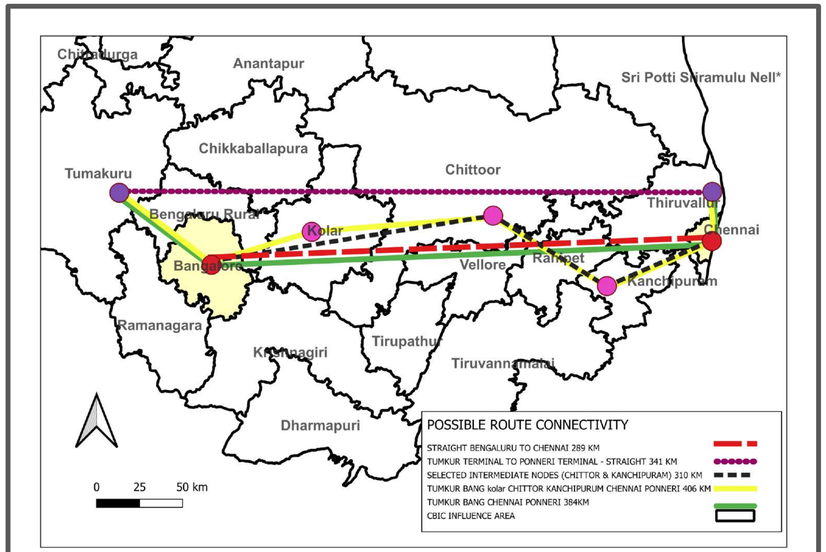
ROUTE ALIGNMENT METHODOLOGY

ROUTE ALIGNMENT ALTERNATIVE

Integrated alternative

Highway Alignment: The guideway is preferred to be configured near ground level on short levelling structures. Roadway barriers would protect hyperloop structures from highway traffic.
Railroad Alignment: parallel to or configured above active railroad.
Greenfield Alignment: along acquired right-of-way between service areas.
Tunnel Alignment

Select, propose and connect Hyperloop freight terminals at key industrial nodes, ensuring connectivity of core CBIC influence area with existing transport networks (rail, road, and ports).



source: author

ROUTE STRAIGHT CONNECTIVITY	DISTANCE
Straight Bengaluru to Chennai	289 km
Straight Tumkuru to Ponneri terminal	341 km
Straight Tumkuru-Bengaluru-Chennai-Ponneri	384 km
Straight Bengaluru-Chittoor-Kanchipuram-Chennai (with selected nodes	310 km
Straight Tumkuru-Bengaluru-Kolar-Chittoor-Kanchipuram-Chennai-Ponneri	406 km

Best potential route connectivity – inclusive of all important terminals and nodes

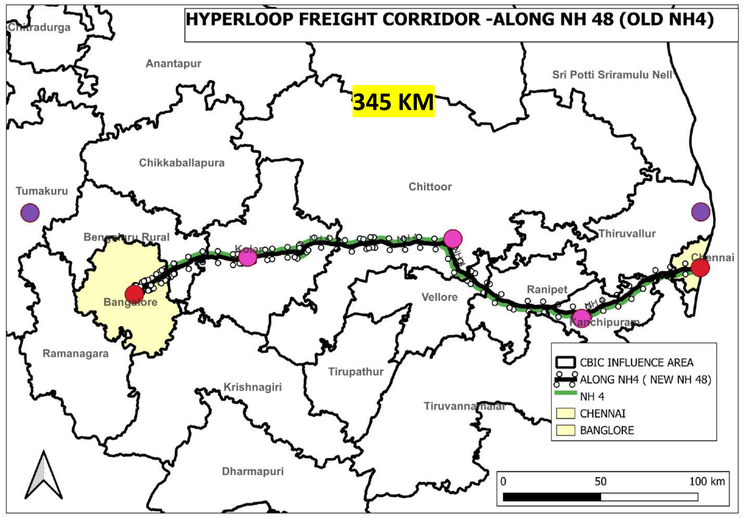
Straight [Tumkuru-Bengaluru-Kolar-Chittoor-Kanchipuram-Chennai-Ponneri]

ROUTE ALIGNMENT (VARIOUS MODE & TYPE)	DISTANCE
Road alignment- Along NH 48	~345 km
Road alignment- Along NE 7 Bengaluru Chennai express way	~260 km (Outside cities)
Rail alignment – Along Bengaluru – Tirupathur – Chennai (include Tumkuru and Ponneri)	~454 km

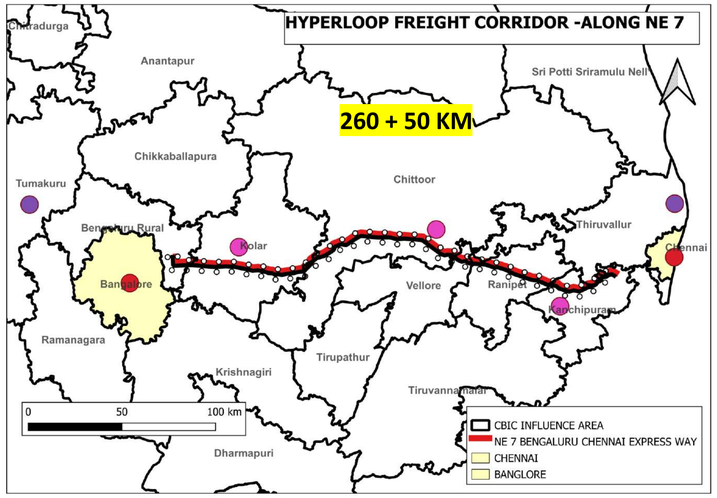
HYPERLOOP FREIGHT CORRIDOR ROUTE PLANNING – CBIC (I)

PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

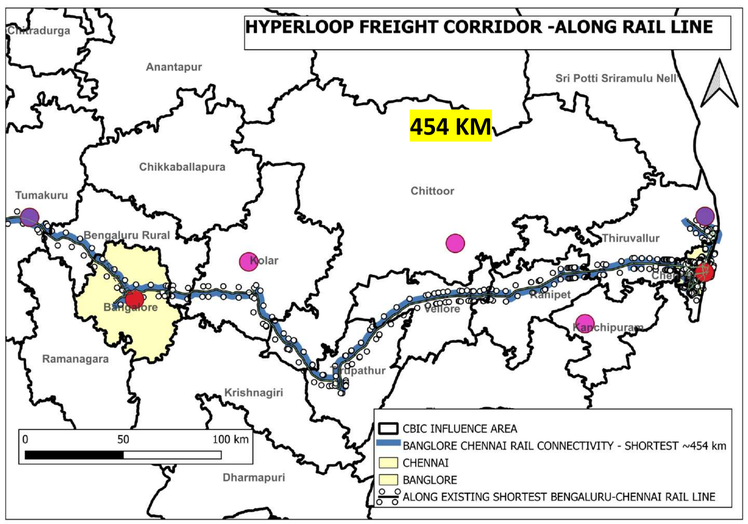
NAME: NITESHKUMAR SK
SCHOLAR NO: 211109041
VIII SEM – B. PLAN



PARAMETERS	QUALITY/ QUANTITY
Align along	NH 48 (OLD NH 4)
Route – terminal/node/intermediate	Bengaluru- Kolar-Mulbagal-Chittoor-Kanchipuram-Chennai
Length	~345 KM
Route quality	Angular bend on Chittoor Chittoor-Kanchipuram-Chennai curve
Constraints	In urban core – less space availability along NH 48 (may need mix and fly over route)
Need for green field add on	Yes – to straighten it
Need for tunnel	No (not outside city)

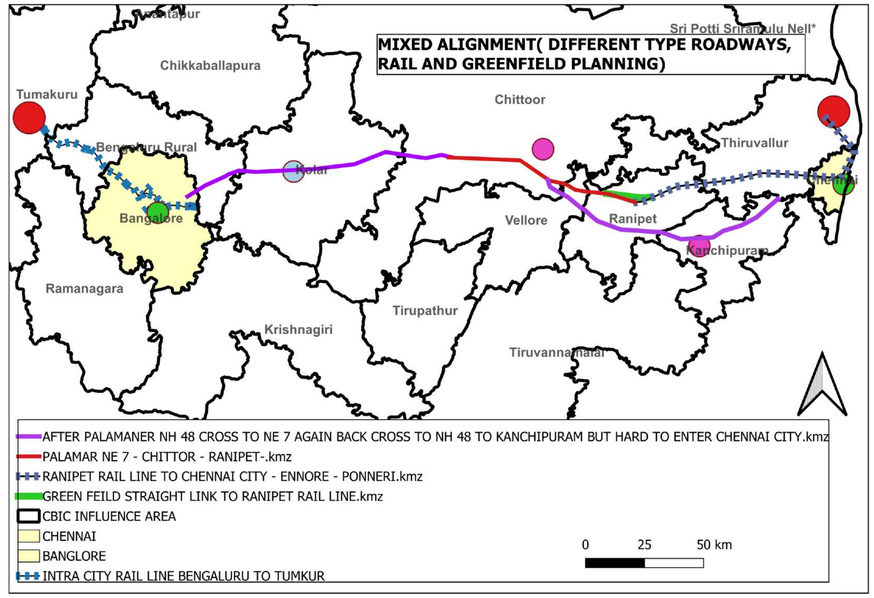


PARAMETERS	QUALITY/ QUANTITY
Align along	NE 7 – BENGALURU CHENNAI EXPRESS WAY
Route	NH 48(Bengaluru) – Hoskote-Venkatagirikota (Kolar)-Palamaner-Ranipet-Sriperumbadur- NH48(Chennai)
Length	260 KM + (20 + 20 NH48) = ~300 KM
Route quality	Short distance angular at nodes – especially in main accelerating point - centre
Constraints	Not straight at accelerating path
Need for green field route	YES
Need for tunnel	NO(not outside city)



PARAMETERS	QUALITY/ QUANTITY
Align along	BENGALURU CHENNAI RAIL LINE
Route	Tumkur-Bengaluru-Kolar-Tirupathur-Vellore-Ranipet-Chennai-Ponneri
Length	~454 km + ~75 tumkur/ponneri = ~530km
Route quality	Not straight, too bendy, terminal connectivity
Constraints	NOT straight
Need for green field route	About 1/4th length of rail line required
Need for tunnel	NO
Advantage	Easy to enter and align in urban area

source: author



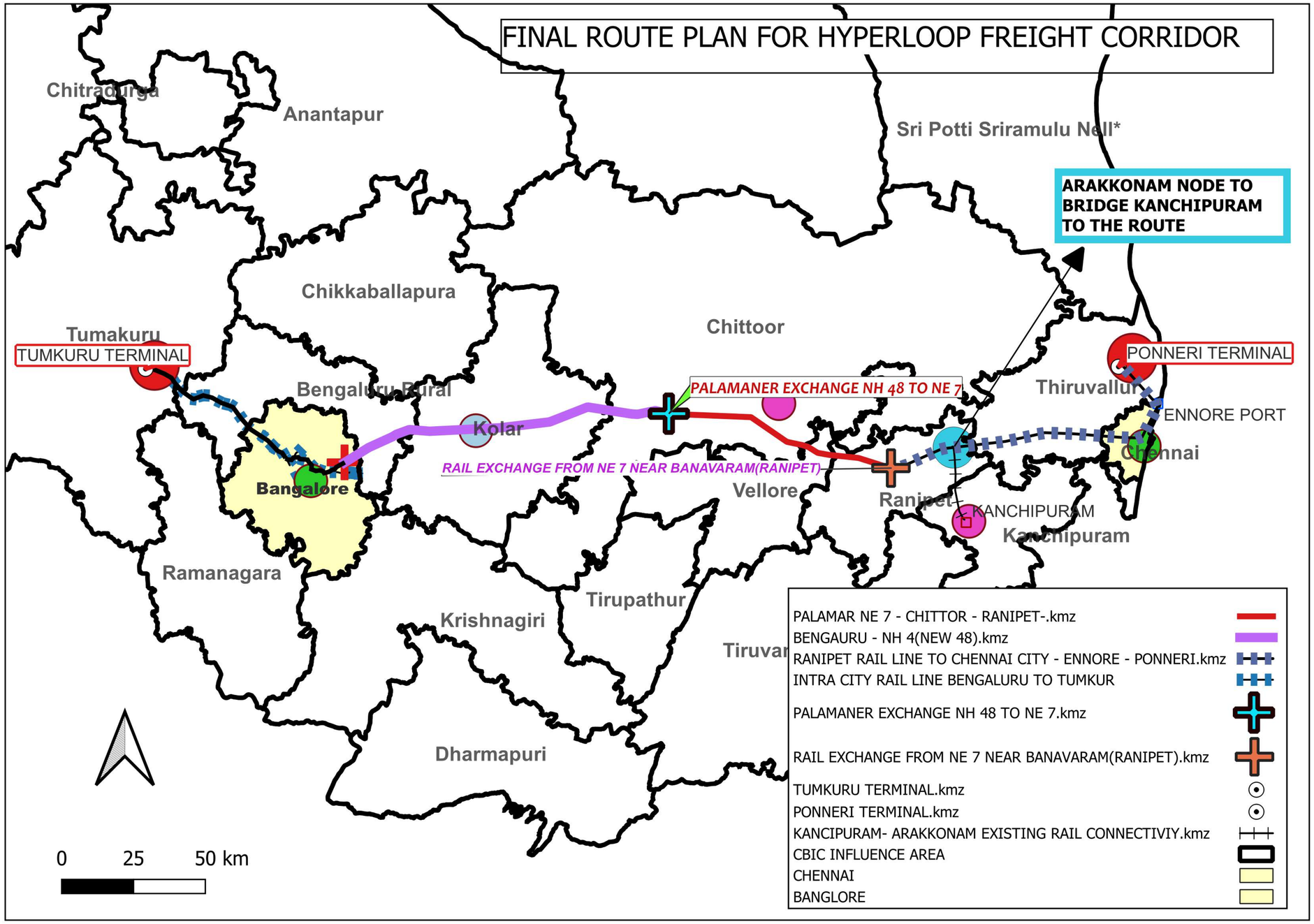
MIX ALIGNMENT COMPONENTS	LENGTH
Tumkuru to Bengaluru (RAIL)	~75 km
Alternative [A] Outside Bengaluru to Chennai via Kanchipuram (NH 48-full)	~345 km
Alternative [B] Outside Bengaluru to till Palamaner exchange (NH 48- part)	~119 km
From [B] Palamaner exchange to RAIL at Ranipet: Banavaram (NE 7)	~83 km
From NE 7 to rail exchange to Chennai-Ponneri via Arakkonam	~121 km
Alternative [A] total	420 + Chennai-Ponneri flyover along NH 48 (50km) = **~470**
Alternative [B]	**398(~400 km)**

HYPERLOOP FREIGHT CORRIDOR ROUTE PLANNING – CBIC (II)

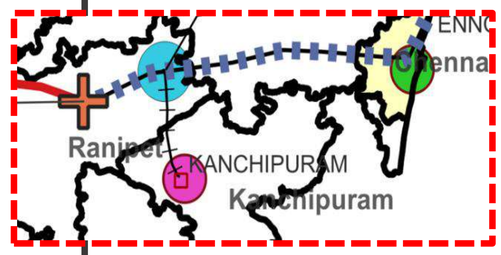
PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

NAME: NITESHKUMAR SK
SCHOLAR NO: 211109041
VIII SEM – B. PLAN

FINAL ROUTE PLAN FOR HYPERLOOP FREIGHT CORRIDOR



Arakkonam and Kanchipuram rail accessory node



TRIP ROUTE AND DISTANCE:

ARAKKONAM TO BENGALURU - ~ 230 KM

KOLAR TO CHENNAI ~ 240 KM

CHITTOR TO CHENNAI ~ 140KM

TUMKUR TO BANG ~ 75 KM

PONNERI TO CHENNAI ~ 30 KM

TOTAL 400 KM

CHENNAI TO BANGLORE ~ 295 KM

source: author

HYPERLOOP FREIGHT CORRIDOR ROUTE PLANNING – CBIC (FINAL)

PLANNING HYPERLOOP FREIGHT CORRIDOR BETWEEN BENGALURU TO CHENNAI

NAME: NITESHKUMAR SK
SCHOLAR NO: 211109041
VIII SEM – B. PLAN

भारत सरकार /GOVERNMENT OF INDIA
रेल मंत्रालय /Ministry of Railways
दक्षिण रेलवे /Southern Railway

पे. नंदलाल, आई.आर.एस.ई.
P. NANDALAL, I.R.S.E.
वरिष्ठ मंडल अभियंता / समन्वय
Senior Divisional Engineer/Co-ordn.

मंडल कार्यालय/Divisional Office
संकर्म शाका/Works Branch
तिरुच्चिराप्पल्ली/Tiruchchirappalli-620001

No.T/W.571/Training/Internship

Date: **28.08.2024**

CERTIFICATE OF INTERNSHIP

This is to certify that **NITESHKUMAR S K** (Scholar No.211109041) student of the Third Year B. Planning (UG 2025) MANIT, Bhopal, has undergone a month long **INTERNSHIP** in this department during July 2024.

He was entrusted with preparation of Parking Plan for the Divisional Office Complex which demands parking space for around 400+ two wheelers and 60+ cars.

He was inquisitive in collection of data and show cased meticulous planning, design and implementation strategies. He was proactive in his approach and the solution provided by him is appreciable.

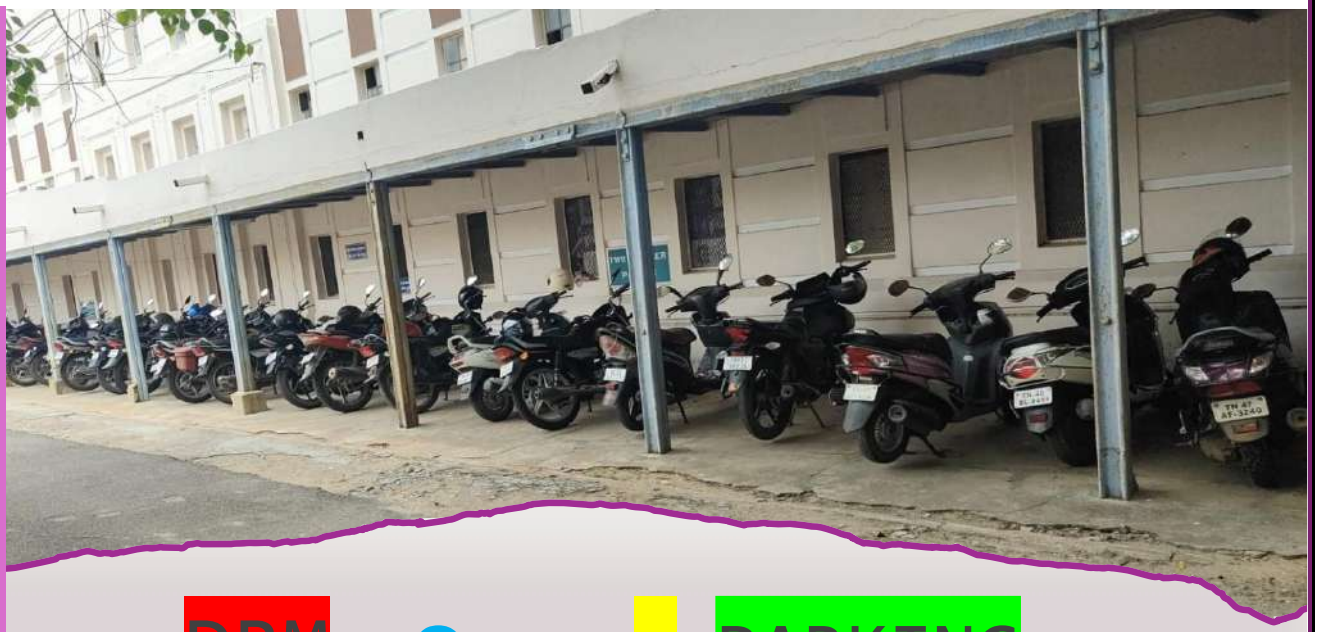
I wish him all the very best.

28/8/24.

Sr.Divisional Engineer/Co-ordn./TPJ

SENIOR DIVISIONAL ENGINEER
Tiruchchirappalli

DIVISIONAL RAILWAY MANAGER
TIRUCHIRAPPALLI JN.
DATE 28/8/24



DRM COMPLEX PARKING

TPJ

DIVISIONAL RAILWAY MANAGER'S COMPLEX PARKING PLAN

REPORT- JULY 2024

TIRUCHCHIRAPPALLI DIVISION

INTERNSHIP PROJECT

FROM: PLANNING STUDENT OF NIT, BHOPAL

ACKNOWLEDGEMENT

I am profoundly grateful to the Southern Railways for providing me with the invaluable opportunity to undertake this internship project on the parking plan for the Divisional Railway Managers Complex. This experience has been a significant learning journey, and I am deeply thankful for the support and guidance I received throughout the project and my internship.

Firstly, I would like to extend my sincere gratitude to **Mr. NANDALAL P** sir, the Senior Divisional Engineer/Co-Ordinator for accepting my request and granting me this opportunity. Your trust and support have been crucial in the successful completion of my summer internship.

I would also like to express my heartfelt thanks to **Mr. VENKATAGANESH A** sir, the Senior Sectional Engineer, for his invaluable mentorship and assistance. His detailed insights and practical knowledge have greatly enhanced the quality of my work in the field.

Special thanks to my professors at **Maulana Azad National Institute of Technology (MANIT), BHOPAL** for equipping me with the necessary skills and knowledge that have been crucial for this project.

THANK YOU.

From:
NITESHKUMAR S K [INTERN]
SCHOLAR NO: 211109041
B. PLANNING (UG 2025), MANIT BHOPAL.
Email: skniteshtn11@gmail.com

ABSTRACT

This report presents a comprehensive parking plan for the Divisional Railway Managers Complex, developed as part of an internship project by a third-year planning student.

The primary objective of this project is to design an efficient and functional parking layout for both two-wheelers and cars, accommodating the needs of staff and visitors.

The report utilizes spatial analysis through GIS software and AutoCAD to assess the parking capacity and optimize the layout within the available space. Various layout options, including perpendicular and angled parking, are explored to maximize space utilization while ensuring ease of access and mobility.

This project underscores the importance of a well-organized, efficient, and user-friendly parking system in enhancing the operational efficiency of the Divisional Railway Managers Complex.

CONTENTS

TOPIC **PAGE NO.**

1- Background

1.1- Need and scope of the work……4

1.2- Aim ……4

1.3- Objective……4

1.4- Methodology……5

1.5- Stakeholder consultation……5

2- Study area & Existing scenario

2.1- Company profile……6

2.2- Area & Land cover……7

2.3- Existing mobility network……8

2.4- Development constraints……8

2.5- Existing parking condition……9

2.6- Issues……10

3- Proposal

3.1- Parking proposal principles……11

3.2- spatial plan……12

3.3- Parking structural plan……13-17

3.4- Supporting amenities, signs and markings ……18

4- Parking policy and post-action plan……19

5- Recommendation & conclusion……20

6- Annexures……21-25

1. BACKGROUND

1.1) NEED AND SCOPE OF THE WORK

The DRM office consists of more than 500 staffs in 15+ department accounting vehicles approximately 400+ 2wheeler and 60+ cars which are parked in the complex during their shift hours.

Therefore, efficient parking with space and accessibility is required to satisfy the demand of parking in the office complex.

Current parking provision and procedure does not produce variable solution to increasing vehicles number as system that exists is fragile/broke and creating chaos in the complex.

So, preparing a parking plan would able to find solution for current need and create adaptable system for future requirements.

1.2) AIM

To prepare an effective parking plan for the staff's vehicles of DRM complex taking into consideration the available capacity, development hurdles, current & future accommodation and ease to use factors as core of the project.

1.3) OBJECTIVES

- To assess the entire DRM complex's spatial situation such as built-up area, built use, land use, mobility network and physical constraints.

- To profile the existing parking scenario, its holding capacity and issues it has & creates in the complex for the users.

- Envisioning the various mitigation on provision of maximum betterment through predicting future projection of staff's population and their vehicles.

- To propose spatio-structural intervention to increase the capacity of parking space by new parking area, extending the existing one and arrangements of bays through technical designs.

- To create uniform and disciplined parking set up that doesn't cause chaos in arrangement, use and mobility within the entire complex.

- Formulating parking policy with action plan for this project with ongoing parallel action taken by RPF (Railway Protection Force).

- Move from passive and weak enforcement of parking regulations to active management of parking.

1.4) METHODOLOGY

1.5) STAKEHOLDER CONSULTATION:

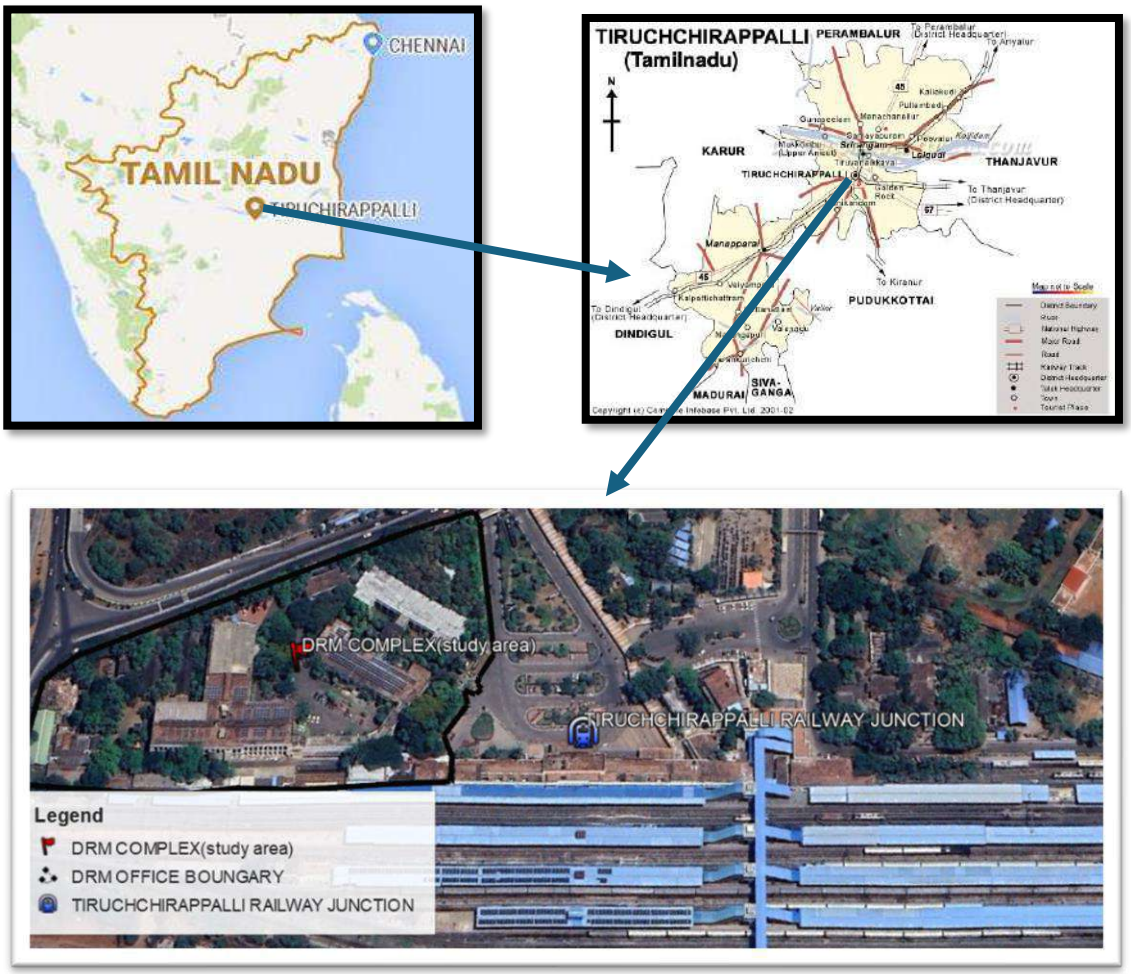
The DRM complex parking is managed and controlled by **RPF (Railway Protection Force) with concerned SSE/W/**TPJ from DRM complex, whom have made a **SOP (Standard Operating Procedure)** for parking. This parking project would be implemented and managed with the combination of this spatial & capacity plan and its effects on RPF's action plan and SOP. This plan accounts all the restrictions mentioned in the SOP (attached in ANNEXURE) provided by RPF and would suggest recommendations for action plan and parking policies.

2. STUDY AREA & EXISTING SCENARIO

2.1) COMPANY PROFILE:

DRM Divisional Railway Manager office, Southern Railway, Tiruchchirappalli

The Divisional Railway Manager (DRM) Office in Trichy (Tiruchirappalli) is a key administrative hub for the Southern Railway zone of Indian Railways. It is responsible for managing and overseeing railway operations within the division, which includes a significant portion of Tamil Nadu.



Office complex: The office is located within Tiruchchirappalli railway junction at Bharathiar Salai, Sangillyandapuram, Tiruchirappalli, Tamil Nadu 620001.

History & significance: The Trichy railway division was established on May 16, 1956, and has a rich history dating back to the construction of the Great Southern of India Railway Company in 1859. The division is known for its extensive network and high traffic, operating around 150 trains per day.

2.2) AREA & LAND COVER

TOTAL COMPLEX AREA:

***Area*:** 32000 sq meter/7.92 acres

CO-ORDINATES:

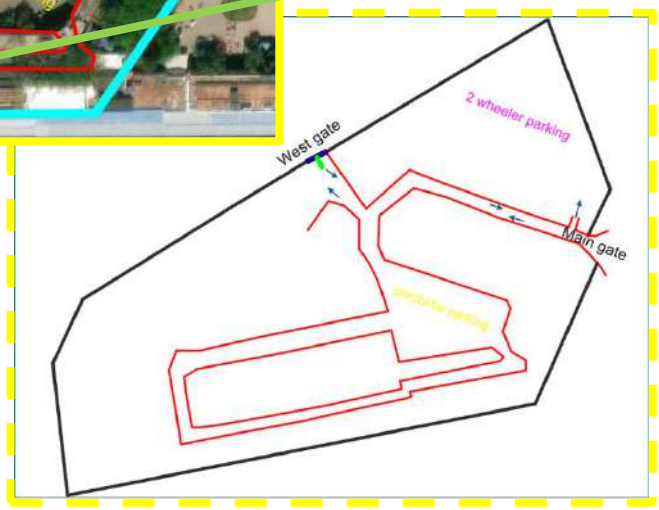
Latitude: 10°47'37.76"N

Longitude: 78°41'1.76"E

LAND COVER:

LAND COVER	AREA	PERCENTAGE
BUILT UP AREA	**11644 SQ.M**	**[36.38%]**
NON-BUILT-UP AREA (inc. open space + vegetation + road + existing parking space)	**20355 SQ.M**	**[63.62%]**
TOTAL EXISTING PARKING SPACE (inc. porch)	**2680 SQ.M**	**[8.3%]**

2.3) EXISTING MOBILTY NETWORK



The existing mobility network consists of 2 entry/exit for entire office complex,

- Main gate

- West gate

Main gate is opened through station entrance whereas west gate is opened out through bharathiar salai (Rd).

Inside the complex the way through main gate leads to the entire complex connecting west gate and ***parking porch***. The first foremost right turn from main gate near post guard cabin leads as the pavement corridor to exchange office and the ***main 2-wheeler parking*** for the staffs & visitors.

The porch is extended into cycled road around DRMs main office connecting the ***other car and 2-wheeler parking locations*** and turns back to porch through building corridor.

2.4) PHYSICAL CONSTRAINTS FOR PARKING DEVELOPMENT

The available open spaces **[63.62%]** in the complex are mostly available as ***undevelopable gaps*** between built-ups and some are ***defined green space*** within the complex and rest are used for various purpose that may not able to support staffs parking, even if used they would be as locations away from the work space buildings which would inaccessible and unease for parking use.

2.5) EXISTING CONDITION

The current parking in divisional railway managers complex is handled and managed by ***RPF (Railway Protection Force)***. They have executed standard operating procedure and specific allocation of locations for various departments.

Though at most cases consignees are not following the regulations strictly and the ***vehicle sticker pass policy*** is yet to be fully initiated (05/07/2024), which forms a non-uniform/undisciplined situation causing issues like congestion, parking space unavailability, etc. without actual spatial plan.

The department-wise allocation of parking space as approved in the SOP is re produced below:

Sl No.	Locations	Department
1	Behind RPF Out Post	Visitors
2	Exchange office side (near boundary wall)	Personnel
3	Opposite of Exchange Office	Personnel
		Commercial
4	Hindi section side	Traffic Account
		Hindi Section
		Stores
5	Near Central Dispatch office	Accounts
6	Government Railway Police side	General branch
		Operating
7	Cash office area	Engineering
		GMT
8	Steel Tank near Canteen	Security
		Electrical
9	Old Divisional Security Commissioner Office	Security
10	Behind Sr. Divisional Personnel Officer office	Safety
		Wayside station employees
11	Near Exchange office	S&T
12	Physically challenged employees	May park vehicle nearby to their concerned department
13	Branch Officers' Four wheeler	Nominated officer's four wheeler shed

Fig: department wise parking allocation by RPF

Fig: locations of existing parking spaces in DRM complex

2.6) Issues

i. Parking requirements is more than the existing capacity
ii. No enough space for new/more proposal.
iii. Overflowing vehicles occupying available space in the current environment- which indeed cause issues in free mobility

(i)

(iii)

iv. Standard regulations and specifications are not followed by the staffs
v. Some Allotted parking spaces are unoccupied due its location

(iv)

(v)

vi. Complains on parking spaces are encroached by visitors, outsiders, non-DRM office staffs.
vii. Rumors on illegal prolonged parking by outsiders
viii. Damaged pavements and cover shed
ix. Lack of supporting amenities.
x. Poor maintenance and control

(viii)

(viii)

3. PROPOSAL

3.1) PARKING PROPOSAL PRINCIPLES

Addition of space – due to less availability of developable space in the complex addition of new space is a big challenge, whereas extension of existing parking place enough to take the load. So, for parking in existing parking locations specifically on 2wheeler parking by introduction of opposite parallel parking on the triangle 2w parking at the side of entry would enable to unleash required extra space.

Efficient use of existing space – through spatial planning and ordering space and parking spots/bays, we could able to bring out the best capability and capacity of a space rather than a disorder random arrangement that eats space. The porch and 2nd entry on road parking are the 2 locations where arrangement plan is imposed to increase capacity without affecting the mobility.

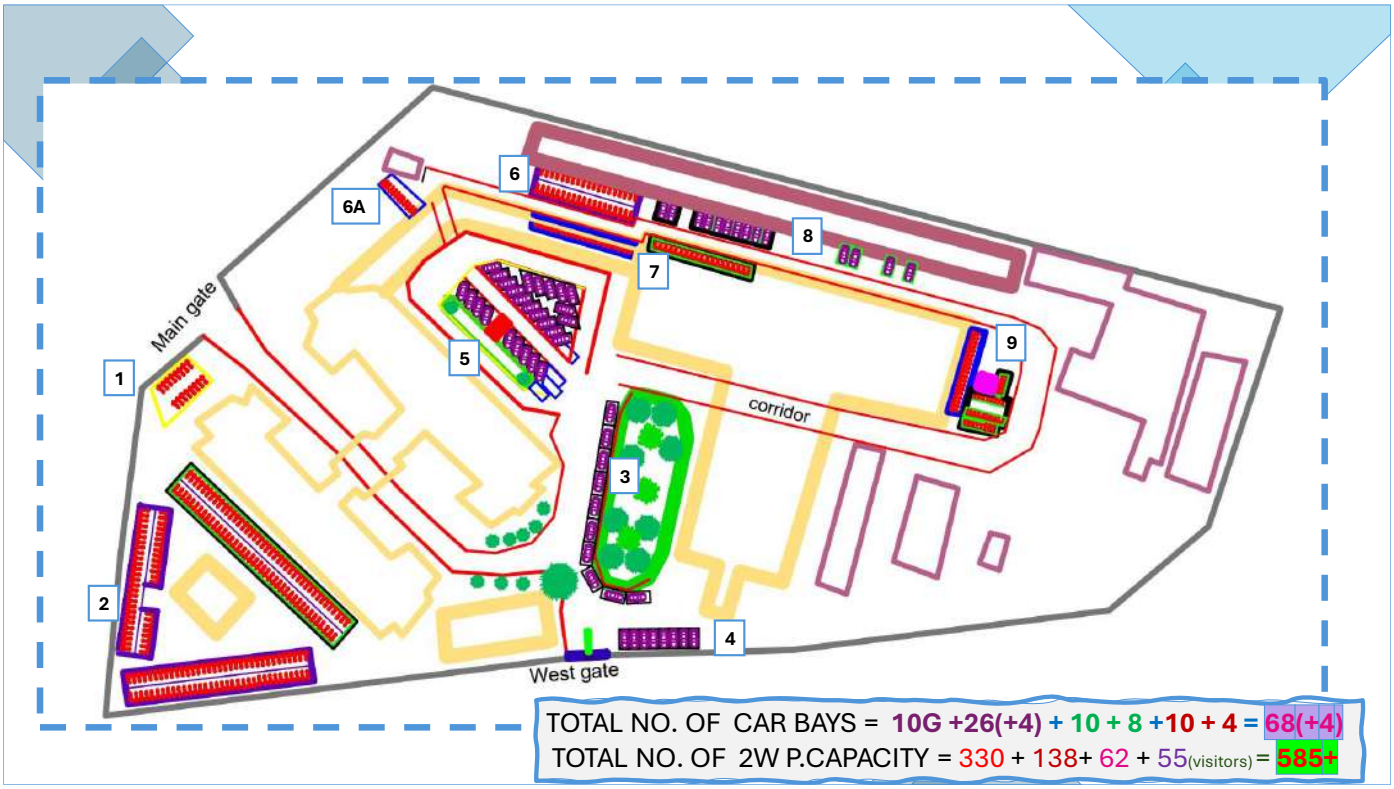
Promoting parking discipline through amenities – shed cover and signage will enable this plan as people tends follows a behavior *(in the complex)* of being disciplined in parking at space with shed cover that looks dedicated and make it formal rather than the space just allotted through just notice. This also accounts on renewing old damaged shed with new ones.

Allotment of locations and space – Though the allocation plan would be handled/revised by the RPF in allotting the location to the staff based on their department and its location/distance from the parking lots, still through this parking plan we able to prepare a better allocation plan with the spatial data and capacity of every parking lot calculated along with the staff's vehicle data and department/building's demographic strength.

Parking restriction and free mobility – by keeping the parking restriction mentioned by RPF in their SOP in mind major location are announced as no parking zone. The major concern that was put on table to solved on this plan was to remove parked vehicles that disturb the flow of traffic and movement of pedestrians inside the complex, this plan tries to solve that through all above-mentioned principles and decided objectives.

Ordered, organized and regulated – to move from a chaos disorganized environment to well ordered & organized system that creates a healthy working ecosystem of people, space, building and vehicles, we have to establish a strict and as well as flexible parking policies and regulations to do so.

3.2) SPATIAL PLAN



This spatial plan represents and shows the overall parking location and space exist, extended and proposed for the entire complex,

1) Visitors 2 wheeler *behind RPF outpost*[55]
2) Main triangle 2 wheeler parking *around exchange office*[330]/[208]
3) Visitors on road parallel 4 wheeler parking *around green patch near west gate*[10]
4) Staff car parking *near west gate boundary wall*[8]
5) Staff and contract car parking *porch* + nominated officers garages[10G+30]
6) Staff 2 wheeler shed parking *near hindi section*[64+20(12-now)]
7) Staff 2 wheeler parking *behind DRM office building*[32+22]
8) Staff car parking *behind DRM office building & near cash office*[10+4]
9) Staff 2 wheeler parking *near steel tank*[32+30]
   - (**Note**: **RED** marked above no.s are based on newly proposed space)

TOTAL PROPOSED PARKING CAPACITY

TOTAL NO. OF CAR BAYS = 10G +26(+4) + 10 + 8 +10 + 4 = 68(+4)

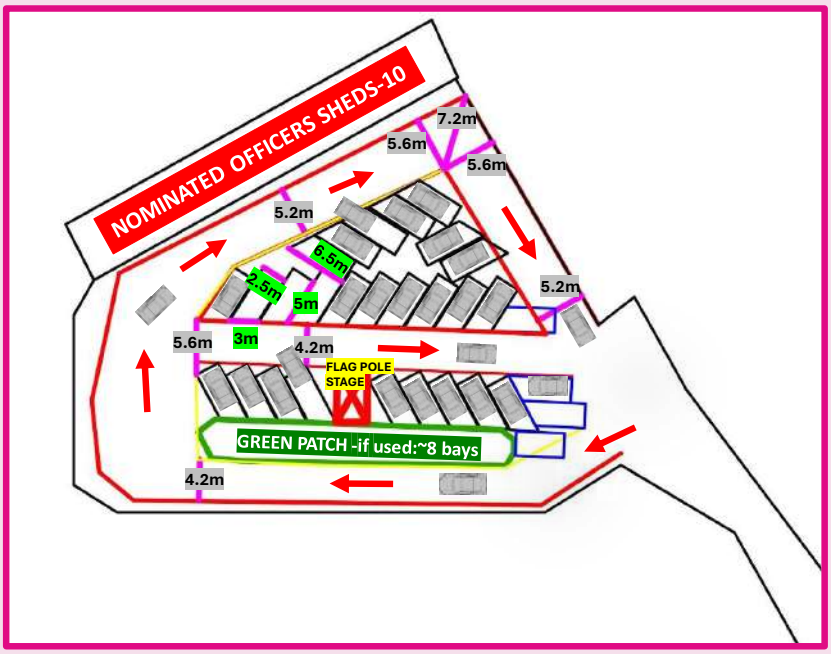
TOTAL NO. OF 2W PARKING CAPACITY = 330 + 118 + 30 + 62 + 55 + 20 = 585+

[including 170 new]

3.3) PARKING STRUCTURAL DESIGNS

60 degree model in porch area

Advantage:
Easy to park in & out
Disadvantage:
less parking bays



TOTAL NO. OF BAYS
= 10S + 26 + 4

LOACATION: DRM complex porch

TYPE: 4-wheeler parking

PARKING CAPACITY:

- Permanent built nominated officers garage = **10G**
- Porch space holding capacity = **26 + 4 bays**
- Total capacity = **36/~40 bays**

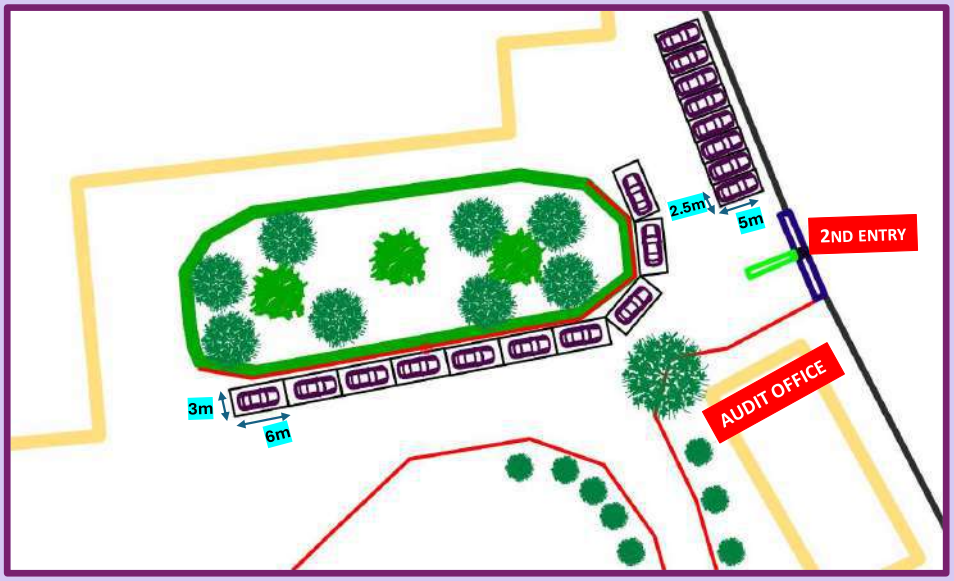
BAY SIZE: **3m * 5.5m (60 degree)**

DESIGN DETAILS: This structural plan is only formulated as efficient design to increase parking bays capacity and mobility to attain uniformity and disciplined system. It is a 60° parking plan that would enable the ease of parking in and out. This also consider the garage users accessibility and mobility in account to produce a complimenting plan. [*alternate 90-degree plan is attached in annexure*]

MOVEMENT: The mobility is facilitated through 3 lane/aisle as one-way directed flow as given in the above plan which ensures no opposite traffic, proper turning radius and inclusive garage parking.

ADDED(PROPOSED) AMENITIES: polycarbonate roof, convex road mirror at turning points, sign boards and name(reserved) board.

PARALLEL PARKING and 2nd ENTRY on road



TOTAL NO. OF BAYS = 10(~11) + 8

LOACATION: Near 2nd entry/ around green patch

TYPE: 4-wheeler parking

PARKING CAPACITY:

- Visitors on road parallel 4 wheeler parking around green patch near west gate = **10(~11)**
- Staff car parking near west gate boundary wall = **8**
- Total capacity = **19 bays**

BAY SIZE:

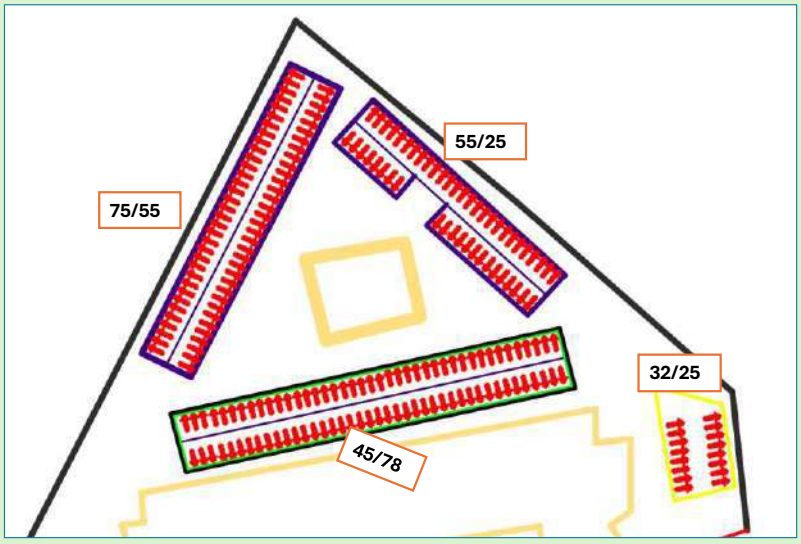
1. **3m * 6m (0 degree/parallel)**
2. **2.5m*5m (90 degree)**

DESIGN DETAILS: Officially bay marked car parking on road, to promote and control parking around office area and road. Parallel parking to reduce road coverage space (for visitors) with time limit restrictions and 90-degree side to promote additional parking for staffs.

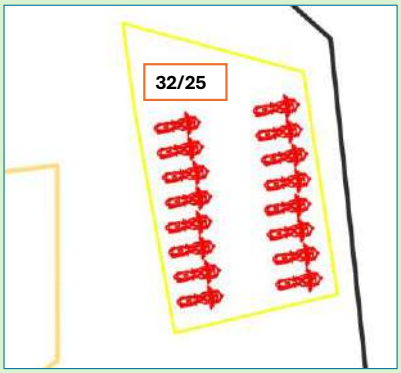
MOVEMENT: parallel side have 1m back and front extra bay size for easy parking movement and directly access the 2nd entry

ADDED(PROPOSED) AMENITIES: polycarbonate roof (only on staff 90-degree side), convex road mirror at turning points, sign boards and name(reserved) board.

2W PARKING near MAIN ENTRY



Visitors 2W parking



NO. OF 2W P.CAPACITY for visitors = **55+**

TOTAL NO. OF 2W P.CAPACITY(excluding visitors) = **330**(+10)

LOACATION: Near main entry/ around exchange office & behind RPF post cabin

TYPE: 2-wheeler parking

PARKING CAPACITY:

- Visitors 2-wheeler parking behind RPF post cabin = **55+**
- Staff 2W parking on behind personal and s&t office building= **78+45**
- Staff 2W parking opposite to exchange office= **55+25(+10)**
- Staff 2W parking near boundary wall against main road= **75+55**
- Total staff parking existing capacity = **208+**
- Total staff parking proposed capacity = **330(+10)**

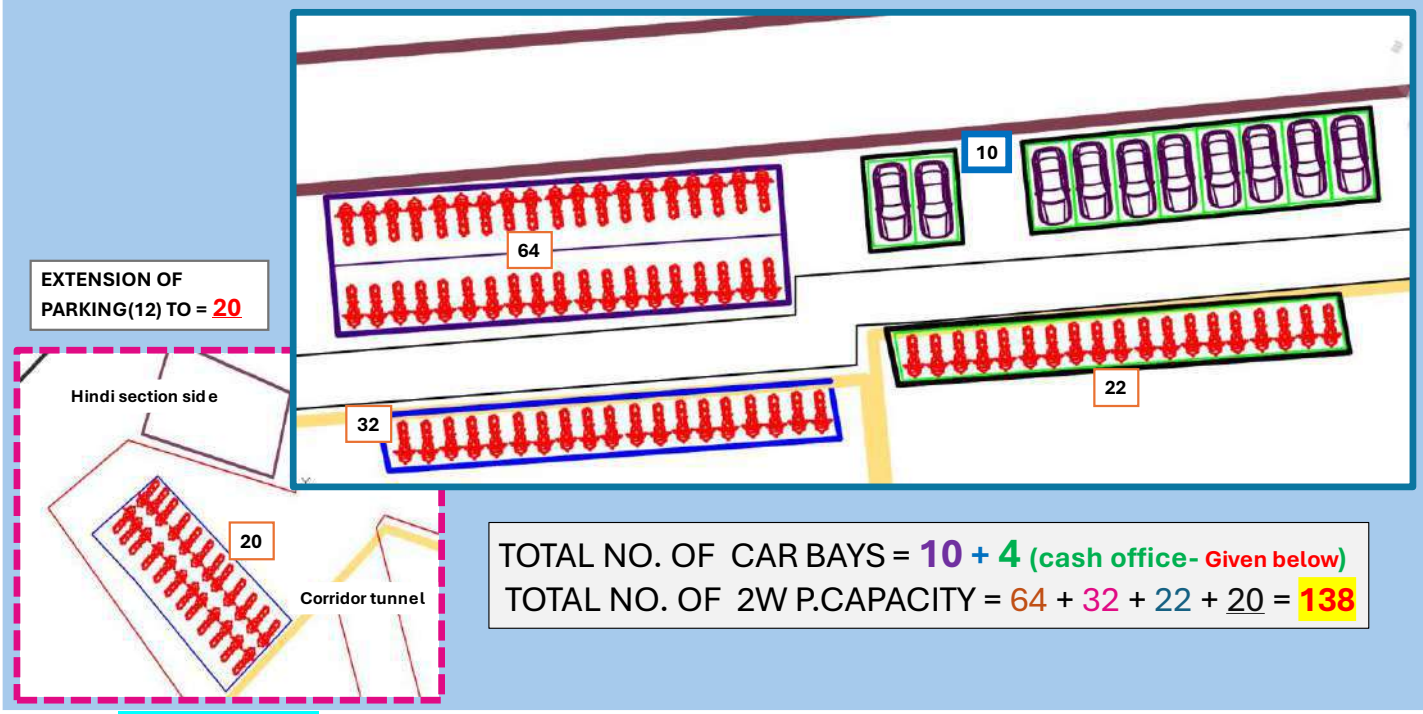
BAY SIZE: **2m*0.75m, aisle b/w row=2m**

DESIGN DETAILS: The provision additional row opposite and parallel to the existing one by extending the pavement on exchange office side with mobility aisle between the rows. Proposal of row control parking on visitors' side with time limit restriction.

MOVEMENT: Through aisle between the opposite rows connected to main gate via pavement lane.

ADDED(PROPOSED) AMENITIES: polycarbonate roof, security camera, sign boards and number markings.

Behind DRM office CAR and 2WHEELER



LOACATION: Behind DRM office & Hindi section side

TYPE: 2 and 4-wheeler parking

PARKING CAPACITY:

- Staffs alloted 90 degree 4 wheeler parking behind DRM office and cash office(*remaing figure is given below along next plan*) = **10 + 4(~5) = 15 bays**
- Staff 2-wheeler parking at old cycle shed and hindi section side shed= **64+20(12)**
- Staff 2-wheeler parking behind car garage/opp. Old cycle shed= **32**
- Staff 2-wheeler parking behind DRM office/ opp. to above mentioned car parking= **22**
- Total 2w parking capacity = **138**

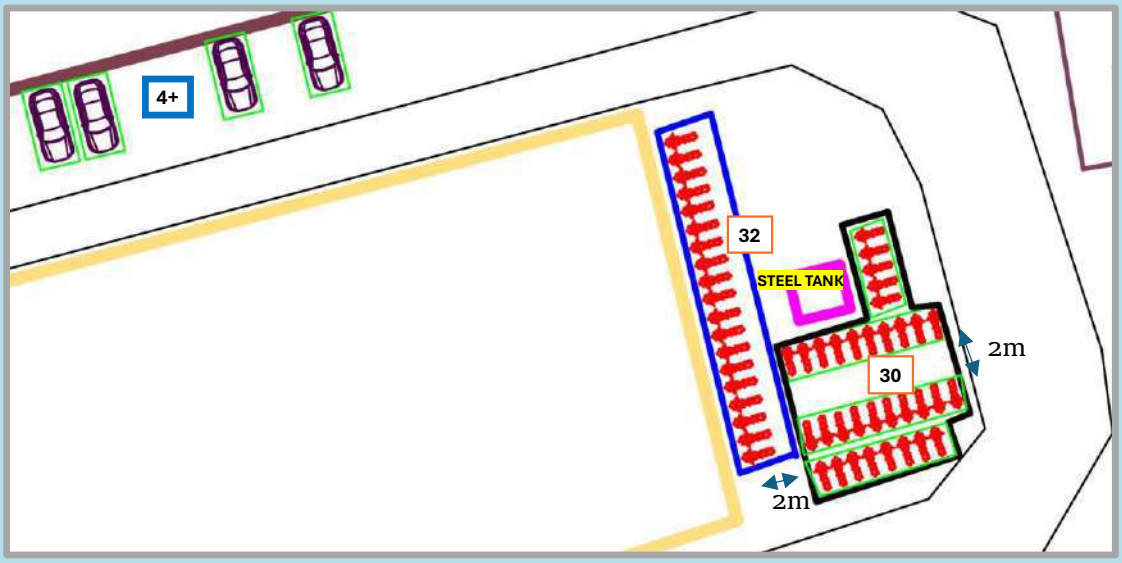
BAY SIZE: **Car=2.5m * 6m (90 degree), 2w=existing**

DESIGN DETAILS: The car parking bay marking to induce order and increase accommodation for enabling usage of this space. For the same reason cover shed mandated in this proposal to enable usage even for 2wheelers and extension of parking space opp. Hindi section to increase capacity from 12 to 20.

MOVEMENT: Connected in a cycled fashion to the porch around DRM office building from Hindi-section tunnel to DRM building corridor.

ADDED(PROPOSED) AMENITIES: polycarbonate roof (on 10 car bays and opp. 2w parking), convex road mirror at turning points, sign boards and name(reserved) board.

2 WHEELER parking NEAR TANK



TOTAL NO. OF 2W P.CAPACITY = 32 + 30 = 62

LOACATION: Near steel tank/ canteen

TYPE: 2-wheeler parking

PARKING CAPACITY:

- Staff 2-wheeler parking near steel tank(existing) = **32**
- Staff 2-wheeler parking infront of steel tank(proposed) = **30**
- Total proposed capacity = **62**

BAY SIZE:

2m * 0.75m (for **new**), **old /new aisle extend=2m**

DESIGN DETAILS: By using available pavement space around steel tank more 2wheeler parking capacity can be inserted for the staffs who are parking in NO parking zone near this spot.

MOVEMENT: The movement is oscillated through DRM office building corridor to front porch

ADDED(PROPOSED) AMENITIES: polycarbonate roof, sign boards and number marking.

3.4) SUPPORTING AMENTIES, SIGNS AND MARKINGS

1. **COVER SHED**: shed cover will help to enable this plan as people tends follows a behavior *(in the complex)* of being disciplined in parking at space with shed cover that looks dedicated and formal rather than just a space allotted through just a notice.

2. **SIGNAGE BOARDS**: to induce strict regulation signage boards indicating the allotment (like STAFF ONLY), restriction (TIME LIMIT-for visitors, NO PARKING ZONE) would mediate to the staffs and public as an information notice.

3. **CONVEX ROAD MIRROR**: as inside the office complex, road is enclosed around by buildings and trees convex road mirror would help and support the traffic in inside the complex by identification of moving vehicles especially at turns and corners.
4. **FIRE EXTINGUISHER**: for the fire safety and precaution.
5. **ROAD MARKINGS**: markings on road to allocate car bays, direction of movement and to manage and control rows of parking.

4. PARKING POLICY AND POST-ACTION PLAN

PARKING POLICY:

Parking Allocation and Zoning- ALLOCATION PLAN would be revised by RPF based on this plan. Whereas zoning will be categorised into,

- Designated Zones
- Reserved Spaces
- Restricted zones

Time-Limited Parking for visitors: Implement time-limited parking spaces for short-term visitors to ensure availability and turnover.

Fee Structure and fine Systems: Establish a fee structure based on extra duration of parking and implement fine system if they don't co-operate with the regulations or time limit

Access Control and Security: Use automated barriers or gates to control entry and exit, ensuring only authorized vehicles can access the parking area. Install CCTV cameras and employ security personnel to monitor the parking area and deter theft or vandalism.

Maintenance and Upkeep: Conduct routine inspections and maintenance of the parking area, including lighting, signage, and pavement conditions. Ensure the parking area is kept clean and free of litter through regular cleaning schedules.

POST-ACTION PLAN

RPF through SOP have proposed and working on vehicle pass for staffs which would specify parking location, special permission that would specify physically abled staffs (As their allotment would be the only exception to park Infront of their office building).

Post guard on security clearance for outsiders and visitors

Sign and parking labeling and along with the old practice of lot numbering which would in turn used to go all the way to allot specific parking bays to every individual.

Continuous Improvement with Conduction of periodic reviews of the parking facilities and policies to ensure they remain effective and efficient. Technological Upgrades-Stay updated with the latest parking management technologies and implement upgrades as needed. Sustainability Measures- Continuously explore and implement new sustainable practices to reduce the environmental impact of the parking area.

5. RECOMMENDATION & CONCLUSION

RECOMMENDATION

1. **Implement Signage and Markings:**
    - Install clear and visible signage for entry, exit, and directional flow within the parking area.
    - Mark individual parking bays and pedestrian walkways distinctly to ensure proper vehicle alignment and pedestrian safety.
2. **Enhance Lighting and Security:**
    - Install adequate lighting throughout the parking area to improve visibility and safety during nighttime.
    - Implement security measures such as CCTV cameras and periodic patrols to deter potential theft and vandalism.
3. **Provide Accessibility Features:**
    - Ensure that designated parking spaces for persons with disabilities are located near the entrance and are compliant with accessibility standards.
    - Include ramps and tactile paving to assist individuals with mobility impairments.
4. **Incorporate Green Spaces:**
    - Integrate landscaped areas with trees and shrubs to enhance the aesthetic appeal and provide shade, reducing the heat island effect.
    - Use permeable paving materials in some sections to improve drainage and reduce stormwater runoff.
5. **Encourage Sustainable Practices:**
    - Install electric vehicle (EV) charging stations when required.
    - Consider implementing a carpooling system to reduce the number of vehicles and promote eco-friendly transportation options.
6. **Regular Maintenance and Monitoring:**
    - Establish a routine maintenance schedule to ensure that the parking area remains in good condition.
    - Conduct periodic assessments to identify and address any emerging issues promptly.

CONCLUSION

The parking plan for the Divisional Railway Managers Complex has been meticulously developed to address the current and anticipated needs of the facility. By implementing the recommended strategies, the Southern Railways can significantly improve the efficiency, safety, and overall functionality of the parking area. Regular maintenance and monitoring will ensure that the parking facilities remain in optimal condition, addressing any issues proactively and maintaining high standards of service.

This project has provided valuable insights and practical experience, contributing to the professional growth of myself.

6) ANNEXURES-I: SOP BY RPF

दक्षिण रेलवे SOUTHERN RAILWAY

वरि. मंडल सुरक्षा आयुक्त का कार्यालय/ रे.सु.ब
Office of the Sr.Divl Security Commissioner/RPF
तिरुच्चिराप्पल्ली/Tiruchchirappalli

संख्या.No. SXC/09/MISC/2024 दिनांक Dated: 17.05.2024

All Branch Officers
TPJ Division

विषय/Sub: Standard Operating Procedure for the Parking of Vehicles at Divisional Railway Manager's Complex, Tiruchchirappalli – reg.

संदर्भ/Ref: SOP approved by DRM/TPJ dated 16.05.2024

****

DRM/TPJ has approved Standard Operating Procedure for the Parking of Vehicles at Divisional Railway Manager's Complex, Tiruchchirappalli regarding procedure to be followed by the employees working at Divisional offices & wayside stations with regard to parking of their vehicles to thwart traffic congestion inside the complex. Copy of the SOP is herewith enclosed for ready reference.

The department-wise allocation of parking space as approved in the SOP is re-produced below:

Sl No.	Locations	Department
1	Behind RPF Out Post	Visitors
2	Exchange office side (near boundary wall)	Personnel
3	Opposite of Exchange Office	Personnel
		Commercial
4	Hindi section side	Traffic Account
		Hindi Section
		Stores
5	Near Central Dispatch office	Accounts
6	Government Railway Police side	General branch
		Operating
7	Cash office area	Engineering
		GMT
8	Steel Tank near Canteen	Security
		Electrical
9	Old Divisional Security Commissioner Office	Security
10	Behind Sr. Divisional Personnel Officer office	Safety
		Wayside station employees
11	Near Exchange office	S&T
12	Physically challenged employees	May park vehicle nearby to their concerned department
13	Branch Officers' Four wheeler	Nominated officer's four wheeler shed

The under mentioned instructions of the SOP for issuing of vehicle pass and parking inside the DRM Complex may be circulated to all employees working in Divisional office and wayside stations for strict adherence.

1. PC/TPJ will issue Vehicle Pass sticker having serial number to railway employee, which has to be pasted on their vehicle. He shall make proper record of issuance of Vehicle Passes.

2. PC/TPJ to coordinate with concerned SSE/W/TPJ for placing of department name boards at all the allotted parking area.

3. Employees intending to park inside the DRM Complex shall give requisition to concerned nominated supervisor in DRM Complex and the same would be forwarded to PC/TPJ to issue of passes. On the pass the location of allotted parking area will also be mentioned by PC/TPJ.

4. Other than Railway Employee, all vehicles without Vehicle Pass should be parked in Visitor's parking area. All employees and concerned department should ensure the Visitors have parked their vehicle in allotted parking area.

5. Vehicle not to be parked by the side of the road in the DRM Complex.

6. Follow instructions of RPF personnel regarding parking regulations.

7. A different kind of sticker and vehicle passes will be issued to Physically challenged employees.

8. Four wheelers of the employees should be parked near the Cash Office.

9. Govt. vehicle and contract vehicles to be parked in surrounding areas of quadrangle and nominated officers' four wheeler shed.

10. Vehicle should not be parked in front of west gate as it will be utilized during any emergency. PC/TPJ should ensure.

11. If any violation, Post Commander/RPF/TPJ can take legal action against the violators duly reflected in their service register.

DA/-02 pages.

वरि. मंडल सुरक्षा आयुक्त/रे.सु.ब
Sr. Divisional Security Commissioner/RPF
तिरुचिरापल्ली/Tiruchchirappalli

Copy to: PC/TPJ for necessary action.
DA/-02 pages.

STANDARD OPERATING PROCEDURE FOR THE PARKING OF VEHICLES AT DIVISIONAL RAILWAY MANAGER'S COMPLEX, TIRUCHCHIRAPPALLI

Ref: DRM/TPJ approval vide note no. 3 on file No. SR-TPJ0SECU(CC)/23/2023-Div.Inspector/CC/RPF/TPJ (Computer No. 397771)

****

A. The SOP is prepared from the suggestions of DRM/TPJ for the proper parking of two wheelers and four wheelers at DRM complex.

B. Staffs from headquarters and wayside stations are parking their vehicles in every nook and corner of DRM complex, thus leading to haphazard parking of vehicle and hindrance to the free movement of vehicles. This haphazard parking also tarnishes the image of railways among the public. The aim is to allot all railway employees earmarked location to park their vehicle.

C. The department-wise and for visitors vehicle parking allotment has already been provided and the same is appended below:

Sl No.	Locations	Department
1	Behind RPF Out Post	Visitors
2	Exchange office side (near boundary wall)	Personnel
3	Opposite of Exchange Office	Personnel
		Commercial
4	Hindi section side	Traffic Account
		Hindi Section
		Stores
5	Near Central Dispatch office	Accounts
6	Government Railway Police side	General branch
		Operating
7	Cash office area	Engineering
		GMT
8	Steel Tank near Canteen	Security
		Electrical
9	Old Divisional Security Commissioner Office	Security
10	Behind Sr. Divisional Personnel Officer office	Safety
		Wayside station employees
11	Near Exchange office	S&T
12	Physically challenged employees	May park vehicle nearby to their concerned department
13	Branch Officers' Four wheeler	Nominated officer's four wheeler shed

D. The under mentioned procedures for issuing of vehicle pass and parking inside the DRM Complex may be circulated to all employees working in Divisional office and wayside stations for strict adherence.

1. PC/TPJ will issue Vehicle Pass sticker having serial number to railway employee, which has to be pasted on their vehicle. He shall make proper record of issuance of Vehicle Passes.

2. PC/TPJ to coordinate with concerned SSE/W/TPJ for placing of department name boards at all the allotted parking area.

3. Employees intending to park inside the DRM Complex shall give requisition to concerned nominated supervisor in DRM Complex and the same would be forwarded to PC/TPJ to issue of passes. On the pass the location of allotted parking area will also be mentioned by PC/TPJ.

4. Other than Railway Employee, all vehicles without Vehicle Pass should be parked in Visitor's parking area. All employees and concerned department should ensure the Visitors have parked their vehicle in allotted parking area.

5. Vehicle not to be parked by the side of the road in the DRM Complex.

6. Follow instructions of RPF personnel regarding parking regulations.

7. A different kind of sticker and vehicle passes will be issued to Physically challenged employees.

8. Four wheelers of the employees should be parked near the Cash Office.

9. Govt. vehicle and contract vehicles to be parked in surrounding areas of quadrangle and nominated officers' four wheeler shed.

10. Vehicle should not be parked in front of west gate as it will be utilized during any emergency. PC/TPJ should ensure.

11. If any violation, Post Commander/RPF/TPJ can take legal action against the violators duly reflected in their service register.

Sr. DCM/TPJ

Sr. DPO/TPJ

16/05/24
Sr. DSC/TPJ

ADRM/TPJ 16/5/24

DRM/TPJ

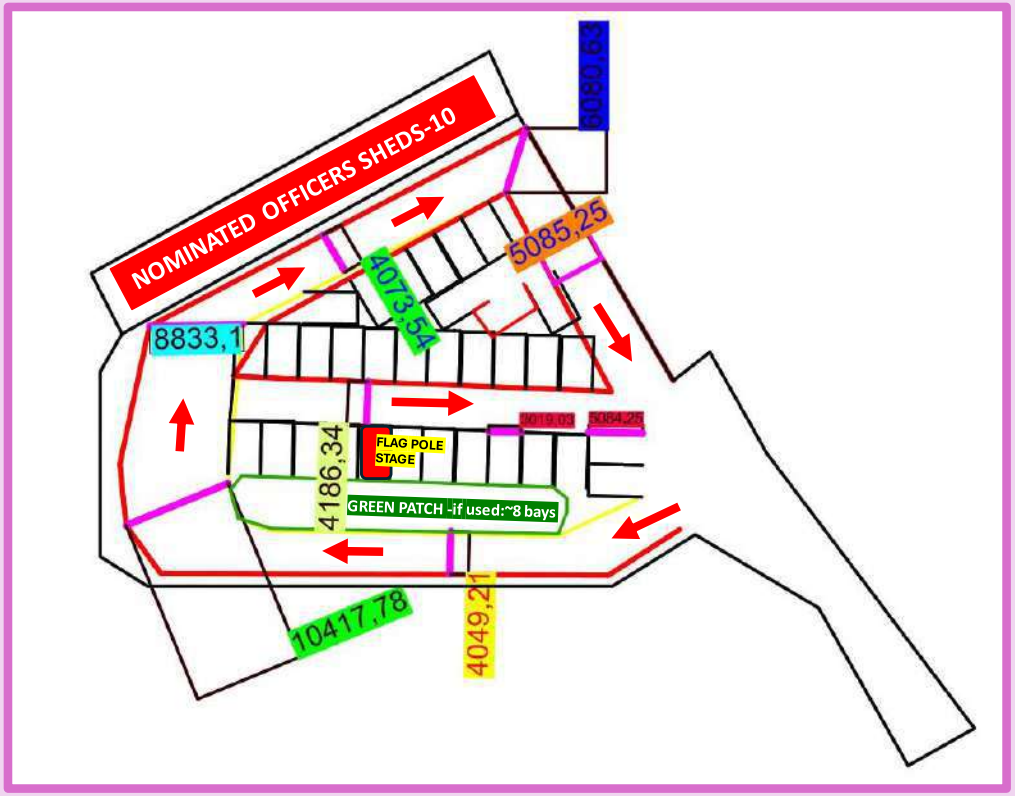
Ok.
16/05/2024.
ADRM

POSSIBLE MITIGATIONS

Spatial planning with effective arrangement of parking bay in porch area to increase car parking bays and induce order.

90 degree model

Advantage:
more parking bays
Disadvantage:
Hard to park & reverse



TOTAL NO. OF BAYS
= 10S + 29 + 3

TEAM ID: U7479QUT

NATION BUILDING

LEAP FROGGING IN EDUCATION

GAJENDRA LODHI
NITESHKUMAR S K
Maulana Azad National Institute of Technology, Bhopal [MANIT]
Central Zone

LEAP FROGGING IN EDUCATION

LEAP FROGGING is a developmental concept in which we leap/skip generation of old, traditional and existing practice to new, latest and advance practice in order to take ahead in the field from others.

"So by enforcing LEAP FROGGING IN EDUCATION we emphasize on promoting advancement in education system and curriculum that *expertise* students on *current & future application* rather than in old existing one to *advance ahead in the field universally*".

Youth unemployment rates by level of education in India, 2012-2018

-Mehrotra, Santosh & Parida, Jajati. (2019). India's Employment Crisis

MAIN OBJECTIVES AND SOLUTIONS

1. **Redefining existing traditional courses** based on current & future applications and trends (not just academic but professional courses too).
2. **Introducing new courses** covering new fields of 21st century for paving its foundation**(pioneering)** to rise and which in turn *give rise to new profession, employment & industry.*
   - Brand new courses
   - Diverging from existing field
   - Converging from existing field(multi-discipline, tailor made by student choice, major-minor[NEP,2020])
3. **SUPER DEGREES(courses)**- as the terminology suggests it is a high-end program that aims to combine the prospects of UG+PG/UG+PG+phD in a single-go course in a *short span* comparatively (it is similar to already in practice dual-degree concept)
   [Expectation: This super degrees is *more valuable/higher-standard than an actual PG* in the same field from different nation/program]
4. **Expertise the field** - Structuring institutions to create specialists in all fields with universal practice based curriculum.
   - Uni-specialist: Expertise in specified subject.
   - Multi-specialist: Expertise in multiple subjects.
   - "We need cardiologists who specializes in *heart of the subject* and general surgeons who can *practice multi-field operations*"
5. **Reducing the gap between graduation and employment**
   - Skill-based academics
   - Research & business based curriculum(encouraging start-up culture)
   - Integrated internship with credit in vocational training equally as academics
6. **Reducing the higher education cost** and create student self-sustaining environment
   - Diversity in courses and choice of language medium in all official Indian languages[NEP,2020]
   - *Low cost/low-zero interest loan on new majors*
   - Scholarship programs, paid-internship & part-time opportunities within campus
7. **Re-systemize the higher education system** for supporting the new generations ideology, workmanship & above objectives. Through,
   - Cancel the rigid system & *enforce a flexible model*
   - Institutional and state guaranty & support
   - Smart learning(tech-based): virtual education[with degree-virtual correspondence], open online resources & materials, etc.
   - Teaching: new teachers(expert in the current trending field & applications), certified short-term lectures on minors.
8. **Life long updating degree**(adapting through times)-breaking this rigid-degree system, we can establish life long updating education with support of degree upgradation through years adapting the change forever.

ISSUES IN EXISTING EDUCATION SYSTEM

1. Academic/exam based.
2. Unaffordability of higher education.
3. Decades old courses.
4. Lack of multidisciplinary courses(before NEP,2020).
5. Lack of choices.
6. Unemployment of graduates.
7. Under-employment.
8. Skill-mismatch after graduation.

"Unemployment rate among graduates in a the age group of 15 years and above is 13.4 per cent in 2022-23"

-source: Periodic Labour Force Survey (PLFS)

THINGS WE'RE GOING TO LEAP

1. Traditional courses and curriculum.
2. Gap between graduation and employment.
3. Rigid and bounded system of education(proposal of life long updating degree).
4. Skill gap between UG and PG/specialist.

9. **Others**: awareness & career guidance, equity & celebration of culture, mental and physical health education.

PROCESS

DATA COLLECTION & ANALYSIS
- CURRENT TRENDS
- STUDENTS PREFERENCES
- SOCIO-ECONOMIC FACTORS & ISSUES

↓

IDENTIFICATION OF GOALS & OBJECTIVES AND ALIGN IT WITH EXISTING EDUCATION POLICY & PROGRAMS

↓

MULTI-DISCIPLINARIES STAKEHOLDER COLLABRATIVE DECISION MAKING

↓

PREPARING THE ACTION PLAN

↓

REVIEW OF ACTION PLAN & FINALISATION

↓

TEST EXECUTION AND REVIEW

↓

FINAL IMPLEMENTAION TOWARDS ACHIEVING OUR GOALS

IMPACTS AND BENIFITS

1) Creates effect on current & future trends
   - ❑ High efficiency and production in economy
   - ❑ Reduce competition and create advantage over other nation
   - ❑ Aligns with new generations trend
2) Pioneering in various new fields in the world scale and advance ahead in existing field
3) Emergence of new industry and new employment opportunities[new economics] due to the effects of new courses and fields.
4) Produce experts, specialists/multi-field specialists in various fields and real life skills universally in short span.
5) Reduce competition over one or few jobs that cause unemployment & under-employment of graduates and increase employment rate.
6) Highly standardise Indian higher education and institution for quality & excellence universally.
7) Offer more choices to students on planning their career and achieve in the self-preferred field(s).
8) More choices, less cost = more participants. Which helps to encourage & increase students to take tertiary education to prospect from newly formed or growing industries.
9) Encourages start-up culture & entrepreneurship to increase home-based corporates and industries.
10) Reduces gap between academic and real skill which in turn reduces gap between graduation and employment.
11) Able to create affordable & self-sustainable environment for students.
12) Life long upgrade and adaptability in knowledge & change.

AREA OF IMPACT	CHANGES
System	Smart, flexible, adaptable & equitable
Curriculum	New generation course, student choice, universal application
Institution	New field, high quality
Resources	Reuse, upgrade, smart management, virtual
Student success	Self-sustainability, guaranteed employment opportunity

FEASIBILITY OF THE MODEL

PROBABLE ISSUES	MITIGATIONS
Curriculum load	Concentrated core essential + reducing old application + focus on current trend + student's preference
Avoidance of new courses by students to avoid risk on new field	Expert formulated courses, affordability & awareness program in high school
Implementation	Institution & government support, pre-planning and test execution
Financial factor & resource management	Low cost implementation by using/redeveloping existing institutes, cross-combine interdisciplinary teaching faculties to support new courses + new teachers who excel in the new field in real life
Slow-pace change[but faster is better]	Focusing more on upcoming generation, starting to make them aware in high school level itself through-mandatory career consultancy by school and government

LINKAGE WITH THE DEFINED IMPACT GOALS

CREATION OF 100 WORLD-CLASS HIGHER EDUCATION INSTITUTIONS (HEIS) BY 2047

- Pioneer the field: In advancement of knowledge and application in the new fields.
- Expertise the field: Produce experts and specialists in various field flourishing higher in the domain universally.
- Highly standardise Indian institutes globally through quality and novelty
- Universal practise based curriculum.
- Encourage researches that change/affect global ecosystem and economy positively.
- Show-case of Indian & local culture, art and talents to the world.
- Special institutions for special courses & specialist cultivation

GROSS ENROLMENT RATIO (GER) OF 75% AT THE TERTIARY LEVEL BY 2047

- More choice-more participants
- Self-sustainable student environment
  - Self affordability for higher education
  - Paid internship within campus
- Employment guarantee and opportunity after graduation
- No language barrier-education will be available in all Indian official language medium
- Pre-college programs-to initiate guidance to enter colleges(free entrance exam preparation guidance, free resources etc.
- Awareness and mandatory career guidance in high school and after graduation

CONCLUSION- FINAL STATEMENT

India as attained its current growth from leap frogging where we skipped & stepped on growing sectors like IT rather than competing in traditional sectors smartly. But at the same time now **we actually lack in conscious Contemporary leap frogging** and instead we try to imitate west/developed countries or follow their path that keeps us behind them continuously. So, in order to improve ourself and pioneer ahead from other developed nation we can use this of leap frogging[in education] as a potential model to achieve **VIKSIT BHART @ 2047.**

"KNOWLEDGE GETS ITS LIFE WHEN IT IS APPLIED"
Viksit Bharat@2047

URBAN DESIGN

BOAT CLUB [UPPER LAKE]

STUDY OF SPACE

211109041-NITESHKUMAR S K

BOAT CLUB-UPPER LAKE

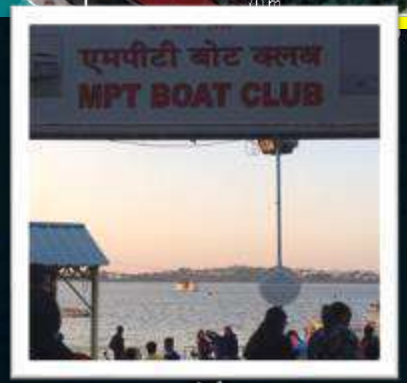
UD

BOAT CLUB

urban design

Legend

- BOAT CLUB
- Boat Club, Bhopal
- COFFEE DAY CAFE
- COMMERCIAL BUILDINGS
- FOOTPATH
- LAKE
- ROAD
- Upper lake Bhopal
- WIND&WAVES-MP TOURISM



Legend

- Boat Club, Bhopal
- Lake
- LAKE VIEW ROAD
- PLACES/STRUCTURES

Google Earth

Image © 2024 Airbus

Lake View Point

Pump Ho

GAMING CENTRE

TANK STRUCTURE

COFFEE DAY CAFE

WIND&WAVES-MP TOURISM

Boat Club, Bhopal

Lake View Road

WIND&WAVES-MP

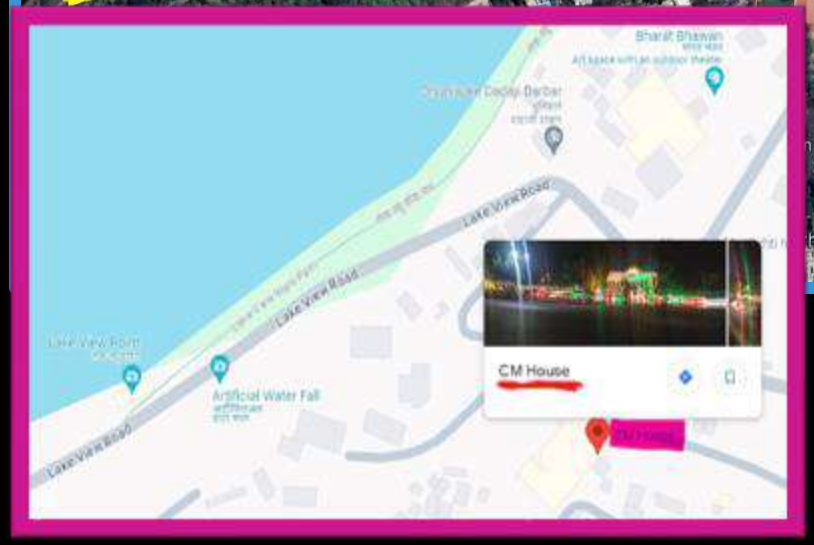
Lake View, Bhopal

STALLS

100 m

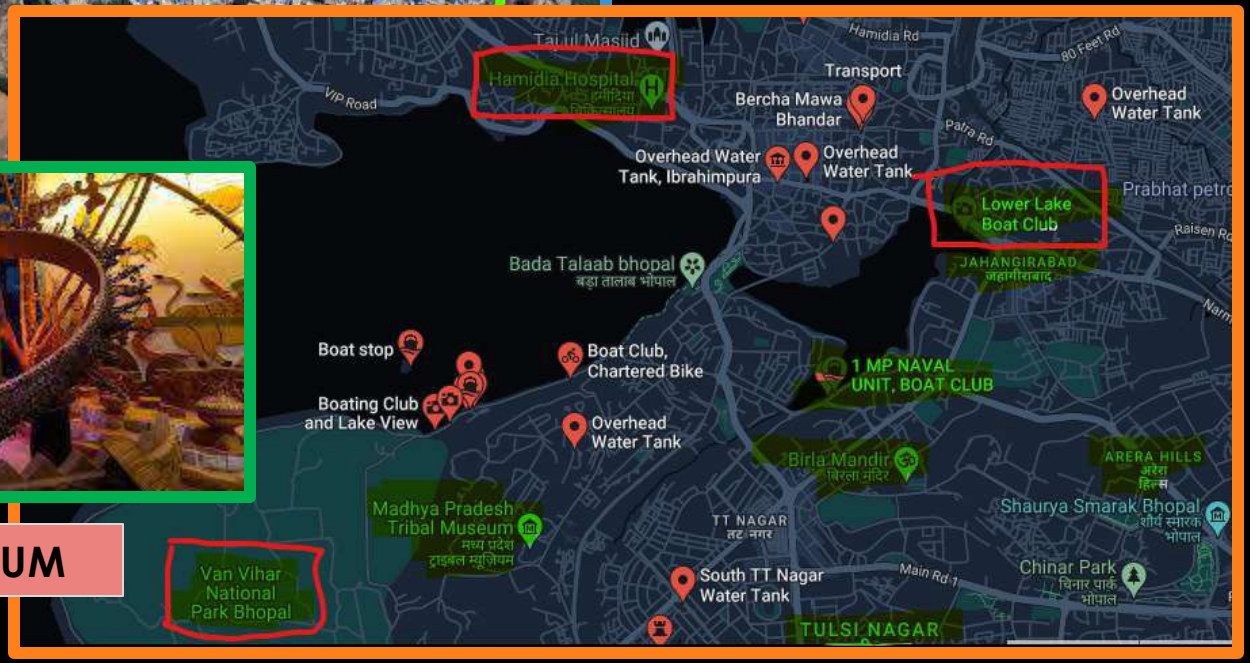
CONNECTIVITY AND SURROUNDING ENVIRONMENT

BHARAT BHAVAN

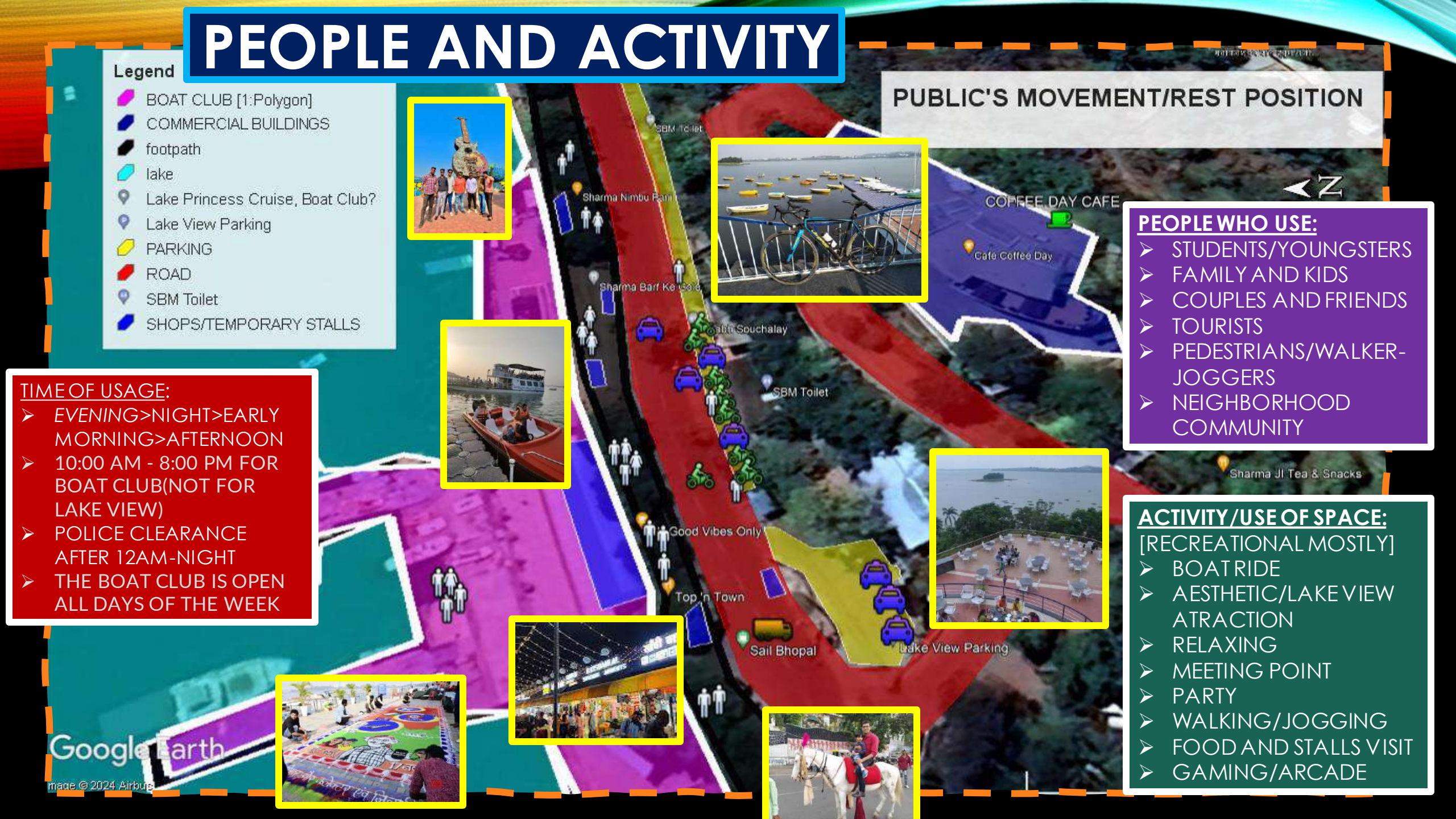


CM HOUSE

MP TRIBAL MUSEUM



PEOPLE AND ACTIVITY



PEOPLE WHO USE:

- STUDENTS/YOUNGSTERS
- FAMILY AND KIDS
- COUPLES AND FRIENDS
- TOURISTS
- PEDESTRIANS/WALKER-JOGGERS
- NEIGHBORHOOD COMMUNITY

TIME OF USAGE:

- *EVENING*>NIGHT>EARLY MORNING>AFTERNOON
- 10:00 AM - 8:00 PM FOR BOAT CLUB(NOT FOR LAKE VIEW)
- POLICE CLEARANCE AFTER 12AM-NIGHT
- THE BOAT CLUB IS OPEN ALL DAYS OF THE WEEK

ACTIVITY/USE OF SPACE:

[RECREATIONAL MOSTLY]

- BOAT RIDE
- AESTHETIC/LAKE VIEW ATRACTION
- RELAXING
- MEETING POINT
- PARTY
- WALKING/JOGGING
- FOOD AND STALLS VISIT
- GAMING/ARCADE

ISSUES IN BOAT CLUB AS A PUBLIC SPACE

STALLS ENCROUGING FOOTPATH AND HIDING VIEW

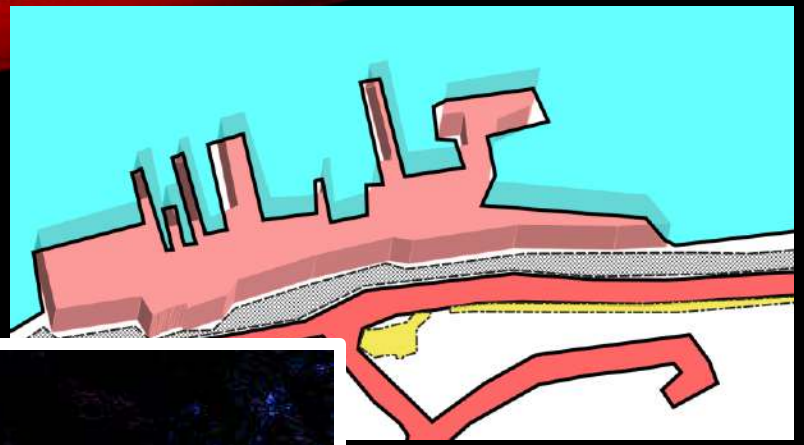
WATER LOGGING AND PAVEMENT DAMAGES

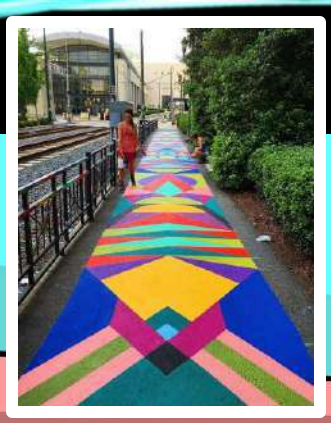
SPITTING & LITTERING

PEDESTRIAN VS VEHICLES CLASH

INACCESIBLE TO WHEEL-CHAIR

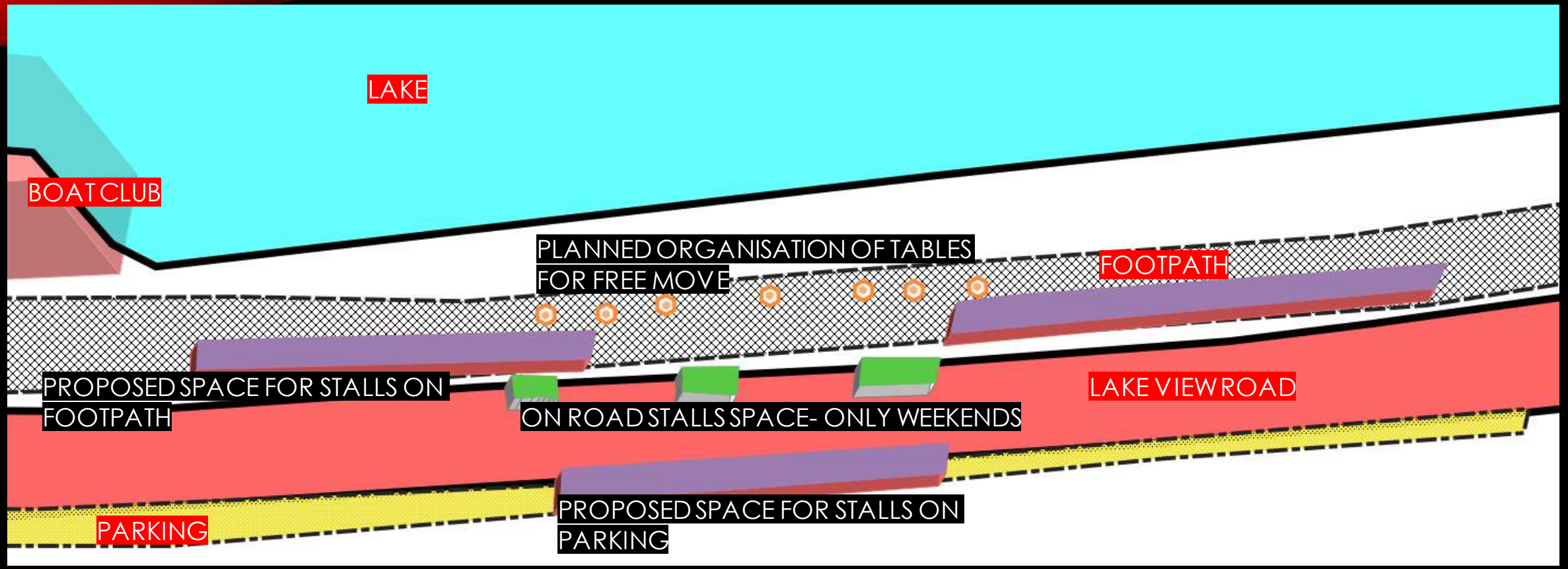
MITIGATIONS AND URBAN DESIGN PROPOSALS





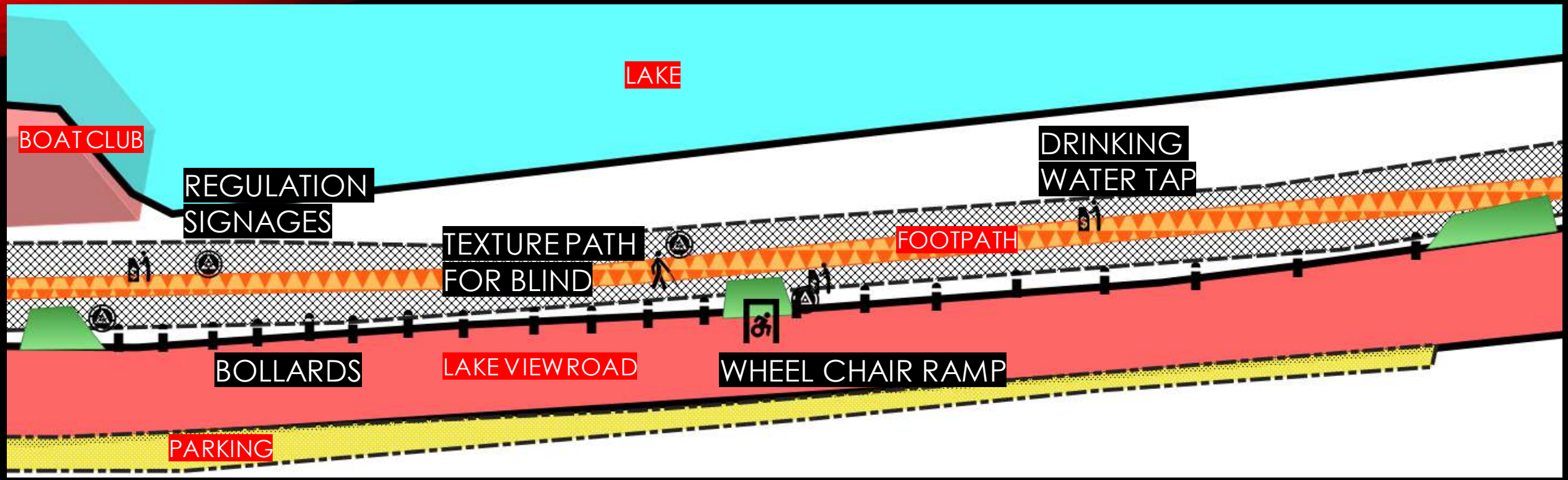
URBAN DESIGN SOLUTIONS TO AVOID TRAFFIC- PEDESTRIAN CHALLENGE AND INCREASE THE AESTHETIC

- HALF-LANE ON WEEKENDS AS THERE ISNO ALTERNATE PATH
- PROPOSAL OF SQUARE TO PROMOTE WALKING AND AESTHETIC
- EXTRAVAGANT LIGHTING ON STREETS BASED ON SEASONS WARM LIGHT IN WINTERS AND COOL LIGHT IN SUMMERS



URBAN DESIGN TO INCREASE FREE MOBILITY AND STALLS COMPROMISE

- ALLOATED SPACE FOR STALLS AND PLANNED ORGANISATION OF STREET FURNITURES



URBAN DESIGN MITIGATION FOR INACCESSIBILTY

- INCLUSIVE DESIGN WITH WHEEL CHAIR RAMP, BLIND TEXTURE PATH, BOLLARDS FOR SAFETY
- AND ACCESSIBLE TO ALL AMENTIES LIKE DRINKING WATER, TOILET
- SIGNAGE GOR PUBLIC REGULATION USAGE OF THE SPACE

SWANS FILLED AREA IN LAKE

WATER-LIGHT SHOW FOUNTAIN

PROPOSAL OF WATER- LIGHT SHOW FOUNTAIN

- AS THERE IS NO COMMON ACTIVITY IN ABSCENSE OF BOATING AT NIGHT
- IT IS CONSIDERE W.R.T SWANS ECOSYSYSTEM BY MAKING IT AWAY FROM THEIR COMMON SPACE

RURAL

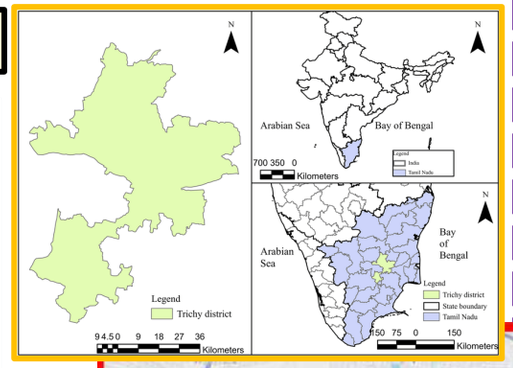
Though obliviously kilakurichy village doesn't comes municipal area boundary- still, it come comes under planning boundary and access all urban plans & benefits

SPECIAL CHARACTER- PERIPHERY VILLAGE

Kilakurichy village, located about 10 kilometers from the urban center of Trichy, offers a remarkable case study for understanding the development of a periphery village and its relationship with urbanization

The village's development showcases how rural areas close to urban centers can experience infrastructural and economic growth without losing their rural identity.

The steady improvement in literacy rates, transportation connectivity, and governance, underpinned by the Panchayati Raj system, illustrates the growing interaction between rural and urban spaces



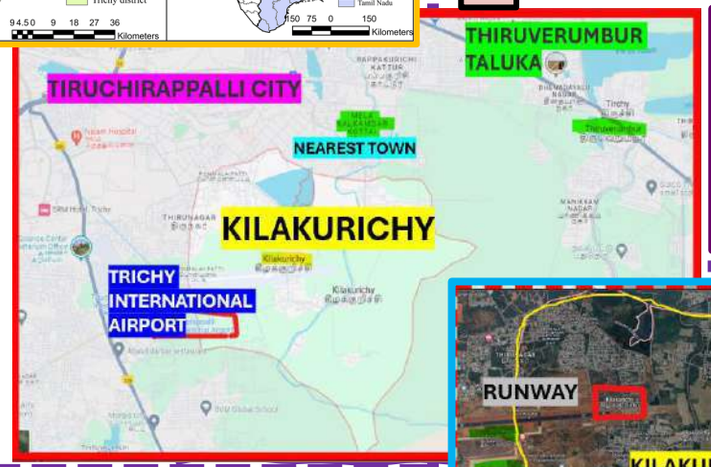
VILLAGE OVERVIEW

KILAKURICHY - VILLAGE OVERVIEW	
Gram Panchayat :	Keela Kuruchi
Block / Taluka :	Thiruverumbur
District :	Tiruchirappalli
State :	Tamil Nadu
Pincode :	620011
Area :	1026.34 hectares
Population :	8,782
Households :	2,232
Assembly Constituency :	Thiruverumbur
Parliament Constituency :	Tiruchirappalli
Nearest Town :	Melakalkondarkottai

DEMOGRAPHICS

Particulars	Total	Male	Female
Total No. of Houses	2,232	-	-
Population	8,782	4,350	4,432
Child (0-6)	725	370	355
Schedule Caste	1,199	601	598
Schedule Tribe	152	84	68
Literacy	92.65 %	96.11 %	89.28 %
Total Workers	3,555	2,550	1,005
Main Worker	2,866	-	-
Marginal Worker	689	408	281
Cultivators	175		
Agriculture labours	405		

LOCATION



- **KILAKURICHY** village is located in **Thiruverumbur taluka of Tiruchirappalli district** in Tamil Nadu, India.
- It is situated 12km away from sub-district headquarter Thiruverumbur (tehsildar office) and 10km away from district headquarter Tiruchirappalli.
- Kilakurichy is the **village panchayat** under Thiruverumbur block

GOVERNANCE & ADMINISTRATION

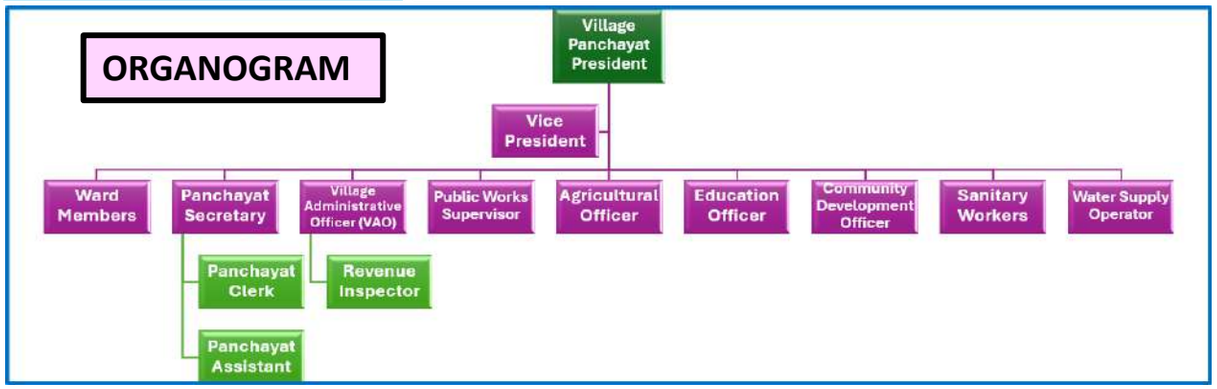
Kilakurichy is administered by a Gram Panchayat headed by an elected village president. It is part of the Thiruverumbur block in Tiruchirappalli district

PHYSICAL CHARACTERISTICS & RURAL ECONOMICS

Total sown/agricultural area is 410.85 ha.. About 410.44 ha is irrigated area.
In Kilakurichy, the Uyyakondan Canal is a vital source of irrigation, supporting the cultivation of various crops

Paddy, Banana and Black Gram are agriculture commodities grow in this village
Kilakurichy is known for its **silk Thread handicrafts**, which are a notable local craft in the village. The production of silk thread involves traditional techniques and plays an important role in the village's economy

ORGANOGRAM



KILAKURICHY VILLAGE – CASE STUDY
Urban & Rural Governance

NITESHKUMAR SK
SCHOLAR NO: 211109041
B. PLAN- SEM VII

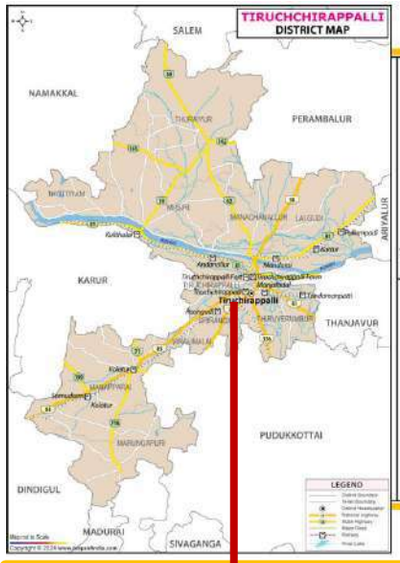
REMARKS

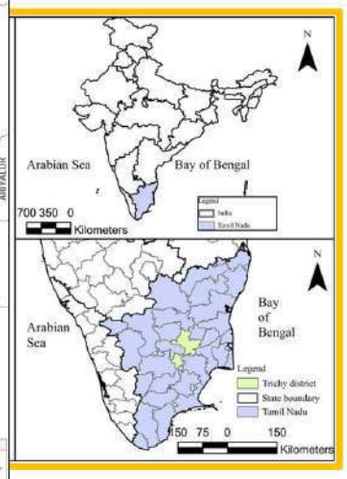
URBAN

CITY OVERVIEW

Tiruchirappalli, (formerly Trichinopoly in English), commonly known as Trichy, is a dynamic and historically significant city in Tamil Nadu. The city serves as a vital economic and cultural hub, due to its robust industrial foundation and an ever-expanding service sector.

Trichy's strategic location on the banks of the Cauvery River has historically made it a focal point of trade and culture and it is the fourth largest city and situated in the geographical center of the State Tamil Nadu





TOURISM & CULTURE

Trichy is a city that offers a rich showcase of **tourism** and **cultural heritage**, blending its ancient history with modern vibrancy. The city's **Rockfort Temple**, perched on a massive rock, and the vast **Sri Ranganathaswamy Temple** in Srirangam, **St. Lourdes Church** are prime examples of Dravidian & Western architecture and cultural significance.

DEMOGRAPHICS

Tiruchirappalli City 2011	Total	Male	Female
City Population	**847,387**	418,400	428,987
Literates	**701,459**	358,276	343,183
Children (0-6)	**79,723**	40,681	39,042
Average Literacy (%)	**91.38 %**	94.85 %	88.01 %
Sexratio	**1025**		
Child Sexratio	**960**		
Schedule Caste	88,833	44,183	44,650
Schedule Tribe	2,353	1,160	1,193

EDUCATION

Tiruchirappall is an important educational centre in the state of Tamil Nadu, and houses nationally recognised institutions such as

- National Institutes of Technology (NIT),
- Indian Institutes of Management (IIM)
- National Law Universities (NLU), and
- Indian Institutes of Information Technology (IIIT) Srirangam.

GOVERNANCE & ADMINISTRATION

The city is divided into five administrative zones—

1. Srirangam Zone,
2. Ariyamangalam Zone,
3. Tiruverumbur Zone,
4. Ponmalai Zone,
5. K. Abishekapuram Zone

—each managed by an Assistant Commissioner, along with their respective engineering and public health officers

- **Tiruchirappalli City Municipal Corporation (TCMC)**: The primary local government body responsible for the city's administration.
- Mayor, **Municipal Commissioner**: The chief executive officer of the TCMC
- Tiruchirappalli City Municipal Corporation (TCMC) is divided into **65 wards**. Each ward is represented by an elected Councilor
- **Tiruchirappalli Urban Development Authority (Tiruchirappalli Metropolitan Development Authority)**: Responsible for planning and regulating land use in the city and surrounding areas

ECONOMY & EMPLOYMENT

In Tiruchirappalli Municipal Corporation out of total population, 302,442 were engaged in work activities. 91.9% of workers describe their work as Main Work (Employment or Earning more than 6 Months) while 8.1% were involved in Marginal activity providing livelihood for less than 6 months. Of 302,442 workers engaged in Main Work, 1,940 were cultivators (owner or co-owner) while 5,118 were Agricultural laborers.

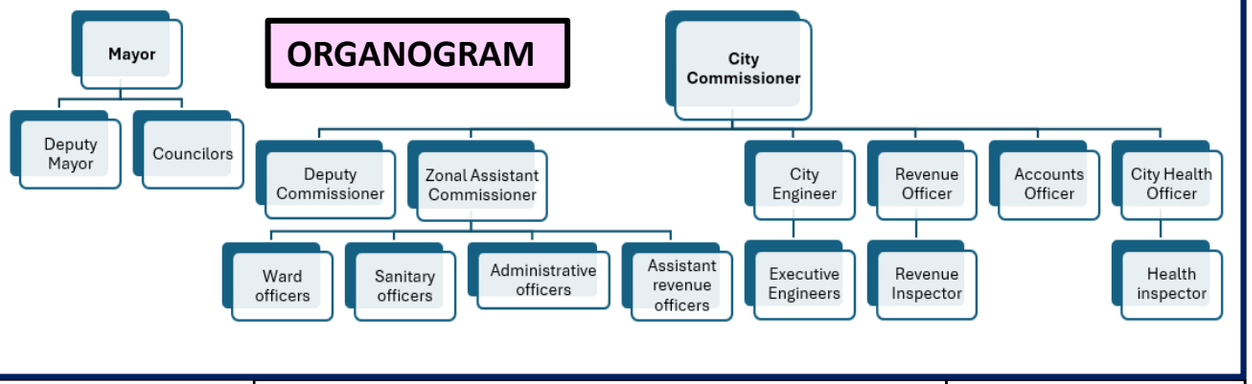
	Total	Male	Female
Main Workers	278,036	222,159	55,877
Cultivators	1,940	1,596	344
Agriculture Labourer	5,118	3,112	2,006
Household Industries	6,459	3,231	3,228
Other Workers	264,519	214,220	50,299
Marginal Workers	24,406	15,011	9,395
Non Working	544,945	181,230	363,715

IMPORTANT INDUSTRIES

1. Golden Rock Railway Workshop
2. Ordnance Factory Tiruchirappalli (OFT)
3. Heavy Alloy Penetrator Project
4. Bharat Heavy Electricals Limited (BHEL)
5. Cement Industry (Dalmia Cement)

- The per capita income for Trichy is estimated to be in the range of ₹1.5 lakh to ₹2 lakh (approximately $1,800 to $2,400 USD) per annum
- Trichy's GDDP has been estimated in the range of ₹30,000 to ₹40,000 crores (approximately $4 billion to $5.5 billion USD)

ORGANOGRAM



TRICHY CITY – CASE STUDY
Urban & Rural Governance

NITESHKUMAR SK
SCHOLAR NO: 211109041
B. PLAN- SEM VII

REMARKS

POTENTIAL

1. To produce variety, high value crops

Datia has potential to produce high value crops and produce that are in demand in market both nationally and internationally

Such sunflower, mulberry, turmeric & exotic fruits that could be easily grown in semi-arid condition and low water requirement

2. To support the growth of secondary sector & have economic communication with other districts

Through agro- produce and available labors, the secondary sector of Datia can flourish in pre-processing, food processing and agro-based industries & transported to other district like Gwalior, Jhansi for further processing.

3. To practice modern organic farming & mixed cropping

with organic products market demand in high rise, Datia agricultural practice could be changed and advanced to enter the organic products market to achieve higher mark up than ordinary crops

Mixed cropping in Datia involves growing two or more crops simultaneously on the same land to maximize resource use and minimize risk. Common combinations might include cereals and legumes (e.g., wheat with chickpeas) to improve soil fertility and reduce pest infestation. This practice enhances yield stability and helps farmers diversify their income while promoting sustainable agriculture.

4. Livestock farming for milk, meat and eggs, Aqua culture & beekeeping

Diary(cow/buffalo & goat) industry as market both inside the district for sweet making and have demand outside district/state/country- with the surplus milk produced and potential to be produced in the district.

Fresh water pisciculture, shrimp culture have government incentive offered production opportunities in the ponds & reservoirs of Datia

Poultry farming and goat farming for exporting in meat market in other districts

Beekeeping on Horticulture farms

	Sun Flower Seeds Rate Today In India.
KG Price	₹ 58.00
10 KG Price	₹ 580.00
Quintal (100 KG) Price	₹ 5,800.00
Ton (1000 KG) Price	₹ 58,000.00
Average Market Price	₹ 5799 / Quintal
Best Market Price	₹ 5799 / Quintal
Lowest Market Price	₹ 5799 / Quintal

Source: kisan deal

Farming system/Enterprises
Crop production + Animal Husbandry + Horticulture
Crop production + Animal husbandry
Crop Production
Fisheries

Farming practice types in Datia

BARRIERS

Irrigation Access & climate

Datia experiences fluctuating rainfall, which lead to drought conditions, affecting crop irrigation of the kharif crops. Larger percentage depend upon canal irrigation – during drought, canals dries affecting a large proportion of cultivation area

Sources	No. of sources	Area covered (ha)	Percentage of total irrigated area
Canal	04	1,02,167	58.89

Livestock Issues

Livestock grazing on farmlands, especially stray cattle, leads to crop damage. This issue of street cattle is common in Datia and impacts agriculture negatively, especially during the crop growing seasons.

Post-Harvest Losses

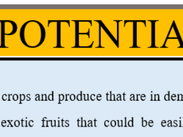
In Datia, post-harvest losses of fruits and vegetables were estimated at ~15-20%, largely due to poor infrastructure, absence of cold storages

Lack of skilled workforce & capacity building

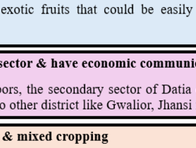
Datia lack skilled human resource - limits the- adoption of modern farming techniques, employment in industries.

Also the lack of capacity building of government entities(both urban & rural)- leads to failure, delay of management, implementation of government initiatives

DISTRICT	Total Stray Cattle
DATIA	14700



Source- NSDC



Source- survey

Mechanization Levels

According to a 2021 report by **NABARD**, only about 35% of farmers in Madhya Pradesh use modern mechanized farming tools. Marginal farmers in Datia primarily rely on manual labor, limiting the productivity potential of the region.

Pump sets	2083
No. of Tractors	6478

Lack of sales infrastructure / system

In Datia, most farmers are small and marginal, which limits their ability to directly access markets and forces them to rely on middlemen. This fragmentation reduces their profit margins, particularly in horticulture and high-value crops.

Policy and Implementation Gaps:

Although several government policies aim to improve the agricultural sector, the actual implementation at the grassroots level is often lacking or delayed, affecting farmers' access to subsidies, schemes, and benefits meant for enhancing production.

Limited double cropping

Despite having a large proportion of agricultural land (about 75% of total land), the utilization for double cropping remains low. Only 31% of the land is used for both Rabi and Kharif crops, while a significant portion remains fallow due to the lack of irrigation and poor water management

Agricultural land use	Area ('000 ha)
Net sown area	195.959
Area sown more than once	34.533

RISKS

1. Economic risk

Risk of Failing to attracting or induce private participation- Datia as only 1 large & medium (subordinate of the large) – i.e. M/S AGRO SOLVENT PRODUCTS PVT. LTD

2. Social risk

- Public misunderstanding towards commercial production & sensitivity towards private involvement
- Protest against farm law 2020- no contract farming with private, no change in the practice of mandi, no change in essential crop list-MSP (Minimum Support Price)

3. Implementation risk (ground level implementation)

Most of the policy is failed to implemented at the level, so benefits are not reached the public and so its estimated impact is not attained. E.g. Soil health card scheme – is not reached all farmers

4. Environmental risk

Unpredictable climatic condition of Datia- inducing drought, waterlogging/flooding

What is the major contingency the district is prone to? (Tick mark)	Regular	Occasional	None
Drought	-	√	-
Flood	-	√	-
Cyclone	-	-	√
Hail storm	-	√	-
Heat wave	-	√	-
Cold wave	-	√	-
Frost	-	√	-
Sea water intrusion	-	-	√
Pests and disease outbreak (specify) Aphid in Mustard and Gram pod Borer	-	√	-

Threshold

On the Basis of

Enablers → Mitigation → Potential ← Restriction ← Barriers / Risks

THRESHOLD

1. AGRI-LAND USE & IRRIGATED AREA

1.3	Land use pattern of the district (latest statistics)	Geographical area	Cultivable area	Forest area	Land under non-agricultural use	Permanent pastures	Cultivable wasteland	Land under Misc. tree crops and groves	Barren and unculturable land	Current fallows	Other fallow lands fallow l.
	Area ('000 ha)	295.9	186.7	29.4	23.8	4.6	10.3	2.8	12.7	16.5	9.1

Source – Directorate of Farmers welfare and Agriculture, Development of Madhya Pradesh. Bhopal, Agriculture Statistics 2009

~19000 ha of potential cultivable land to be added to the threshold of ~190000 ha cultivable land available & 16500 ha gross cultivable land would be available from the fallow land.

~30000 ha of forest area that supports the district wood based MSMEs & 46000 ha of pasture to livestock.

Agricultural land use	Area ('000 ha)	Cropping intensity %
Net sown area	195.959	117.6
Area sown more than once	34.533	
Gross cropped area	230.492	

~34000 ha is sown more than once contributing 230000 ha of gross cropped area

Irrigation	Area ('000 ha)
Net irrigated area	172.430
Gross irrigated area	175.732
Rainfed area	58.062

Irrigation by Different Sources	No.	Area irrigated (km²)
Dug wells	24856	596.0
Tube wells/Bore wells	414	54.69
Tanks/Ponds	2	5.74
Canals	3	1059.69
Other Sources		13.91
Net Irrigated Area		1724.30
Gross Irrigated Area		1757.32

Almost 75% of total cultivable land is irrigated & remaining is rainfed. The major part of irrigated area uses *canals* followed by *dug(open) wells* as their main source of irrigation.

Source: Agriculture Contingency Plan for Datia

2. SOIL TYPE, HYDROLOGY & CLIMATIC CONDITION

Soil:

The region predominantly has alluvial soils mixed with some clay loam (heavy soil). These soils are fertile and support a variety of crops, especially vegetables, cereals and pulses. Followed by medium & light soil.

Soil Types		
Type	Area (000 ha)	Percentage of total area
Light *(Ranker)*	12.15	06.20
Medium *(Padua)*	49.20	25.20
Heavy *(Kabar/Mar)*	133.92	68.60

Source: KRISHI VIGYAN KENDRA, DATIA

Hydrology:

Datia is hydrated by 2 major seasonal rivers Sindh & Pahuj.

- **Sindh River**: A major tributary of the Yamuna River, flowing through parts of Datia, providing water for irrigation and supporting agriculture in the region
- **Pahuj River**: Flows in proximity to Datia and serves as a source of water for nearby villages and agricultural activities.

Major drainage	Sindh river and its tributaries
	Pahuj river and its tributaries
Ground water level (2006)	Pre monsoon – 8.34 to 27.12 mbgl
	Post monsoon – 7.80 to 27.78 mbgl

Source: KRISHI VIGYAN KENDRA, DATIA

Climatic condition:

- Datia has uncertain climatic condition that cause drought & waterlogging
- Monsoon -favorable conditions for the sowing of kharif crops
- Winter - favorable conditions for the sowing of rabi crops
- The region relies heavily on monsoon rains for irrigation- which activates the seasonal rivers and canals that irrigates the major part of total irrigated area.

Climate	
Particulars	Characteristics
Temperature	Minimum 2.2°C (Dec), 24.9°C (May)
	Maximum 25.2°C (Dec), 42.7°C (May)
Average Annual Rainfall	742.9 mm
Wind Velocity	6.2 Km/hr (Jun), 1.0 Km/hr (Dec)
Relative Humidity	Minimum 31.0 %,
	Maximum 90.8%

3. EXISTING CROP PRODUCTION

Source: Agriculture Contingency Plan for Datia

- Black gram (urad), wheat
- New crops: soybean, cluster bean, kharif onion, peppermint
- Horticulture: lime, guava, anola, table pea, potato, tomato, brinjal, chilli, coriander & marry gold- max produce
- Oilseeds-40000 tons
- Reidue fodders – more than 1000000 tons

Ll	Name of crop	Kharif		Rabi		Summer		Total		Crop residue as fodder ('000 tons)
		Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	
	Major Field crops (Crops to be identified based on total acreage)									
1	Black Gram	17.31	780					17.31	780	13.84
2	Groundnut	13.38	1325					13.38	1325	16.19
3	Sesamum	24.80	800					24.80	800	72.00
4	Wheat			385.80	2858			385.80	2858	482.25
5	Gram			47.10	1253			47.10	1253	35.33
6	Pea			18.30	850			18.30	850	9.15
7	Mustard			20.50	1053			20.50	1053	43.05
	Major Horticultural crops (Crops to be identified based on total acreage)									
	Fruits	(t/ha)								
1	Guava	37.00	176.19					37.00	176.19	5.60
2	Lime	95.75	170.69					95.75	170.69	10.11
3	Aonla	58.00	193.30					58.00	193.30	6.20
	Vegetables									
1	Tomato	146.50	114.00					146.50	114.00	90.45
2	Potato	501.00	123.70					501.00	123.70	125.50
3	Brinjal	59.90	70.47					59.90	70.47	40.80
4	Bitter gourd	655.00	93.57					655.00	93.57	200.00
5	Cauliflower	277.80	200.00					277.80	200.00	50.00
	Spices crops									
1	Coriander	5.10	3.89					5.10	3.89	1.56
2	Chilli	4.74	22.50					4.74	22.50	0.95
3	Garlic	167.28	100.00					167.28	100.00	12.36
	Flower crops									
1	Marry gold	750.00	50.00					750.00	50.00	11.34
2	Rose	4.60	4.80					4.60	4.80	0.67

4. LIVESTOCK DEMOGRAPHICS & PISCICULTURE

Livestock is the second highest economic sector in Datia after agriculture - **Rs.51000 lakhs net value** added for 2020-21 (Estimates of district domestic product Madhya Pradesh)

- Cattle, buffalo – are exceptionally high in numbers, including a visible share of stray castles.
- Goat population – are significant in numbers & under-utilized due to low meat-eating population & goat milk consumers.
- Poultry is comparatively lower but still enough to create thrust for future potential

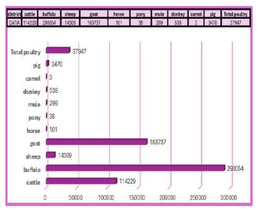
Fisheries:

Capture inland- **9 private ponds, 30 reservoirs & 30 village tanks**

Fresh water culture- in **106.23** ha of water spread area & produce **68800 tons of fresh water fish (1.16 ton/ha)**

Source: Agriculture Contingency Plan for Datia

Livestock	cattle	buffalo	sheep	goat	horse	pony	mules	donkey	camel	pig	Total poultry
(2012)	114258	260304	14359	163737	921	38	299	1101	3	3470	37947



Fisheries					
A. Capture	Boats		Nets		
(Data Source: Fisheries Department)	No. of fishermen	Mechanized	Non-mechanized	Mechanized (Trawl nets, Gill nets)	Non-mechanized (Shore Seines, Stake & trap nets)
	-	-	-	-	-
a) Inland (Data Source: Fisheries Department)	No. Farmer owned ponds		No. of Reservoirs		No. of village tanks
	09		30		30
B. Culture	Water Spread Area (ha)		Yield (t/ha)		Production ('000 tons)
a) Brackish water (Data Source: MPEDA/ Fisheries Department)					
b) Fresh water (Data Source: Fisheries Department)	106.230		1.16		68.80
c) Others					

5. INDUSTRIES & HUMAN RESOURCES

S. No.	Name of Ind. Area	Land acquired (In hectare)	Land developed (In hectare)	Prevailing Rate Per Sqm. (In Rs.)	No. of allotted Plots	No of Vacant Plots	No. of Units in Production	Remark
1	Industrial area, Gandhari	24.09	8.00	20/	121	113	08	03	Six (6) no them are only no. 8
2	Semi Urban Industrial Estate Datia	4.18	4.18	20/	59	50	Nil	20	-
3	Gramin karmashala Indergarh	1.10	1.10	20/	03 Shed	Nil	03 Shed	Nil	Six (6) shed are area 6 ft
4	Total	29.37	13.28	--	--	---	------	------	

Datia as only 1 large & medium (subordinate of the large) – i.e M/S AGRO SOLVENT PRODUCTS PVT. LTD

Remaining all are MSMEs – which enable districts manufacturing sector & providing employment to urban population of Datia (around 2000 employment in registered industries).

3.3 Details of Existing Micro & Small Enterprises and Artisan Units in the District Datia

NIC CODE NO.	TYPE OF INDUSTRY	NUMBER OF UNITS	INVESTMENT (Lakh Rs.)	EMPLOYMENT
20	Agro based	180	92.12	567
22	Soda water	--	--	--
23	Cotton textile	--	--	--
24	Woolen, silk & artificial Thread based clothes.	--	--	--
25.	Jute & jute based	--	--	--
26.	Ready-made garments & embroidery	151	7.55	160
27.	Wood/wooden based furniture	120	6.12	365
28.	Paper & Paper products	--	--	--
29.	Leather based	--	--	--
31.	Chemical/Chemical based	02	2.00	8
30.	Rubber, Plastic & petro based	--	--	--
32.	Mineral based	17	255.00	192
33.	Metal based (Steel Fab.)	--	--	--
35.	Engineering units	4	2.50	10
36.	Electrical machinery and transport equipment	--	--	--
97.	Repairing & servicing	513	23.75	530
01.	Others	8	5.71	18

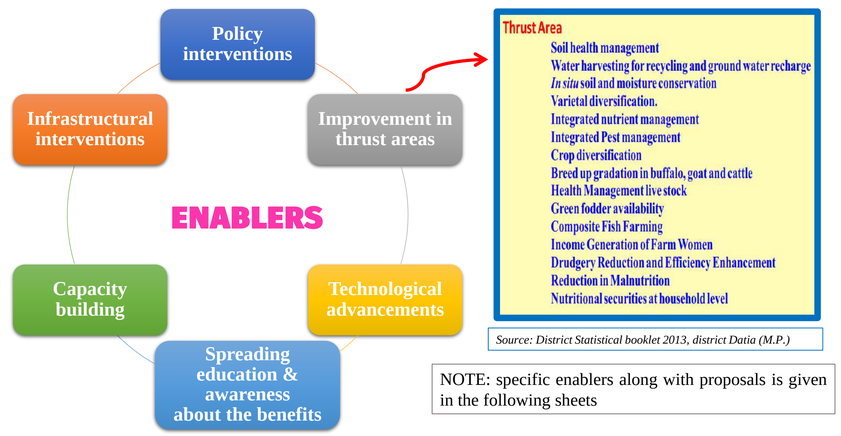
Source: DIC Datia

Human resources:

- Datia as work force that could enable industries
- For rural areas, the unemployment rate was **2.4%**, while for urban areas, it stood at **5.4%**.
- The rural labor force participation rate (LFPR) also showed an increase, indicating that more people in rural areas are seeking employment, likely due to economic pressures or reverse migration.(source: NSSO)

For Datia district in the 2011 Census:

Total Population: 786,754
Total Workers: 314,577
Main Workers: 219,968
Marginal Workers: 94,609
Non-Workers: 472,177



Thrust Area

- Soil health management
- Water harvesting for recycling and ground water recharge
- *In situ* soil and moisture conservation
- Varietal diversification.
- Integrated nutrient management
- Integrated Pest management
- Crop diversification
- Breed up gradation in buffalo, goat and cattle
- Health Management live stock
- Green fodder availability
- Composite Fish Farming
- Income Generation of Farm Women
- Drudgery Reduction and Efficiency Enhancement
- Reduction in Malnutrition
- Nutritional securities at household level

Source: District Statistical booklet 2013, district Datia (M.P.)

NOTE: specific enablers along with proposals is given in the following sheets

DISTRICT DEVELOPMENT PLAN DATIA

POTENTIAL ANALYSIS

Group Members:

Name	Scholar No.
Shivansh	211109004
Riya Narvariya	211109005
Sharad Yadav	211109007
Gwala LB	211109008
Aashish Patel	211109017
Aayushi Chourasia	211109036
Rutuja Ghode	211109038
Nitesh SK	211109041
Lenine Shende	211109045

SHEET NUMBER

VIII

OIL SEED CROP PRODUCTION

To support the domestic use of the state and nation that helps in reducing the needs for cooking oil imports. The Indian government has been promoting oilseed cultivation, through various schemes such as the **National Mission on Oilseeds and Oil Palm (NMOOP)**.

Farmers in Datia can take advantage of subsidies on seeds, fertilizers, and equipment to reduce input costs and improve yields.

- **Groundnut**- exist as large share in the current cultivation
- **Sesame**- exist as large share in the current cultivation
- **Cotton seed** – presence of existing cotton (govt. proposed export item for Datia)
- **Sunflower seed** – has potential with suitable condition to be commercially cultivated to supply Domestic demand and as Export Potential to other states. it can be a valuable crop for diversification and rotation in Datia's traditional wheat and pulse cultivation systems.
- **Castor beans** - The oil extracted from castor beans is used in various industrial applications, from cosmetics to lubricants, making it a valuable cash crop

Promotion of establishing oil mills near the raw materials – MSME oil extraction will be also supported

Existing prominent crops are further supported to increase its production to push them into larger market

Increase the cultivation area of these crops to increase production it is efficient as it be grown seasonal crop by sharing the year with major crops in **Crop Rotation**

Higher quality- yield varieties are promoted

Training Program

Offline Training (In-Person)
- Workshops & Field Demonstrations
- Local Extension Services

Online Training (Digital)
- Webinars & Virtual Workshops
- Online Courses
- Mobile App-Based Learning

1. Physical Support
   - Distribution of High-Quality Seeds
   - Access to Modern Agricultural Equipment
   - Extension Services & Training

3. Environmental Support
   - Soil Health Programs
   - Water Conservation Initiatives
   - Promotion of Renewable Energy

The proposal will be initiated and supported through various government and private sector schemes and programs to ensure successful implementation at every stage.

2. Financial Support
   - Subsidies & Grants
   - Low-Interest Loans
   - Crop Insurance Program

4. Economic Support
   - Market Infrastructure Development
   - Market Linkages & Partnerships
   - Agri-Tech Platforms

HORTICULTURE CROP PRODUCTION (INCREASING EXISTING MAJOR PRODUCE+NEW CROP)

More emphasis on higher value produce & raw material to pre-processing industry

FRUITS- lime, aonla(amla), guava

(lime production can be increased to use as raw material for other product production- citric acid, beverages and transported to districts that are need of lime products for industrial use & Guava large produce export)

VEGETABLES- table peas, potato, cauliflower, brinjal & tomato, onion

(Tables peas, cauliflower & tomato are the highest horticulture produce in the district- we should promote its market to the surrounding districts- considered to be potential export product)

FLOWER – marigold, rose, sunflower, Mulberry

(Natural Insect Repellents) Companion Planting and Organic Farming, Natural pesticides industry development, cosmetic industry raw material, medical use & decorative use.

establishment of processing units to produce products out of them

RECOMMENDED CULTIVATION MODELS

Cluster-Based Approach

Farms in a specific geographic area are grouped into clusters based on the type of crop produced.

Agri-Business Park Model

Establishing dedicated zones where agricultural production is integrated with processing, packaging, and distribution.

Agroforestry and Integrated Farming Model

Combines crops with livestock and forestry to diversify income streams, improve land productivity, and promote sustainability. This model is especially useful for farmers growing horticultural and stable crops alongside other agri-practices.

Recommendation to establish Stevia processing units after certain threshold of production

Local retail market, restaurants, food processing units supply chain establishment through private involvement.

SPECIAL CROPS PRODUCTION

Introduce and promote the cultivation of high-value, low-maintenance crops to diversify agricultural income and utilize local resources efficiently. Target both domestic consumption and niche export markets.

Mushroom
- Promotion of both cottage/house level production & mass production
- Focus on cultivation of high-demand varieties like button, oyster, and shiitake mushrooms.
- Training program both online & offline
- Provide starter kits and composting facilities for quality substrate production.

Stevia
- A natural sweetener in high demand, stevia is used widely in the food and beverage industry
- It enables new industry in the region- local processing units for drying and extracting stevia
- Promotion of stevia use in Datia local sweet industry along with tourism
- Provide high-yield stevia seedlings to farmers, along with guidance on propagation techniques-Training program both online & offline

Creation of market linkage and value chain development

Physical Sales Channels
1. Local Markets & Farmers' Markets
2. Wholesale to Retailers & Distributors
3. Cooperative Sales
   - Farmer Cooperatives
   - Regional Produce Hubs
4. Export Market

Online Sales Channels
1. Through E-mandi program
2. E-Commerce Platforms
3. Subscription & Delivery Services
4. B2B Online Platforms (Virtual Wholesale Auctions)

Preparation of export plan along with logistics plan
(In the following sections)

STABLE CROPS PRODUCTION INCREMENTATION

1. Wheat (govt. proposed export item for Datia)
2. Sugarcane (govt. proposed export item for Datia)
3. Chickpeas
4. Gram & black gram

Promotion of cultivation of higher quality stable crops to increase the surplus for export market, by
- Introducing higher quality-higher yield crop seeds
- Organic farming
- Smart farming – using drip irrigation system, regular soil testing, etc.

DISTRICT DEVELOPMENT PLAN DATIA

COMMERCIAL PRODUCTION PROPOSALS

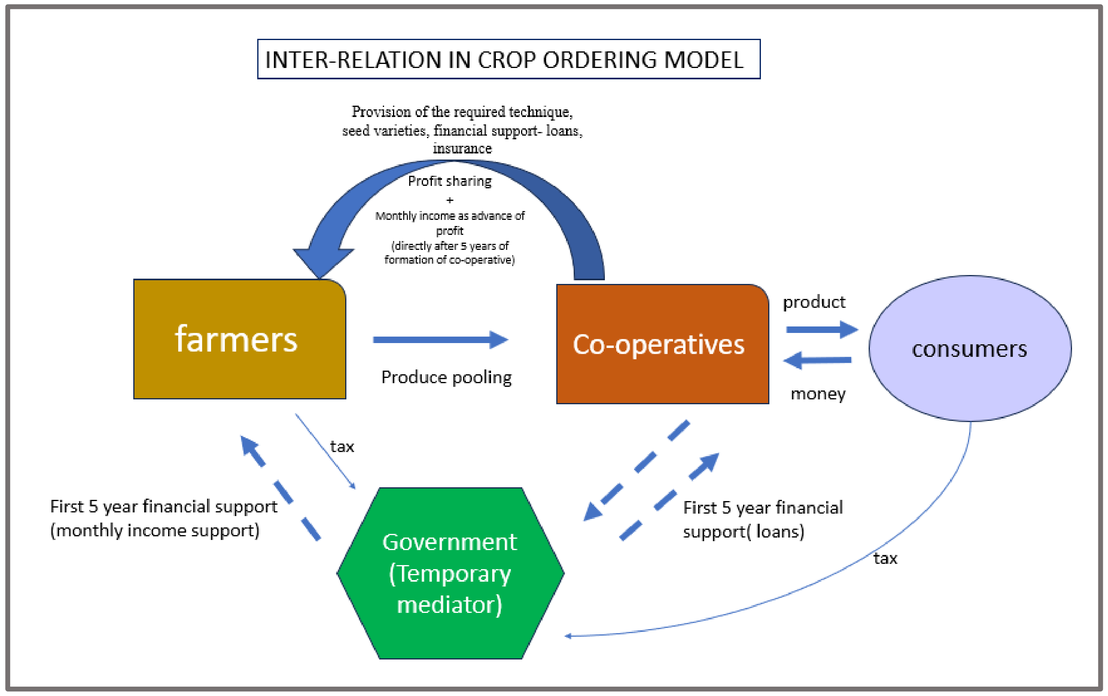
Group Members:

Name	Scholar No.
Shivansh	211109004
Riya Narvariya	211109005
Sharad Yadav	211109007
Gwala LB	211109008
Aashish Patel	211109017
Aayushi Chourasia	211109036
Rutuja Ghode	211109038
Nitesh SK	211109041
Lenine Shende	211109045

AGRICULTURE- FARMERS PRODUCE PROPOSALS

Government crop ordering model

- Certain crops are promoted to be cultivated and sold through government mediating cooperatives by providing the required technique, seed varieties, financial support- monthly salary as advance from final selling value- subsidies, loans on cultivating the requested crops
- Government would work as a temporary mediator between co-operatives & associated farmers to get an inertia with financial support
- This would help both market driven production, produce pooling and provide economic support to the farmers.



RECOMMENDED FARMING PRACTICES

Organic farming practice

Supplying the commercial market demand

Organic farming practices

Dung based domestically produced manure, marigold pesticides

An **Integrated Agriculture Practice (IAP)**

involves combining various agricultural components—crops, livestock, poultry, aquaculture, and agroforestry—into a cohesive and sustainable farming system.

The aim is to optimize resources, increase productivity, reduce risks, and promote ecological balance

A **Mixed Farming Practice**

Aims to enhance agricultural productivity and sustainability by integrating crop cultivation and livestock rearing on the same farm.

This approach maximizes resource use efficiency, diversifies income sources for farmers, and improves soil fertility through natural recycling of nutrients.

A **Sustainable & Smart Farming Practice**

focuses on adopting environmentally friendly and technologically advanced agricultural practices to ensure long-term productivity and resilience.

This approach integrates precision farming techniques, such as the use of sensors, and data analytics, to optimize resource use, minimize waste, and enhance crop yields

also encourages renewable energy adoption, such as solar-powered irrigation systems, and provides training for farmers on using smart technologies

FARMERS WELFARE DEVELOPMENT

AGRICULTURE DEVELOPMENT PROPOSALS

AGRICULTURE INFRASTURCTURE DEVELOPMENT

INFRASTRUCTURE	REQUIREMENT	PROPOSALS	IMPLEMENTATION PLAN
Irrigation Systems	Development of efficient and sustainable water management	Drip irrigation, water harvesting tanks	Launch community-based projects and promote micro-irrigation systems.
Storage and Warehousing	Infrastructure to reduce post-harvest losses and store produce	Cold storage units, rural warehouses, grain silos	Build strategically located storage units near farming regions.
Transportation and Roads	Improved connectivity for farm-to-market access	Rural road networks, transportation hubs, bridges	Upgrade existing rural roads and develop new transport links.
Market Infrastructure	Establishment of facilities to help farmers sell their produce	Farmers' markets, auction centers, wholesale markets	Set up markets in high-production areas and offer logistical support.
Renewable Energy	Power infrastructure to support farming activities	Solar-powered pumps, biogas units, wind energy projects	Install renewable energy systems in farming areas to reduce costs.
Digital Infrastructure	Technology to improve access to information and markets	Internet connectivity, mobile apps, digital kiosks	Expand digital connectivity and provide training on using agri-tech tools.
Research and Development	Centers for innovation in agricultural practices	Agricultural research labs, seed development centers	Partner with universities and research institutes for new agricultural solutions.

NEW-POLICY RECOMMENDATION

POLICING POLICY

Policy reforms that initiate and monitor the complete implementation of other policies

This policy would cover the evaluation of local governance in

- Initiation of the policies
- Implementing the scheme & programs as per the planned timeline
- Transparency and Accountability
- Resource Allocation
- Illegality & corruption

FARMERS MARKET

Farmers market- local farmers market where they sell their own production/ sometimes through government facilitated auctions.

1.Direct Sales and Fair Prices: Farmers can sell their crops without intermediaries, ensuring they receive fair market value, which boosts their income and reduces price manipulation.

2.Community Engagement: These markets foster a sense of community by connecting farmers with local consumers, promoting the consumption of fresh, locally grown produce.

3.Seasonal and Diverse Produce: Farmers have the opportunity to showcase a wide variety of seasonal fruits, vegetables, and other farm products, encouraging dietary diversity and supporting sustainable practices.

4.Government-Facilitated Auctions: In cases of surplus production, the government may organize auctions,

DISTRICT DEVELOPMENT PLAN DATIA

AGRICULTURE DEVELOPMENT PROPOSALS

Group Members:

Name	Scholar No.
Shivansh	211109004
Riya Narvariya	211109005
Sharad Yadav	211109007
Gwala LB	211109008
Aashish Patel	211109017
Aayushi Chourasia	211109036
Rutuja Ghode	211109038
Nitesh SK	211109041
Lenine Shende	211109045

SHEET NUMBER

X

DIARY & LIVESTOCK PRODUCTION

Cattles & buffalo milk production and management

Introduction of Tehsil-Level Milk Collection Centers

Set up milk collection centers in each tehsil to streamline the collection for produce pooling.

Components:

- **Infrastructure**: Build well-equipped centers with cooling tanks, milk testing kits, and automated collection systems.
- **Establish collection network:** produce from farmers various villages in the tehsil is pooled with daily collection & transportation system(milk van)
- Involve both private & co-operatives management

District-Level Milk Cooperative

Establish a district-level cooperative to oversee milk collection, pricing, and distribution.:

- **Central Management**: Coordinate between tehsil-level centers and processing units.
- **Price Stability**: Ensure fair pricing and protect farmers from market fluctuations by negotiating contracts with buyers.
- **Training Programs**: Organize training in dairy management, feed optimization, and breeding practices.

Surplus Milk Processing and Export

Milk Processing Units: Set up facilities to process surplus milk into value-added products such as cheese, butter, ghee, and milk powder.

Export Potential: Develop export strategies for value-added products to national and international markets, emphasizing quality and organic certification.

Storage & Distribution: Invest in logistics, such as refrigerated transport and warehouses, to maintain product freshness and extend shelf life.

Variety Breeding for High-Yield Milk Production

Selective Breeding Programs, Artificial Insemination Centers, Disease Management

Public Milk Production Industry Utilizing Stray Cattle

Address the stray cattle issue by integrating them into a structured public milk production system-Stray Cattle Shelters, Milk Production Units, Manure and Biogas Production

Goat diary export market establishment

The global goat milk products market was valued at approximately USD 11.4 billion in 2022 and is expected to grow at a compound annual growth rate (CAGR) of around 6% to 8% over the next few years.

So, a **chance to enter and establish market** leadership – with **prominent goat population** in the district.

sheep	goat
14309	163737

Increase **goat milk collection** from livestock owners through the same collection system proposed for cow/buffalo milk

Establish of food processing units – to convert goat milk to the products that are in demand in market – infant formula, milk powder, cheese, etc.,

Market

- Goat milk-based infant formula is an area of high growth potential, especially in markets like China & other Asian countries.
- Europe is the largest consumer of goat milk products, particularly in Mediterranean countries like Spain, Greece, and France. Goat cheese (chèvre) is a staple in many European diets.

Major enabler

Dairy Entrepreneurship Development Scheme (DEDS): Provides financial assistance to establish small dairy farms and dairy infrastructure like milk chilling plants and cold storage.

Export of poultry and Live Goat Stock

1.Breeding and Rearing-Select fast-growing and high-meat-yield breeds for both poultry and goats to optimize productivity

2. Infrastructure Development: Establish modern goat farms with proper shelters, grazing areas, and feeding programs for healthy livestock

3.Develop efficient transport logistics

4.Quality Assurance & Certification

5.Health and Veterinary Care

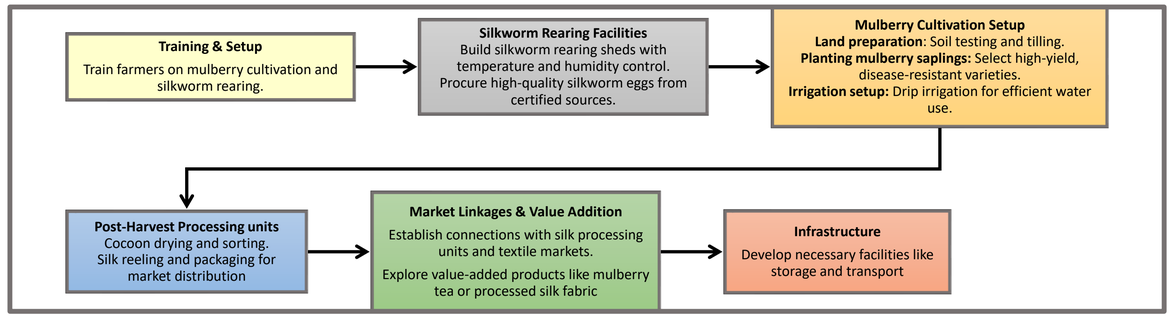
6.Market Development-Collaborate with international importers, trade associations

SERI CULTURE

- For silk production - culturing of silk worms.
- Datia is prominent in horticulture - as for the integration of sericulture - cultivation of mulberry plantation for its leaves is the major part of sericulture- which is the feedstock for silk worms.
- Datia's winter, soil and horticulture background creates threshold for sericulture industry and the mulberry -high value fruit cultivation is giving additional market value.
- With long winter of Datia even two-time cultivation is also possible.

ENABLERS

1. Silk Samagra Yojana
2. Central Silk Board (CSB) Support for Infrastructure Development such as financial aid for developing rearing houses, chawki (young silkworm) rearing centers, and equipment needed for silk reeling and processing
3. National Sericulture Development Programme (NSDP)



ANIMAL HUSBANDRY

AQUACULTURE-PISCICULTURE & SHRIMP BREEDING

Pisciculture

- To increase the production of existing fresh water fish cultures (from 70k tons to annual exponential increase of 20%).
- To add more species variety in the market
- Integrated production along with poultry & paddy (semi – aqua crop)

Shrimp breeding

- Development of shrimp cultivation industry in Datia
- In 2024 budget - financial support for shrimp breeding centers & facilitate export through NABARD And cut in custom duty for fish & shrimp feeds.
- Manufacturing of fish and shrimp feeds -duty exemption

Enablers

1. Artificial ponds & tanks to increase water spread area (current area- 106 ha)
2. Secure water through harvesting infrastructure, treatment plants etc.
3. Training & skill development
4. Marine Products Export Development Authority (MPEDA)

Training & Capacity Building

- Conduct workshops and hands-on training for farmers on pond management, species selection, and disease control.
- Set up demonstration units and partner with fisheries experts for ongoing support.

↓

Establishment of Hatcheries

Develop local fish and shrimp hatcheries to ensure a steady supply of quality seed varieties

↓

Supporting Infrastructure

- Build water reservoirs, rainwater harvesting systems, and establish cold storage units.
- Develop an ice factory and improve logistics for efficient transport and storage.

↓

Integrated Aquaculture Model

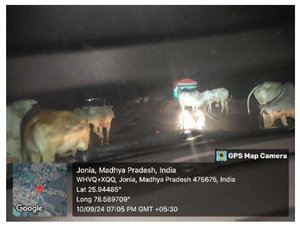
- Combine fish farming with livestock rearing (e.g., ducks, goats) to create a closed-loop system.
- Use pond silt as organic fertilizer for crops and adopt crop rotation practices.

↓

Market Establishment and Promotion

1. **Local & export Market Linkages**: Develop a network of local markets and restaurants that prioritize sourcing fresh fish and shrimp from Datia-based farmers.
2. **Fish and Shrimp Cooperatives**: Form cooperatives to collectively market the produce, negotiate better prices, and access bulk buyers, such as hotels, supermarkets, and export firms.
3. **Branding and Promotion**: Launch a local brand emphasizing fresh, sustainably farmed produce.

CATTLES ON ROAD



Cattle blocking causing traffic

DISTRICT	Total Stray Cattle
DATIA	14700

REASONs

- Warmth from Roads
- Safety from insects
- Poor Fencing
- Inactive Grazing Period
- Comfort from Hard Surfaces

Comprehensive cattle Management plan

Productive cattle

Cows/cattle are distributed to public through schemes for livelihood improvement. (socio-economic intervention)

Others included in public milk farming industries (Economic intervention)

Un-productive cattle

Establishing dedicated cattle retirement sanctuaries in rural areas on unused waste land.

Setting up small-scale biogas plants or organic fertilizer production units using cow dung and urine from non-productive cattle (Economic intervention)

Forming cattle management co-operatives where local communities or panchayats can manage, supported by government grants and crowdfunding campaigns. This could generate Temporary employment for local people.

Proposal plan

Spatial-location are selected and proposed based on grazing land and unused waste land – land cover area in Datia → Cattle rest sheds would be build on fenced environment with warm floors, fodder-water provision, dung/waste collection facilities(physical intervention) → Relocating/transporting the cattle to the allotted place → Management system & strategies, policies are made to guide & support

DISTRICT DEVELOPMENT PLAN DATIA

ANIMAL HUSBANDRY DEVELOPMENT PROPOSALS

Group Members:

Name	Scholar No.
Shivansh	211109004
Riya Narvariya	211109005
Sharad Yadav	211109007
Gwala LB	211109008
Aashish Patel	211109017
Aayushi Chourasia	211109036
Rutuja Ghode	211109038
Nitesh SK	211109041
Lenine Shende	211109045

SHEET NUMBER

XI

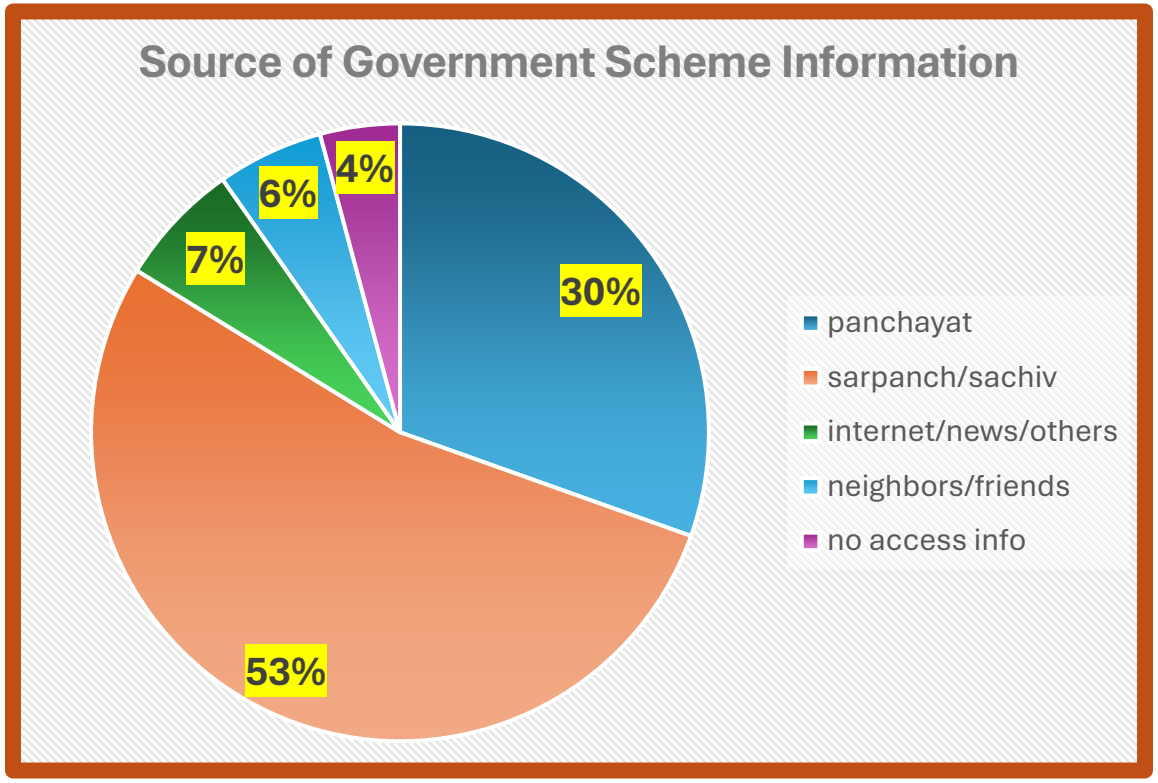
SCHEMES IMPLEMENTATION & SUCCESS

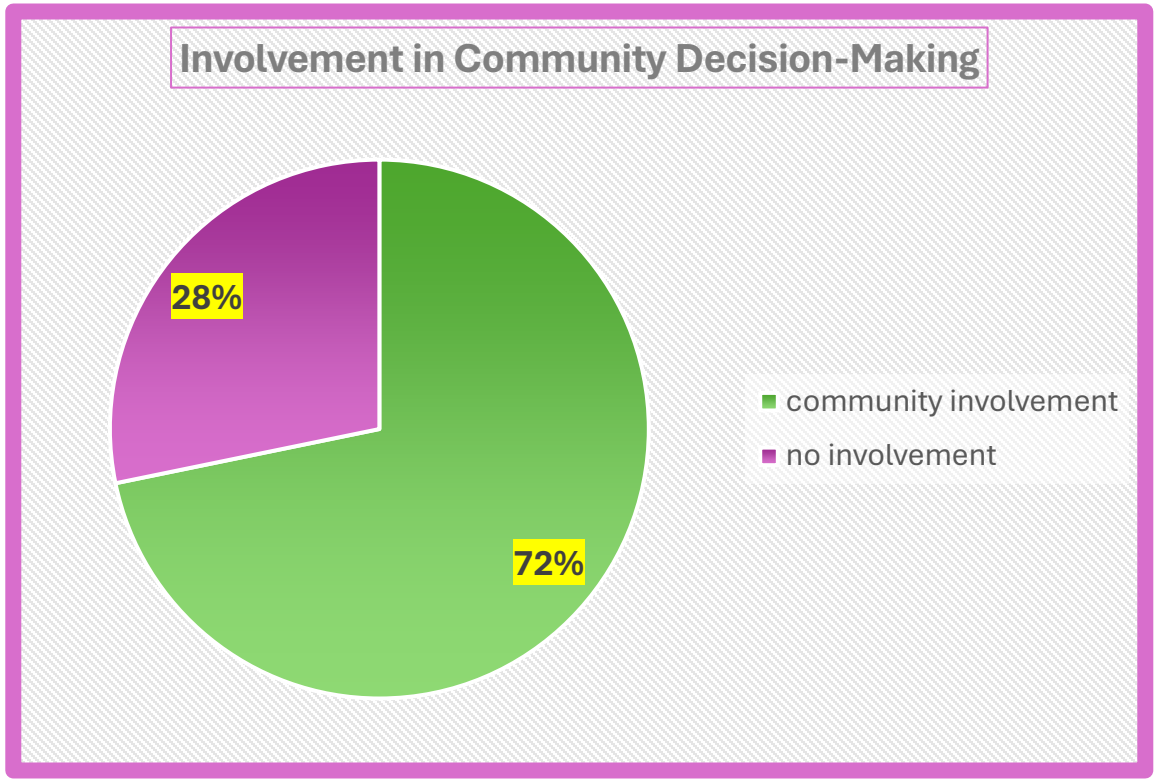
79 VILLAGES

752 SAMPLES

(may slightly differ for various data type – due to data cleaning)

COMMUNITY INVOVEMENT & INFO- SOURCE





Village Panchayat(notice, announcements, from sarpanch/sachiv/other panchayat executives

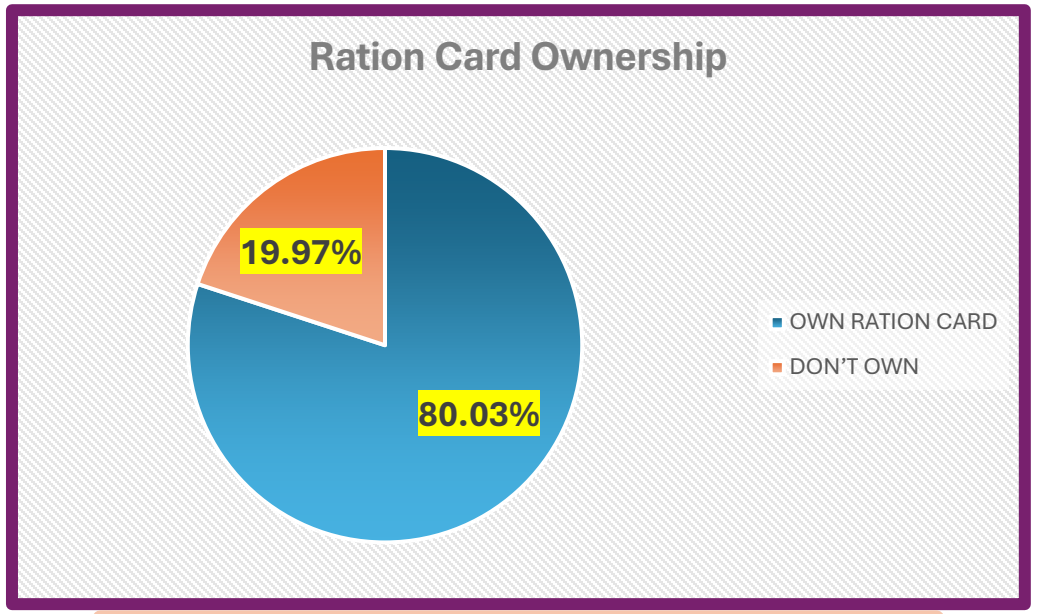
↓

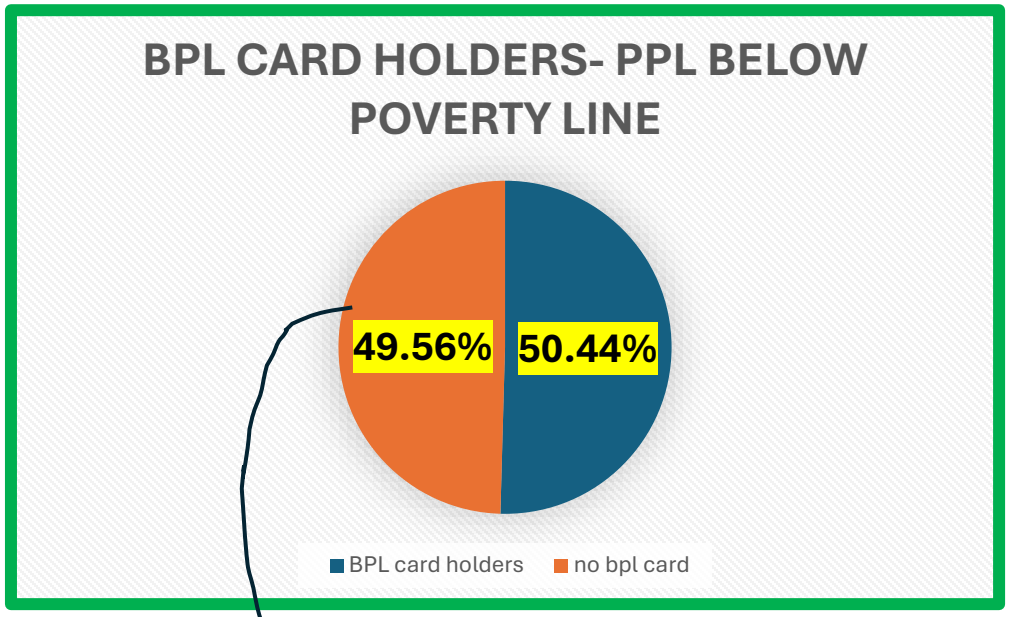
ACTIVE IN SPREADING INFORMATION

ALSO →

ACTIVE PUBLIC PARTICIPATION

RATION AND BPL CARD



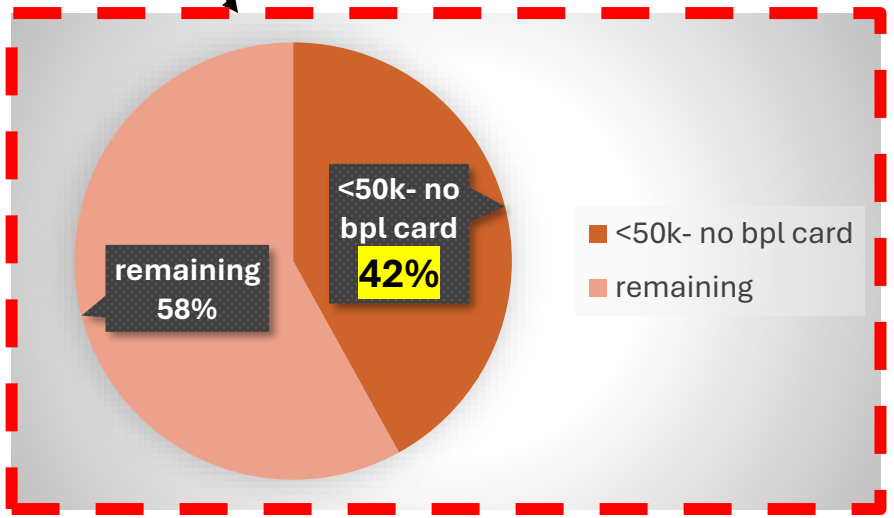


~20% HH don't own Ration card- which is **negative** clash – in providing benefits of policies/schemes

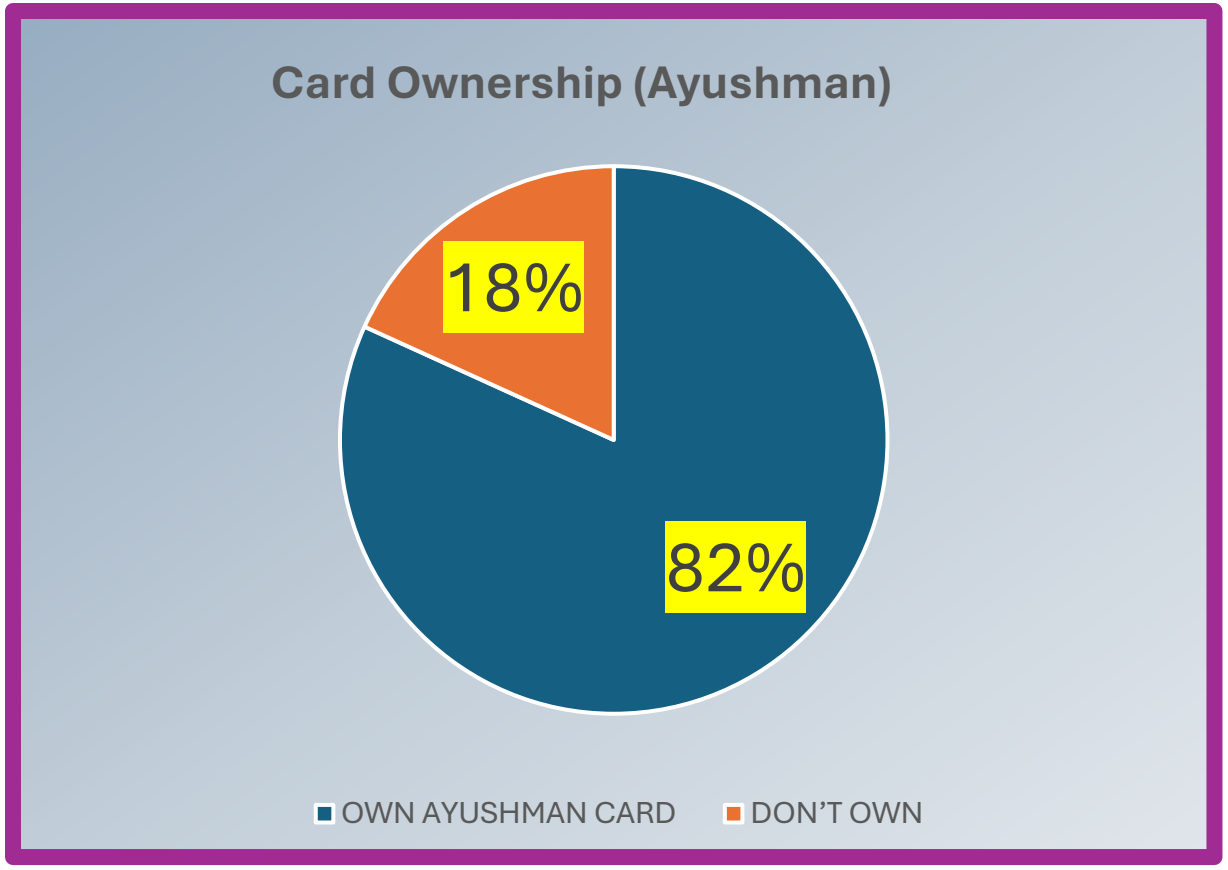
We also have to make almost all schemes to be access through just Aadhar card

MORE THAN HALF – of rural population is below poverty line

So, primary objective of our regional plan should be UPLIFTMENT OF RURAL POP. FROM POVERTY



Ayushman Card Ownership



Covering SECC-Socio-Economic Caste Census – IDENTIFIED POPULATION

Availed by more than half of the sample HH in all villages in average

Scheme awareness & information is spread actively

Soil Health Card

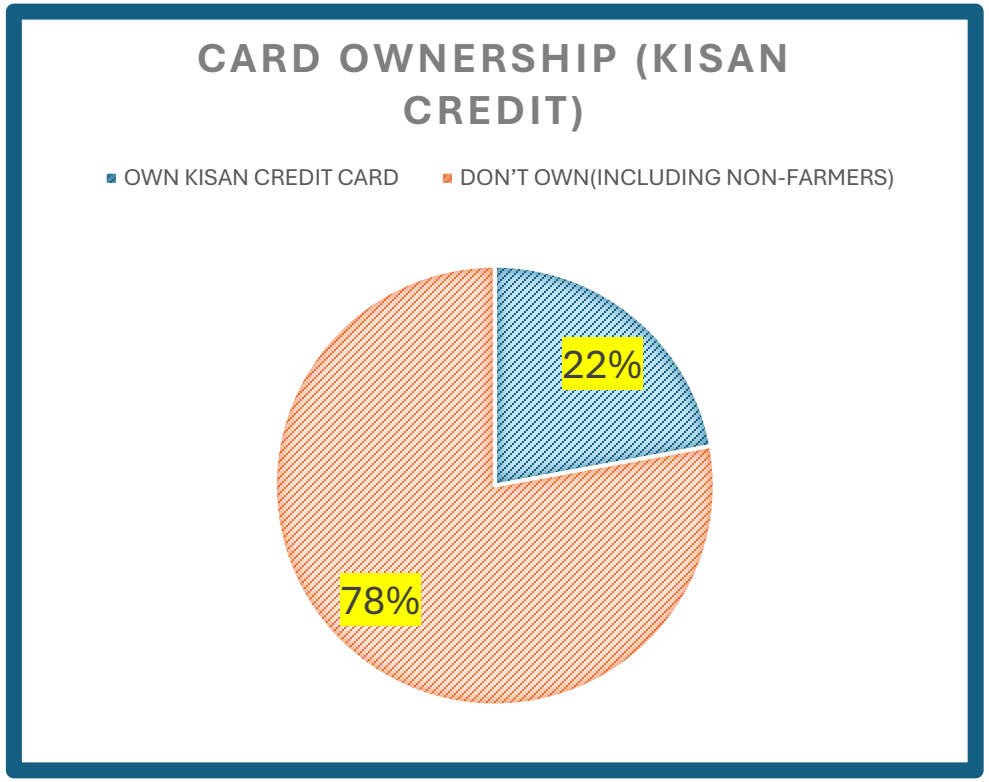
ONLY 7

OUT TOTAL CULTIVABLE LAND OWNERS- OWN SOIL HEALTH CARD

UN AWARE

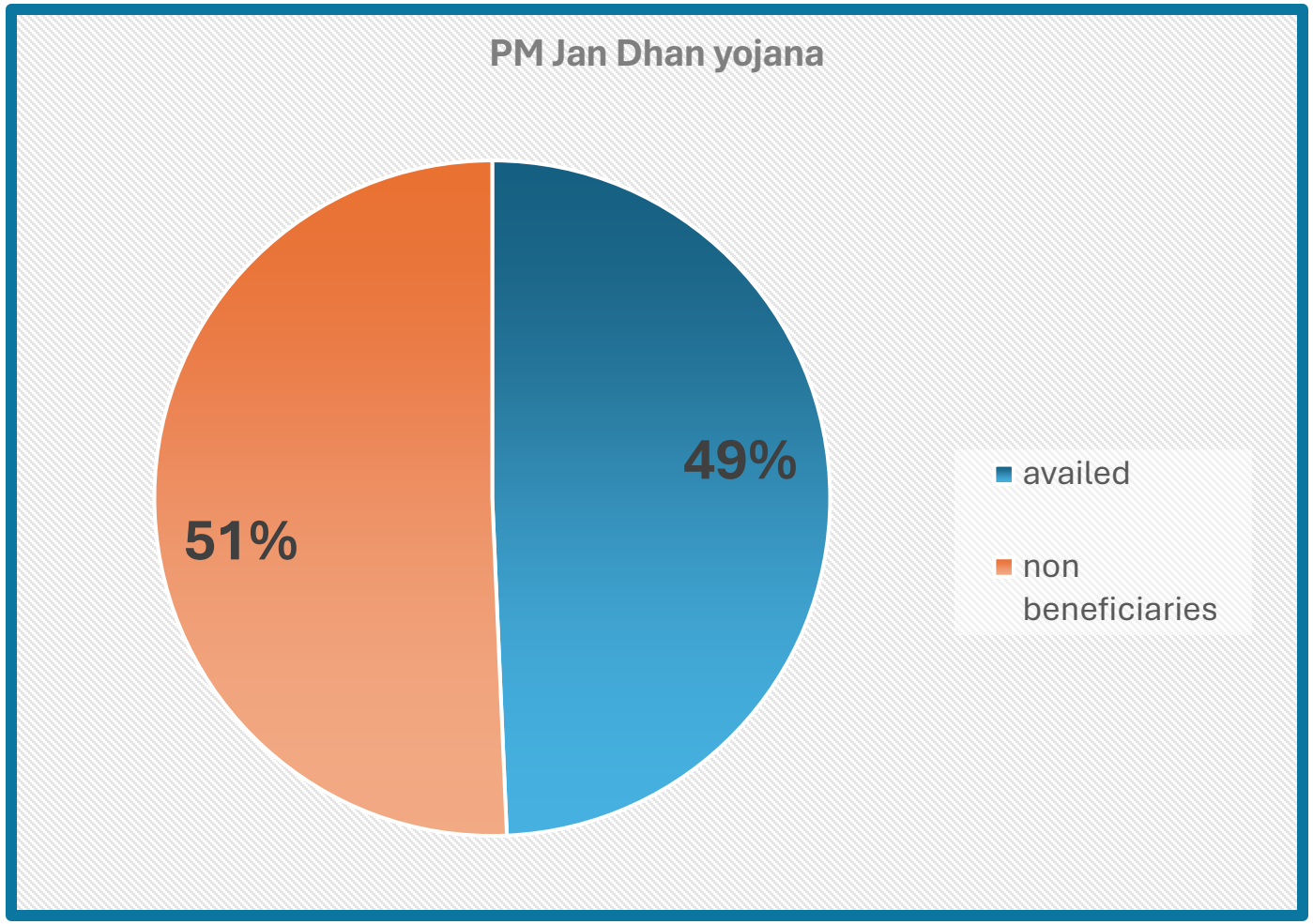
NOT INTERESTED

Kisan Credit Card



AFFECTED BY CREDIT HISTORY/ACCOUNT ACCESS

PM Jan Dhan yojana



AROUND 50%

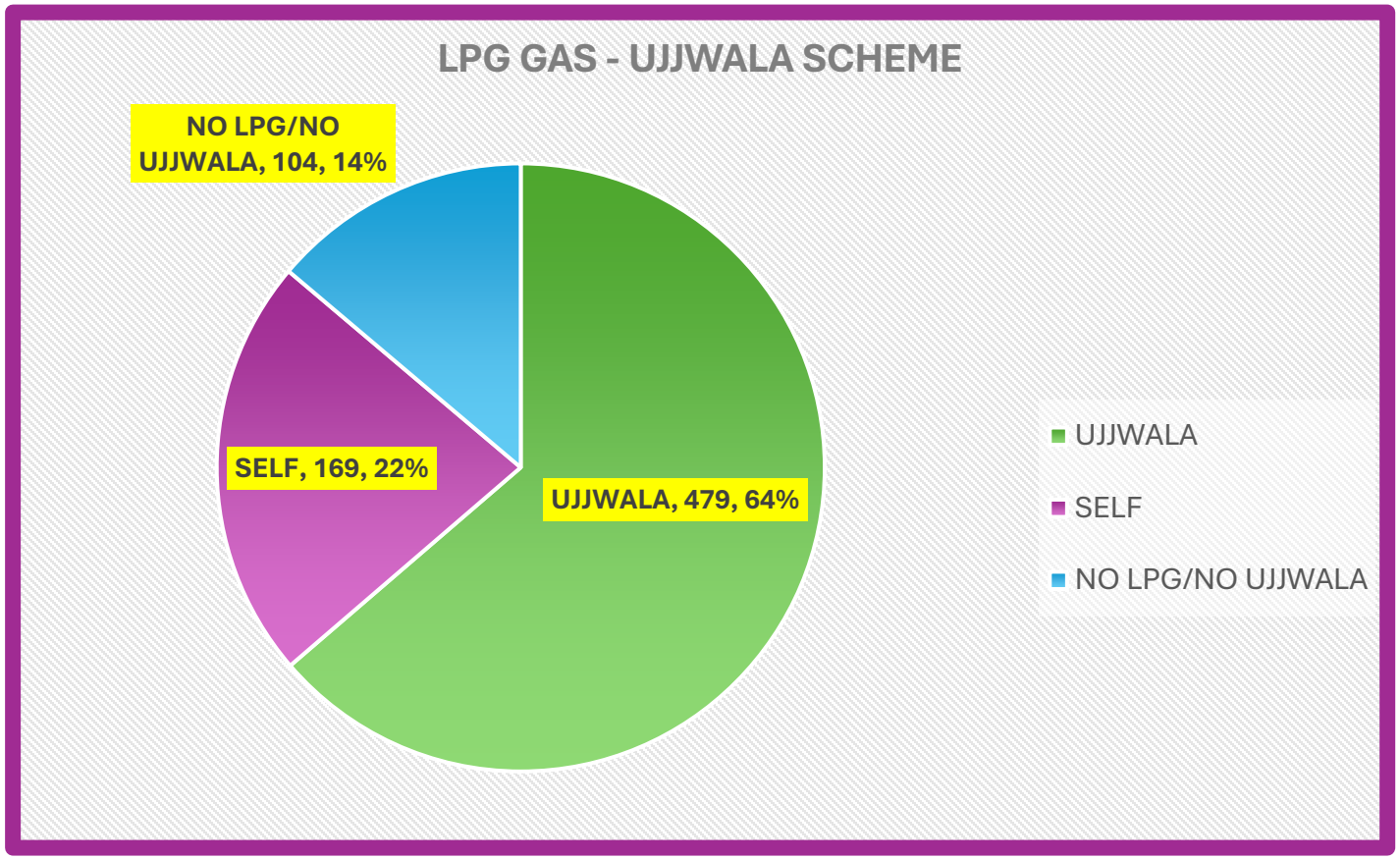
Have access to banking/have bank account

Through jan dhan yojana

REMAINING THROUGH SELF
&
HAVE TO COVER MUCH MORE HH-PMJD

UJJWALA SCHEME COVERAGE

COVERAGE	LOW COVERAGE VILLAGES
0 COVERAGE-LOW LPG	BHARSULA
0 COVERAGE-LOW LPG	Maliapura
mostly self- 0U	Kaserua
mostly self- 0U	Nichroli
mostly self- 0U	Agora
mostly self- 0U	Bijora
mostly self- 0U	Mangrol
mostly self- 0U	Laryata
mostly self- 0U	Chirula
mostly self- 0U	Gyara
low coverage	BHARDHUWAN
low coverage	SAKULI

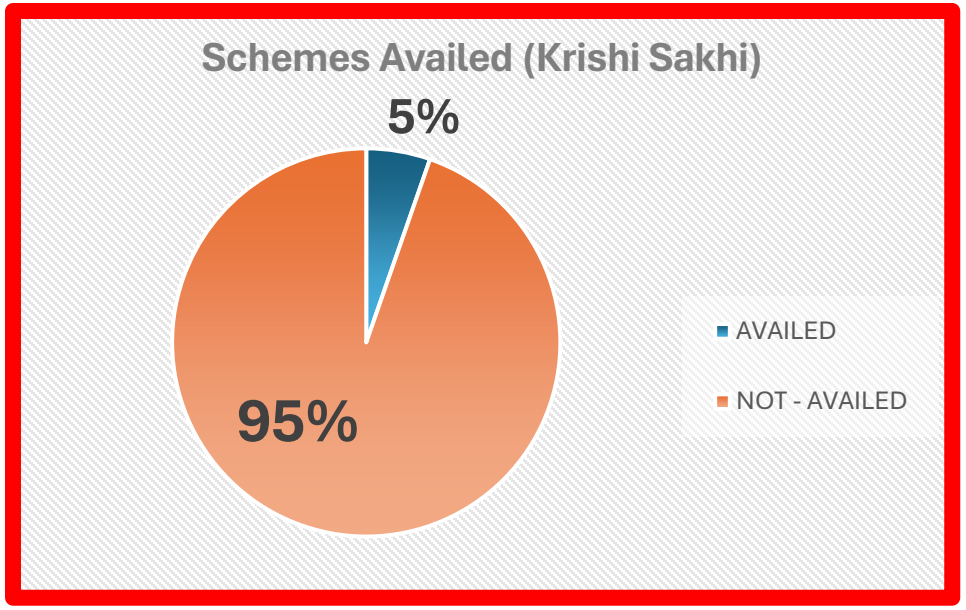


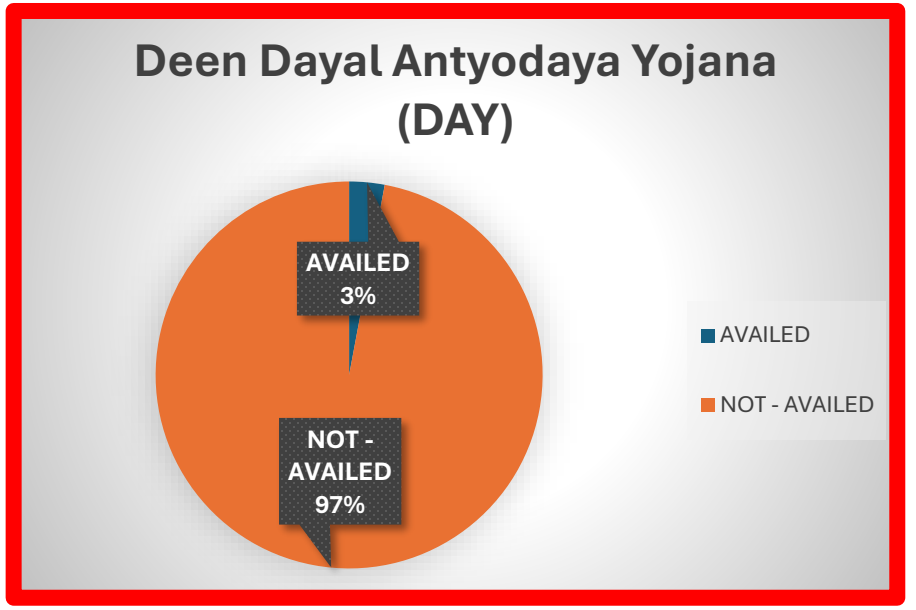
14%-NO LPG-NO UJJWALA

POPULATION who doesn’t own BPL CARD- STILL LIVE IN BPL

DON’T HAVE ACCESS TO UJJWALA

Krishi Sakhi Scheme & Deen Dayal Antyodaya Yojana (DAY)





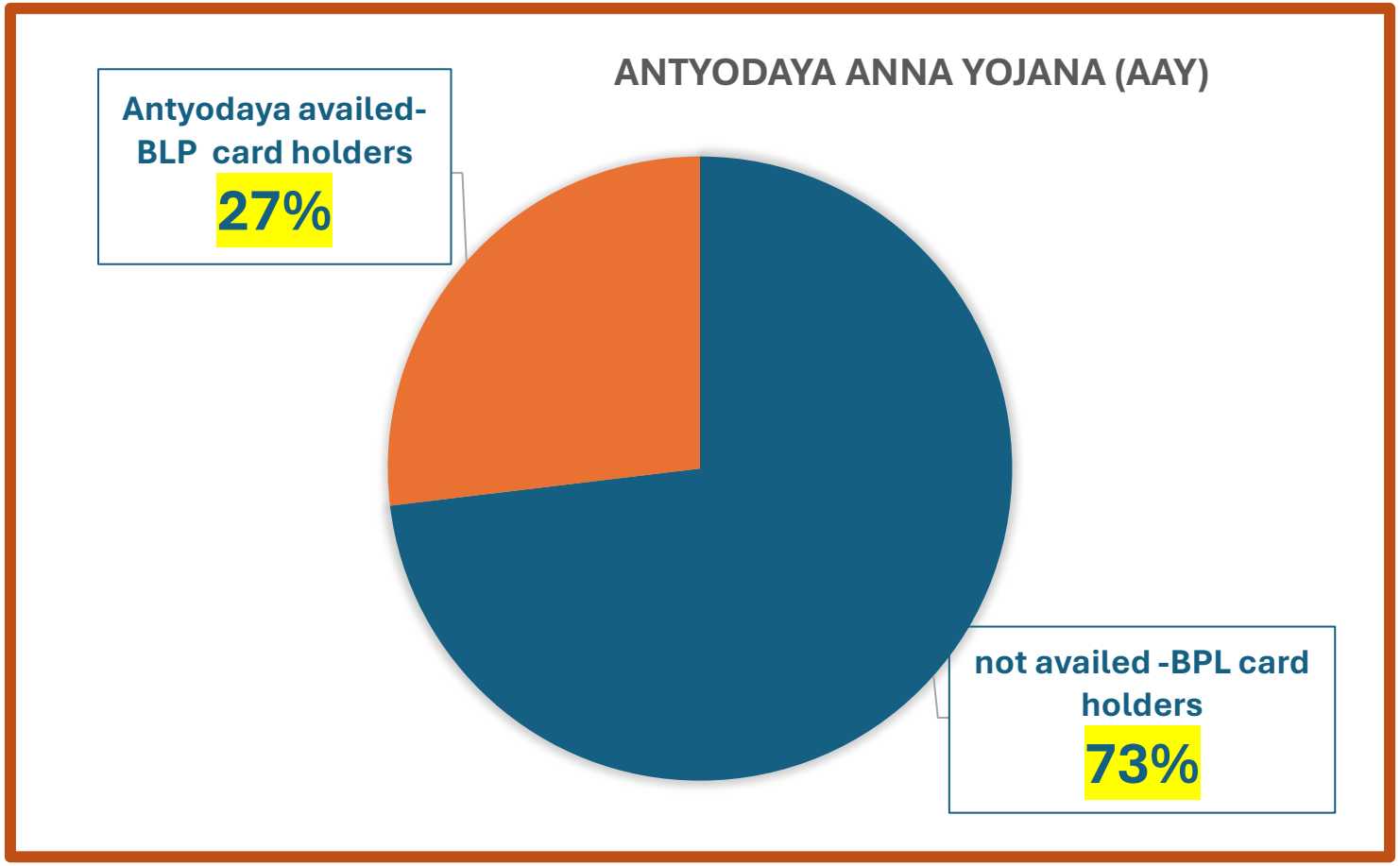
UN AWARE

NOT INTERESTED

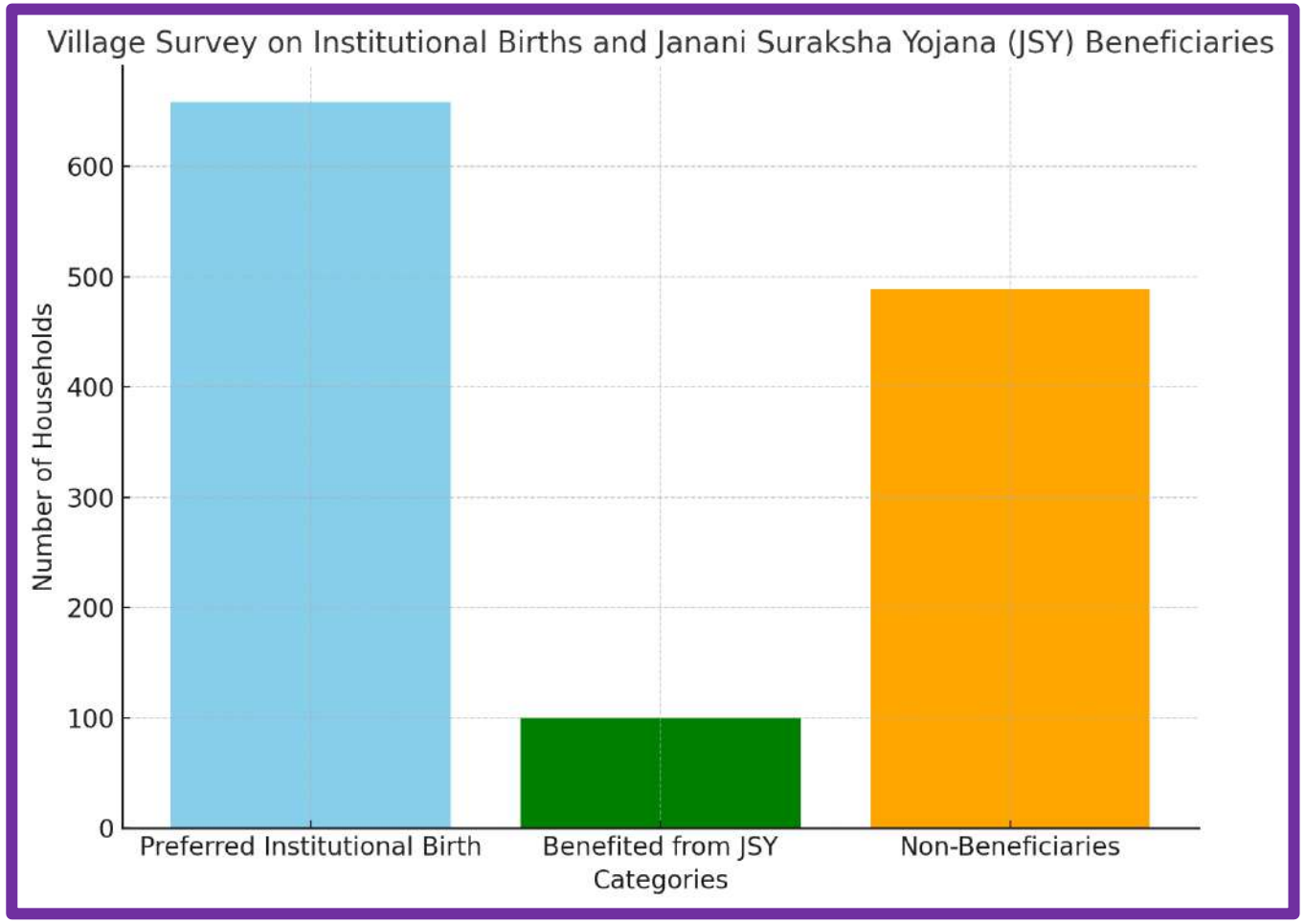
WHY?

NO MONEY
NO FREEBIES

Antyodaya Anna Yojana (AAY)- beneficiaries



Janani Suraksha Yojana (JSY)



More prefer institutional birth
but

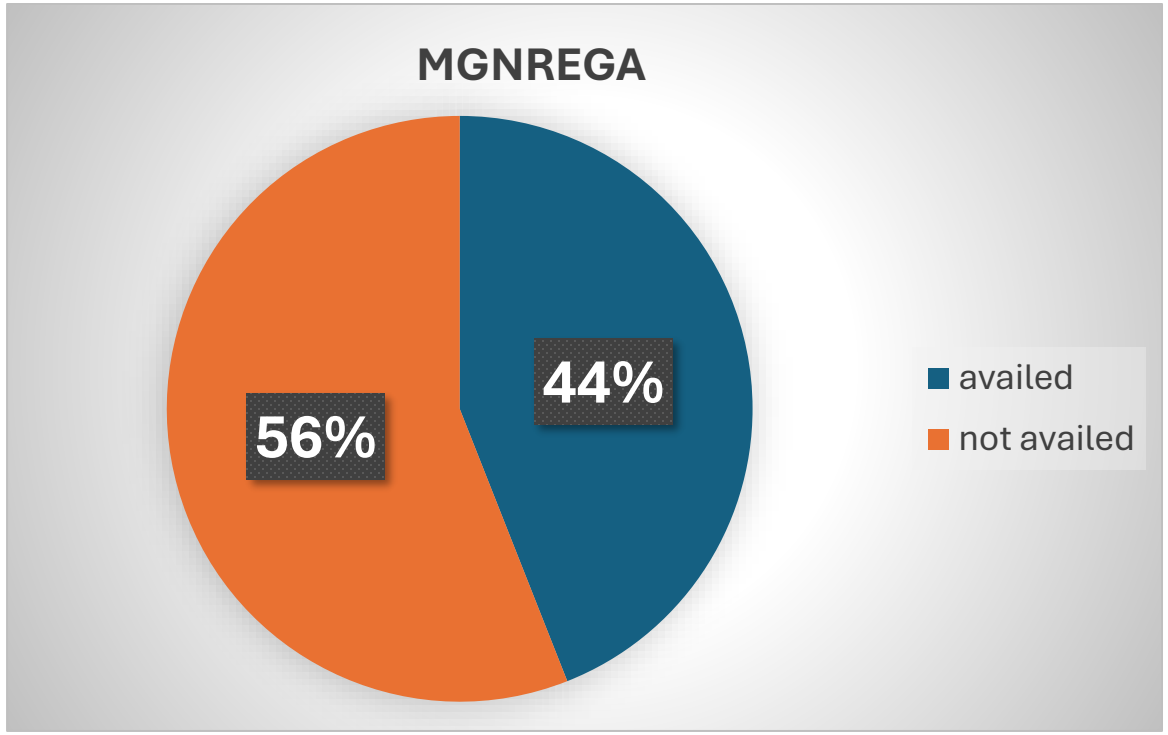
↓

NO PRENATAL CARE

↓

UNAWARE OF
Janani Suraksha Yojana

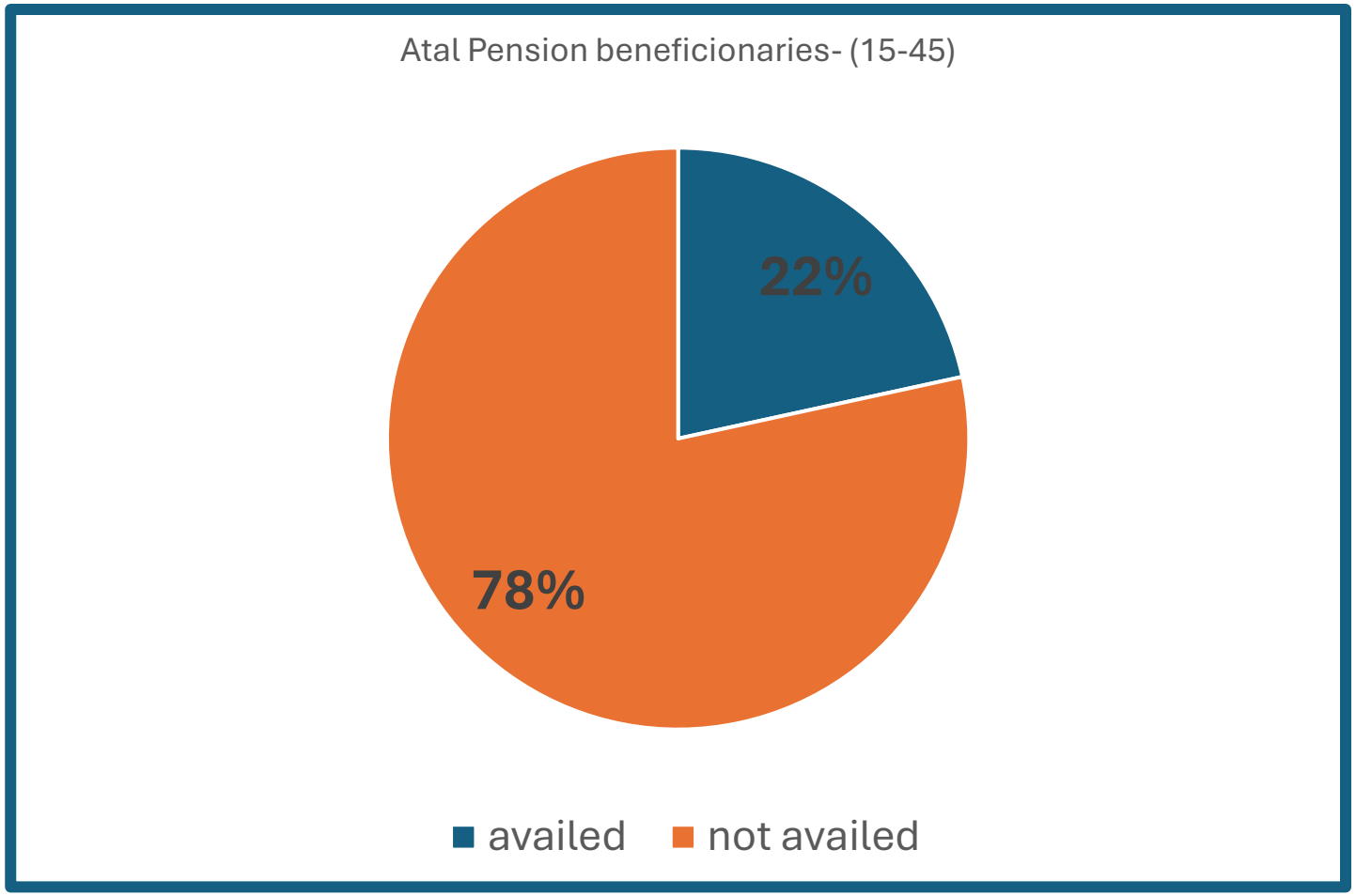
MGNREGA



MORE THAN HALF- SEASONAL/MARGINAL WORKING POPULATION

block	21-22 Number of works taken up	22-23 Number of works undertaken	23-24 Number of works undertaken
2	3	7	11
Total	48184	3479	3300
BHANDER	10294	727	607
DATIA	23308	1719	1728
SEONDHA	14582	1033	965
Total	48184	3479	3300

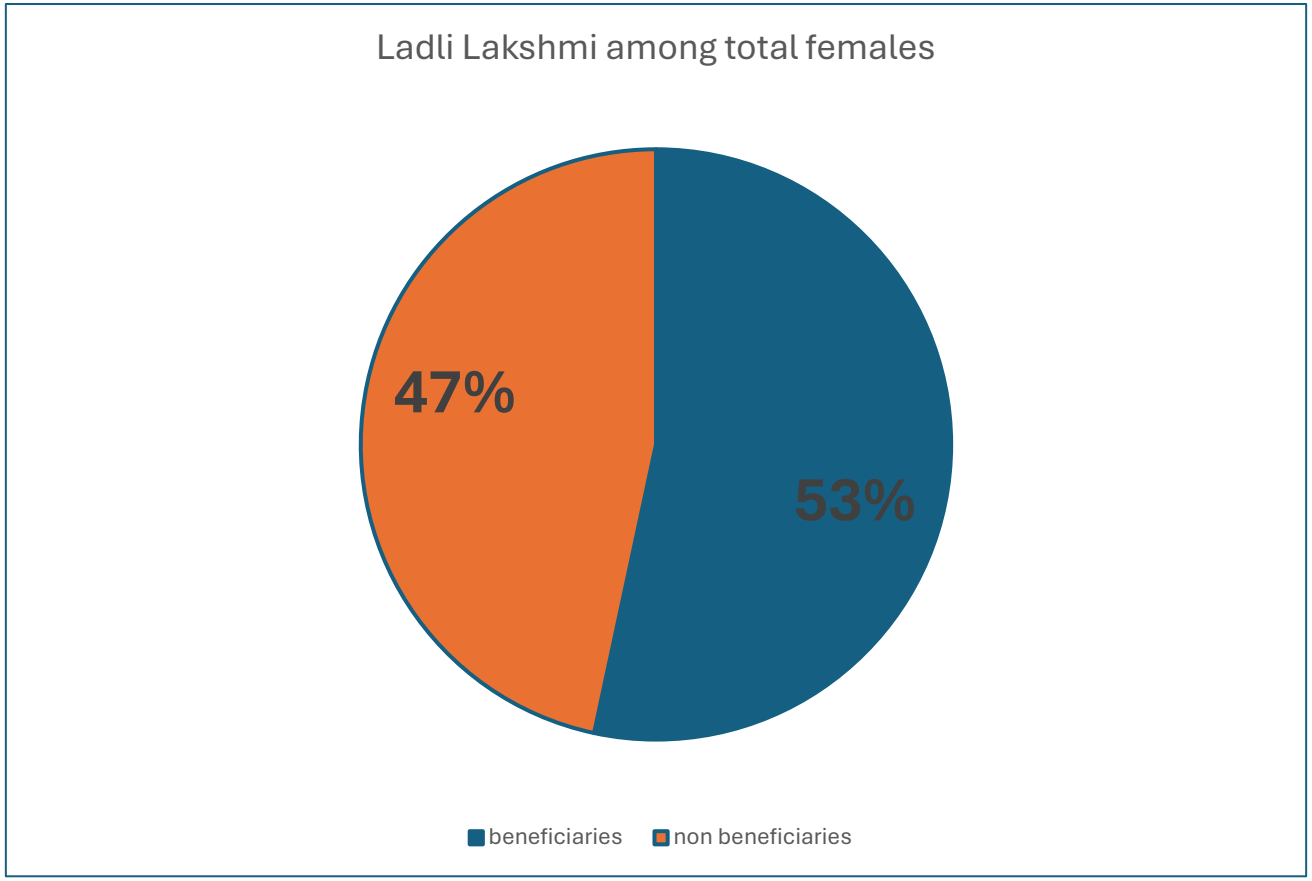
Atal Pension Yojana (APY)



Less aware – less knowledge

May be not ready for 20yr prior commitment to deposit the part of their unsteady income

Ladli Lakshmi Yojana



CASE STUDY ON STUDIO VILLAGE

KASEURA VILLAGE PANCHAYAT

NITESHKUMAR SK

211109041

B. PLAN-VIITH SEM

KASEURA VILLAGE PANCHAYAT

Kaseura village is part of the Seondha block in the Datia district of Madhya Pradesh. It falls under the jurisdiction of the Seondha block's Gram Panchayat.

Local Name	कसेरूआ
Type	Village Panchayat
Villages	Kaserua
Inter Panchayat	Seondha
Block	Seondha
District Panchayat	Datia
State	Madhya Pradesh

- Public hearing – every Tuesday (11-1 am)
- Gram panchayat meeting 15th of the month

VILLAGE INFO

Source: village survey

Category	Details
Total Accommodation	86
Total Population	2362
Female Population	1072
Male Population	1290
Voters/18+ Population	1510
Total Number of Posts	15
Total BPL Families	65
Total Schools	4
Primary Schools (Upper Primary)	04 (1)
Middle and High Schools	0
Total Anganwadi Centers	3
Hospital	1
Number of Self-Help Groups	7
Total MNREGA Job Cards	176
Total Pensioners	134
Total Sambal Registrations	450

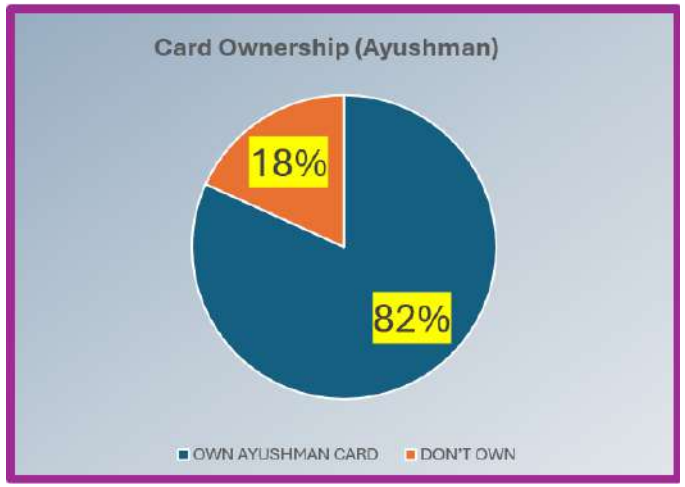
FUNCTIONARIES & ORGANISATION

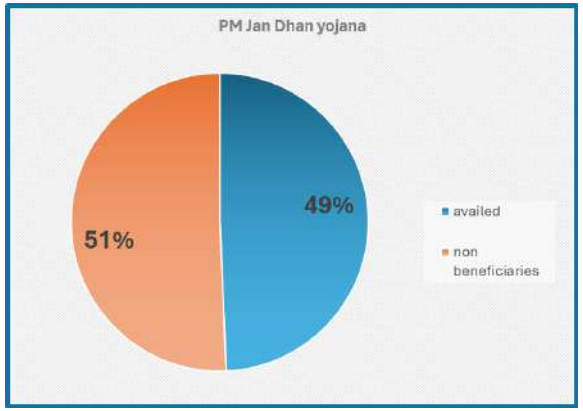
Name	Designation	Department
Arun Kumar	Patvari	Department Of Revenue
Arun Kumar Divoliya	Patwari	Department Of Revenue
Ashok Kushwaha	Teacher	Department Of Education
Gangaram Baghel	Head Master	Department Of Education
Hari Ganesh Sharma	Gau Sevak	Department Of Animal Husbandry And Dairying
Hemlata Rajpoot	Anm	Department Of Health
Kishun Lal Goyal	Head Master	Department Of Education
Mala Taygi	Asha	Department Of Health & Family Welfare
Manju Devi	Aganbadi Karykarta	Department Of Women & Child Development
Narayan Parihar	Schiv	Department Of Panchayati Raj
Ramumar	Krashak Friend	Department Of Agriculture
Smt. Hemlata Rajput	Anm	Department Of Health & Family Welfare
Sunita Sharma	Aganbadhi Workers	Department Of Women & Child Development
Virendra Goad	Pco	Department Of Rural Development

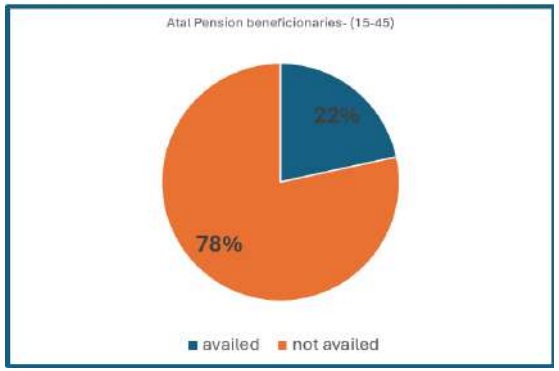
ELECTED REPRESENTATIVES

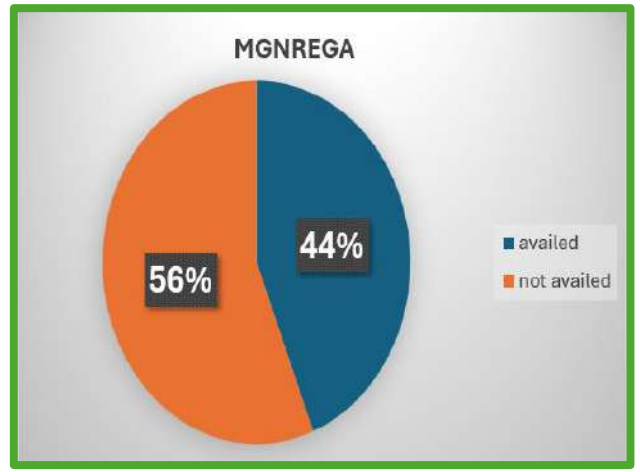
Sarpanch Name	Vitolidevi Baghel
Secretary Name	Naransingh Parihar

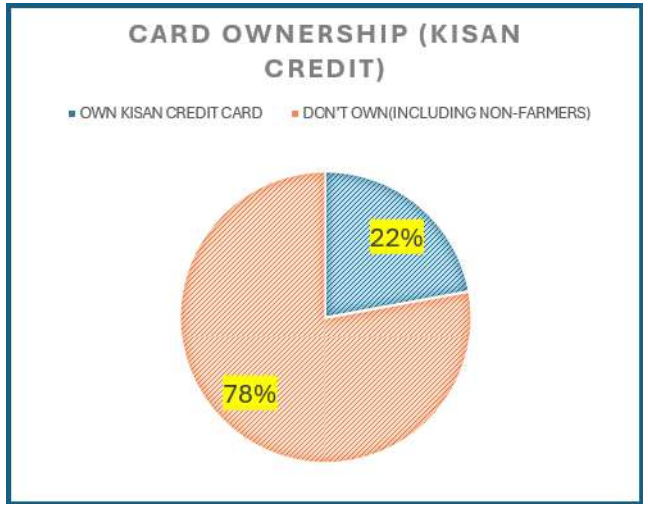
SCHEMES & COVERAGE

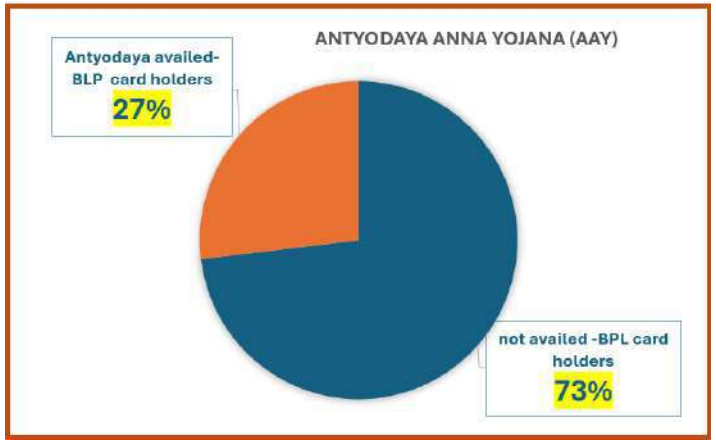


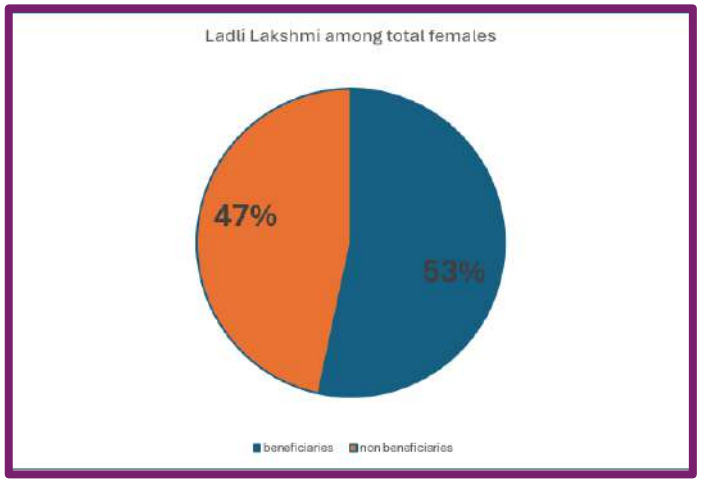








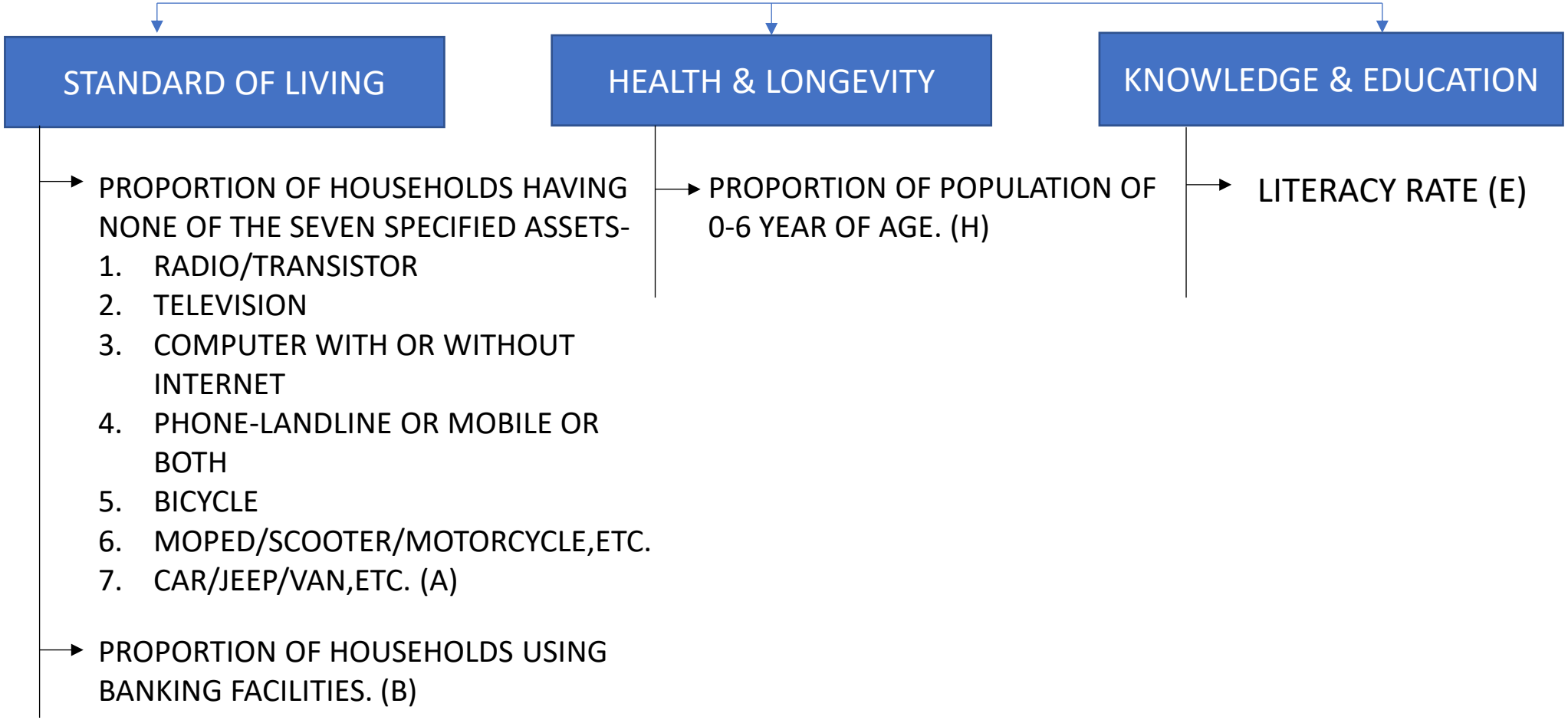




HUMAN DEVELOPMENT INDEX

FOR MADHYA PRADESH AND ITS CONSTITUENT DISTRICTS

INDICATORS


STANDARD OF LIVING
HEALTH & LONGEVITY
KNOWLEDGE & EDUCATION
PROPORTION OF HOUSEHOLDS HAVING NONE OF THE SEVEN SPECIFIED ASSETS-
1. RADIO/TRANSISTOR
2. TELEVISION
3. COMPUTER WITH OR WITHOUT INTERNET
4. PHONE-LANDLINE OR MOBILE OR BOTH
5. BICYCLE
6. MOPED/SCOOTER/MOTORCYCLE,ETC.
7. CAR/JEEP/VAN,ETC. (A)
PROPORTION OF HOUSEHOLDS USING BANKING FACILITIES. (B)
PROPORTION OF POPULATION OF 0-6 YEAR OF AGE. (H)
LITERACY RATE (E)

MINIMUM & MAXIMUM VALUES

0
MINIMUM

1
MAXIMUM

Indicator		Minimum	Maximum
Standard of living	Proportion of households having none of the assets	8.09	64.9
	Proportion of households using banking facilities	30.38	77.53
Health & Longevity	Proportion of population in 0-6 age group	11	21
Knowlegde & education	Literacy Rate	0	100

$$\text{Dimension index} = \frac{\text{actual value} - \text{minimum value}}{\text{maximum value} - \text{minimum value}}$$

PROCEDURE OF CALCULATION

First, all the variables were normalized. The goal posts used for normalization were 0 and 1 for all indicators.

The HDI has been calculated as the product of the index of health (ih), index of education (ie) and index of household standard of living (is).

In other words

$$HDI = (ih * ie * is)1/3$$

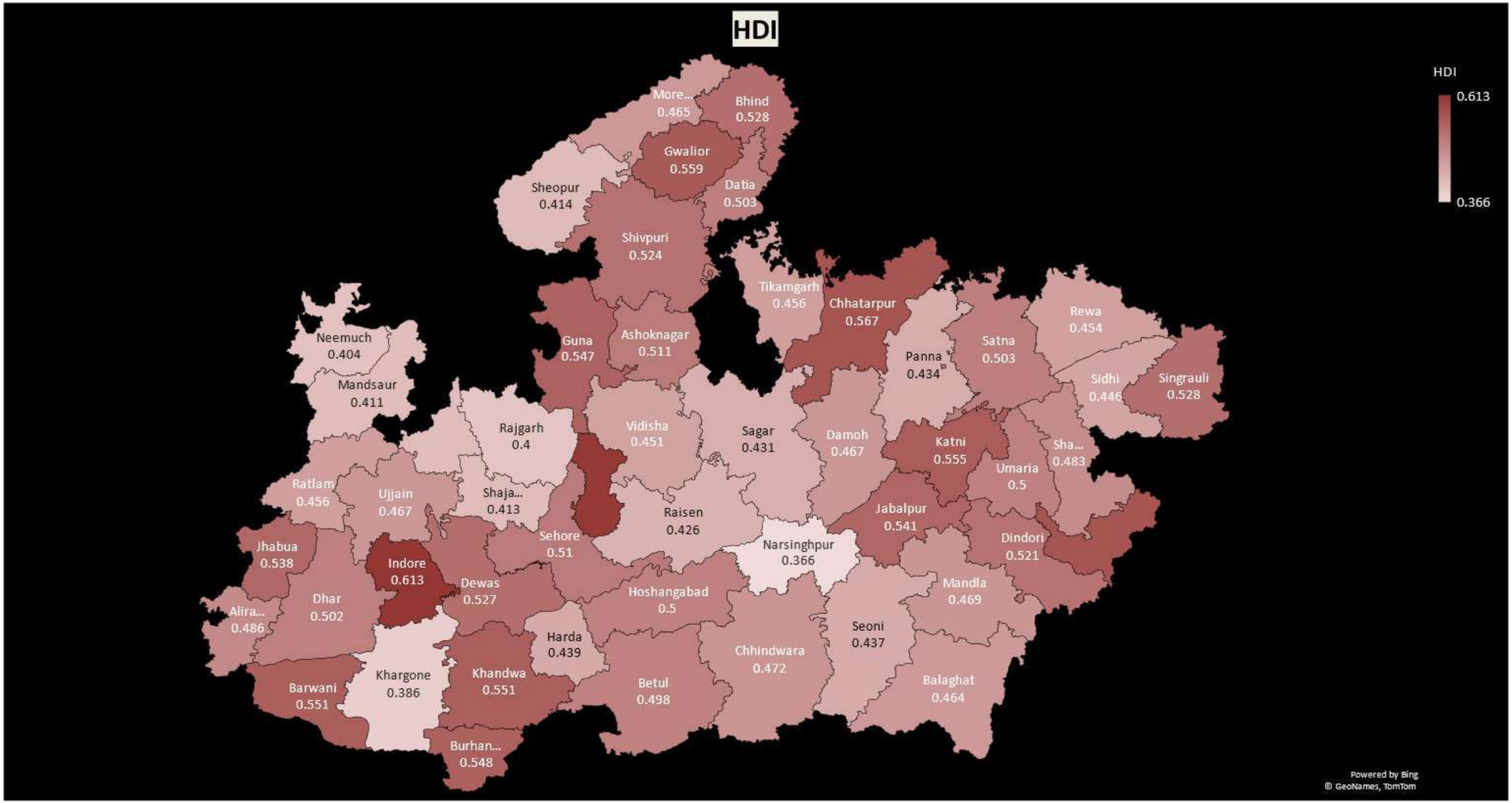
where

$$s = a+b/2$$

and a, b, e, h, and s are the normalized values of indicators A, B, E, H and S respectively.

HDI in Madhya Pradesh and constituent districts, 2011					
State/District	**Standard Of Living**	**Health & longevity**	**Knowledge & Education**	**HDI Score =(Is+Ih+Ie)/3**	**Rank**
	Index	Index	Index		
Madhya Pradesh	0.456	0.394	0.706	0.519	
Alirajpur	0.126	0.961	0.372	0.486	26
Anuppur	0.691	0.319	0.691	0.567	3
Ashoknagar	0.241	0.556	0.736	0.511	18
Balaghat	0.420	0.147	0.824	0.464	33
Barwani	0.176	0.84	0.637	0.551	7
Betul	0.551	0.242	0.701	0.498	25
Bhind	0.428	0.362	0.794	0.528	13
Bhopal	0.795	0.184	0.846	0.608	2
Burhanpur	0.339	0.536	0.77	0.548	9
Chhatarpur	0.454	0.516	0.731	0.567	4
Chhindwara	0.491	0.2	0.724	0.472	28
Damoh	0.211	0.426	0.763	0.467	30
Datia	0.466	0.302	0.742	0.503	20
Dewas	0.505	0.372	0.705	0.527	15
Dhar	0.341	0.56	0.606	0.502	22
Dindori	0.411	0.498	0.654	0.521	17
Guna	0.325	0.584	0.732	0.547	10
Gwalior	0.715	0.184	0.779	0.559	5
Harda	0.463	0.12	0.734	0.439	39
Hoshangabad	0.553	0.225	0.723	0.500	23
Indore	0.834	0.183	0.823	0.613	1
Jabalpur	0.672	0.097	0.855	0.541	11
Jhabua	0.169	1	0.444	0.538	12
Katni	0.467	0.404	0.794	0.555	6
Khandwa	0.391	0.5	0.763	0.551	8
Khargone	0.249	0.269	0.64	0.386	49
Mandla	0.615	0.111	0.681	0.469	29
Mandsaur	0.447	0.046	0.741	0.411	46
Morena	0.392	0.225	0.777	0.465	32
Narsinghpur	0.303	0.002	0.792	0.366	50
Neemuch	0.498	0	0.715	0.404	47
Panna	0.378	0.276	0.648	0.434	41
Raisen	0.304	0.232	0.743	0.426	43
Rajgarh	0.275	0.179	0.746	0.400	48
Ratlam	0.425	0.179	0.763	0.456	34
Rewa	0.461	0.168	0.734	0.454	36
Sagar	0.338	0.179	0.775	0.431	42
Satna	0.564	0.164	0.781	0.503	21
Sehore	0.562	0.183	0.785	0.510	19
Seoni	0.466	0.009	0.836	0.437	40
Shahdol	0.616	0.149	0.684	0.483	27
Shajapur	0.392	0.124	0.723	0.413	45
Sheopur	0.297	0.353	0.592	0.414	44
Shivpuri	0.516	0.318	0.739	0.524	16
Sidhi	0.276	0.38	0.681	0.446	38
Singrauli	0.410	0.429	0.745	0.528	14
Tikamgarh	0.399	0.245	0.724	0.456	35
Ujjain	0.569	0.058	0.774	0.467	31
Umaria	0.556	0.271	0.673	0.500	24
Vidisha	0.338	0.28	0.736	0.451	37

State/District	HDI	Rank
Indore	0.613	1
Bhopal	0.608	2
Anuppur	0.567	3
Chhatarpur	0.567	4
Gwalior	0.559	5
Katni	0.555	6
Barwani	0.551	7
Khandwa	0.551	8
Burhanpur	0.548	9
Guna	0.547	10
Jabalpur	0.541	11
Jhabua	0.538	12
Bhind	0.528	13
Singrauli	0.528	14
Dewas	0.527	15
Shivpuri	0.524	16
Dindori	0.521	17
Ashoknagar	0.511	18
Sehore	0.51	19
Datia	0.503	20
Satna	0.503	21
Dhar	0.502	22
Hoshangabad	0.5	23
Umaria	0.5	24
Betul	0.498	25
Alirajpur	0.486	26
Shahdol	0.483	27
Chhindwara	0.472	28
Mandla	0.469	29
Damoh	0.467	30
Ujjain	0.467	31
Morena	0.465	32
Balaghat	0.464	33
Ratlam	0.456	34
Tikamgarh	0.456	35
Rewa	0.454	36
Vidisha	0.451	37
Sidhi	0.446	38
Harda	0.439	39
Seoni	0.437	40
Panna	0.434	41
Sagar	0.431	42
Raisen	0.426	43
Sheopur	0.414	44
Shajapur	0.413	45
Mandsaur	0.411	46
Neemuch	0.404	47
Rajgarh	0.4	48
Khargone	0.386	49
Narsinghpur	0.366	50


HDI
HDI
0.613
0.366
More...
0.465
Bhind
0.528
Gwalior
0.559
Sheopur
0.414
Datia
0.503
Shivpuri
0.524
Tikamgarh
0.456
Chhatarpur
0.567
Rewa
0.454
Satna
0.503
Panna
0.434
Sidhi
0.446
Singrauli
0.528
Neemuch
0.404
Guna
0.547
Ashoknagar
0.511
Mandsaur
0.411
Rajgarh
0.4
Vidisha
0.451
Sagar
0.431
Damoh
0.467
Katni
0.555
Sha...
0.483
Umaria
0.5
Ratlam
0.456
Ujjain
0.467
Shaja...
0.413
Raisen
0.426
Jabalpur
0.541
Dindori
0.521
Jhabua
0.538
Sehore
0.51
Narsinghpur
0.366
Indore
0.613
Dewas
0.527
Hoshangabad
0.5
Mandla
0.469
Dhar
0.502
Alira...
0.486
Harda
0.439
Seoni
0.437
Chhindwara
0.472
Khargone
0.386
Khandwa
0.551
Barwani
0.551
Betul
0.498
Balaghat
0.464
Burhan...
0.548
Powered by Bing
© GeoNames, TomTom

MADHYA PRADESH DISTRICTS RANKED BASED ON ASPIRATIONAL DISTRICT INDICATORS

ASPIRATIONAL DISTRICT RANKING

CHOSEN INDICATORS

HEALTH AND NUTRITION

1. Percentage of ANC registered within the first trimester against total ANC registrations+++++
2. Percentage of institutional deliveries out of total estimated deliveries+++++
3. Sex Ratio at birth+++++
4. Proportion of live babies weighed at birth
5. Percentage of low birth weight babies (Less than 2500 gms) ++++
6. Percentage of new-borns breastfed within one hour of birth
7. Percentage of children fully immunized (9-11 months) (BCG+ DPT3 + OPV3 + Measles1)++++++

EDUCATION

1. Transition Rate - Primary to Upper Primary level++++
2. Transition Rate - Upper Primary to Secondary level+++++
3. Female literacy rate (15+ Age group)++++++

SKILL DEVELOMENT

no of youth in district in age group 15-29++++

BASIC INFRA

1. Percentage of households with electricity connection+++++++
2. Percentage of households with Individual Household Latrines+++++
3. Percentage of Rural habitations with access to adequate quantity of potable water - 40 lpcd drinking water in rural+++++

FINANCE

Atal Pension Yojana (APY): Number of Beneficiaries per 1 Lakh population++++

AGRICULTURE

Number of Soil Health Cards distributed+++++

OUR METHODOLOGY

We are using weightage index :

1. Divide your data into 3 class of score 1-3-5 or 4 or 5 classes 12345 based on the variation difference from average value in data.

2. Least class should start on least data value not zero

3. If data are in numbers divide them by their respective population OR reasonable facotrs and multiply by 100 to get percentile

4. Also find average value of all parameters of mp districts

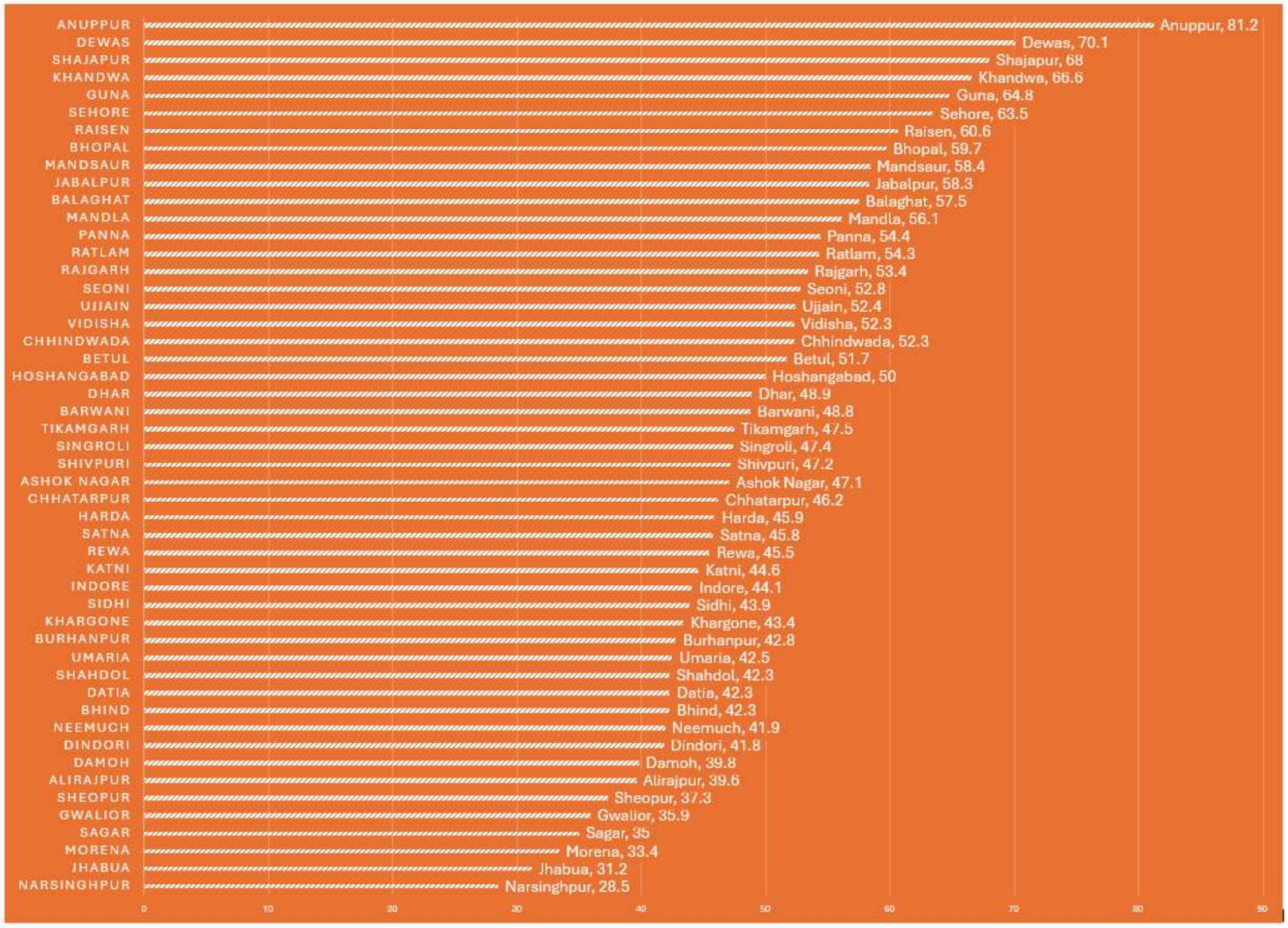
DISTRICTS	% 1st Trimester registration to Total ANC Registrations	SCORING
	2011-12	score(weightage index)
Madhya pradesh	49.1(average)	
Narsinghpur	28.5	1
Jhabua	31.2	1
Morena	33.4	1
Sagar	35	1
Gwalior	35.9	1
Sheopur	37.3	1
Alirajpur	39.6	1
Damoh	39.8	1
Dindori	41.8	3
Neemuch	41.9	3
Bhind	42.3	3
Datia	42.3	3
Shahdol	42.3	3
Umaria	42.5	3
Burhanpur	42.8	3
Khargone	43.4	3
Sidhi	43.9	3
Indore	44.1	3
Katni	44.6	3
Rewa	45.5	3
Satna	45.8	3
Harda	45.9	3
Chhatarpur	46.2	3
Ashok Nagar	47.1	3
Shivpuri	47.2	3
Singroli	47.4	3
Tikamgarh	47.5	3
Barwani	48.8	3
Dhar	48.9	3
Hoshangabad	50	3
Betul	51.7	3
Chhindwada	52.3	3
Vidisha	52.3	3
Ujjain	52.4	3
Seoni	52.8	3
Rajgarh	53.4	3
Ratlam	54.3	3
Panna	54.4	3
Mandla	56.1	3
Balaghat	57.5	3
Jabalpur	58.3	3
Mandsaur	58.4	3
Bhopal	59.7	3
Raisen	60.6	5
Sehore	63.5	5
Guna	64.8	5
Khandwa	66.6	5
Shajapur	68	5
Dewas	70.1	5
Anuppur	81.2	5

CLASS	WEIGHT
28-39.9	1
40-59.9	3
>60	5

HEALTH

% 1st Trimester registration to Total ANC Registrations

LEAST	28.5	Narsinghpur
HIGHEST	81.2	Anuppur
AVG	49.1	

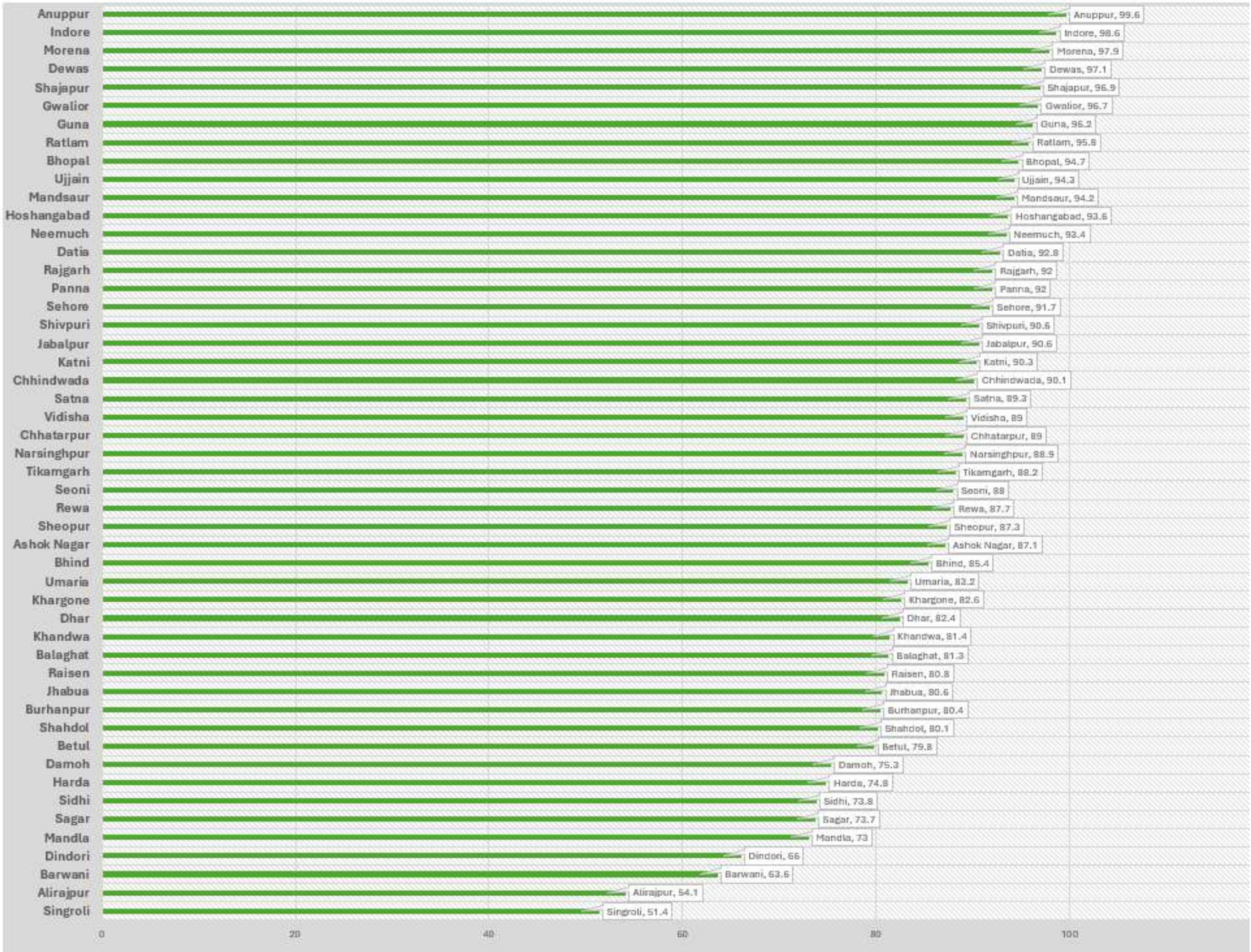


DISTRICTS	% Institutional deliveries to Total Reported Deliveries	score(weightage index)
	2011-12	
Madhya pradesh	86.3(average)	
Singroli	51.4	1
Alirajpur	54.1	1
Barwani	63.6	1
Dindori	66	1
Mandla	73	1
Sagar	73.7	1
Sidhi	73.8	1
Harda	74.8	1
Damoh	75.3	3
Betul	79.8	3
Shahdol	80.1	3
Burhanpur	80.4	3
Jhabua	80.6	3
Raisen	80.8	3
Balaghat	81.3	3
Khandwa	81.4	3
Dhar	82.4	3
Khargone	82.6	3
Umaria	83.2	3
Bhind	85.4	3
Ashok Nagar	87.1	3
Sheopur	87.3	3
Rewa	87.7	3
Seoni	88	3
Tikamgarh	88.2	3
Narsinghpur	88.9	3
Chhatarpur	89	3
Vidisha	89	3
Satna	89.3	3
Chhindwada	90.1	4
Katni	90.3	4
Jabalpur	90.6	4
Shivpuri	90.6	4
Sehore	91.7	4
Panna	92	4
Rajgarh	92	4
Datia	92.8	4
Neemuch	93.4	4
Hoshangabad	93.6	4
Mandsaur	94.2	4
Ujjain	94.3	4
Bhopal	94.7	4
Ratlam	95.8	5
Guna	96.2	5
Gwalior	96.7	5
Shajapur	96.9	5
Dewas	97.1	5
Morena	97.9	5
Indore	98.6	5
Anuppur	99.6	5

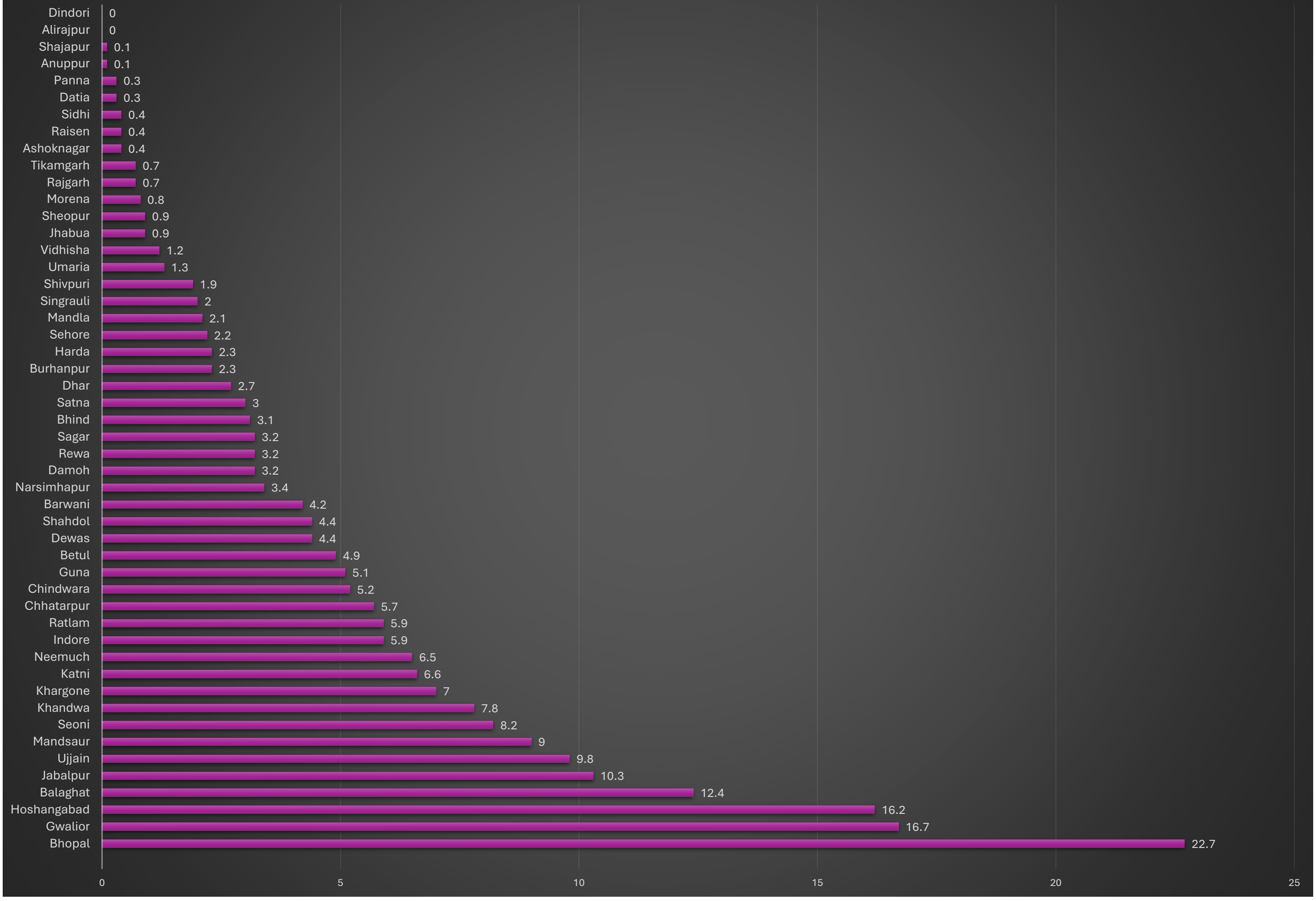
CLASS	WEIGHT
<75	1
75-89.9	3
90-94.9	4
>95	5

LEAST	51.4	Singroli
HIGHEST	99.6	Anuppur
AVG	86.3	

% Institutional deliveries to Total Reported Deliveries



% C-section deliveries (Public + Pvt.) to reported institutional (Public + Pvt.) deliveries

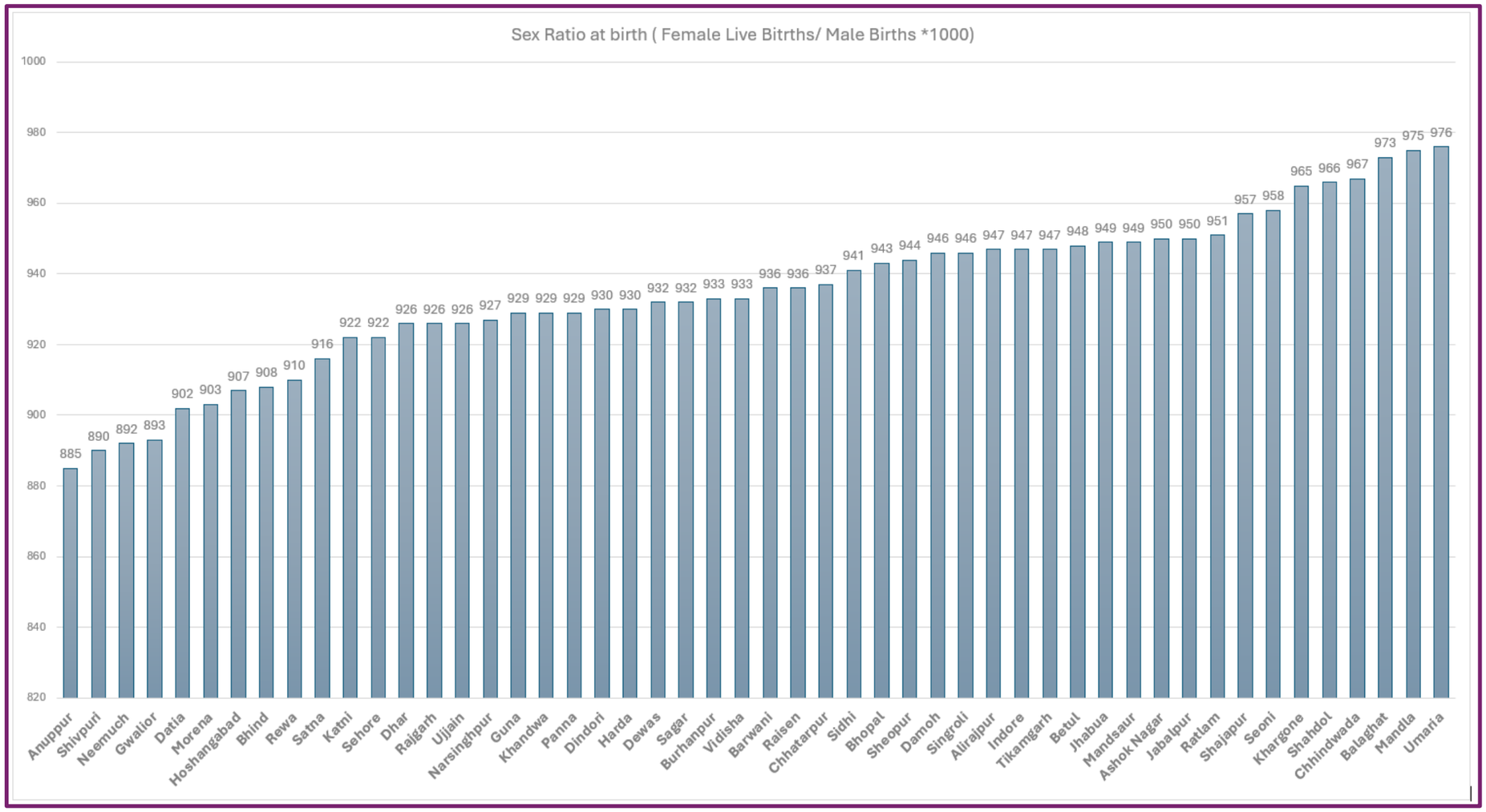


DISTRICTS	Sex Ratio at birth (Female Live Bitrths/ Male Births *1000)	SCORE
	2011-12	2010-11
Madhya pradesh	934(average)	
Anuppur	885	1
Shivpuri	890	1
Neemuch	892	1
Gwalior	893	1
Datia	902	2
Morena	903	2
Hoshangabad	907	2
Bhind	908	2
Rewa	910	2
Satna	916	2
Katni	922	2
Sehore	922	2
Dhar	926	3
Rajgarh	926	3
Ujjain	926	3
Narsinghpur	927	3
Guna	929	3
Khandwa	929	3
Panna	929	3
Dindori	930	3
Harda	930	3
Dewas	932	3
Sagar	932	3
Burhanpur	933	3
Vidisha	933	3
Barwani	936	3
Raisen	936	3
Chhatarpur	937	3
Sidhi	941	3
Bhopal	943	3
Sheopur	944	3
Damoh	946	3
Singroli	946	3
Alirajpur	947	3
Indore	947	3
Tikamgarh	947	3
Betul	948	3
Jhabua	949	3
Mandsaur	949	3
Ashok Nagar	950	3
Jabalpur	950	3
Ratlam	951	5
Shajapur	957	5
Seoni	958	5
Khargone	965	5
Shahdol	966	5
Chhindwada	967	5
Balaghat	973	5
Mandla	975	5
Umaria	976	5

CLASS	WEIGHT
<900	1
900-924	2
925-950	3
>950	5

LEAST	885	Anuppur
HIGHEST	976	Umaria
AVG	934	

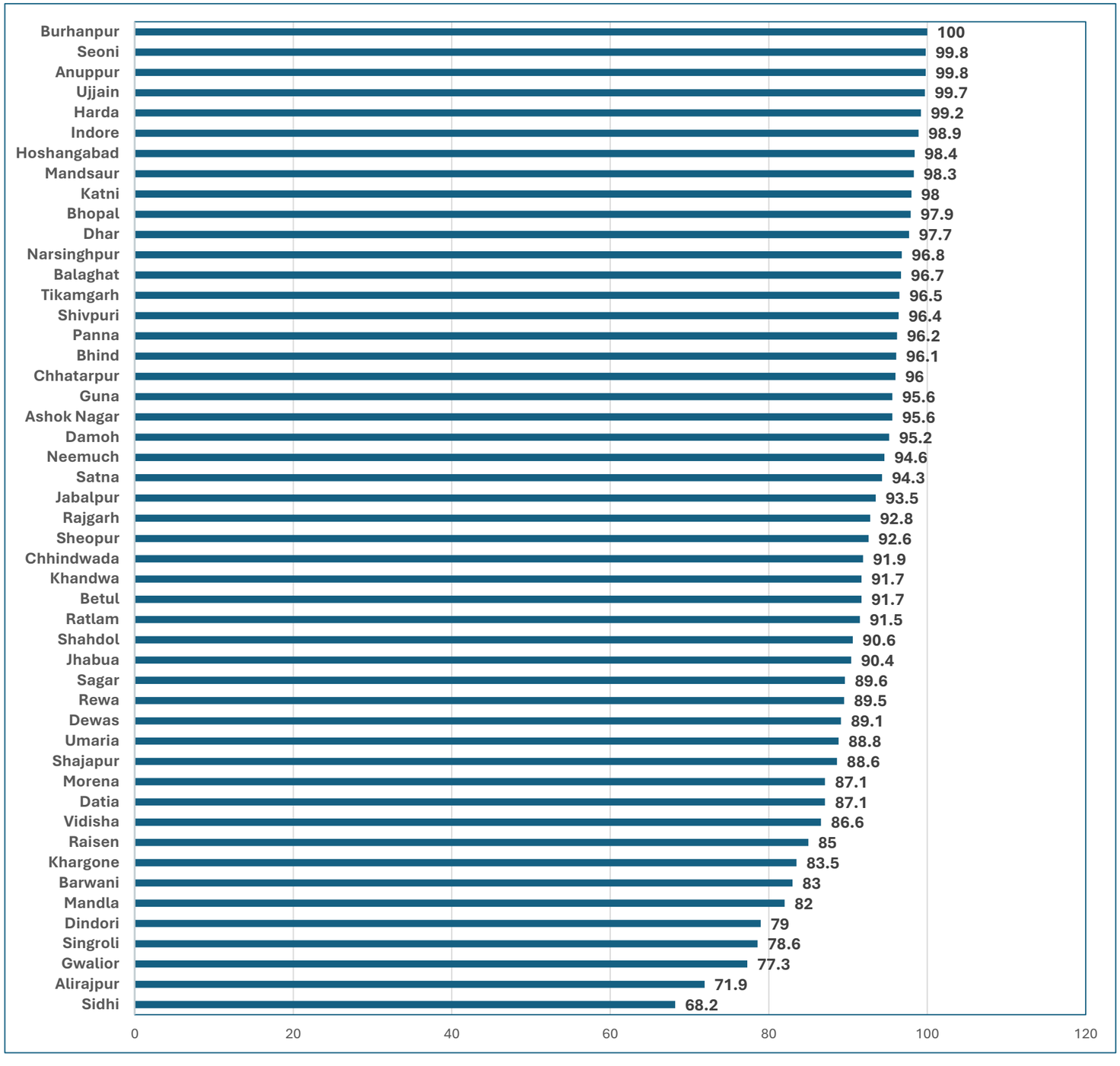
Sex Ratio at birth



	% Newborns weighed at birth to live birth	
		score(weightage index)
	2011-12	
Madhya pradesh	91.2(average)	1
Sidhi	68.2	1
Alirajpur	71.9	1
Gwalior	77.3	1
Singroli	78.6	1
Dindori	79	1
Mandla	82	1
Barwani	83	1
Khargone	83.5	1
Raisen	85	1
Vidisha	86.6	2
Datia	87.1	2
Morena	87.1	2
Shajapur	88.6	2
Umaria	88.8	2
Dewas	89.1	2
Rewa	89.5	2
Sagar	89.6	2
Jhabua	90.4	2
Shahdol	90.6	2
Ratlam	91.5	2
Betul	91.7	2
Khandwa	91.7	2
Chhindwada	91.9	2
Sheopur	92.6	3
Rajgarh	92.8	3
Jabalpur	93.5	3
Satna	94.3	3
Neemuch	94.6	3
Damoh	95.2	3
Ashok Nagar	95.6	3
Guna	95.6	3
Chhatarpur	96	4
Bhind	96.1	4
Panna	96.2	4
Shivpuri	96.4	4
Tikamgarh	96.5	4
Balaghat	96.7	4
Narsinghpur	96.8	4
Dhar	97.7	4
Bhopal	97.9	4
Katni	98	4
Mandsaur	98.3	4
Hoshangabad	98.4	4
Indore	98.9	4
Harda	99.2	5
Ujjain	99.7	5
Anuppur	99.8	5
Seoni	99.8	5
Burhanpur	100	5

CLASS	WEIGHT	
<86	1	
86-92	2	
92-95.9	3	
96-98.9	4	
>99	5	
LEAST	63.7	Anuppur
HIGHEST	100	Umaria
AVG	91.2	

% Newborns weighed at birth to live birth

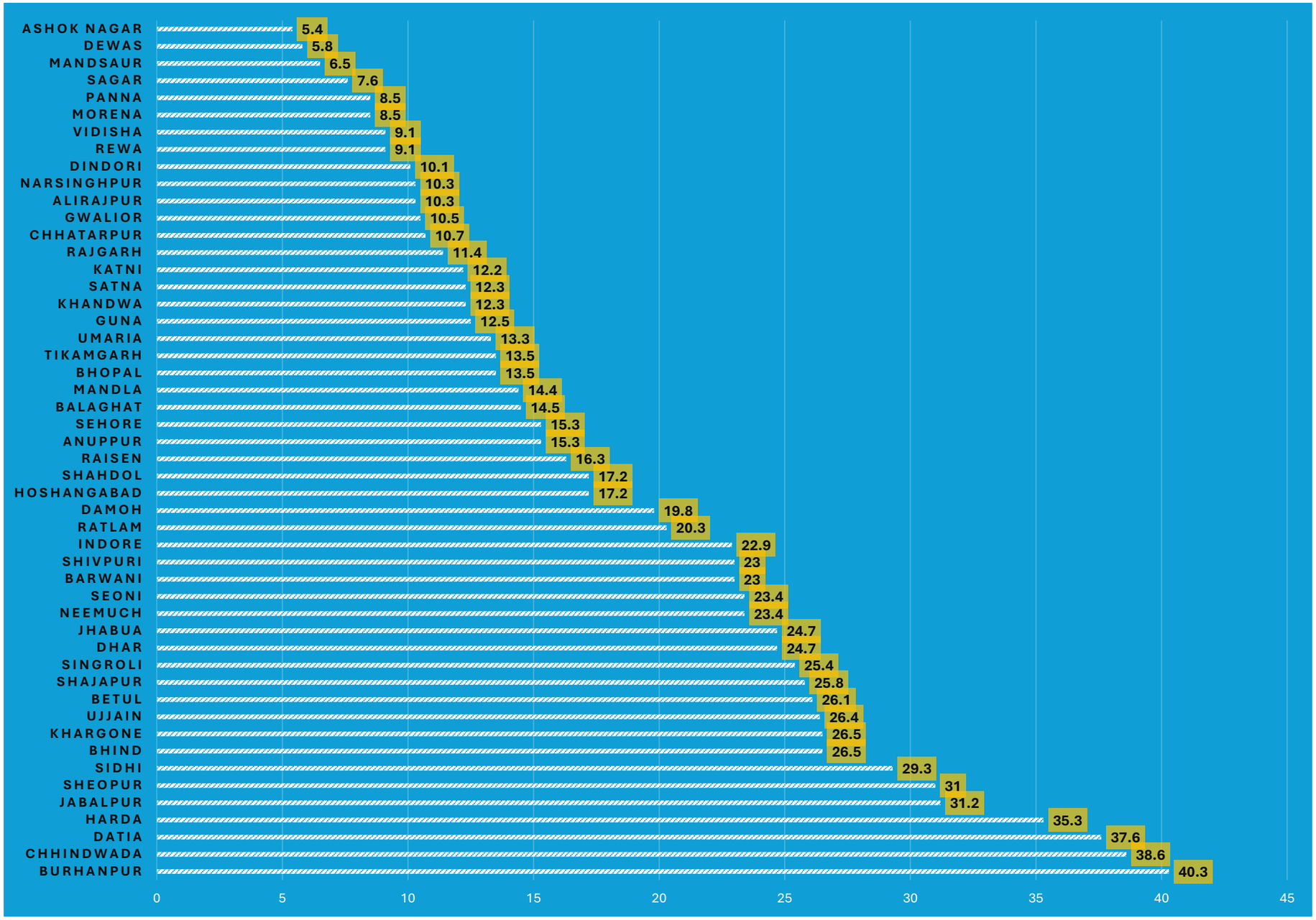


	% Newborns having weight less than 2.5 kg to Newborns weighed at birth	
	2011-12	score(weightage index)
Alirajpur	10.3	-2
Anuppur	15.3	-3
Ashok Nagar	5.4	-1
Balaghat	14.5	-2
Barwani	23	-4
Betul	26.1	-5
Bhind	26.5	-5
Bhopal	13.5	-2
Burhanpur	40.3	-5
Chhatarpur	10.7	-2
Chhindwada	38.6	-5
Damoh	19.8	-3
Datia	37.6	-5
Dewas	5.8	-1
Dhar	24.7	-4
Dindori	10.1	-2
Guna	12.5	-2
Gwalior	10.5	-2
Harda	35.3	-5
Hoshangabad	17.2	-3
Indore	22.9	-4
Jabalpur	31.2	-5
Jhabua	24.7	-4
Katni	12.2	-2
Khandwa	12.3	-2
Khargone	26.5	-5
Mandla	14.4	-2
Mandsaur	6.5	-1
Morena	8.5	-1
Narsinghpur	10.3	-2
Neemuch	23.4	-4
Panna	8.5	-1
Raisen	16.3	-3
Rajgarh	11.4	-2
Ratlam	20.3	-4
Rewa	9.1	-1
Sagar	7.6	-1
Satna	12.3	-2
Sehore	15.3	-3
Seoni	23.4	-4
Shahdol	17.2	-3
Shajapur	25.8	-5
Sheopur	31	-5
Shivpuri	23	-4
Sidhi	29.3	-5
Singroli	25.4	-5
Tikamgarh	13.5	-2
Ujjain	26.4	-5
Umaria	13.3	-2
Vidisha	9.1	-1

CLASS	WEIGHT
>25	-5
20-24.9	-4
15-19.9	-3
10 to 15	-2
<10	-1

LEAST(BETTER)	5.4	Ashok Nagar
HIGHEST(BAD)	40.3	Burhanpur
AVG	18.2	

% Newborns having weight less than 2.5 kg to Newborns weighed at birth

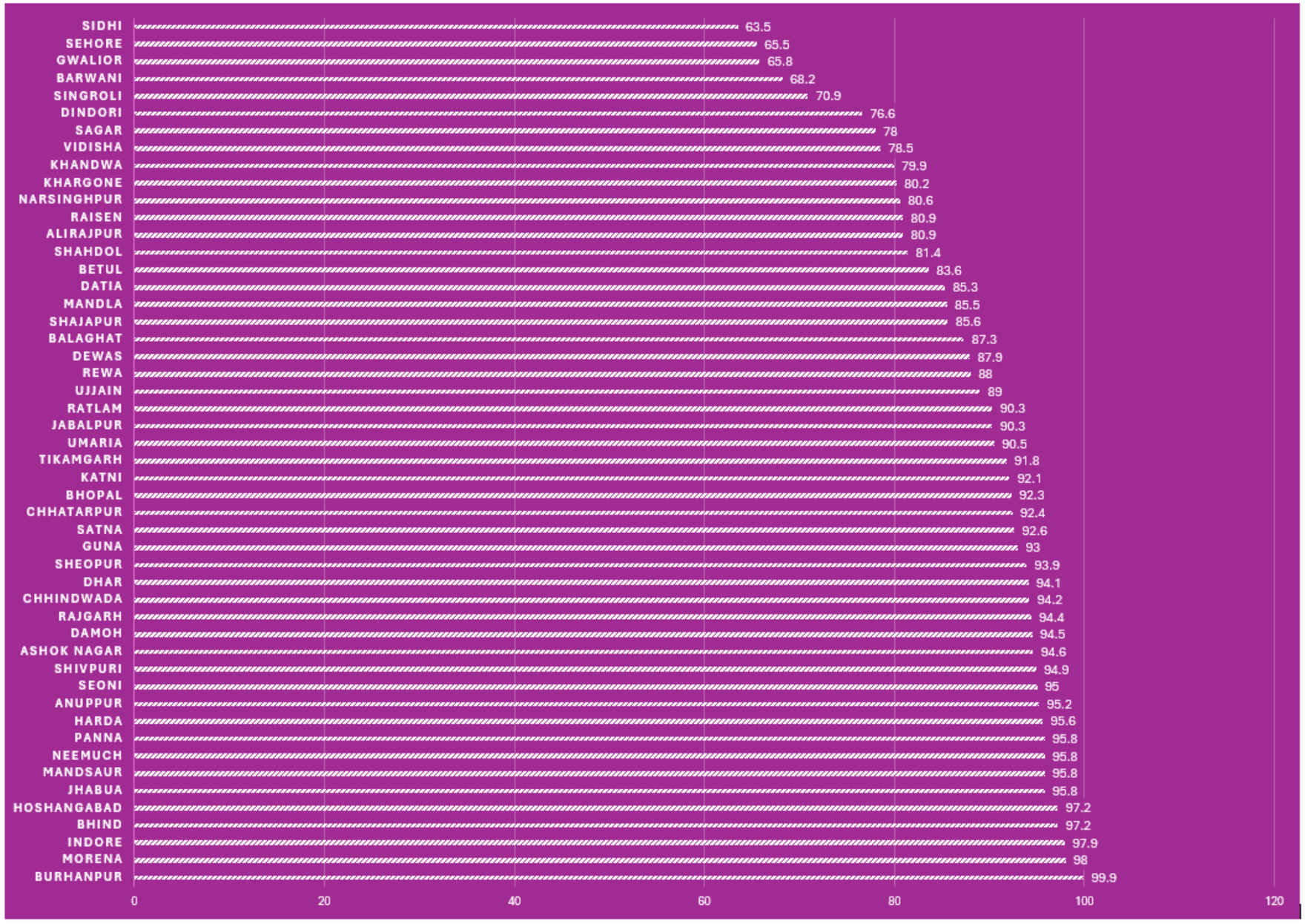


	% Newborns breast fed within 1 hour of birth to Total live birth	
	2011-12	score(weightage index)
Burhanpur	99.9	5
Morena	98	5
Indore	97.9	5
Bhind	97.2	5
Hoshangabad	97.2	5
Jhabua	95.8	5
Mandsaur	95.8	5
Neemuch	95.8	5
Panna	95.8	5
Harda	95.6	5
Anuppur	95.2	5
Seoni	95	5
Shivpuri	94.9	4
Ashok Nagar	94.6	4
Damoh	94.5	4
Rajgarh	94.4	4
Chhindwada	94.2	4
Dhar	94.1	4
Sheopur	93.9	4
Guna	93	4
Satna	92.6	4
Chhatarpur	92.4	4
Bhopal	92.3	4
Katni	92.1	4
Tikamgarh	91.8	4
Umaria	90.5	4
Jabalpur	90.3	4
Ratlam	90.3	4
Ujjain	89	3
Rewa	88	3
Dewas	87.9	3
Balaghat	87.3	3
Shajapur	85.6	3
Mandla	85.5	3
Datia	85.3	3
Betul	83.6	3
Shahdol	81.4	3
Alirajpur	80.9	3
Raisen	80.9	3
Narsinghpur	80.6	3
Khargone	80.2	3
Khandwa	79.9	2
Vidisha	78.5	2
Sagar	78	2
Dindori	76.6	2
Singroli	70.9	1
Barwani	68.2	1
Gwalior	65.8	1
Sehore	65.5	1
Sidhi	63.5	1

CLASS	WEIGHT
<75	1
75-80	2
80-89.9	3
90-94.9	4
>95	5

LEAST	63.5	Sidhi
HIGHEST	99.9	Burhanpur
AVG	87.7	

% Newborns breast fed within 1 hour of birth to Total live birth

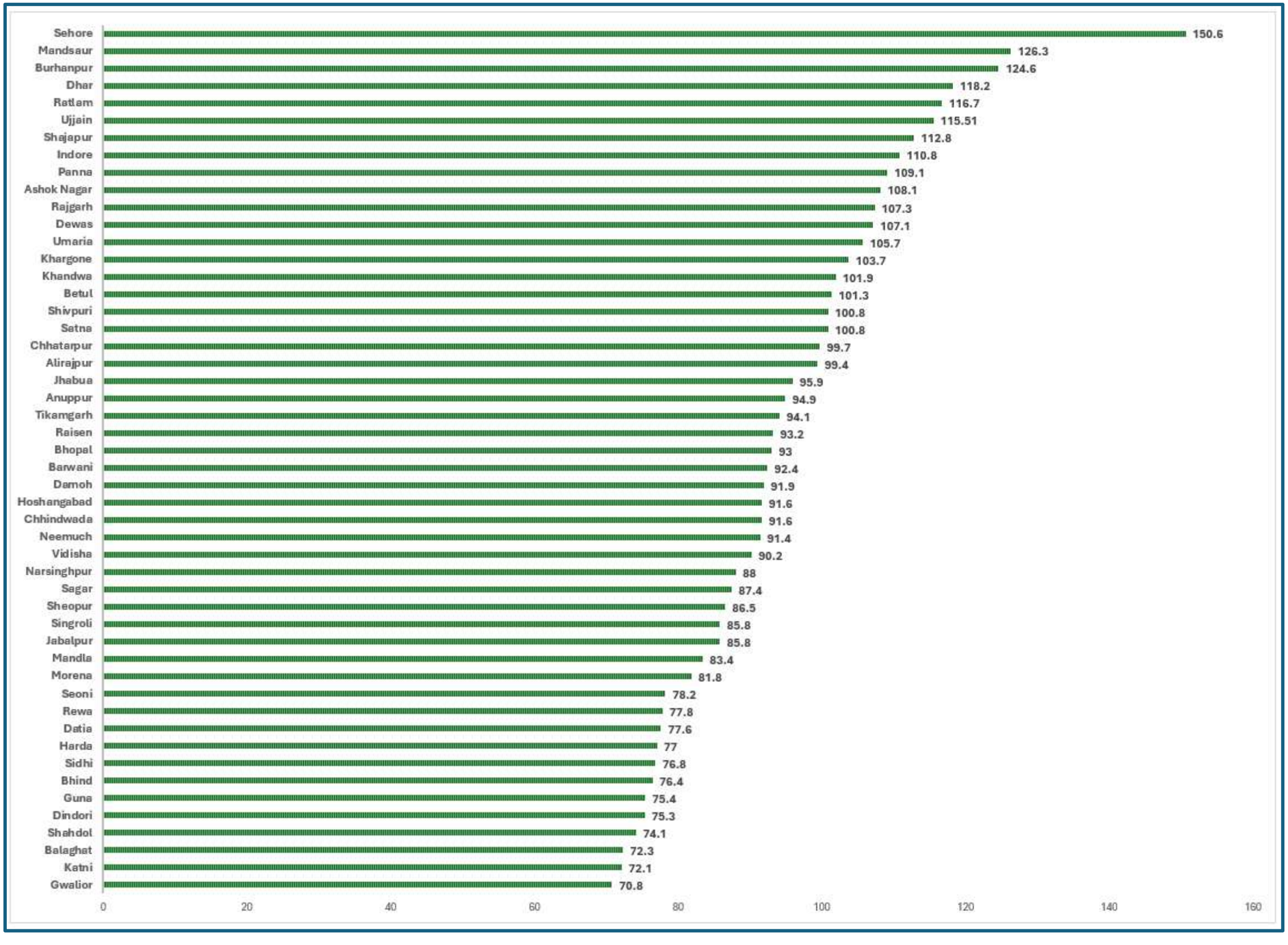


Number of fully immunized children (9-11 months)

	Number of fully immunized children (9-11 months) 2011-12	score(weightage index)
Gwalior	70.8	1
Katni	72.1	1
Balaghat	72.3	1
Shahdol	74.1	1
Dindori	75.3	1
Guna	75.4	1
Bhind	76.4	1
Sidhi	76.8	1
Harda	77	1
Datia	77.6	1
Rewa	77.8	1
Seoni	78.2	1
Morena	81.8	1
Mandla	83.4	1
Jabalpur	85.8	1
Singroli	85.8	1
Sheopur	86.5	1
Sagar	87.4	1
Narsinghpur	88	1
Vidisha	90.2	3
Neemuch	91.4	3
Chhindwada	91.6	3
Hoshangabad	91.6	3
Damoh	91.9	3
Barwani	92.4	3
Bhopal	93	3
Raisen	93.2	3
Tikamgarh	94.1	3
Anuppur	94.9	3
Jhabua	95.9	3
Alirajpur	99.4	3
Chhatarpur	99.7	3
Satna	100.8	5
Shivpuri	100.8	5
Betul	101.3	5
Khandwa	101.9	5
Khargone	103.7	5
Umaria	105.7	5
Dewas	107.1	5
Rajgarh	107.3	5
Ashok Nagar	108.1	5
Panna	109.1	5
Indore	110.8	5
Shajapur	112.8	5
Ujjain	115.51	5
Ratlam	116.7	5
Dhar	118.2	5
Burhanpur	124.6	5
Mandsaur	126.3	5
Sehore	150.6	5

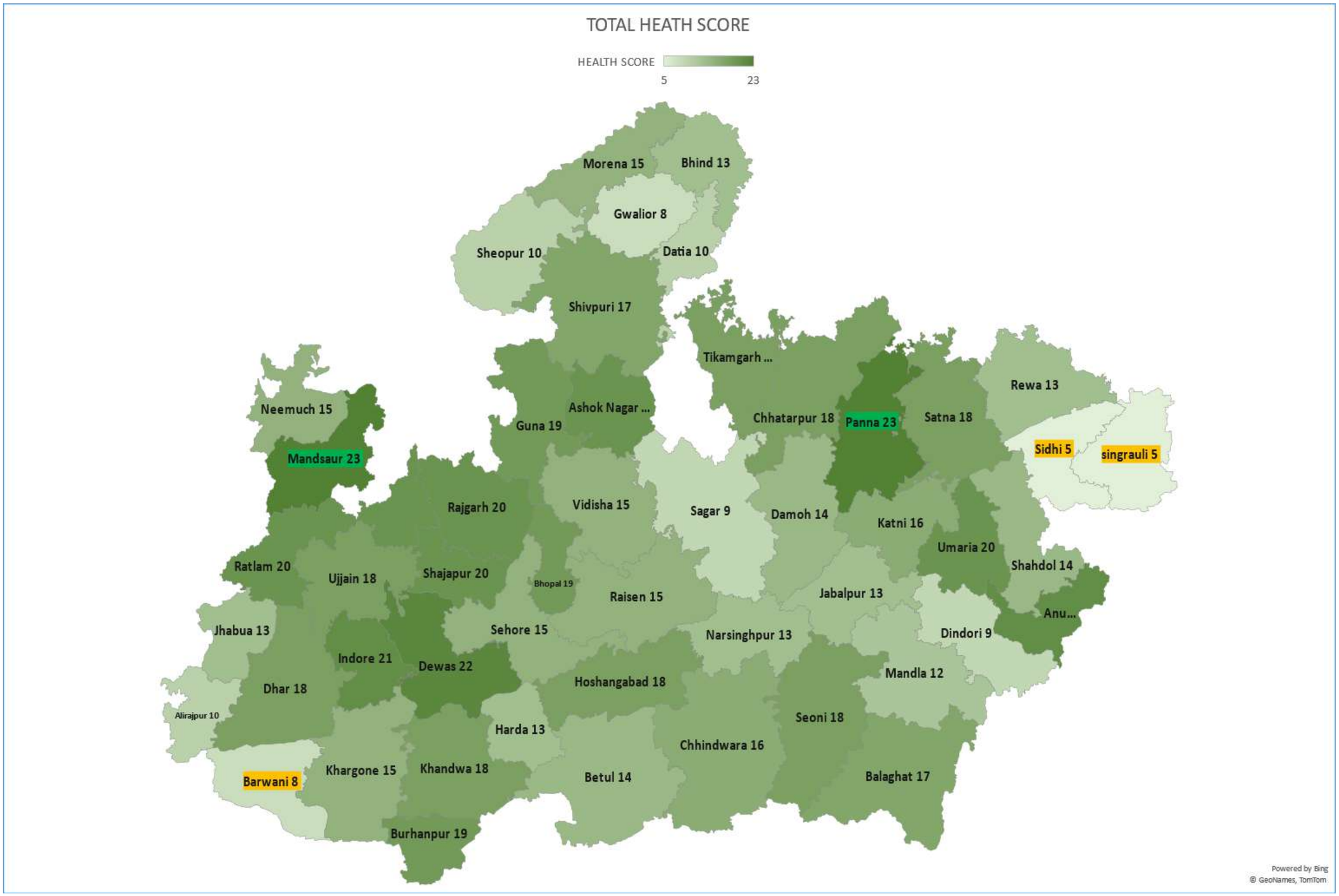
CLASS	WEIGHT	
<90	1	
90-99	3	
>=100	5	
LEAST	70.8	Gwalior
HIGHEST	150.8	Sehore
AVG	94.9	

Reason for having more than 100 percent could be due to accountancy of children from past year who may have turned 9month in the accounted year vs accounted year birth



DISTRICT	HEALTH SCORE
Mandsaur	23
Panna	23
Dewas	22
Anuppur	21
Indore	21
Ashok Nagar	20
Rajgarh	20
Ratlam	20
Shajapur	20
Umaria	20
Bhopal	19
Burhanpur	19
Guna	19
Chhatarpur	18
Dhar	18
Hoshangabad	18
Khandwa	18
Satna	18
Seoni	18
Tikamgarh	18
Ujjain	18
Balaghat	17
Shivpuri	17
Chhindwara	16
Katni	16
Khargone	15
Morena	15
Neemuch	15
Raisen	15
Sehore	15
Vidisha	15
Betul	14
Damoh	14
Shahdol	14
Bhind	13
Harda	13
Jabalpur	13
Jhabua	13
Narsinghpur	13
Rewa	13
Mandla	12
Alirajpur	10
Datia	10
Sheopur	10
Dindori	9
Sagar	9
Barwani	8
Gwalior	8
Sidhi	5
singrauli	5

TOTAL HEATH SCORE

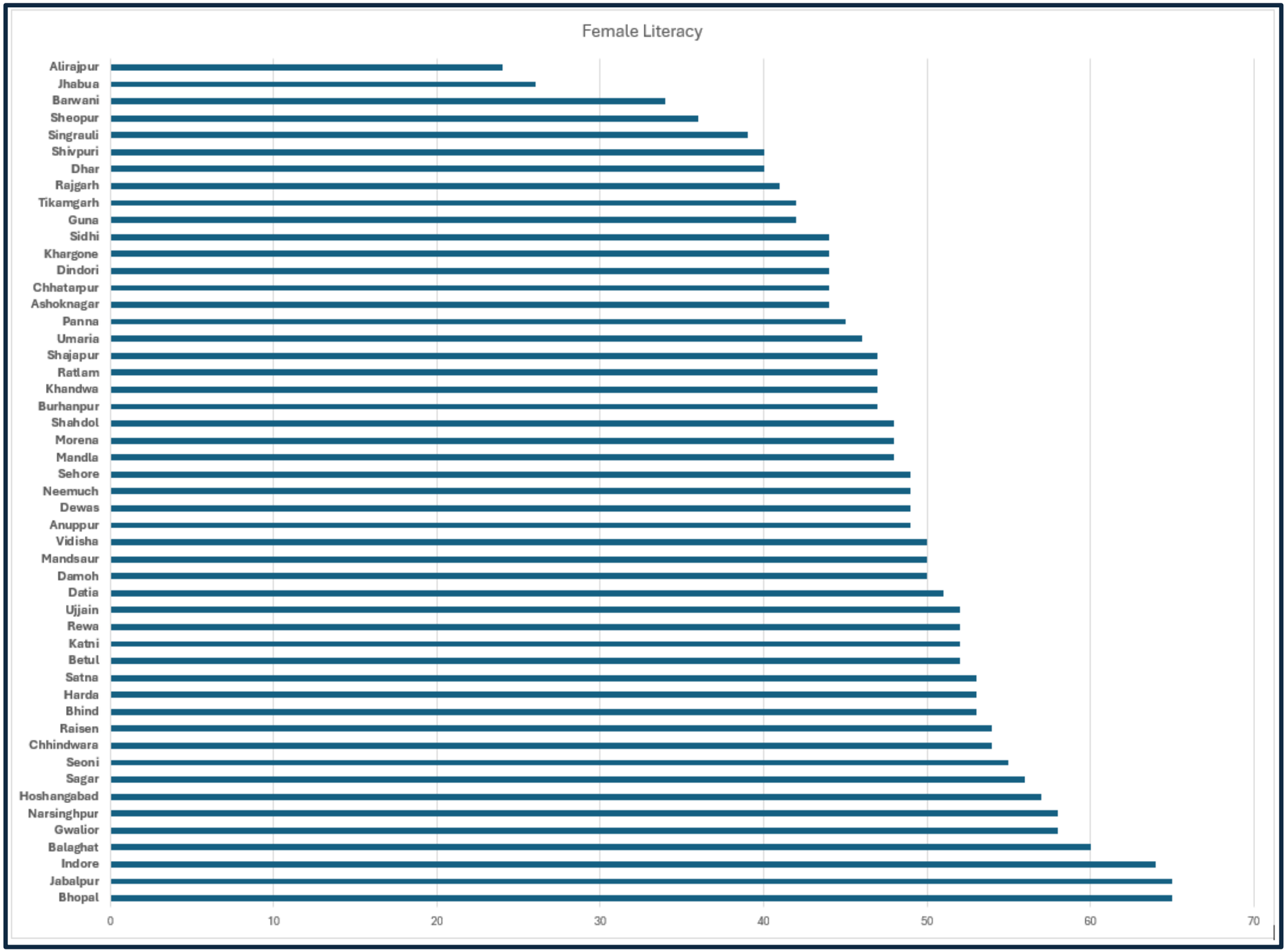


Name of District	Female Literacy(15+ AGE)	Weitage
Alirajpur	24	1
Anuppur	49	2
Ashoknagar	44	1
Balaghat	60	3
Barwani	34	1
Betul	52	2
Bhind	53	2
Bhopal	65	3
Burhanpur	47	2
Chhatarpur	44	1
Chhindwara	54	2
Damoh	50	2
Datia	51	2
Dewas	49	2
Dhar	40	1
Dindori	44	1
Guna	42	1
Gwalior	58	3
Harda	53	2
Hoshangabad	57	3
Indore	64	3
Jabalpur	65	3
Jhabua	26	1
Katni	52	2
Khandwa	47	2
Khargone	44	1
Mandla	48	2
Mandsaur	50	2
Morena	48	2
Narsinghpur	58	3
Neemuch	49	2
Panna	45	2
Raisen	54	3
Rajgarh	41	1
Ratlam	47	2
Rewa	52	2
Sagar	56	3
Satna	53	2
Sehore	49	2
Seoni	55	3
Shahdol	48	2
Shajapur	47	2
Sheopur	36	1
Shivpuri	40	1
Sidhi	44	1
Singrauli	39	1
Tikamgarh	42	1
Ujjain	52	2
Umaria	46	2
Vidisha	50	2

65-55	3
55-45	2
<45	1

EDUCATION

Female Literacy (15+ AGE)



Name of District	Primary to Upper Primary	Weitage
Bhind	111	4
Rewa	104	4
Morena	92	4
Gwalior	90	4
Jabalpur	86	4
Bhopal	84	4
Indore	84	4
Satna	83	4
Sidhi	83	4
Umaria	76	3
Katni	75	3
Hoshangabad	74	3
Raisen	74	3
Tikamgarh	74	3
Anuppur	73	3
Narsinghpur	73	3
Sehore	73	3
Shahdol	73	3
Damoh	72	3
Datia	72	3
Panna	72	3
Dindori	71	3
Sagar	71	3
Dewas	70	3
Betul	69	3
Balaghat	68	3
Mandla	64	3
Shivpuri	64	3
Singrauli	64	3
Chhatarpur	63	3
Chhindwara	61	3
Rajgarh	61	3
Ujjain	61	3
Ratlam	60	2
Harda	59	2
Neemuch	59	2
Vidisha	59	2
Dhar	58	2
Mandsaur	58	2
Seoni	58	2
Sheopur	58	2
Ashoknagar	56	2
Guna	55	2
Khargone	53	2
Burhanpur	51	2
Jhabua	50	2
Barwani	49	2
Khandwa	49	2
Shajapur	49	2
Alirajpur	46	2

Name of District	Upper Primary to Secondry	Weitage
Alirajpur	48	3
Anuppur	46	3
Ashoknagar	35	2
Balaghat	54	3
Barwani	48	3
Betul	58	3
Bhind	49	3
Bhopal	80	4
Burhanpur	65	4
Chhatarpur	41	3
Chhindwara	64	4
Damoh	38	2
Datia	40	3
Dewas	53	3
Dhar	55	3
Dindori	38	2
Guna	46	3
Gwalior	66	4
Harda	55	3
Hoshangabad	62	4
Indore	76	4
Jabalpur	70	4
Jhabua	46	3
Katni	50	3
Khandwa	50	3
Khargone	48	3
Mandla	46	3
Mandsaur	46	3
Morena	54	3
Narsinghpur	55	3
Neemuch	55	3
Panna	41	3
Raisen	40	3
Rajgarh	48	3
Ratlam	49	3
Rewa	56	3
Sagar	45	3
Satna	52	3
Sehore	43	3
Seoni	48	3
Shahdol	45	3
Shajapur	26	2
Sheopur	42	3
Shivpuri	37	2
Sidhi	44	3
Singrauli	48	3
Tikamgarh	36	2
Ujjain	53	3
Umaria	39	2
Vidisha	41	3

PRIMARY TO UPPER PRIMARY &

UPPER PRIMARY TO SECONDRY

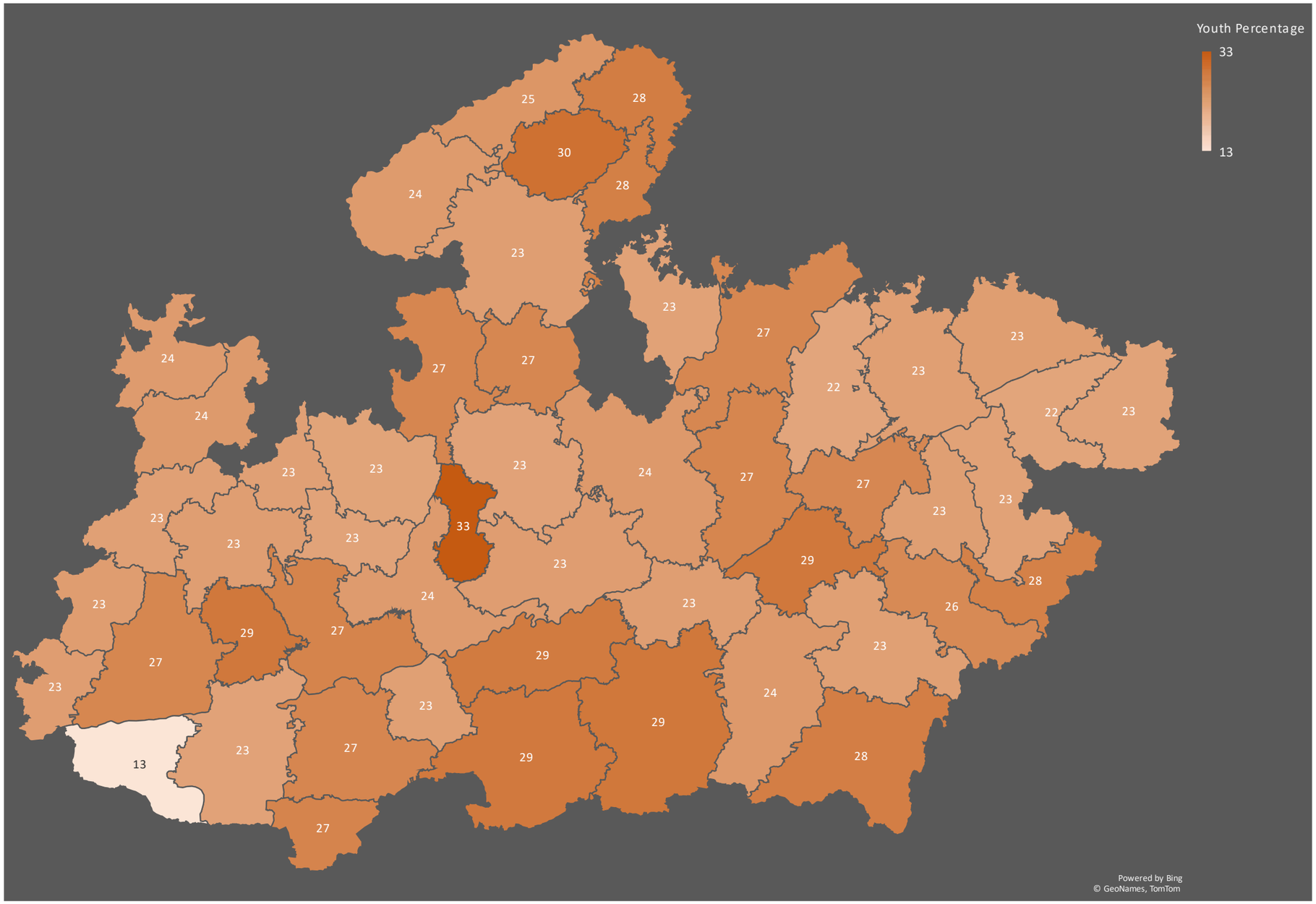
40-60	2
60-80	3
80+	4

20-40	2
40-60	3
60-80	4

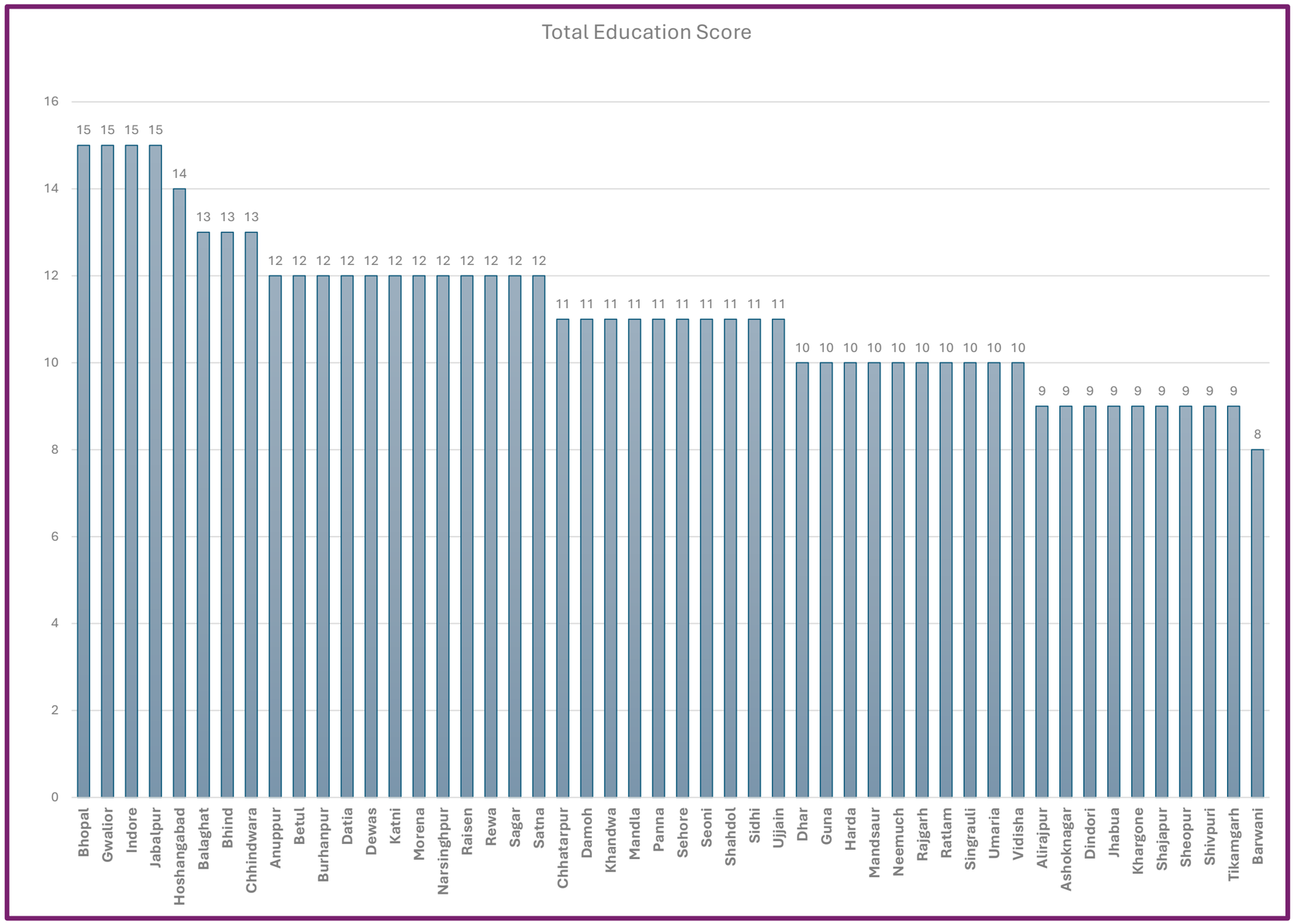
120
100
80
60
40
20
0
Bhind
Rewa
Morena
Gwalior
Jabalpur
Bhopal
Indore
Satna
Sidhi
Umaria
Katni
Hoshangabad
Raisen
Tikamgarh
Anuppur
Narsinghpur
Sehore
Shahdol
Damoh
Datia
Panna
Dindori
Sagar
Dewas
Betul
Balaghat
Mandla
Shivpuri
Singrauli
Chhatarpur
Chhindwara
Rajgarh
Ujjain
Ratlam
Harda
Neemuch
Vidisha
Dhar
Mandsaur
Seoni
Sheopur
Ashoknagar
Guna
Khargone
Burhanpur
Jhabua
Barwani
Khandwa
Shajapur
Alirajpur
Primary to Upper Primary
Name of District
Upper Primary to Secondry

Name of District	Youth Percentage	Weitage
Alirajpur	23	3
Anuppur	28	4
Ashoknagar	27	4
Balaghat	28	4
Barwani	13	2
Betul	29	4
Bhind	28	4
Bhopal	33	4
Burhanpur	27	4
Chhatarpur	27	4
Chhindwara	29	4
Damoh	27	4
Datia	28	4
Dewas	27	4
Dhar	27	4
Dindori	26	3
Guna	27	4
Gwalior	30	4
Harda	23	3
Hoshangabad	29	4
Indore	29	4
Jabalpur	29	4
Jhabua	23	3
Katni	27	4
Khandwa	27	4
Khargone	23	3
Mandla	23	3
Mandsaur	24	3
Morena	25	3
Narsinghpur	23	3
Neemuch	24	3
Panna	22	3
Raisen	23	3
Rajgarh	23	3
Ratlam	23	3
Rewa	23	3
Sagar	24	3
Satna	23	3
Sehore	24	3
Seoni	24	3
Shahdol	23	3
Shajapur	23	3
Sheopur	24	3
Shivpuri	23	3
Sidhi	22	3
Singrauli	23	3
Tikamgarh	23	3
Ujjain	23	3
Umaria	23	3
Vidisha	23	3

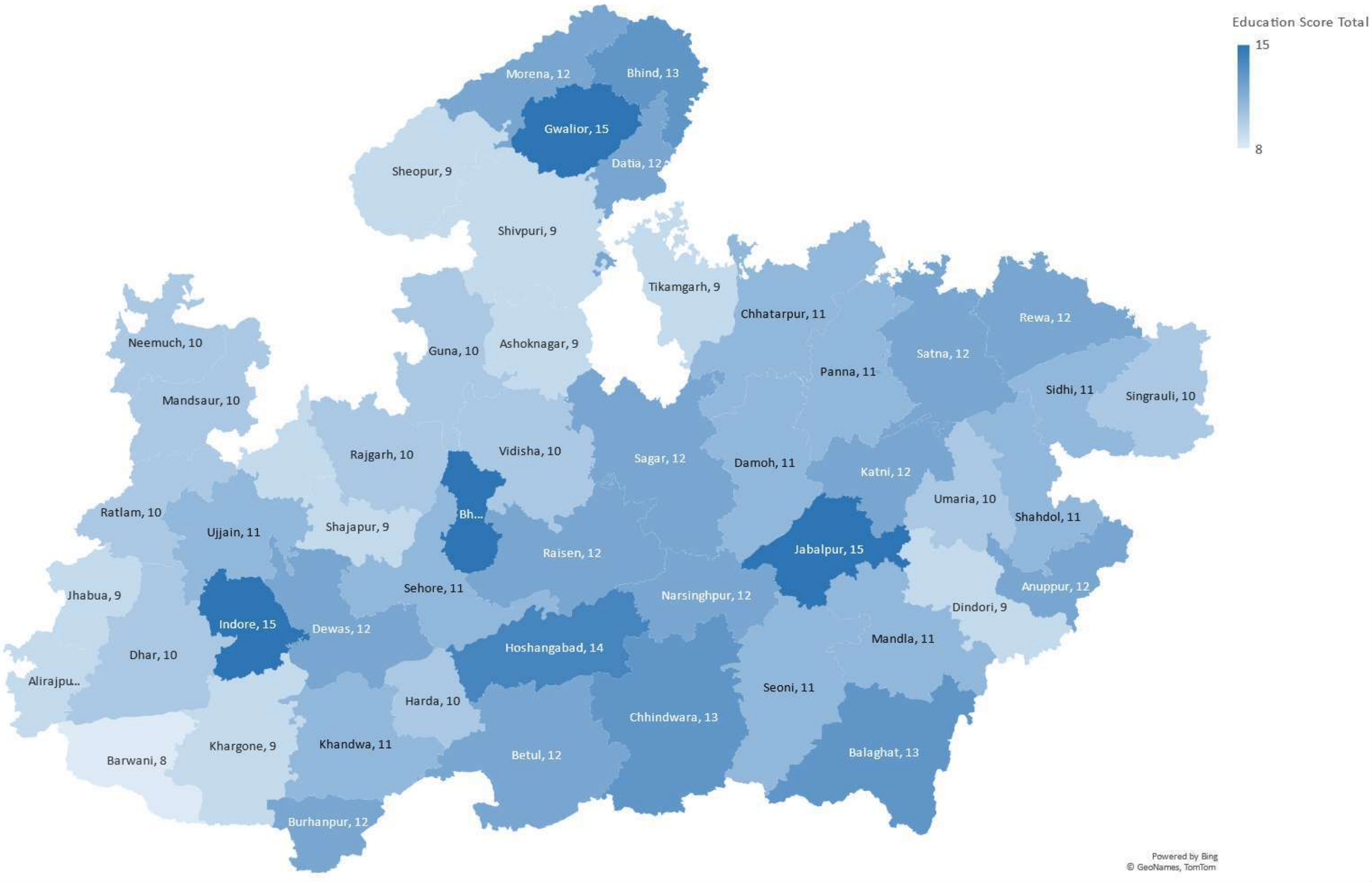
YOUTH PERCENTAGE
(15-29)



Name of District	Education Score Total
Bhopal	15
Gwalior	15
Indore	15
Jabalpur	15
Hoshangabad	14
Balaghat	13
Bhind	13
Chhindwara	13
Anuppur	12
Betul	12
Burhanpur	12
Datia	12
Dewas	12
Katni	12
Morena	12
Narsinghpur	12
Raisen	12
Rewa	12
Sagar	12
Satna	12
Chhatarpur	11
Damoh	11
Khandwa	11
Mandla	11
Panna	11
Sehore	11
Seoni	11
Shahdol	11
Sidhi	11
Ujjain	11
Dhar	10
Guna	10
Harda	10
Mandsaur	10
Neemuch	10
Rajgarh	10
Ratlam	10
Singrauli	10
Umaria	10
Vidisha	10
Alirajpur	9
Ashoknagar	9
Dindori	9
Jhabua	9
Khargone	9
Shajapur	9
Sheopur	9
Shivpuri	9
Tikamgarh	9
Barwani	8



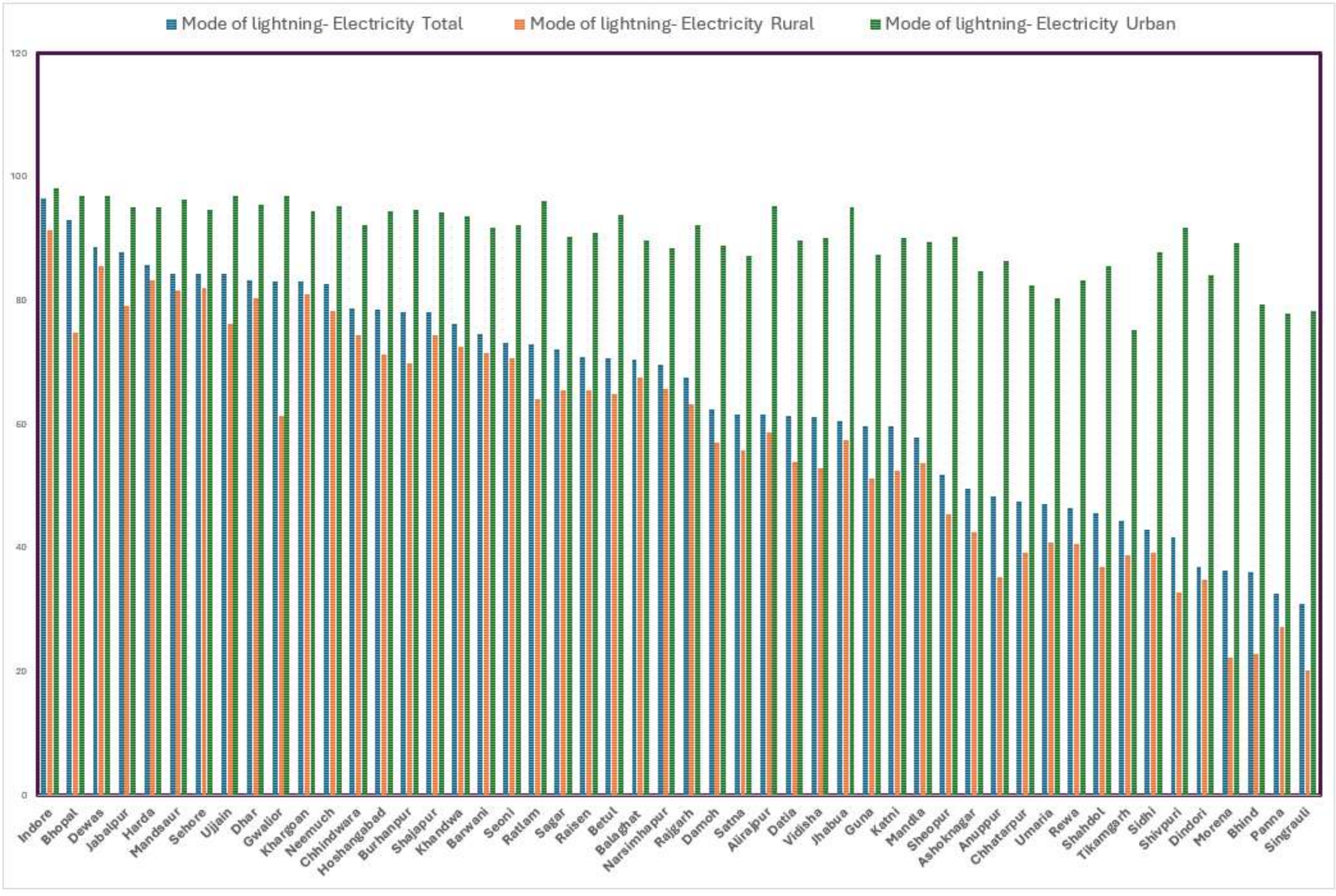
total educational score



DISTRICTS	Mode of lightning-Electricity			
	Total	Rural	Urban	ELECTRICITY SCORE
Alirajpur	61.5	58.6	95.2	3
Anuppur	48.2	35.3	86.3	1
Ashoknagar	49.6	42.5	84.7	1
Balaghat	70.5	67.5	89.6	4
Barwani	74.5	71.5	91.8	4
Betul	70.6	64.8	93.7	4
Bhind	36.1	22.7	79.4	1
Bhopal	92.9	74.7	97	5
Burhanpur	78	69.8	94.6	4
Chhatarpur	47.5	39.1	82.5	1
Chhindwara	78.7	74.3	92.1	4
Damoh	62.3	57	88.9	3
Datia	61.2	53.8	89.6	3
Dewas	88.7	85.6	97	5
Dhar	83.3	80.4	95.4	5
Dindori	36.9	34.8	84.1	1
Guna	59.7	51.1	87.3	2
Gwalior	83	61.4	96.9	5
Harda	85.7	83.3	95	5
Hoshangabad	78.4	71.2	94.5	4
Indore	96.5	91.4	98.1	5
Jabalpur	87.8	79.2	95	5
Jhabua	60.4	57.3	95.1	3
Katni	59.7	52.5	90	2
Khandwa	76.3	72.4	93.5	4
Khargoan	83	80.9	94.5	5
Mandla	57.7	53.6	89.5	2
Mandsaur	84.3	81.5	96.2	5
Morena	36.3	22.2	89.2	1
Narsimhapur	69.5	65.6	88.5	3
Neemuch	82.7	78.3	95.2	5
Panna	32.6	27.2	77.8	1
Raisen	70.9	65.4	90.8	4
Rajgarh	67.6	63.2	92.1	3
Ratlam	72.9	64	96	4
Rewa	46.5	40.7	83.3	1
Sagar	72	65.5	90.3	4
Satna	61.6	55.7	87.2	3
Sehore	84.3	82.1	94.7	5
Seoni	73	70.6	92.1	4
Shahdol	45.5	36.8	85.6	1
Shajapur	78	74.4	94.3	4
Sheopur	51.7	45.4	90.3	2
Shivpuri	41.6	32.7	91.7	1
Sidhi	42.9	39.2	87.8	1
Singrauli	30.8	20.2	78.2	1
Tikamgarh	44.4	38.8	75.1	1
Ujjain	84.2	76.3	96.8	5
Umaria	47.1	40.8	80.3	1
Vidisha	61	52.8	90.1	3

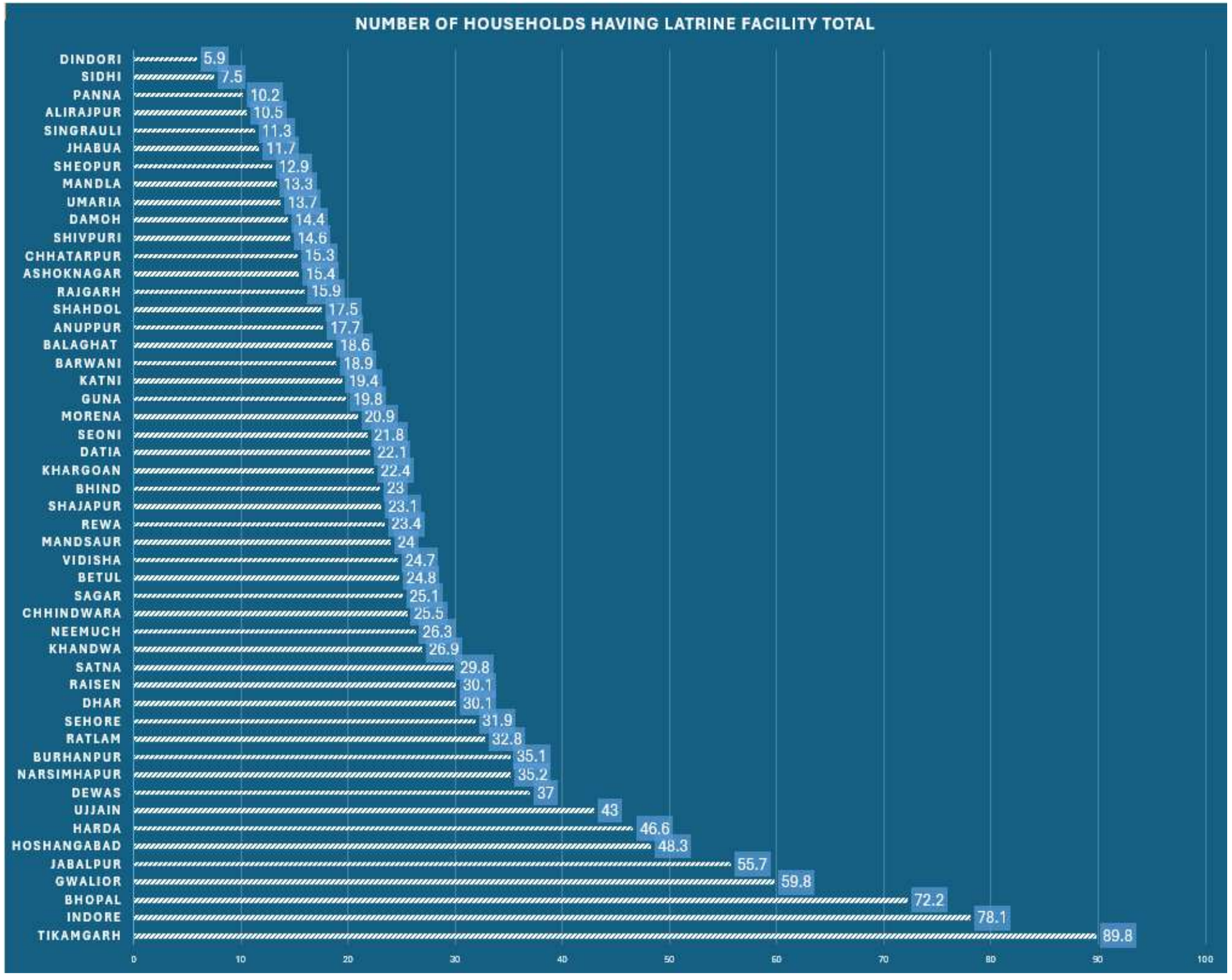
BASIC INFRASTRUCTURE

Mode of lighting- Electricity



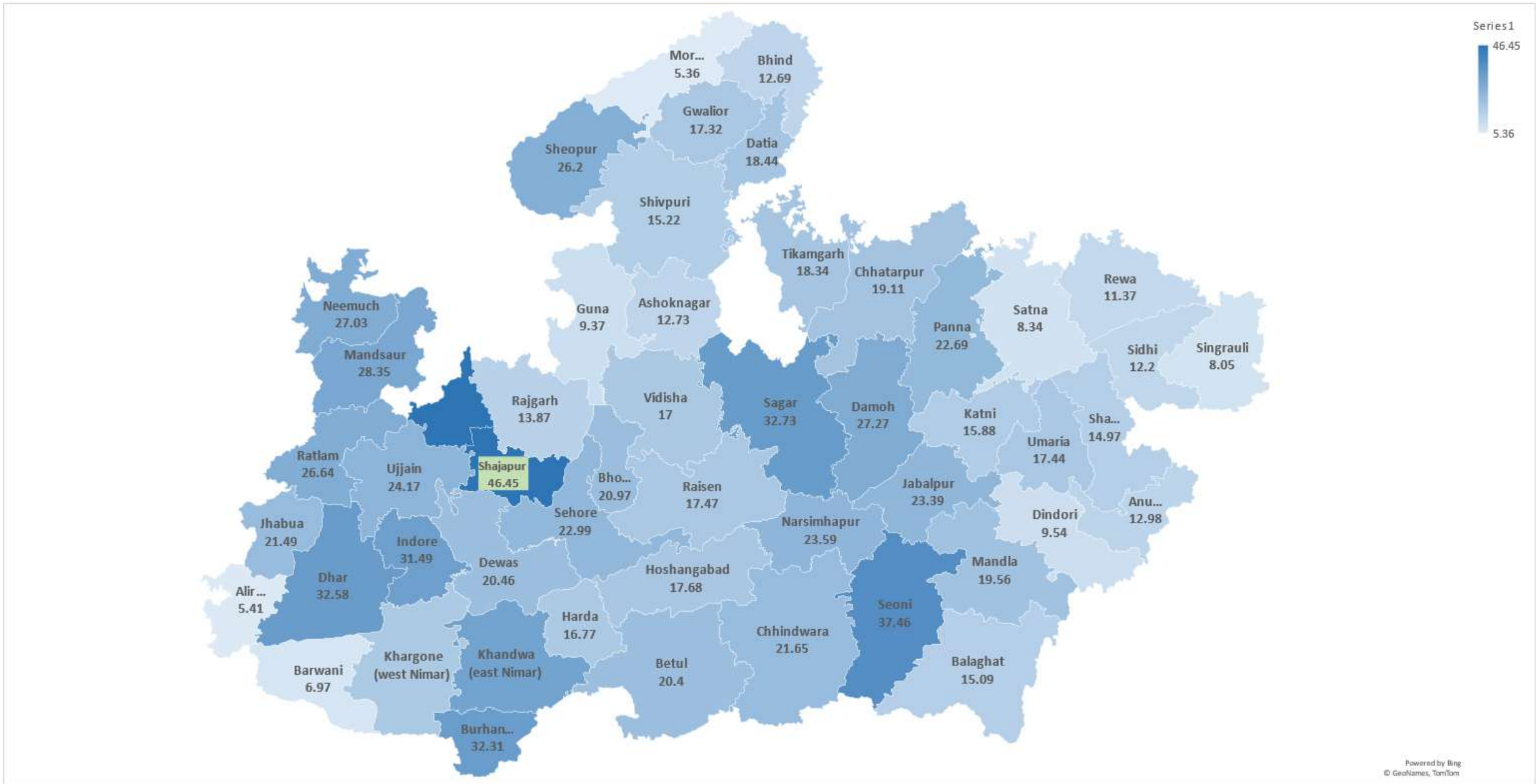
Number of households having latrine facility		
	Total	LATERINES SCORE
Tikamgarh	89.8	5
Indore	78.1	5
Bhopal	72.2	4
Gwalior	59.8	4
Jabalpur	55.7	4
Hoshangabad	48.3	3
Harda	46.6	3
Ujjain	43	3
Dewas	37	3
Narsimhapur	35.2	3
Burhanpur	35.1	3
Ratlam	32.8	3
Sehore	31.9	3
Dhar	30.1	3
Raisen	30.1	3
Satna	29.8	3
Khandwa	26.9	3
Neemuch	26.3	3
Chhindwara	25.5	3
Sagar	25.1	3
Betul	24.8	1
Vidisha	24.7	1
Mandsaur	24	1
Rewa	23.4	1
Shajapur	23.1	1
Bhind	23	1
Khargoan	22.4	1
Datia	22.1	1
Seoni	21.8	1
Morena	20.9	1
Guna	19.8	1
Katni	19.4	1
Barwani	18.9	1
Balaghat	18.6	1
Anuppur	17.7	1
Shahdol	17.5	1
Rajgarh	15.9	1
Ashoknagar	15.4	1
Chhatarpur	15.3	1
Shivpuri	14.6	1
Damoh	14.4	1
Umaria	13.7	1
Mandla	13.3	1
Sheopur	12.9	1
Jhabua	11.7	1
Singrauli	11.3	1
Alirajpur	10.5	1
Panna	10.2	1
Sidhi	7.5	1
Dindori	5.9	1

Number of households having latrine facility

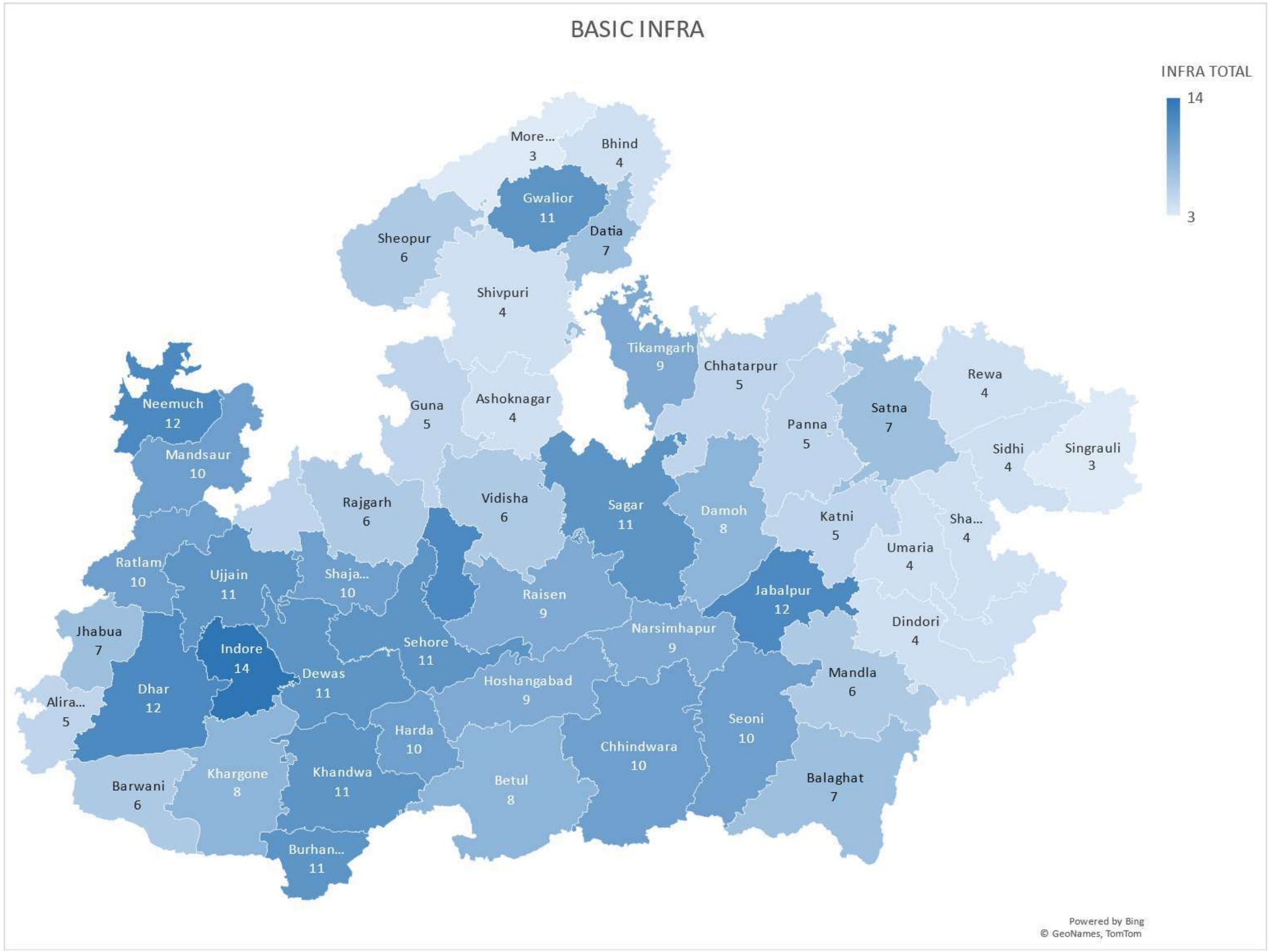


DISTRICTS- 2018	Percentage of Rural habitations with access to adequate quantity of potable water - 40 lpcd drinking water in rural	SCORE
Shajapur	46.45	5
Seoni	37.46	5
Sagar	32.73	4
Dhar	32.58	4
Burhanpur	32.31	4
Indore	31.49	4
Khandwa	30.2	4
Mandsaur	28.35	4
Damoh	27.27	4
Neemuch	27.03	4
Ratlam	26.64	3
Sheopur	26.2	3
Ujjain	24.17	3
Narsimhapur	23.59	3
Jabalpur	23.39	3
Sehore	22.99	3
Panna	22.69	3
Chhindwara	21.65	3
Jhabua	21.49	3
Bhopal	20.97	3
Dewas	20.46	3
Betul	20.4	3
Mandla	19.56	3
Chhatarpur	19.11	3
Datia	18.44	3
Tikamgarh	18.34	3
Hoshangabad	17.68	2
Raisen	17.47	2
Umaria	17.44	2
Gwalior	17.32	2
Vidisha	17	2
Harda	16.77	2
Khargone	16.62	2
Katni	15.88	2
Shivpuri	15.22	2
Balaghat	15.09	2
Shahdol	14.97	2
Rajgarh	13.87	2
Anuppur	12.98	2
Ashoknagar	12.73	2
Bhind	12.69	2
Sidhi	12.2	2
Rewa	11.37	2
Dindori	9.54	2
Guna	9.37	2
Satna	8.34	1
Singrauli	8.05	1
Barwani	6.97	1
Alirajpur	5.41	1
Morena	5.36	1

Percentage of Rural habitations with access to adequate quantity of potable water - 40 lpcd drinking water in rural


Series1
46.45
5.36
Mor...
5.36
Bhind
12.69
Gwalior
17.32
Datia
18.44
Sheopur
26.2
Shivpuri
15.22
Tikamgarh
18.34
Chhatarpur
19.11
Rewa
11.37
Satna
8.34
Panna
22.69
Sidhi
12.2
Singrauli
8.05
Guna
9.37
Ashoknagar
12.73
Neemuch
27.03
Mandsaur
28.35
Rajgarh
13.87
Vidisha
17
Sagar
32.73
Damoh
27.27
Katni
15.88
Sha...
14.97
Umaria
17.44
Ratlam
26.64
Ujjain
24.17
Shajapur
46.45
Bho...
20.97
Raisen
17.47
Jabalpur
23.39
Anu...
12.98
Dindori
9.54
Sehore
22.99
Narsimhapur
23.59
Jhabua
21.49
Indore
31.49
Dewas
20.46
Hoshangabad
17.68
Mandla
19.56
Dhar
32.58
Alir...
5.41
Seoni
37.46
Harda
16.77
Chhindwara
21.65
Khargone
(west Nimar)
Khandwa
(east Nimar)
Betul
20.4
Balaghat
15.09
Barwani
6.97
Burhan...
32.31
Powered by Bing
© GeoNames, TomTom

	INFRA TOTAL
Indore	14
Bhopal	12
Dhar	12
Jabalpur	12
Neemuch	12
Burhanpur	11
Dewas	11
Gwalior	11
Khandwa	11
Sagar	11
Sehore	11
Ujjain	11
Chhindwara	10
Harda	10
Mandsaur	10
Ratlam	10
Seoni	10
Shajapur	10
Hoshangabad	9
Narsimhapur	9
Raisen	9
Tikamgarh	9
Betul	8
Damoh	8
Khargone	8
Balaghat	7
Datia	7
Jhabua	7
Satna	7
Barwani	6
Mandla	6
Rajgarh	6
Sheopur	6
Vidisha	6
Alirajpur	5
Chhatarpur	5
Guna	5
Katni	5
Panna	5
Anuppur	4
Ashoknagar	4
Bhind	4
Dindori	4
Rewa	4
Shahdol	4
Shivpuri	4
Sidhi	4
Umaria	4
Morena	3
Singrauli	3



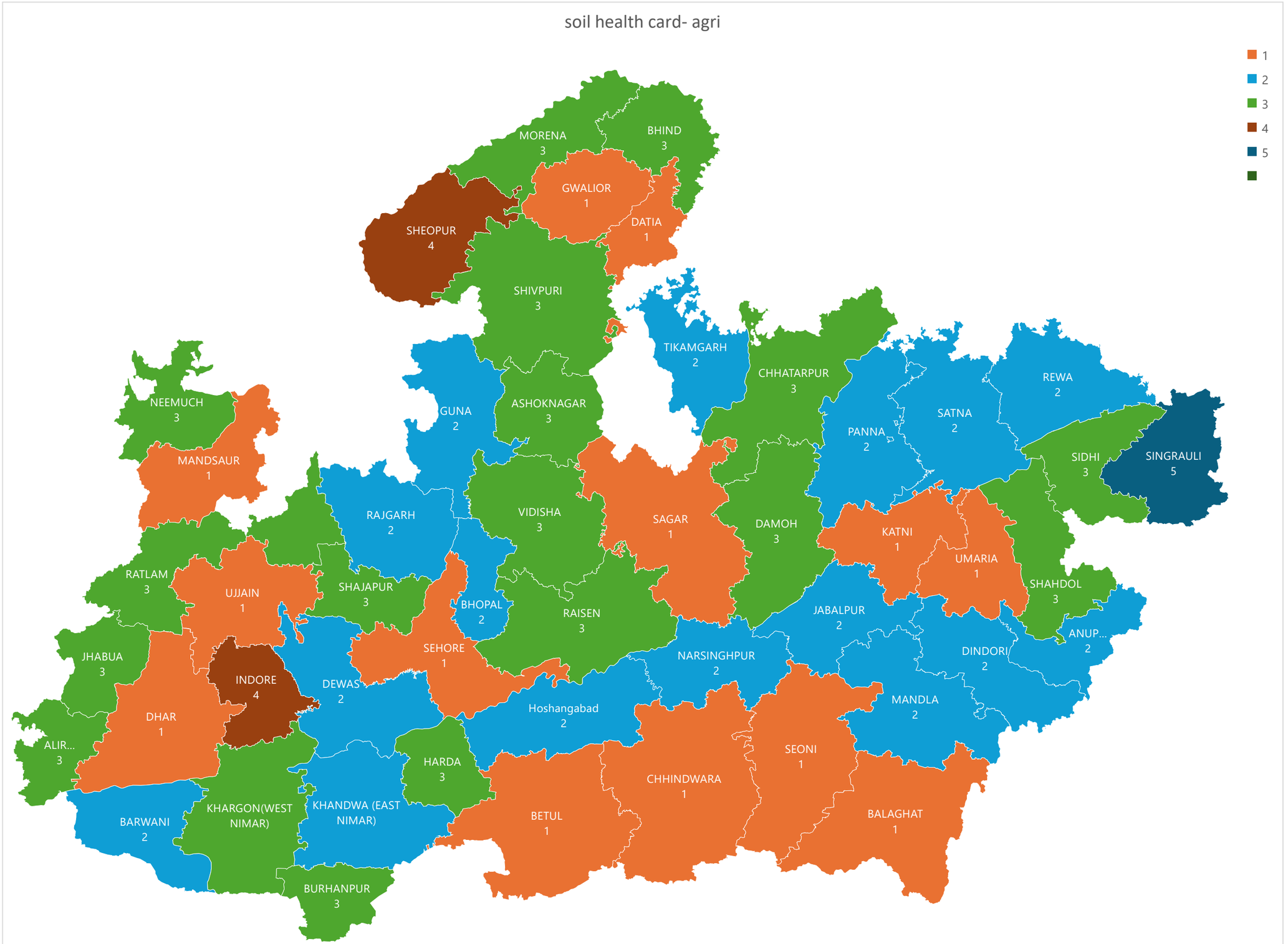
District	Total Test Alloted	No. of Samples Collected	Test Completed & SHC Generated	percent :SHCgranted/sample collected	score
DATIA	7033	6628	356	5.371152686	1
BETUL	25735	23772	1655	6.961972068	1
BALAGHAT	25735	25054	2026	8.086533089	1
CHHINDWARA	28291	29627	2478	8.363992304	1
KATNI	15336	15327	1450	9.460429308	1
MANDSAUR	12780	12779	1260	9.859926442	1
DHAR	33227	31744	3246	10.22555444	1
SAGAR	28116	23571	2478	10.51291842	1
UJJAIN	15381	15364	1719	11.18849258	1
GWALIOR	9077	7378	902	12.22553538	1
SEONI	20622	17762	2297	12.93210224	1
SEHORE	12780	12663	1764	13.93034826	1
UMARIA	7667	7667	1149	14.98630494	1
RAJGARH	15336	14613	2475	16.93697393	2
DEWAS	15336	15273	2693	17.63242323	2
GUNA	12780	12275	2206	17.97148676	2
NARSINGHPUR	15336	15533	2901	18.67636645	2
DINDORI	18068	18048	3427	18.98825355	2
SATNA	20447	18400	3500	19.02173913	2
PANNA	12780	10096	2096	20.76069731	2
Hoshangabad	18068	18685	4007	21.44500937	2
REWA	23004	15398	3326	21.60020782	2
KHANDWA	17893	17491	3790	21.66828655	2
TIKAMGARH	10224	10225	2272	22.2200489	2
MANDLA	23179	23179	5222	22.52901333	2
ANUPPUR	10224	10224	2392	23.39593114	2
BARWANI	18068	18070	4263	23.59158827	2
JABALPUR	17893	16823	3981	23.66403139	2
BHOPAL	5113	4851	1154	23.7889095	2
MORENA	17885	17547	4564	26.01014418	3
SHAHDOL	12780	12780	3377	26.42410016	3
ASHOKNAGAR	10224	10216	2705	26.47807361	3
BURHANPUR	5113	4723	1282	27.14376456	3
DAMOH	17893	17888	5020	28.06350626	3
SHAJAPURinc agar	20448	20433	5838	28.57142857	3
CHHATARPUR	20447	19040	5762	30.26260504	3
BHIND	15336	13998	4287	30.62580369	3
RAISEN	17893	18199	5922	32.54024946	3
JHABUA	15511	15453	5360	34.68582152	3
SIDHI	12780	11150	3942	35.35426009	3
KHARGONE	23004	23877	8576	35.91741006	3
NEEMUCH	7667	7652	2831	36.99686357	3
RATLAM	15511	15507	6028	38.87276714	3
VIDISHA	17893	17823	6992	39.23020816	3
ALIRAJPUR	15336	15335	6037	39.36746006	3
SHIVPURI	20447	19632	8444	43.01140994	3
HARDA	7667	7669	3471	45.26013822	3

AGRICULTURE- GRANTED SOIL HEALTH CARD

CLASS	WEIGHT
<15	1
15-25	2
25-49.9	3
50-75	4
>75	5

LEAST	5.3	DATIA
HIGHEST	88.3	SINGRAULI
AVG	25.3657	

SHEOPUR	7632	7570	3934	51.9682959	4
INDORE	10224	9216	5085	55.17578125	4
SINGRAULI	7667	7668	6773	88.32811685	5

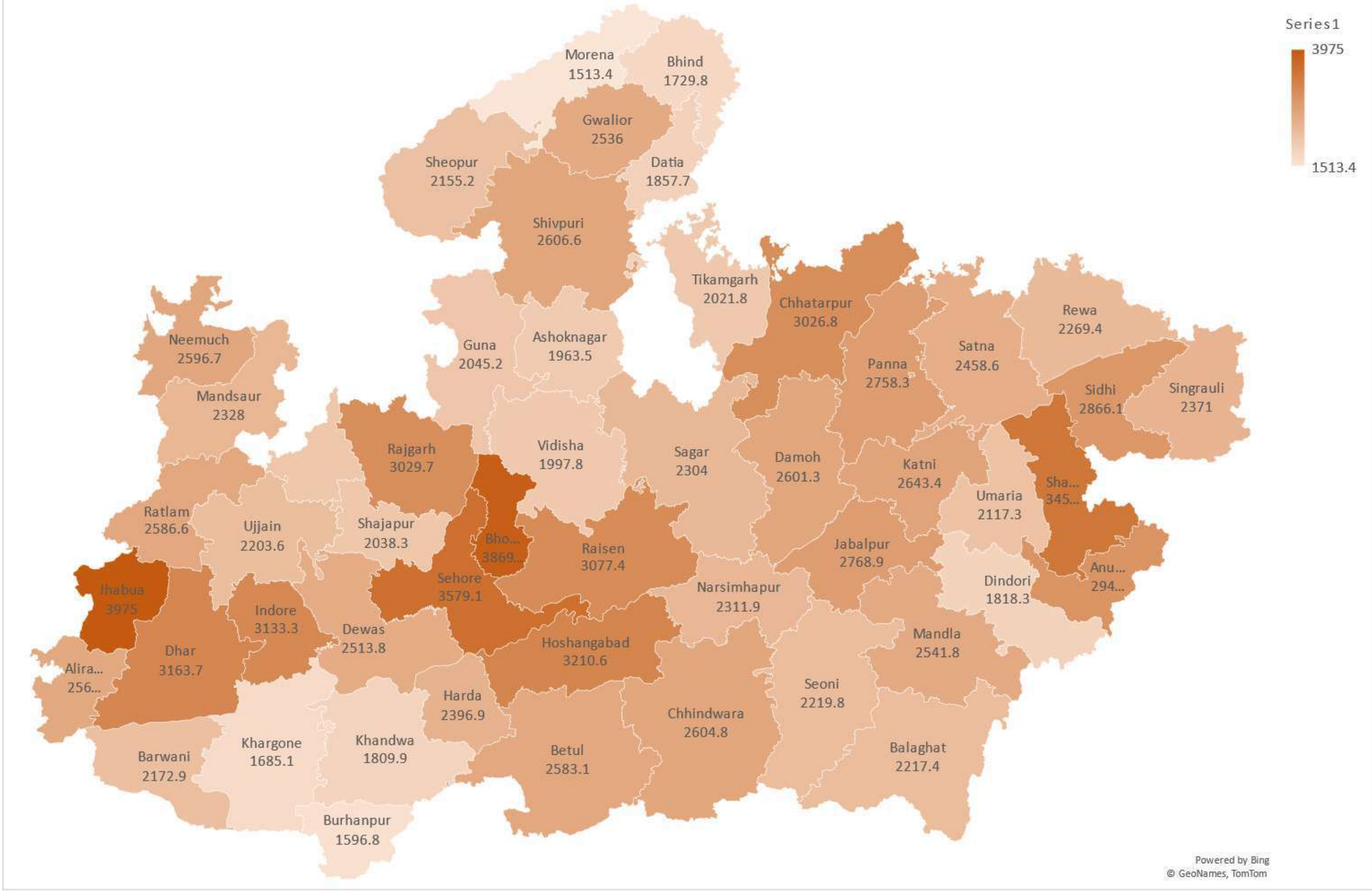


DISTRICTS	Atal Pension Yojana (APY): Number of Beneficiaries per 1 Lakh pop.	SCORE
Morena	1513.4	1
Burhanpur	1596.8	1
Khargone	1685.1	1
Bhind	1729.8	1
Khandwa	1809.9	1
Dindori	1818.3	1
Datia	1857.7	1
Ashoknagar	1963.5	1
Vidhisha	1997.8	1
Tikamgarh	2021.8	2
Shajapur	2038.3	2
Guna	2045.2	2
Umaria	2117.3	2
Sheopur	2155.2	2
Barwani	2172.9	2
Ujjain	2203.6	2
Balaghat	2217.4	2
Seoni	2219.8	2
Rewa	2269.4	2
Sagar	2304	2
Narsimhapur	2311.9	2
Mandsaur	2328	2
Singrauli	2371	2
Harda	2396.9	2
Satna	2458.6	2
Dewas	2513.8	3
Gwalior	2536	3
Mandla	2541.8	3
Alirajpur	2567.3	3
Betul	2583.1	3
Ratlam	2586.6	3
Neemuch	2596.7	3
Damoh	2601.3	3
Chindwara	2604.8	3
Shivpuri	2606.6	3
Katni	2643.4	3
Panna	2758.3	3
Jabalpur	2768.9	3
Sidhi	2866.1	3
Anuppur	2940.5	3
Chhatarpur	3026.8	4
Rajgarh	3029.7	4
Raisen	3077.4	4
Indore	3133.3	4
Dhar	3163.7	4
Hoshangabad	3210.6	4
Shahdol	3452.6	4
Sehore	3579.1	5
Bhopal	3869.4	5
Jhabua	3975	5

FINANCE- Atal Pension Yojana (APY): Number of Beneficiaries per 1 Lakh population

	CLASS	WEIGHT
	<2000	1
	2000-2500	2
	2500-3000	3
	3000-3500	4
	>3500	5
LEAST	1513.4	Morena
HIGHEST	3975	Jhabua
AVG	2496.7	

APY benfi- FINANCE

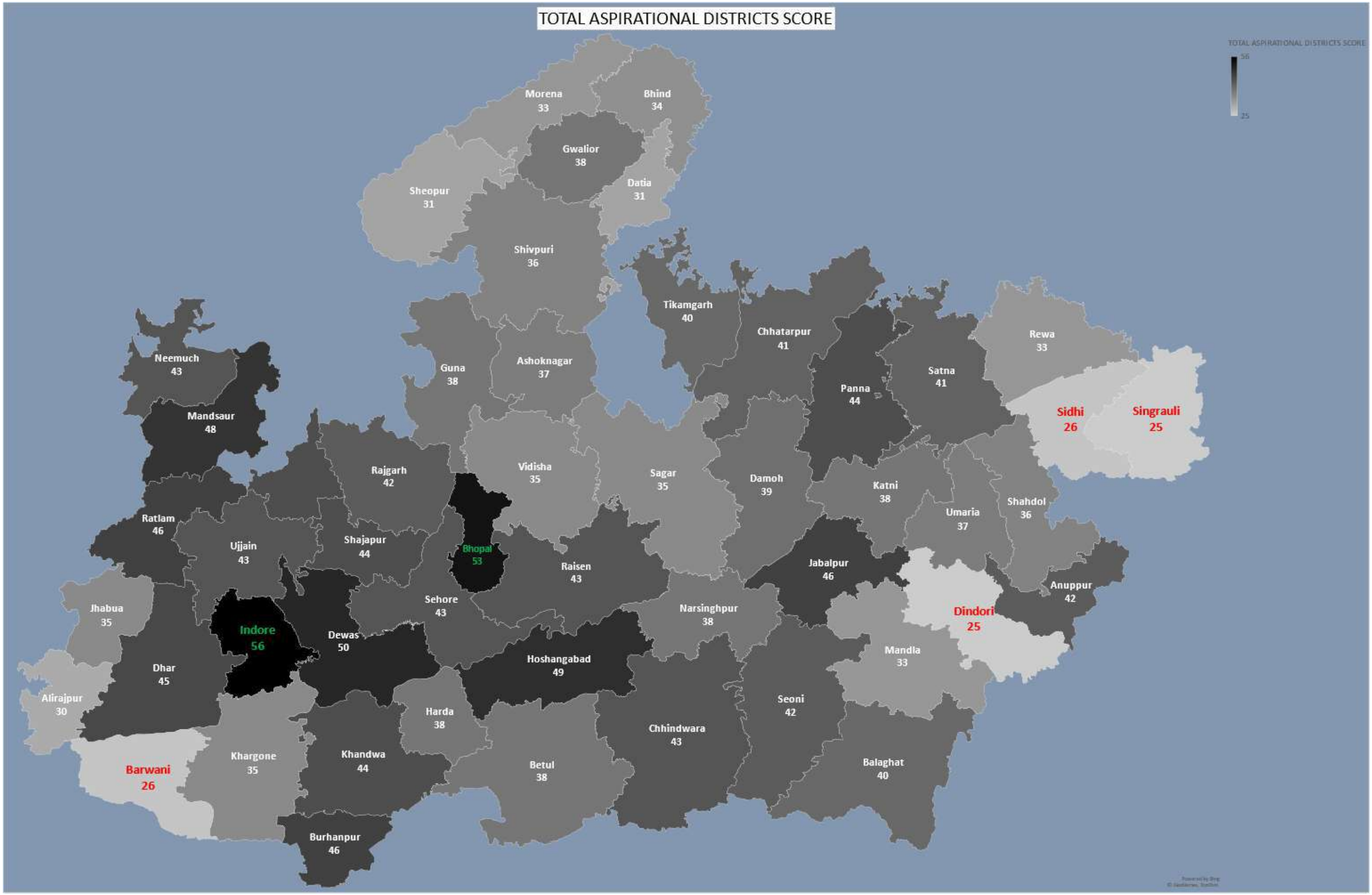


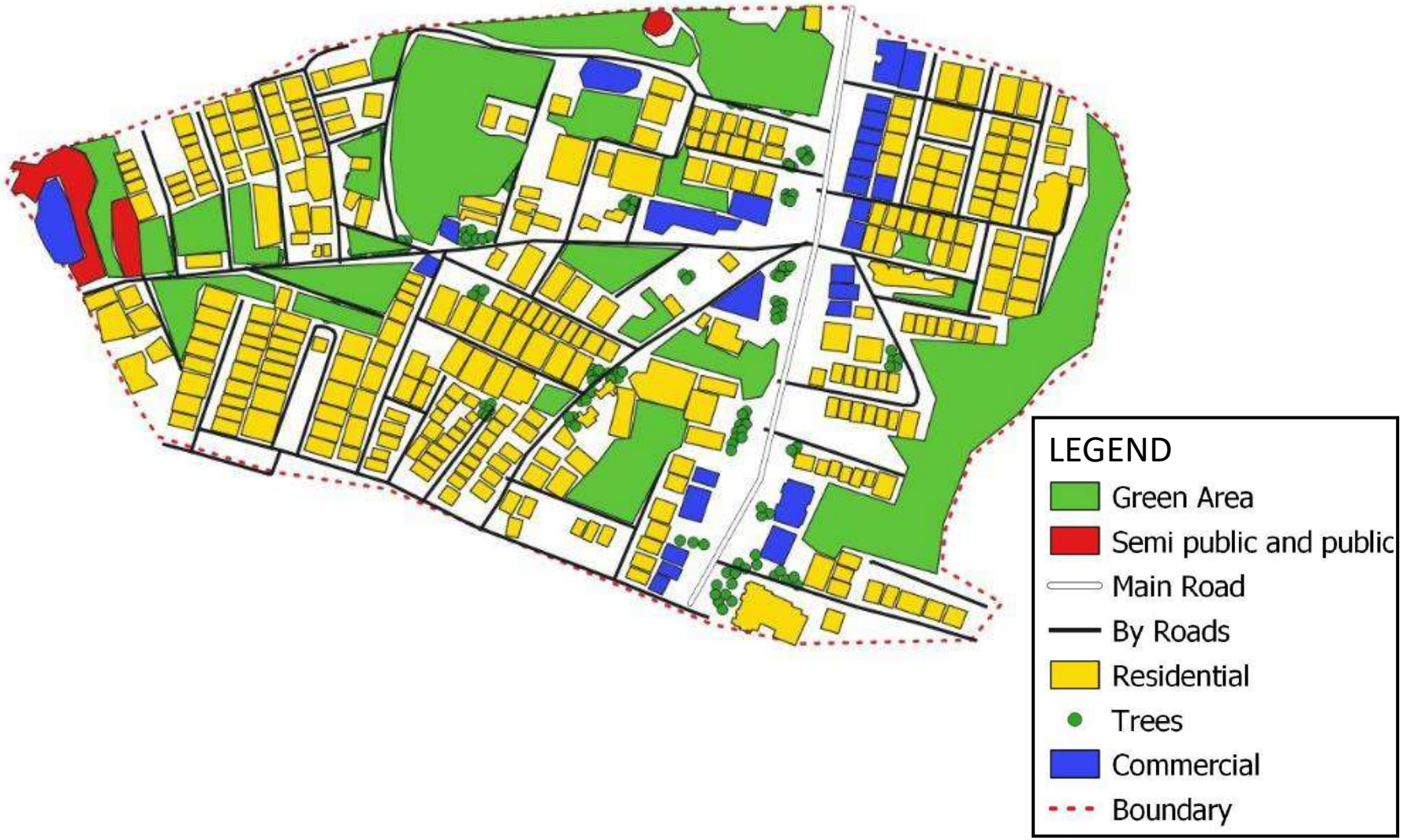
ASPIRATIONAL DISTRICTS RANKING

DISTRICTS	TOTAL ASPIRATIONAL DISTRICTS SCORE	RANK
Indore	56	1
Bhopal	53	2
Dewas	50	3
Hoshangabad	49	4
Mandsaur	48	5
Burhanpur	46	6
Jabalpur	46	6
Ratlam	46	6
Dhar	45	7
Khandwa	44	8
Panna	44	8
Shajapur	44	8
Chhindwara	43	9
Neemuch	43	9
Raisen	43	9
Sehore	43	9
Ujjain	43	9
Anuppur	42	10
Rajgarh	42	10
Seoni	42	10
Chhatarpur	41	11
Satna	41	11
Balaghat	40	12
Tikamgarh	40	12
Damoh	39	13
Betul	38	14
Guna	38	14
Gwalior	38	14
Harda	38	14
Katni	38	14
Narsinghpur	38	14
Ashoknagar	37	15
Umaria	37	15
Shahdol	36	16
Shivpuri	36	16
Jhabua	35	17
Khargone	35	17
Sagar	35	17
Vidisha	35	17
Bhind	34	18
Mandla	33	19
Morena	33	19
Rewa	33	19
Datia	31	20
Sheopur	31	20
Alirajpur	30	21
Barwani	26	22
Sidhi	26	22
Dindori	25	23
Singrauli	25	24

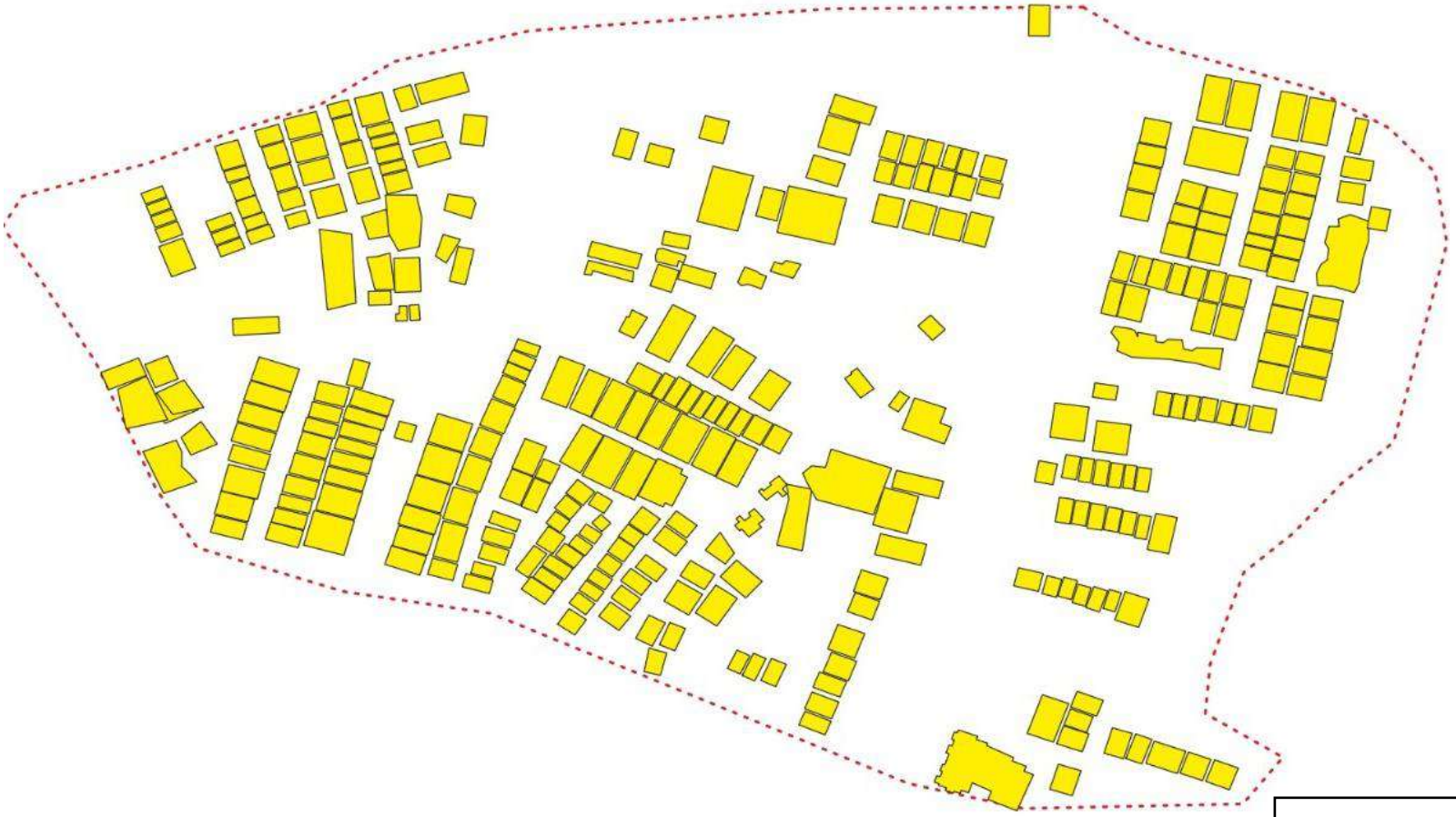
Madhya Pradesh	Barwani
Madhya Pradesh	Chhatarpur
Madhya Pradesh	Damoh
Madhya Pradesh	Guna
Madhya Pradesh	Khandwa
Madhya Pradesh	Rajgarh
Madhya Pradesh	Singrauli
Madhya Pradesh	Vidisha

TOTAL ASPIRATIONAL DISTRICTS SCORE





DIMENSIONS		REMARKS		SHEET NO.
SCALE: AS SHOWN IN THE MAP	LAND USE MAP - CHUNNABHATTI	N ↑	SUBMITTED BY: ECONOMICAL SUBJECT: PLANNING STUDIO – 3 SUBJECT CODE – PLN211 2nd YEAR III SEMESTER B. PLANNING	1

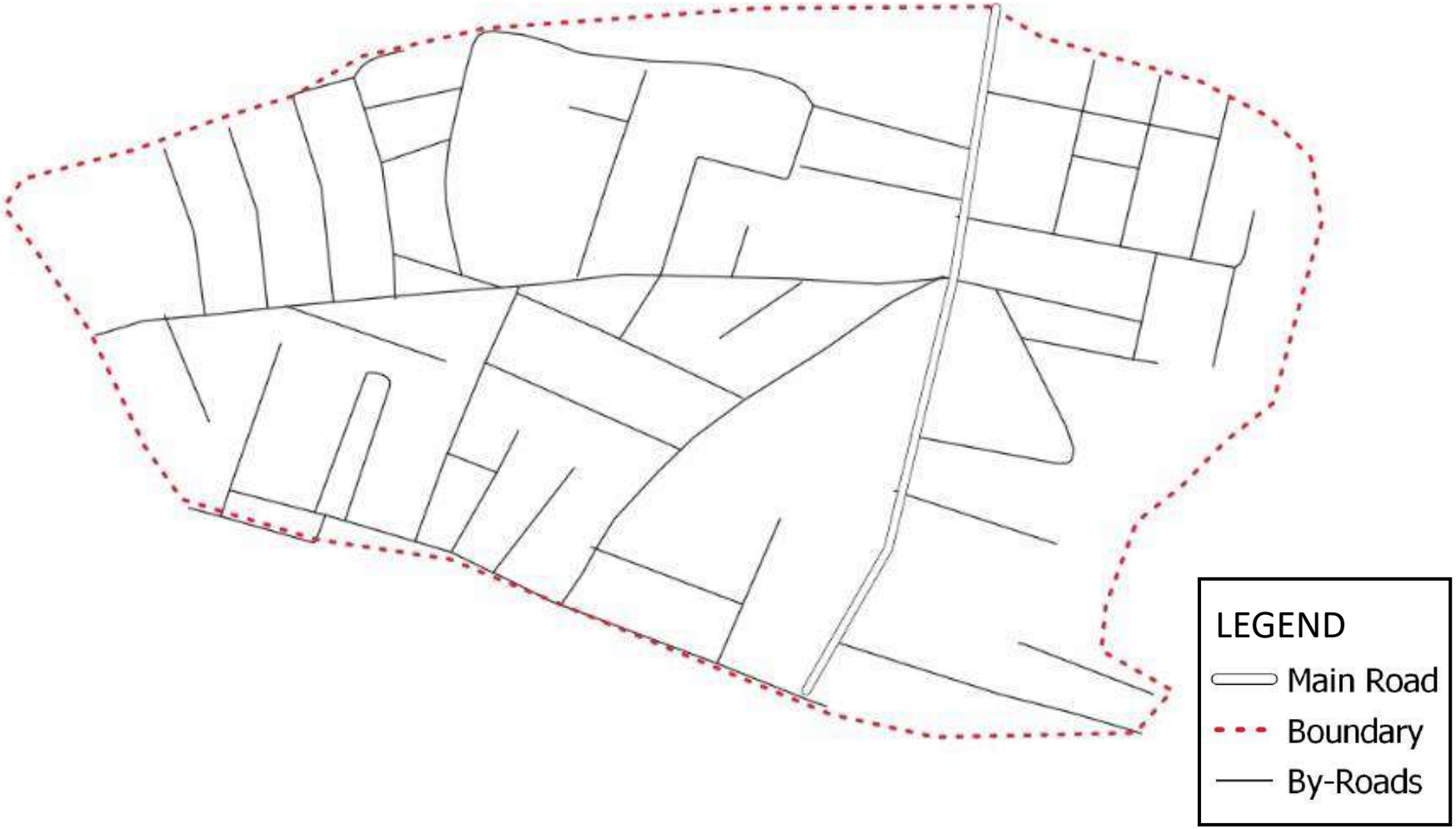


DIMENSIONS		REMARKS		SHEET NO.
SCALE: AS SHOWN IN THE MAP	RESIDENTIAL- CHUNNABHATTI	N	SUBMITTED BY: ECONOMICAL SUBJECT: PLANNING STUDIO – 3 SUBJECT CODE – PLN211 2nd YEAR III SEMESTER B. PLANNING	2

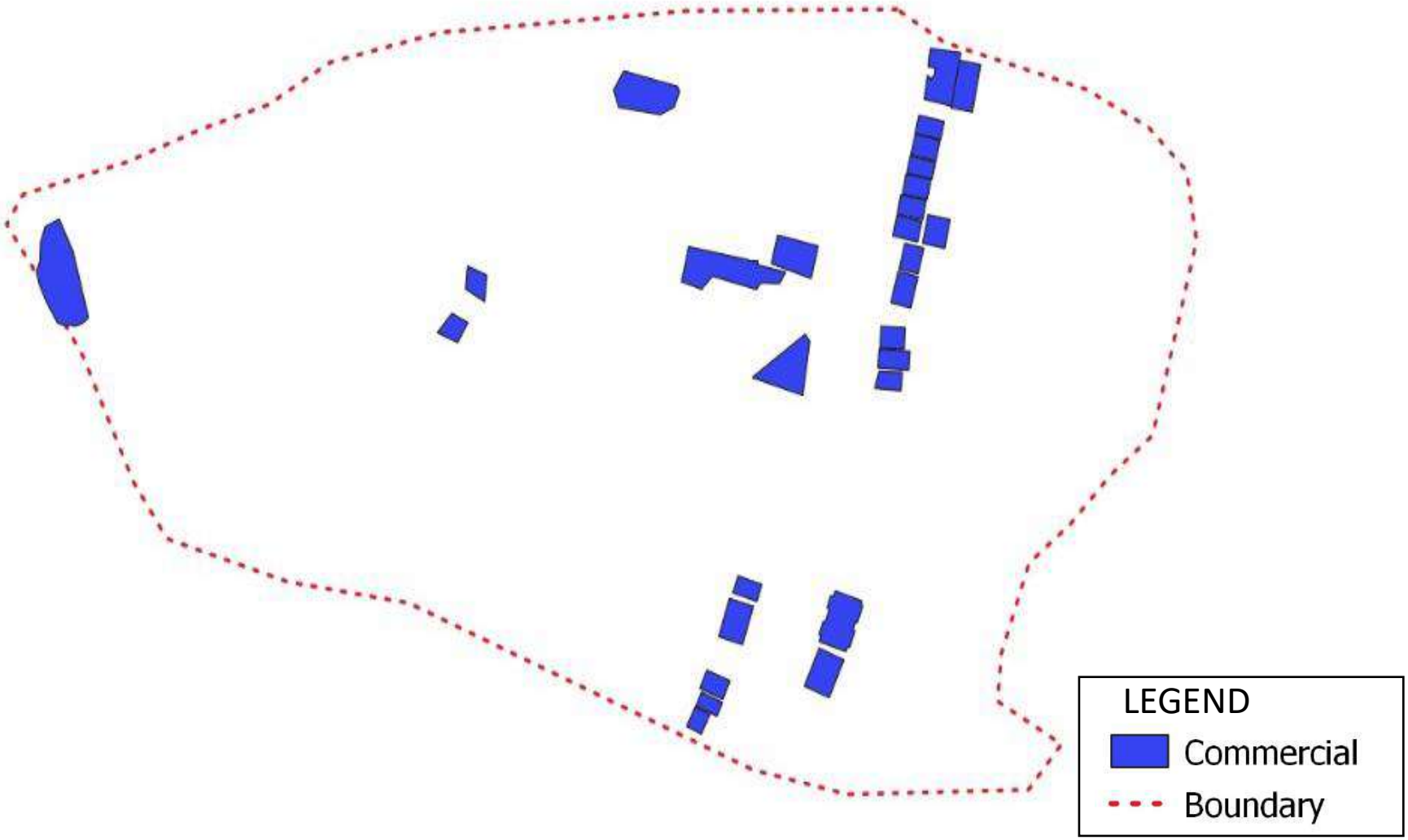
LEGEND

- G+4
- G+3
- G+2
- G+1
- Main Road
- By Roads
- Boundary

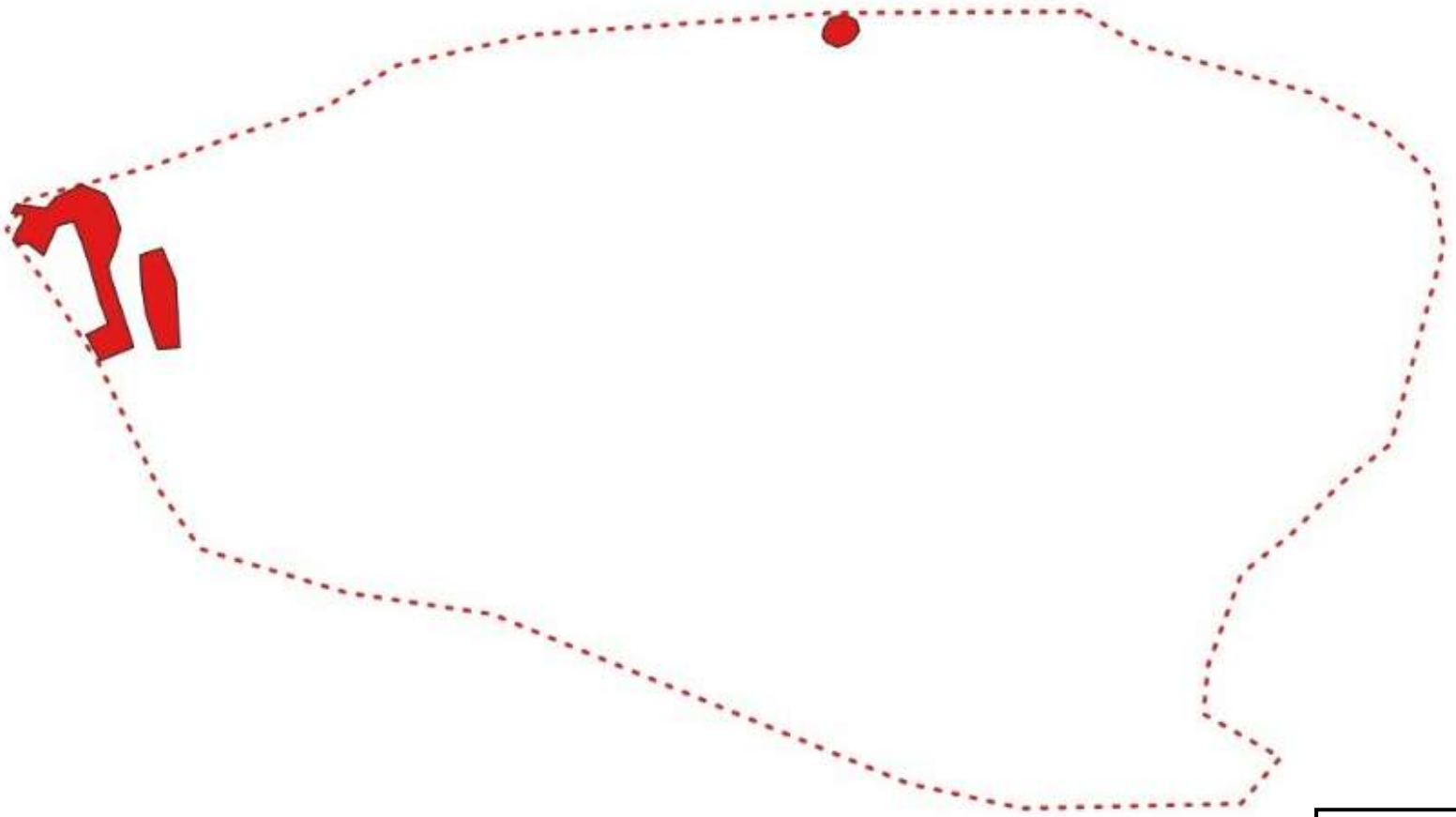
DIMENSIONS		REMARKS		SHEET NO.
SCALE: AS SHOWN IN THE MAP	HIGH RISE BUILDINGS- CHUNNABHATTI	N	SUBMITTED BY: ECONOMICAL SUBJECT: PLANNING STUDIO – 3 SUBJECT CODE – PLN211 2nd YEAR III SEMESTER B. PLANNING	3



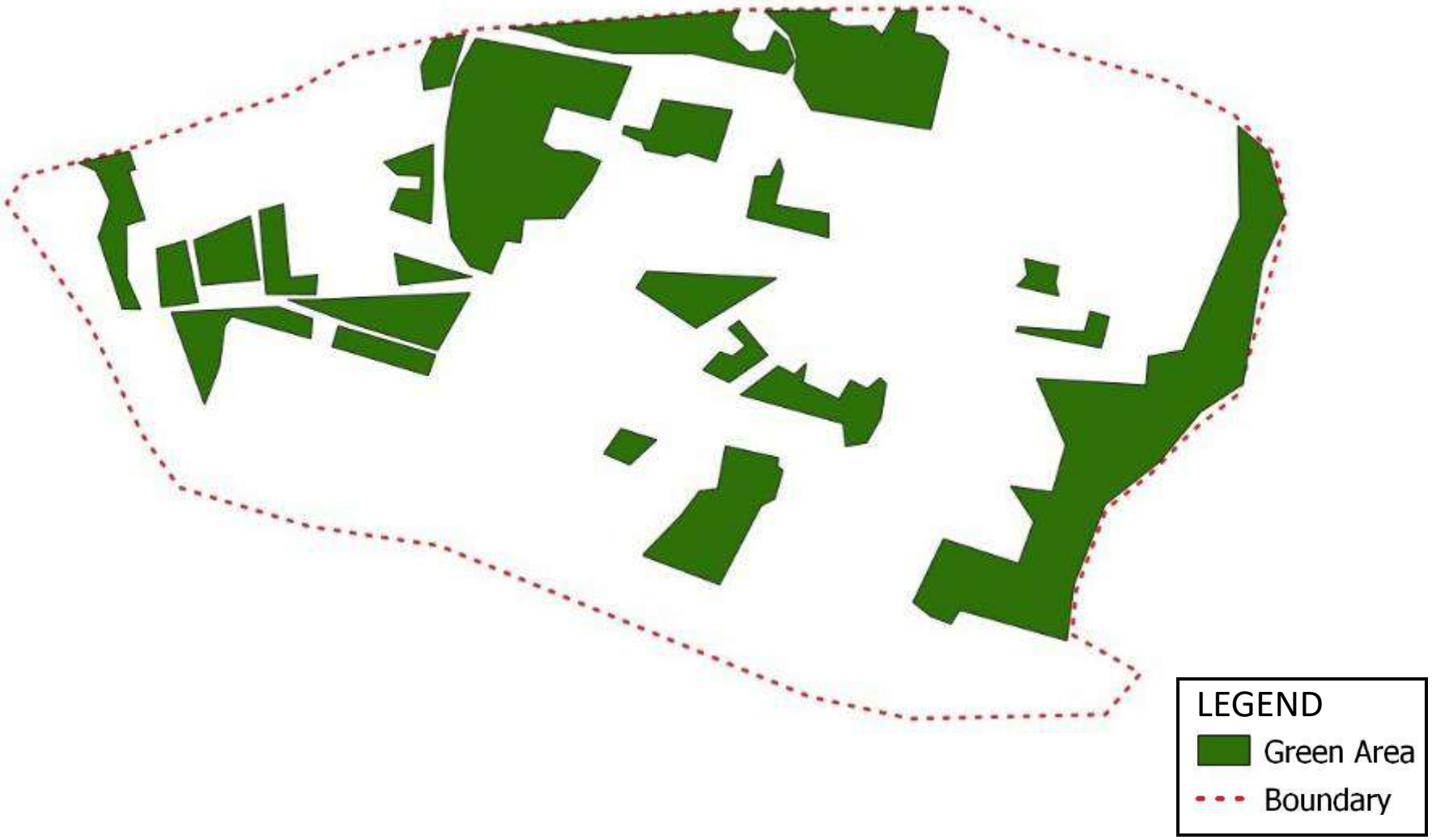
DIMENSIONS		REMARKS		SHEET NO.
SCALE: AS SHOWN IN THE MAP	TYPES OF ROADS – CHUNNABHATTI	N	SUBMITTED BY: ECONOMICAL SUBJECT: PLANNING STUDIO – 3 SUBJECT CODE – PLN211 2nd YEAR III SEMESTER B. PLANNING	4



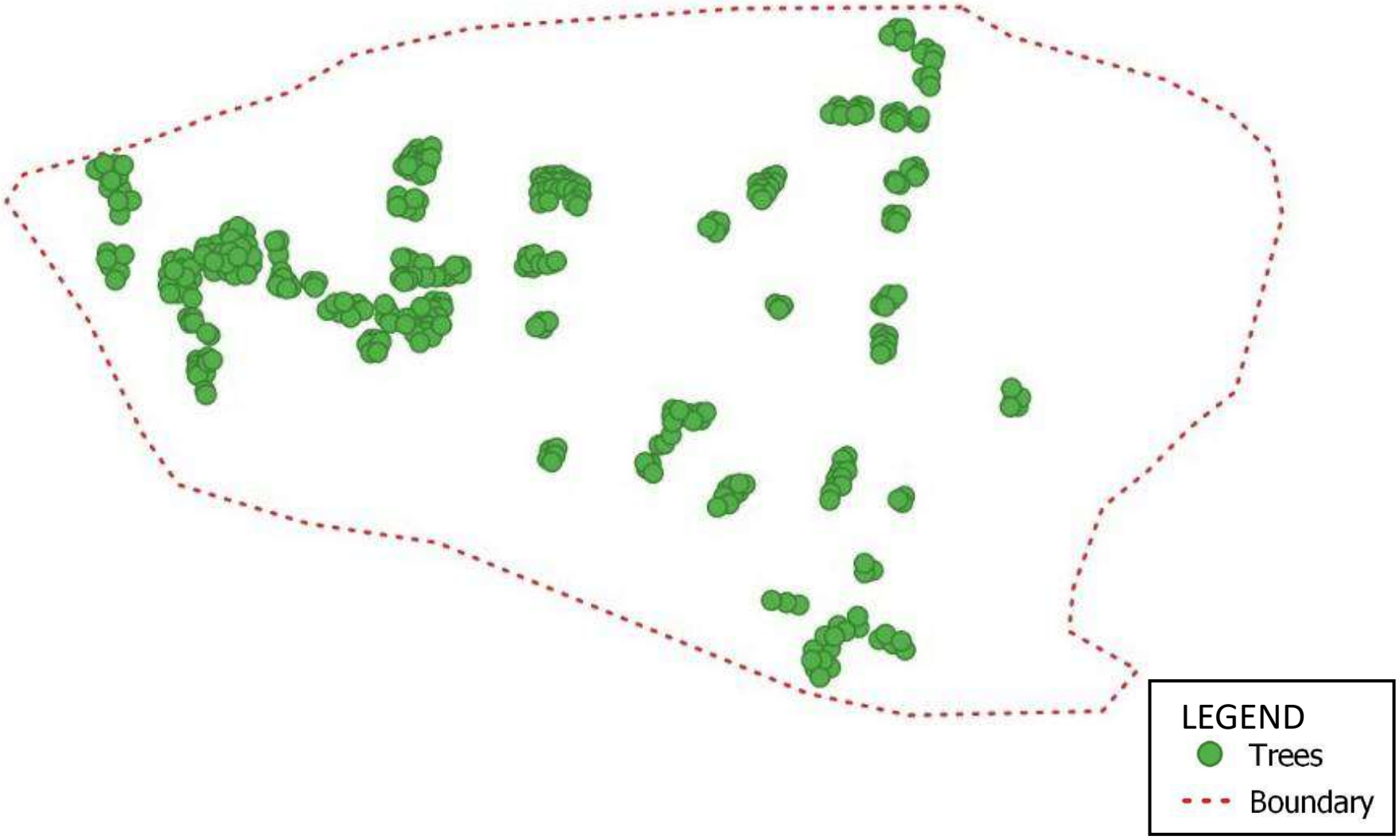
DIMENSIONS		REMARKS		SHEET NO.
SCALE: AS SHOWN IN THE MAP	COMMERCIAL- CHUNNABHATTI	N	SUBMITTED BY: ECONOMICAL SUBJECT: PLANNING STUDIO – 3 SUBJECT CODE – PLN211 2nd YEAR III SEMESTER B. PLANNING	5



DIMENSIONS		REMARKS		SHEET NO.
SCALE: AS SHOWN IN THE MAP	PUBLIC BUILDINGS- CHUNNABHATTI	N	SUBMITTED BY: ECONOMICAL SUBJECT: PLANNING STUDIO – 3 SUBJECT CODE – PLN211 2nd YEAR III SEMESTER B. PLANNING	6



DIMENSIONS		REMARKS		SHEET NO.
SCALE: AS SHOWN IN THE MAP	GREEN AREA- CHUNNABHATTI	N	SUBMITTED BY: ECONOMICAL SUBJECT: PLANNING STUDIO – 3 SUBJECT CODE – PLN211 2nd YEAR III SEMESTER B. PLANNING	7



DIMENSIONS		REMARKS		SHEET NO.
SCALE: AS SHOWN IN THE MAP	TREES- CHUNNABHATTI	N	SUBMITTED BY: ECONOMICAL SUBJECT: PLANNING STUDIO – 3 SUBJECT CODE – PLN211 2nd YEAR III SEMESTER B. PLANNING	8

Climate change induced water borne diseases and vector breeding in flood vulnerable region: A case of Kolkata city

PRESENTED BY:

NITESHKUMAR S K – CC25MTECH11002

UNDER SUPERVISION OF:

DR. AMBIKA S

FACULTY:

DEPT. OF ENVIRONMENTAL ENGINEERING &
CLIMATE CHANGE, CIVIL ENGINEERING,
IIT HYDERABAD

BACKGROUND

CLIMATE CHANGE INDUCED WATER BORNE DISEASES

(Stagnant) WATER → contamination → **DIRECT WATER BORNE**

WATER BORNE (WATER RELATED)

(Stagnant) WATER → Breeding ground → **VECTOR BREEDING**

- **Cholera**
- **Acute Diarrheal Disease (ADD)**
- **Typhoid**
- **Leptospirosis**

vector borne - mosquitoes, flies

- **Dengue**
- **Malaria**
- **Chikungunya**

"Bacterial concentration increases by a **factor of 1.5 for every 1°C increase** in water temperature."

Source: Climate Change-Related Temperature Impact on Human Health Risks of Vibrio Species (PMC, 2025)

- Warmer temperatures shorten the Extrinsic Incubation Period (EIP) of the virus inside the mosquito.
- **13% increase in dengue incidence per 1°C rise** in temperature. risk rises **29.5%** per 1°C increase in minimum temperature.

Source: A Systematic Review and Meta-Analysis of Dengue Risk with Temperature Change (PMC, 2023)

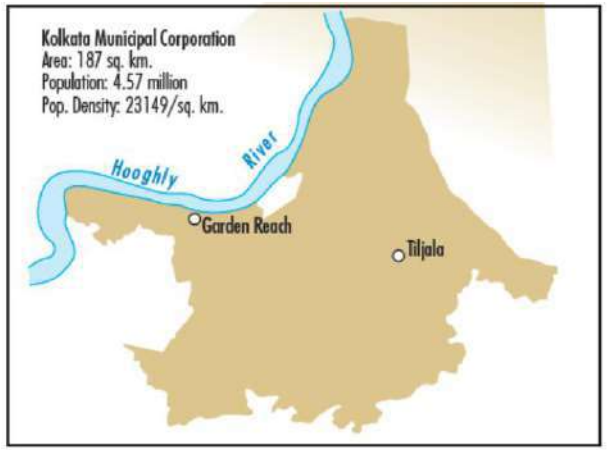
❖ Flood/extreme climate events damages water and sewage structure – contaminate water

❖ Post flood warm stagnant water – enable the proliferation of pathogens

Post flood/rain warm stagnant water – act as breeding ground for vectors

CASE AREA- KOLKATA

- Kolkata lies on the low-lying Ganga-Brahmaputra delta of state west Bengal along Hooghly River and has an elevation of 1.5 to 9 meters above sea level (5– 30 ft)
- Kolkata has a Tropical wet-and-dry climate with an annual mean temperature is 24.8 °C
- Rains are brought by the Bay of Bengal branch of South-West monsoon between June and September
- More than one third of the total population lives in slum – who are highly vulnerable to extreme weather related events, especially flooding.



Source: https://openknowledge.worldbank.org/handle/10986/2818

WATER-BORNE AND VECTOR-BORNE DISEASES IN KOLKATA

Cholera & Acute Diarrheal Disease (ADD)

- West Bengal consistently reports the highest burden of water-borne diseases in India, with **3.35 crore cumulative cases** recorded between **2005 and 2022**.
- Additionally, research by ICMR-NICED confirms a distinct seasonal trend in Kolkata, where cholera outbreaks **exhibit a "bi-annual peak"** occurring during the **pre-monsoon** (April-May) and **post-monsoon** (August-October) periods.

Source: Integrated Disease Surveillance Programme (IDSP)

Typhoid

- Clinical surveillance indicates that *Salmonella Typhi* accounts for approximately **54% of positive blood culture isolates** in Kolkata, with recent 2023-24

Source: Surveillance for Enteric Fever in Asia Project (SEAP) and ICMR-NICED

Malaria

- Kolkata Municipal Corporation (KMC) reported **17,002 malaria cases** in 2023, KMC alone accounted for **18.5% of India's total *P. vivax* malaria cases** for that year.

Source: NCVBDC Annual Report 2023

Dengue

- West Bengal recorded **30,683 confirmed dengue cases** in 2023, classifying it as a severe outbreak year.
- Furthermore, recent spatial analysis published in *Emerging Microbes & Infections (2025)* identified specific Kolkata wards, such as **Ward 106 and 109**, as high-risk zones where case density exceeded **30 cases per 10,000 population**.

Source: NCVBDC Annual Report 2023

CLIMATE CHANGE AND EXTREME CLIMATE EVENTS IN KOLKATA

TEMPERATURE RISE

Between 1950 and 2018, Kolkata's surface air **temperature rose by 2.6°C**.

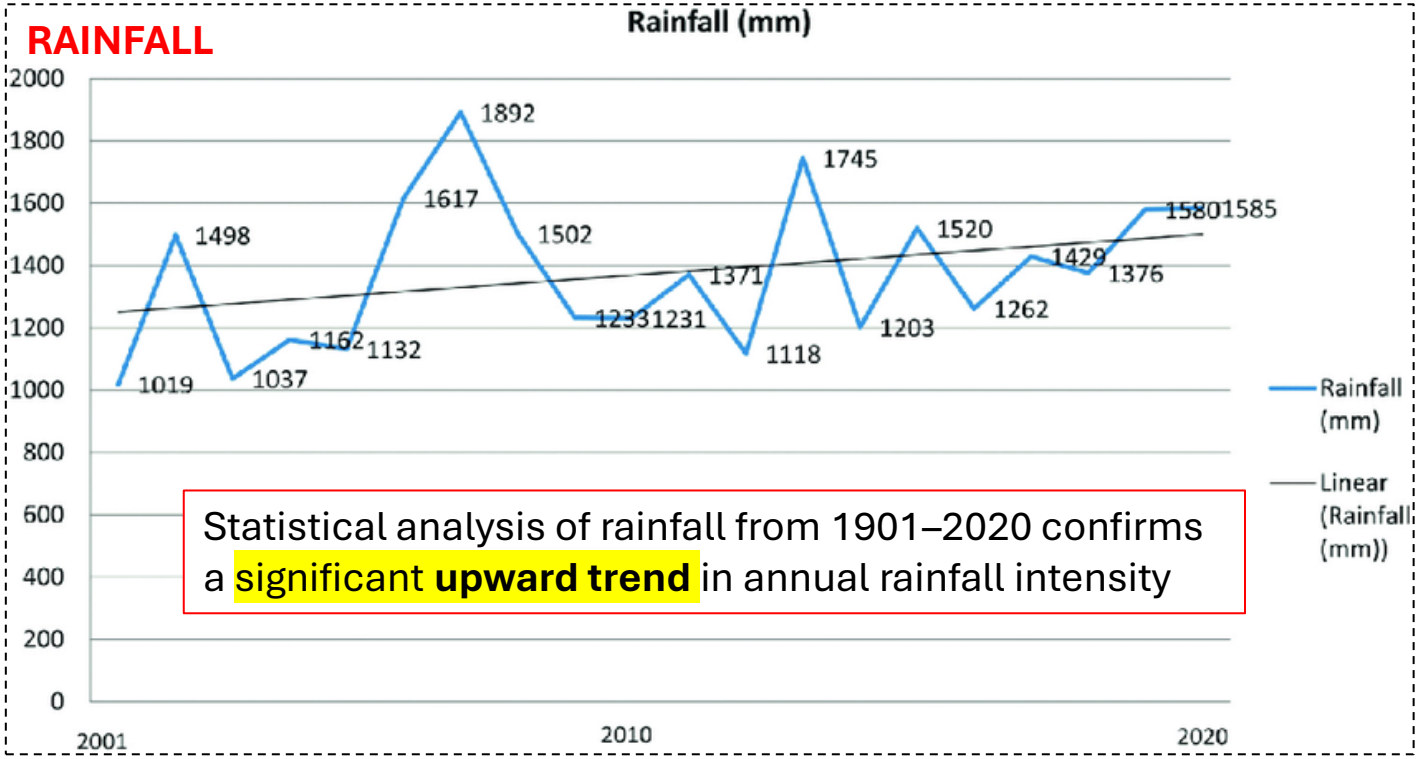
Source: Kolkata Climate Action Plan (KCAP).

SEA LEVEL RISE

The sea level in the northern Bay of Bengal (near Sundarbans and Kolkata) is rising at **4.5 mm/year (**higher than the global average 3.3 mm/year)

Source: INCOIS (Indian National Centre for Ocean Information Services).

RAINFALL



Statistical analysis of rainfall from 1901–2020 confirms a significant **upward trend** in annual rainfall intensity

Source: Mukherjee, Alokananda & Bardhan, Suchandra. (2021).

FLOOD AND WATERLOGGING

- Kolkata is one of the worst hit metropolises in India in terms of urban floods.
- **Floods occur every year during the monsoon** when the city receives large amount of rainfall.
- In recent years the severity of flooding in certain parts of the city has increased.
- Certain parts remain water logged for over three days and the depth of water logging varies between 1.5- 3 feet.
- Over the years, not only has there been an **increase in the depth and duration of water logging** in these areas, there has also been a significant rise in the extent of area flooded

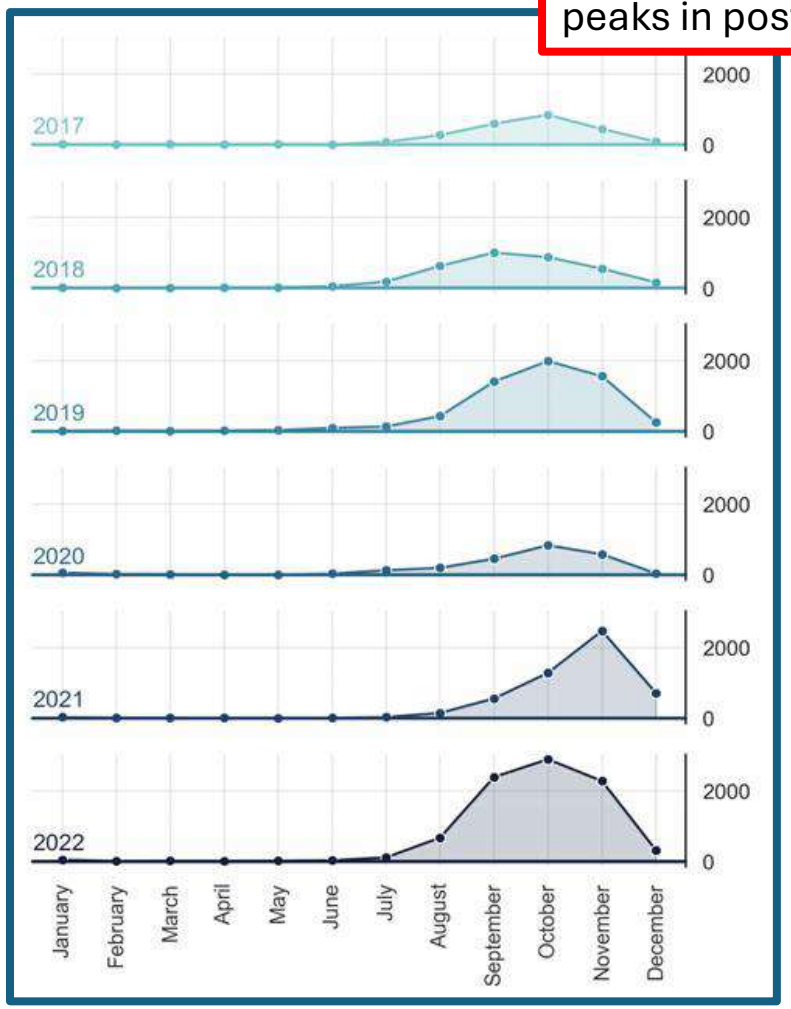
Source: Mukherjee, Alokananda & Bardhan, Suchandra. (2021).

waterlogged street after heavy rainfall in Kolkata. September 24, 2025. REUTERS/Sahiba

Ground for waterborne pathogens and vector breeding

CLIMATE and VECTOR BORNE DISEASE

peaks in post-monsoon months



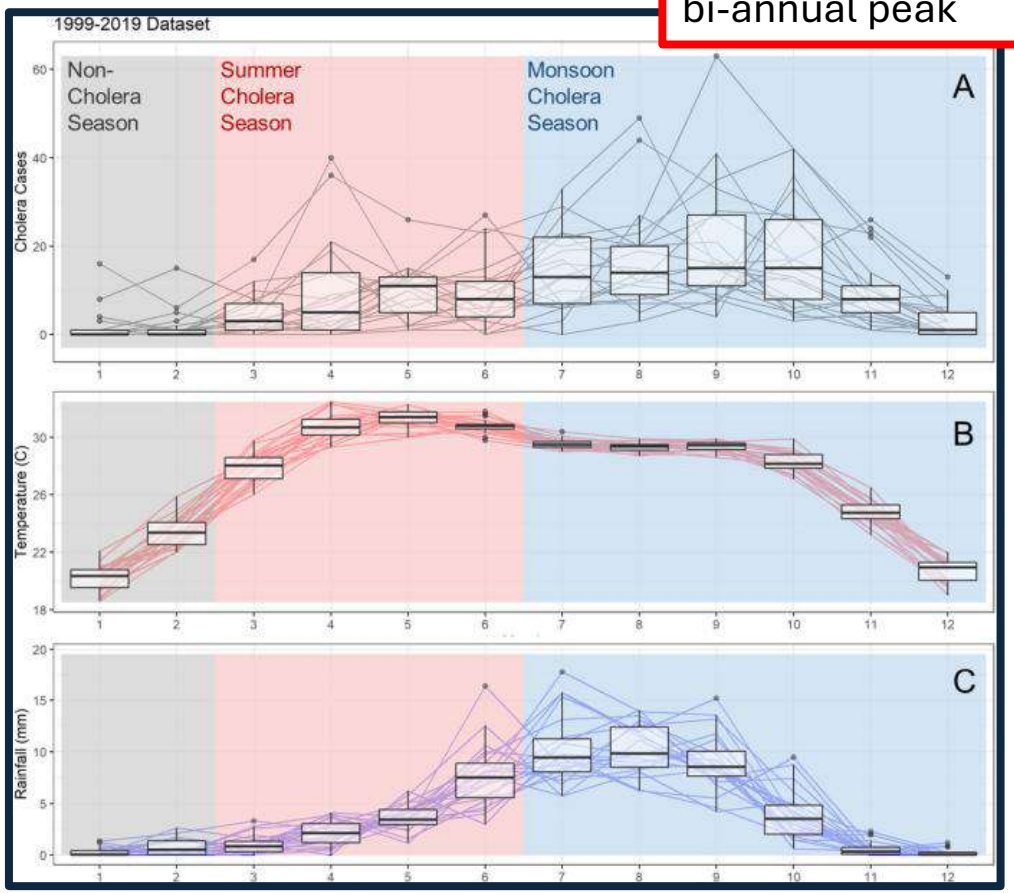
magnitude of the dengue has increased every year

Source: S. Rani jena et al (2025)

- **Seasonal distribution of dengue** in Kolkata, increasing from August and reaching the peak in post-monsoon months across the years.
- The graph also shows that the magnitude of the disease has increased every year since 2017.

CLIMATE and WATER BORNE DISEASE

bi-annual peak



Shackleton, D., Economou, T., Memon, F.A. *et al.* (2023

Seasonal distribution of cholera in Kolkata Variation in monthly values between 1999–2019 for **A** confirmed cholera cases in Kolkata, **B** mean daytime temperature in the Kolkata, **C** total precipitation in the Kolkata. Each line represents values for one year.

METHODOLOGY

INDENTIFICATION
-problem statement
-case area(Kolkata)

STUDY AREA ANALYSIS
-data collection and analysis
-literature reviews
(in references)
-existing scenarios and strategies

GAP INDENTIFICATION
(based on existing system)
AND RESPECTIVE STRATEGY DEVELOPMENT
(based on international case studies)

PROPOSAL & **IMPLEMENTATION FRAMEWORK**

EXAMPLE PROJECT AND **ITS RESULTS**

TOWARDS THE STRATEGY – EXISTING SYSTEM

"KOLKATA has two very advanced, world-class systems that operate independently"

CLIMATE SYSTEM

Kolkata's
Flood Forecasting and Early Warning System (FFEWS)

is an **IoT-driven sensor network** designed for real-time operational response – street level analysis

Measures,
Hydrometric data – water level, street flood depth
Climate and meteorological data – rainfall, temperature, humidity

Decision Support System (DSS)- with hydrological stimulation – forecast flood at street level

Does not predict disease outbreaks with climate data

HEALTH SYSTEM

Integrated Disease Surveillance Programme (IDSP)
with
Integrated Health Information Platform (IHIP)

Tracks medical/epidemiological cases from hospitals and health workers (through APP)

Real-Time Data
Geotagging
Mobile App

Alert outbreaks

Does not predict outbreaks

PROPOSAL & IMPLEMENTATION FRAMEWORK

SO, INTEGRATED HEALTH AND CLIMATE SYSTEM

THE INTEGRATED ADAPTATION STRATEGY!

AIM

To move from “ALERTING OUTBREAKS" to "PREVENTING OUTBREAKS“
By
developing and modelling disease outbreak prediction system based on integrated climate and health data system

TO build,

A DATA BRIDGE

INTEGRATED SURVEILLANCE

INTER-DEPARTMENTAL COMMITTEE

Human Health
Integrated Health Information Platform (IHIP)

Environmental Health
SCADA – supervisory control and data acquisition
Real-time monitoring of sewage, drainage, water quality and air quality

Climate Data
IMD
FFEWS

=

COMMON HEALTH AND CLIMATE CENTER

Application:

PREDICTING-PREVENTING-PREPARING for
Vector-Borne Diseases
Water-Borne Diseases
Zoonoses(animal borne)
Heat Stress

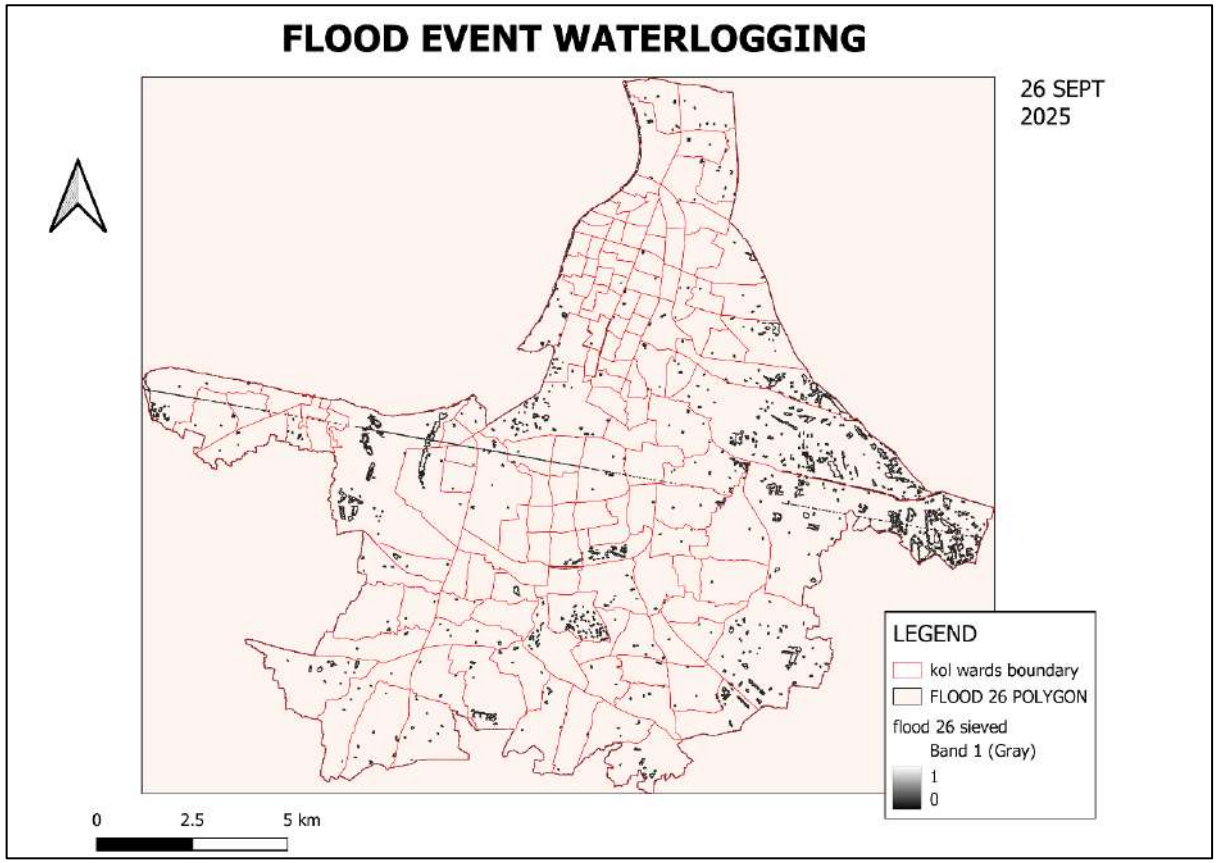
Tools:

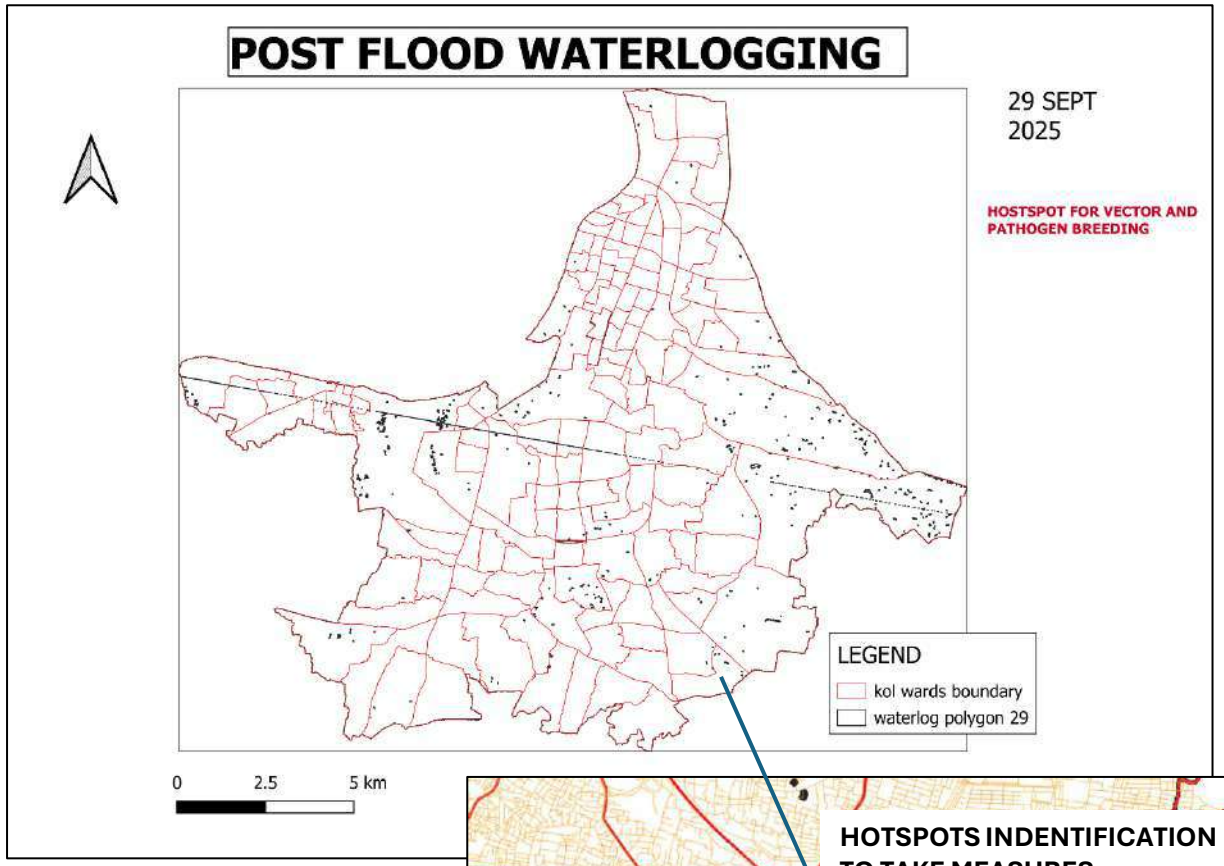
AI & ML models

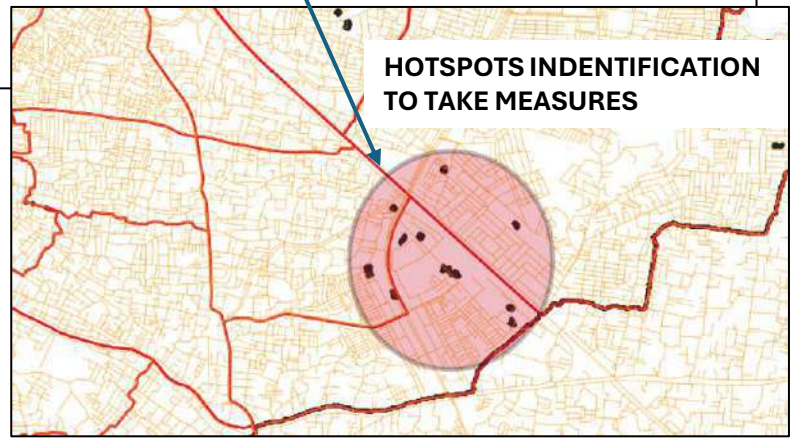
Comprehensive climate and health resilience, prevention and reaction plan

ACTION DEPATMENTS KMC- Public health, water and sanitation, PWD

CLIMATE-HEALTH INTEGRATED STRATEGY EXAMPLE PROJECT AND ITS RESULTS







PROJECT: PATHOGEN AND VECTOR BREEDING HOTSPOTS INDENTIFICATION – POST FLOOD CONDITION
Based on 23 sept 2025 – high intensity rainfall – caused Kolkata flood

DATA USED: SENTINAL 1 – SAR (VV,VH) – from copernicus
METHOD: BACKSCATTER BASED CHANGE DETECTION (water is < 17 db, permanent water removed to identify flood)
REFERENCE DATE (DRY) – 17 SEPT 2025
FLOOD EVENT – 26 SEPT 2025 (there was no data for 23rd available, SO I used 26th) – so limitation
POST FLOOD – 29 SEPT 2025

DISCUSSION AND CLAIMS

EXAMPLE PROJECT RESULT AND REAL WORLD POTENTIAL CLAIM

With satellite data of past flood we can able to identify potential hotspots that could cause waterborne disease outbreaks

CLAIM: Then by integrating existing independent advanced systems **FFEWS and IHIP** which can produce real time data and street level forecast of waterlogging

We would have the most advanced CLIMATE-HEALTH PREDICTION AND WARNING SYSTEM

IMPLEMENTAION ADVANTAGES

- ❑ Working with already existing well established systems
- ❑ Can be enabled and supported with existing SPVs(Special Purpose Vehicles) and programs

Like

The proposed

COMMON HEALTH AND CLIMATE CENTER

Can be established as part of

ICCC(Integrated Command And Control Center)

NOVELTY OF OUR PROJECT

Retrofitting the existing technology and system to provide solution to our problem statement

(whereas most case examples we studied either used simple satellite data or established a completely new system)

LESSONS LEARNT

Finding and understanding the gap in existing systems, policies will give a clear solution to the problem by itself

Integrating and coordinating various related domains (HEALTH, CLIMATE, ENVIRONMENT) could give greater advantage than working independently

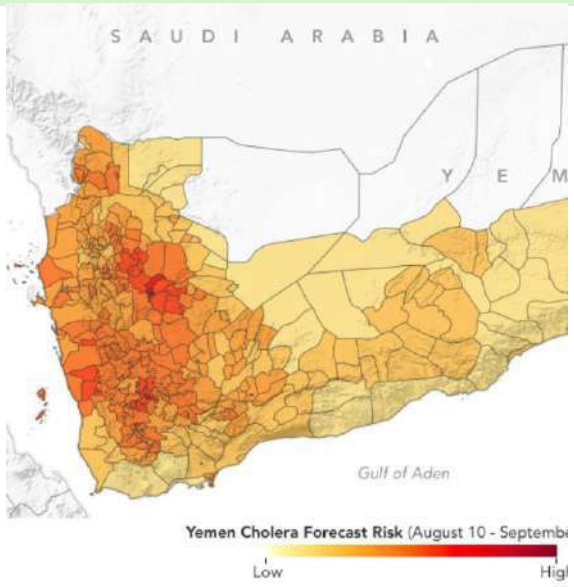
CASE STUDIES ON INTEGRATED HEALTH AND CLIMATE STRATEGY

Case study 1: Predicting Cholera Risk in Yemen

During 2017 with War, heavy rainfall and collapsed WASH systems caused a massive cholera outbreak but conflict made ground health data collection nearly impossible.

Cholera Risk was predicted with climate data using satellite variables air temperature and precipitation to detects bacterial bloom conditions.

The Outcome: For the outbreak of 2017, predicted with 92% accuracy, even in the inland areas. Detected risk in unusual inland regions & now used by UNICEF for hotspot-based aid distribution. Shift from reactive to anticipatory humanitarian action.



Source: Predicting cholera risk in Yemen. NASA Earth Observatory (2021, November 15).

Case study 2: Dengue Emergency Operations Center (COE), Rio de Janeiro

In need for cross-sector urban climate resilience, Integrated data platform was established combining climate and health service data and epidemiological surveillance.

With which georeferencing of dengue cases, Hotspot detection for vector breeding and 4-week predictive planning window is enabled

The Outcome: Predicted dengue spike 2 months earlier than traditional models, 90 percent of cases geolocated within days and 2.8 million properties inspected, 447,000 breeding sites eliminated (first 3 months of 2024).

Soruce: ClimaHealth. *Dengue in Rio de Janeiro*.

Case study 3: Early Warning and Response System (EWARS), Ethiopia

In need to embed climate resilience into the health system, Real-time climate–health risk monitoring was established.

With Sentinel sites across eco-climatic zones and integrated data collection of weather, Local disease incidence Early Warning and Response System (EWARS) was enabled

The Outcome: In 2020, Bahir Dar malaria risk crossed endemic threshold but with early alert, emergency plan activated before peak which reduced outbreak intensity.

Source: ClimaHealth. *Developing early warning alert and response systems (EWARS) to combat climate-sensitive diseases in Ethiopia*.

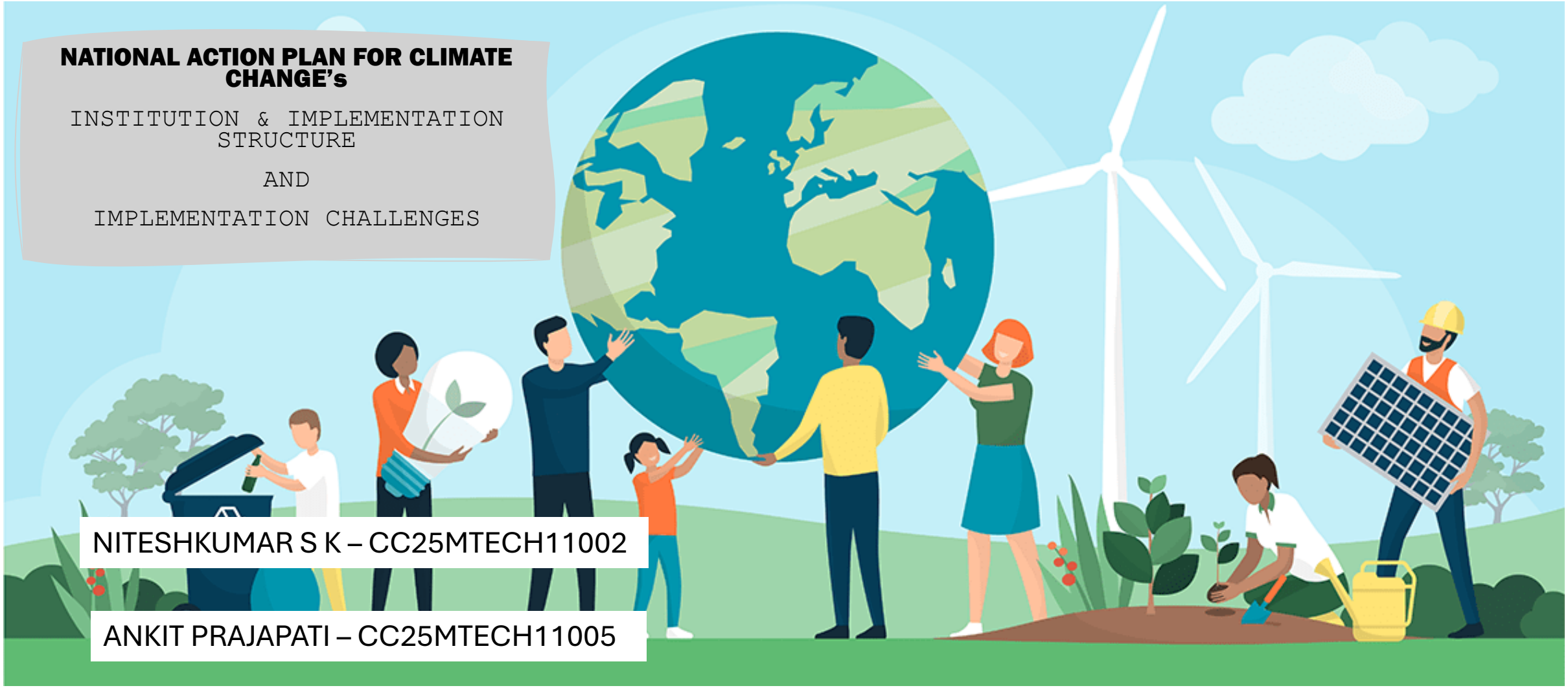
REFERENCES

1. Sasmita Rani Jena, C. Elanchezhiyan, Avik Kumar Sam, Ujjal Gogoi, Varun Shende, Srimoyee Basu, Saikat Bandyopadhyay, Bithika Mandal, Baishakhi Biswas, Atanu Banerjee, Kalpana Baruah, Harish C. Phuleria & Ipsita Pal Bhowmick (2025) Spatio temporal distribution of dengue cases and vectors along with the interrelationship of environmental and climatic factors in the metropolitan city, Kolkata, India, for 2017–2022: calling implications for vector control, Emerging Microbes & Infections, 14:1, 2493924, DOI: 10.1080/22221751.2025.2493924
2. Shackleton, D., Economou, T., Memon, F.A. *et al.* Seasonality of cholera in Kolkata and the influence of climate. *BMC Infect Dis* **23**, 572 (2023). https://doi.org/10.1186/s12879-023-08532-1
3. Mukherjee, Alokananda & Bardhan, Suchandra. (2021). Flood vulnerability and slum concentration mapping in the Indian city of Kolkata: A post-Amphan analysis. Water Science. 35. 109-126. 10.1080/23570008.2021.1957641.
4. National Aeronautics and Space Administration. (2021, November 15). *Predicting cholera risk in Yemen*. NASA Earth Observatory. https://science.nasa.gov/earth/earth-observatory/predicting-cholera-risk-in-yemen-147101/
5. ClimaHealth. *Developing early warning alert and response systems (EWARS) to combat climate-sensitive diseases in Ethiopia*. https://climahealth.info/resource-library/developing-early-warning-alert-and-response-systems-ewars-to-combat-climate-sensitive-diseases-in-ethiopia/
6. ClimaHealth. *Dengue in Rio de Janeiro*. https://climahealth.info/resource-library/dengue-in-rio-de-janeiro/
7. *Climate Change-Related Temperature Impact on Human Health Risks of Vibrio Species (PMC, 2025)*
8. *A Systematic Review and Meta-Analysis of Dengue Risk with Temperature Change (PMC, 2023)*
9. *Surveillance for Enteric Fever in Asia Project (SEAP) and ICMR-NICED*

Thank You

CLIMATE GOVERNANCE

Indian Institute of Technology Hyderabad



NATIONAL ACTION PLAN FOR CLIMATE CHANGE's

INSTITUTION & IMPLEMENTATION STRUCTURE

AND

IMPLEMENTATION CHALLENGES

NITESHKUMAR S K – CC25MTECH11002

ANKIT PRAJAPATI – CC25MTECH11005

INSTITUTION AND IMPLEMENATION

The **institutional structure and implementation system** is the “hidden infrastructure” of climate governance

Climate action policies do **not fail because the science is weak**, but because the **institutions responsible for translating policy into action are fragmented, under-resourced, or poorly coordinated**

Climate change or any governance is **multi-level, multisectoral and long-term**, so no single ministry or government layer can manage it alone

INSTITUTIONAL AND IMPLEMENTATION STRUCTURE

National Level (NAPCC)

Institutional Structure

PRIME MINISTER'S COUNCIL ON CLIMATE CHANGE (PMCCC)

Apex Body

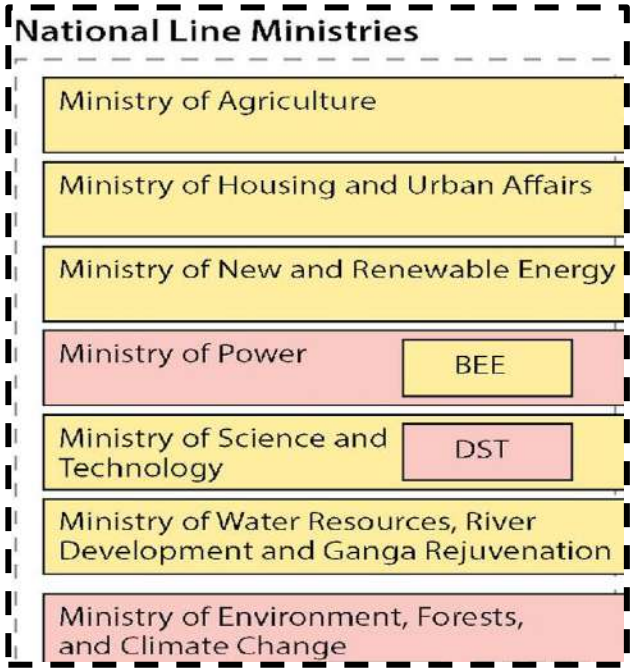
serves as the highest policy-making body
It is responsible for setting national targets and coordinating the national response

EXECUTIVE COMMITTEE

A committee under the Principal Secretary to the Prime Minister monitors the implementation of missions.

NODAL MINISTRIES

Implementation is decentralized across specific Union Ministries responsible for the eight National Missions



IMPLEMENTATION FRAMEWORK

The NAPCC operates through a **"Mission Mode"** approach

Each of the eight missions functions as a standalone vertical with its own budget, targets, and timeline.

- **National Solar Mission** (Ministry of New & Renewable Energy)
- **National Water Mission** (Ministry of Jal Shakti)
- **Green India Mission (GIM)** MoEFCC
- **Sustainable Habitat** MoUD / MoEFCC
- **Sustainable Agriculture** (Ministry of Agriculture)
- **Sustaining the Himalayan Ecosystem** (Dept. of Science & Tech)
- **Enhanced Energy Efficiency** Ministry of Power (via BEE)
- **Strategic Knowledge for Climate Change** (Dept. of Science & Tech)

The framework relies on "co-benefits," integrating climate goals into existing development programs (e.g., MGNREGA for water conservation) rather than creating entirely new legal frameworks.

State Level (SAPCC)

Institutional Structure

Nodal Department

⇩

MoEFCC's

NATIONAL STEERING COMMITTEE (NSCCC)

acts as the secretariat

→

STATE STEERING COMMITTEE

Headed by the

Chief Minister or Chief Secretary

this body approves the SAPCC and oversees progress.

STATE CLIMATE CHANGE CELL (SCCC)

↔

LINE DEPARTMENTS

The actual implementing agencies

- A designated **Nodal Department** (often Environmental Department / Planning / Special Cell) to coordinate.
- Example: Tamil Nadu's SAPCC sets up a State Climate Change Cell under the Dept. of Environment as nodal agency.
- A technical unit (often funded by DST or international agencies) responsible for research, data collection, and drafting plans.

Departments like

- Agriculture,
- Water Resources, and
- Urban Development

IMPLEMENTATION FRAMEWORK

States utilize a **"Common Framework"** provided by the

MoEFCC
to draft
SAPCC

The framework requires states to:

1. Analyse observed climate trends.
2. Develop sectoral vulnerability assessments.
3. Propose adaptation/mitigation strategies.
4. Align state budgetary allocations with these strategies ("Climate Budgeting").

Sectoral Working Groups

Line departments
(Agriculture, Water, Energy)
Form
WORKING GROUPS
to propose adaptation actions which are then compiled into the SAPCC

DISTRICT AND LOCAL LEVEL

Institutional Structure

DISTRICT CLIMATE CHANGE COMMITTEES (DCCCS)

Headed by the District Collector

District Planning Committee (DPC)

Constitutional body responsible for integrating rural and urban plans

CITY LEVEL

Urban Local Bodies (ULBs)
Municipal Corporations.

Special Purpose Vehicles
Specific "Smart City" SPVs often bypass elected municipal councils.

RURAL LEVEL

Gram panchayat

Biodiversity Management Committees (BMCs)
Mandated under the Biological Diversity Act, relevant for local adaptation

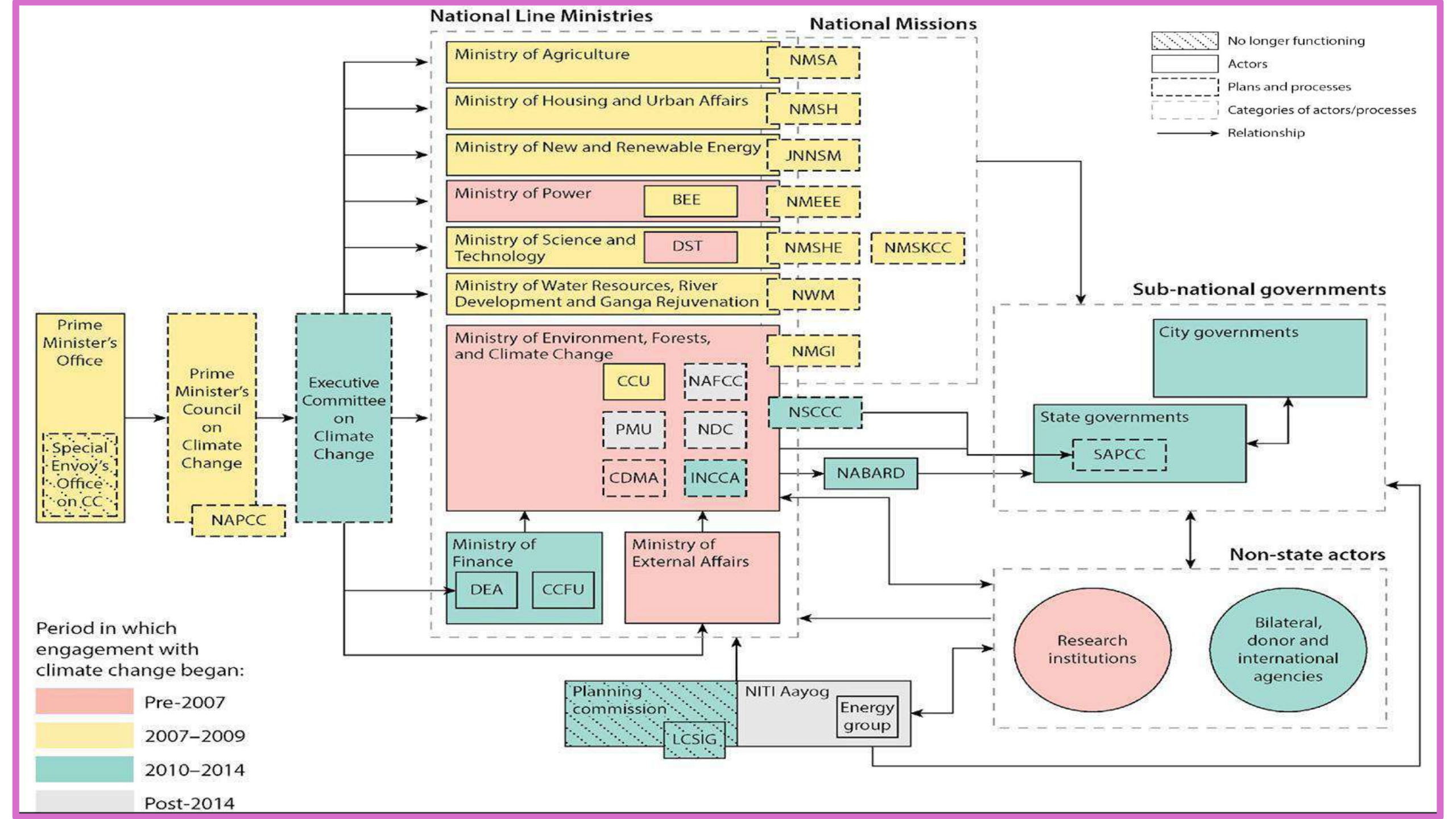
COMMUNITY BODIES

Joint Forest Management Committees (JFMCs)
Village Forest Committees
And
Women's Self-Help Groups (SHGs) envisaged for local adaptation works.

IMPLEMENTATION FRAMEWORK

CONVERGENCE
Implementation relies on converging national schemes (AMRUT, Smart Cities Mission, Swachh Bharat) **with local development plans**

MASTER PLANS
and
CITY CLIMATE ACTION PLANS (CCAPS)
Cities (e.g., Mumbai, Ahmedabad, Chennai) draft specific CAPs focusing on heat, flooding, and emissions.


National Line Ministries
National Missions
No longer functioning
Actors
Plans and processes
Categories of actors/processes
Relationship
Ministry of Agriculture
NMSA
Ministry of Housing and Urban Affairs
NMSH
Ministry of New and Renewable Energy
JNNSM
Ministry of Power
BEE
NMEEE
Ministry of Science and Technology
DST
NMSHE
NMSKCC
Ministry of Water Resources, River Development and Ganga Rejuvenation
NWM
Ministry of Environment, Forests, and Climate Change
NMGI
CCU
NAFCC
PMU
NDC
CDMA
INCCA
NSCCC
NABARD
Prime Minister's Office
Special Envoy's Office on CC
Prime Minister's Council on Climate Change
NAPCC
Executive Committee on Climate Change
Ministry of Finance
DEA
CCFU
Ministry of External Affairs
Sub-national governments
City governments
State governments
SAPCC
Non-state actors
Research institutions
Bilateral, donor and international agencies
Planning commission
LCSIG
NITI Aayog
Energy group
Period in which engagement with climate change began:
Pre-2007
2007–2009
2010–2014
Post-2014

IMPLEMENTAION ISSUES AND CHALLENGES

THE COMMON FRAMEWORK AND THE COPY-PASTE SYNDROME

G, D., Gupta, D., & Prakash, A. (2023)

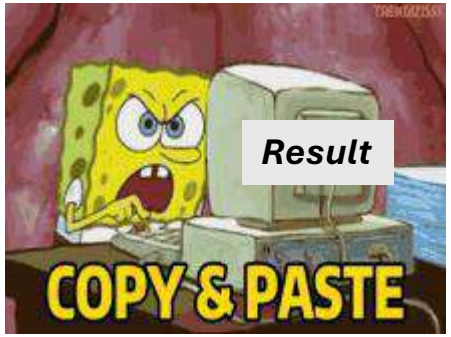
To guide the states, the Ministry of Environment, Forest and Climate Change (**MoEFCC**), in partnership with agencies like the **UNDP**

provided a →

"**Common Framework**" for the preparation of SAPCCs While intended to ensure **consistency**

But, this framework often encouraged a
"TEMPLATE-DRIVEN" APPROACH
lacking internal technical capacity, engaged external consultants who borrowed heavily from national documents like the *India Network for Climate Change Assessment* (INCCA) report.

The result
Homogenization of climate strategies that failed to capture local nuances



For instance, an analysis of 15 SAPCCs revealed significant weaknesses in action plans due to the **limited availability of reliable climate information at the state and local scale.**

States often utilized broad regional climate projections provided by national bodies rather than conducting granular, district-level vulnerability assessments.

CLIMATE WASHING

The Centre for Science and Environment (CSE)
noted that
states often repackaged
EXISTING DEVELOPMENT SCHEMES as **CLIMATE ACTION**
to attract central funding.

Irrigation projects, rural employment schemes (MGNREGA), and afforestation targets

→ were retroactively labeled as

"adaptation" or "mitigation" measures
(without any substantive change in their design or implementation to account for future climate risks)

This phenomenon converted the SAPCCs into "documents of intent" rather than rigorous investment plans

creating a disconnect between the stated climate goals and the actual developmental trajectory of the states

Coping with climate change Vol II (SAPCC) report- (CSE)

The "Unfunded Mandate" and the NAFCC Failure

When the SAPCCs were first mandated, there was an implicit expectation among states that the central government would provide the necessary financial support

The primary vehicle for this support was →

The National Adaptation Fund for Climate Change (NAFCC)
launched in 2015 to fund concrete adaptation activities.

HOWEVER
THE SCALE OF THE NAFCC HAS PROVEN TO BE GROSSLY INADEQUATE RELATIVE TO THE NEED

Recent budget analyses reveal a drastic reduction in central support. The allocation for the NAFCC plummeted **from INR 115.36 crore in 2017-18 to a mere INR 34 crore in 2022-23**. This reduction is particularly jarring given the escalating climate risks. Furthermore, the fund operates with a capping mechanism, limiting projects to approximately INR 25 crore.

State	Annual Adaptation Investment Need (INR Billion)	Annual Funding Gap (INR Billion)	Fiscal Deficit (2022-23) as % of GSDP
Tamil Nadu	176.6 (USD 2.2 bn)	39.4 (USD 492.5 mn)	4.1% (Limit: 3.5%)
Odisha	198.9 (USD 2.49 bn)	9.5 (USD 118.75 mn)*	Lower Gap due to assumed central transfers
Himachal Pradesh	High (Combined Gap INR 70bn**)	Not Specified separately	5.0% (Limit: 3.5%)

Note: *Odisha's lower reported gap relies on optimistic assumptions of future central transfers.
***Himachal Pradesh gap includes mitigation.*
Source: Financing Adaptation in India (2024) - *Climate Policy Initiative*

Access Barrier to International Finance
While international climate finance mechanisms like the Green Climate Fund (GCF) and the Global Environment Facility (GEF) exist, **accessing them requires complex technical capacities that many state governments lack**.

Coping with climate change Vol II (SAPCC) report- (CSE)

OTHER GAPS

- No effective institutional framework for monitoring and evaluation
- Lack of comprehensive climate impact vulnerability assessment at local, regional and state levels
- Lack of local, regional or state and sector-specific reliable future climate projections with specific time periods
- Lack of clarity on climate change
- Additionality of climate action not identified

CASE STUDIES OF IMPLEMENTAION FAILURES

AT 3 GOVERNANCE LEVELS

National Level

Case Study: National Solar Mission – The Solar Park Fiasco (CAG Report 2025)

The Mandate: The mission aimed to install 40,000 MW of solar power capacity through Solar Parks by 2019-20.
The Failure: A performance audit by the Comptroller and Auditor General (CAG) revealed that by March 2024, only **28%** of the targeted capacity was achieved.

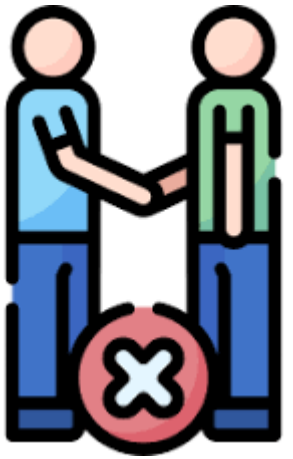
Structural Flaw – COORDINATION FAILURE

The Ministry of New and Renewable Energy (MNRE)
failed to coordinate with
State Transmission Utilities (STUs) and **land revenue departments**

Causing issues in

Land Acquisition: Inadequate feasibility studies led to the selection of unsuitable land (e.g., ecological hotspots or disputed territories), causing projects to be stalled or cancelled after initial expenditure.

Grid Connectivity: Solar parks were built without simultaneous development of external transmission systems, leaving generated power stranded.



State Level

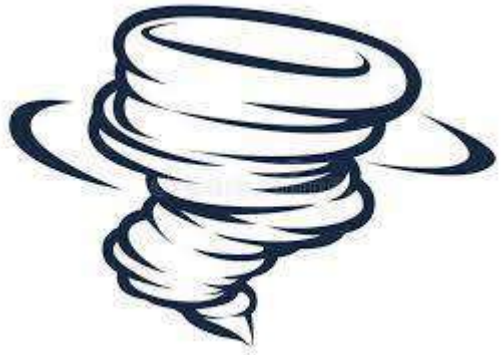
Case Study: Odisha – Cyclone Shelters and Coastal Resettlement

Financial Maladaptation and Maintenance Gaps
States cite a lack of funds ("unfunded mandate"), but CAG audits reveal that even available funds are mismanaged or used for infrastructure that is not climate-resilient.

A performance audit found severe neglect in **Multipurpose Cyclone Shelters (MCS)**.
Critical approach roads were missing, making shelters inaccessible during floods.
Water supply and sanitation facilities in these shelters were non-functional due to lack of maintenance funds.

Maladaptation
The resettlement of sea-erosion victims from **Satabhaya** to the **Bagapatia** colony failed the resilience test.

The resettlement site itself was inundated by tidal waters during Cyclone Dana (2024), proving that the vulnerability assessment for the new site was technically flawed.



City Level

Case Study: Chennai – The "Man-Made" Flood (CAG Report 2017)

Context
Chennai has a City Disaster Management Plan and acts under the Tamil Nadu SAPCC
.
The Disaster
The 2015 floods were labeled a "man-made catastrophe" by the CAG.

Specific Institutional Failure

Government Encroachment: The **Officers Training Academy (OTA)**, a government entity, constructed a bridge across the Adyar River that blocked 50% of the river's width. The state government (PWD) permitted this despite the flood risk

Drainage Failure: The CAG noted that 8 storm water drainage projects under JNNURM were abandoned or incomplete due to encroachments. The Chennai Metropolitan Development Authority (CMDA) failed to enforce zoning laws, allowing construction on wetlands.

AREA OF FAILURE	SPECIFIC AUDIT FINDING	CONSEQUENCE
Reservoir Management	The Water Resources Department (WRD) failed to create two new upstream reservoirs recommended by the Nucleus Cell for flood mitigation.	Inability to store excess inflow; mismanagement of discharge from Chembarambakkam tank exacerbated flooding.
Encroachment	Government bodies themselves were encroachers. The Officers Training Academy (OTA) constructed a bridge blocking the Adyar River.	The OTA bridge and earthen ramps blocked 50% of the river width, causing backflow and inundation of Nungambakkam and Manapakkam areas.
Drainage Infrastructure	Storm water drains were ill-designed and disconnected. 8 JNNURM drainage projects were left incomplete due to encroachments.	"Non-execution of works before monsoon hindered the free flow of floodwater," directly contributing to the inundation of residential areas.
Lake Restoration	Tardy implementation of restoration projects led to the abandoning of lakes.	Significant reduction in the natural water storage capacity of the city, increasing surface runoff.

INNOVATIONS & WAY FORWARD

Tech & Transparency

Digital tools can aid oversight

e.g., geotagged plantations (CAMPA), online climate scorecards, and real-time dashboards for mission progress. Use of big data (satellite observations) for monitoring forests, cropland health, water status.

National Adaptation Communications

India's next NAC (under UNFCCC) will strengthen national adaptation planning and tracking, aligning with SAPCCs and NAPCC.

Climate Finance Architecture

Proposal to enhance dedicated climate funds (e.g. stronger NAFCC, State-level Climate Funds) and leverage private finance (green bonds, ESG budgets). Improved budgeting (tagging climate in finance bills) is being piloted.

Convergence Mechanisms

Strengthen multi-level coordination

e.g., joint central–state project approval committees, district climate cells linking DRR and development plans, and stronger role for NITI Aayog (mandating state NDC roadmaps).

Emphasize bottom-up planning by building capacities in panchayats and municipalities.

Press Release-Press Information Bureau

CC5283: CLIMATE DATA ANALYSIS AND METHODS

PROJECT

"FLOOD VULNERABILTY INDEX USING PCA: CASE OF KERALA DISTRICTS"

CC25MTECH11002

FVI

FLOOD VULNERABILITY INDEX

A **Flood Vulnerability Index (FVI)** is an index that measures the susceptibility of a community or region to the adverse impacts of flooding. It is not a single, universally fixed formula but rather a tool built by selecting, weighting, and combining a set of indicators *(Pathak, Kulshrestha, & Kulshreshtha, 2021)*

"Old" IPCC framework– Vulnerability based
(vulnerability itself was the main outcome, and it included exposure)

Vulnerability = f(Exposure, Sensitivity, Adaptive Capacity)

(Source: IPCC AR4 Glossary)

"Current" IPCC framework- Risk based
(where vulnerability is a component of overall risk)

Risk = Hazard × Exposure × Vulnerability

(Source: IPCC AR6 Glossary)

IN INDIA

various institution uses various IPCC framework based on the applications

The Council on Energy, Environment and Water (CEEW) developed a comprehensive, district-level CVI for India. Based on the **classic IPCC Vulnerability Framework**

(Vulnerability = Exposure + Sensitivity - Adaptive Capacity)

DST-Supported District-Level Climate Risk Assessment conducted by institutions like IIT Mandi, IIT Guwahati, and CSTEP explicitly follows the modern IPCC AR5/AR6 Risk Framework

(Risk = Hazard x Exposure x Vulnerability)

APPLICATION OF FVI

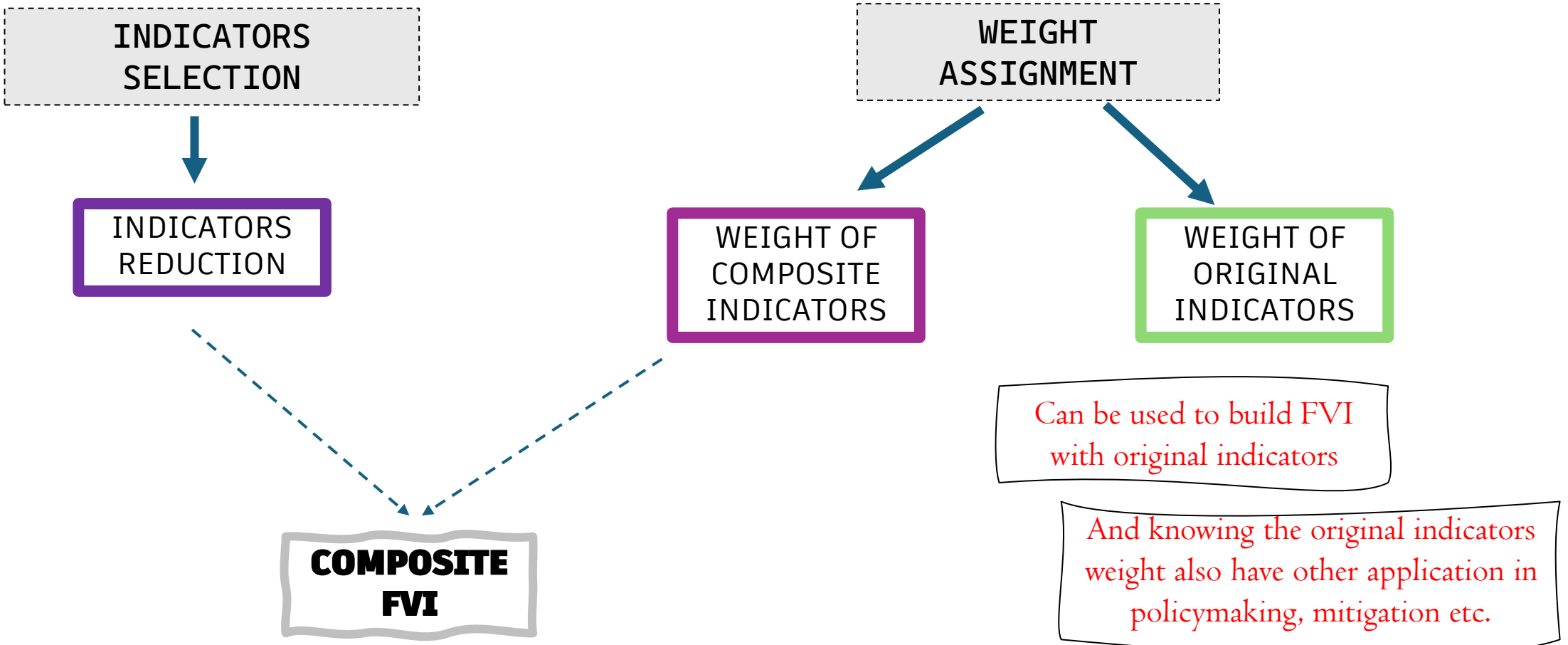
- Flood risk assessment
- Flood-prone area identification and mapping
- Disaster management planning
- Prioritization of high-risk districts or regions
- Urban and regional planning
- Land-use planning and zoning regulations
- Climate change vulnerability and adaptation studies
- Policy making for flood mitigation
- Community-level preparedness and evacuation planning

FVI *using* PCA

PCA is used in FVI building to address and determine the two important factors of index

Biased
SUBJECTIVITY
vs
OBJECTIVITY
It would have local spatial and temporal quality
EXPERT OPINION
DATA DRIVEN

FVI using PCA



WEIGHT PROBLEM

PCA is actually a popular technique in INDEX building

Some use equal weights
(even to composite indicators)

BUT the methodology used
OR
the way PCA is used varies

Sometimes the methods even contradict the other-one's principle

Some follow one or other method or a combination

THE METHODS

VARIANCE AS WEIGHT

More variance = more important pattern

Explain spatial and temporal variation

Index = (PC1_Score * %_Var_PC1) + (PC2_Score * %_Var_PC2) +...

INVERSE VARIANCE AS WEIGHT

1/variance

Less variance = more stable/reliable indicator (LESS NOISY)

Focus on Reliability

Index = (PC1_Score / %Var_PC1) + (PC2_Score / %Var_PC2)+...

VARIANCE AS WEIGHT WITH VARIMAX ROTATION

Used to interpret and label composite indicators
and
Determine original variable weightage

ORTHOGONAL ROTATION

each variable or covarying variables loads strongly on one principle component

Like
pc1 is socioeconomic,
pc2 is physical

Intra component rotation without affecting the variance of the principle component

Make higher loadings higher and lower loadings lower

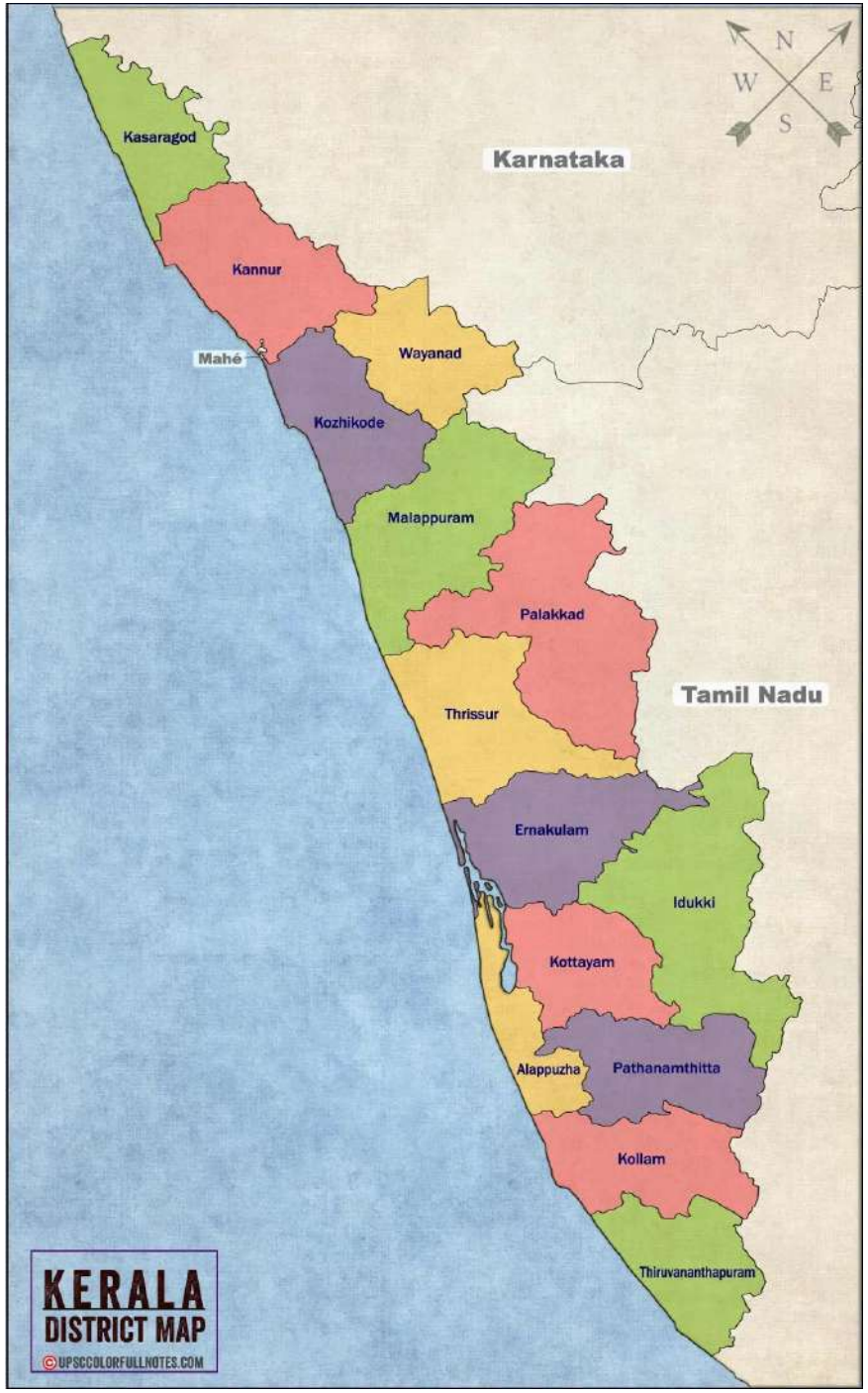
WHAT I AM DOING – building 2 different index

BUILD COMPOSITE FVI WITH TRANSFORMED COMPOSITE INDICATORS WITH ITS VARIANCE AS WEIGHT

DETERMINE THE WEIGHTAGE OF ORIGINAL INDICATORS
BUILD FVI WITH DETERMINED WEIGHT USING PCA

GOING FOR VARIANCE AS WEIGHT

PERFORMING PCA USING “STATSMODEL”
EIGENVAL = % VARIANCE
EIGENVECS = EIGEN VECTOR
SCORE = PC



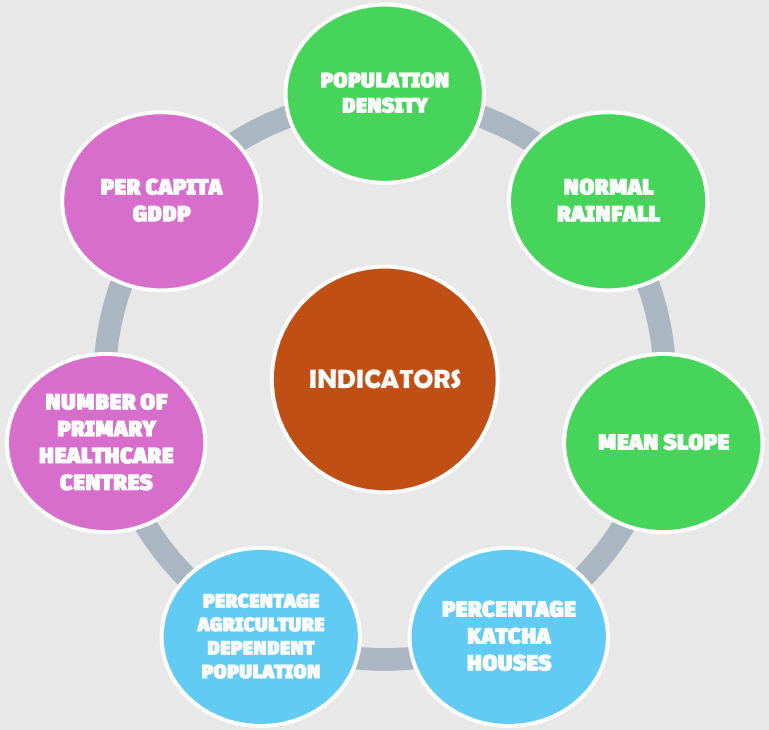
STUDY AREA

Districts of Kerala

1.Thiruvananthapuram
2.Kollam
3.Pathanamthitta
4.Alappuzha
5.Kottayam
6.Idukki
7.Ernakulam
8.Thrissur
9.Palakkad
10.Malappuram
11.Kozhikode
12.Wayanad
13.Kannur
14.Kasaragod

Source: upsc colorfull notes

INDICATORS



DISTRICT	POPULATION DENSITY	MEAN SLOPE (DEGREES)	NORMAL RAINFALL SW MONSOON (MM)	PERCENTAGE KATCHA HOUSES	PERCENTAGE AGRICULTURE DEPENDENT POPULATION	NUMBER OF PRIMARY HEALTHCARE CENTRES	PER CAPITA GDDP
Alappuzha	1504	1.1	1643	3.8	16.6	69	421460
Ernakulam	1072	3.8	2116.9	2.2	8.8	83	423751
Idukki	255	12.5	2574.3	10.7	47.8	52	355980
Kannur	852	5.1	2623	6.6	15.1	67	281339
Kasaragod	657	4.9	2846.2	11.2	24.9	39	400504
Kollam	1061	3.5	1257.6	3	13	69	292073
Kottayam	895	5.8	1905.3	2.3	17.8	69	254373
Kozhikode	1316	6.2	2561.6	4	11.9	72	373132
Malappuram	1157	6	1956.5	6.1	14	92	211798
Palakkad	627	7.5	1556.1	8	35.2	88	260664
Pathanamthitta	452	7	1572.7	2.1	16.7	49	256476
Thiruvananthapuram	1508	4.2	844.6	4.3	11.4	87	349421
Thrissur	1031	4	2132.1	3.1	15.1	83	326734
Wayanad	384	9	2464.7	16	49.7	24	233119

COMPOSITE FVI

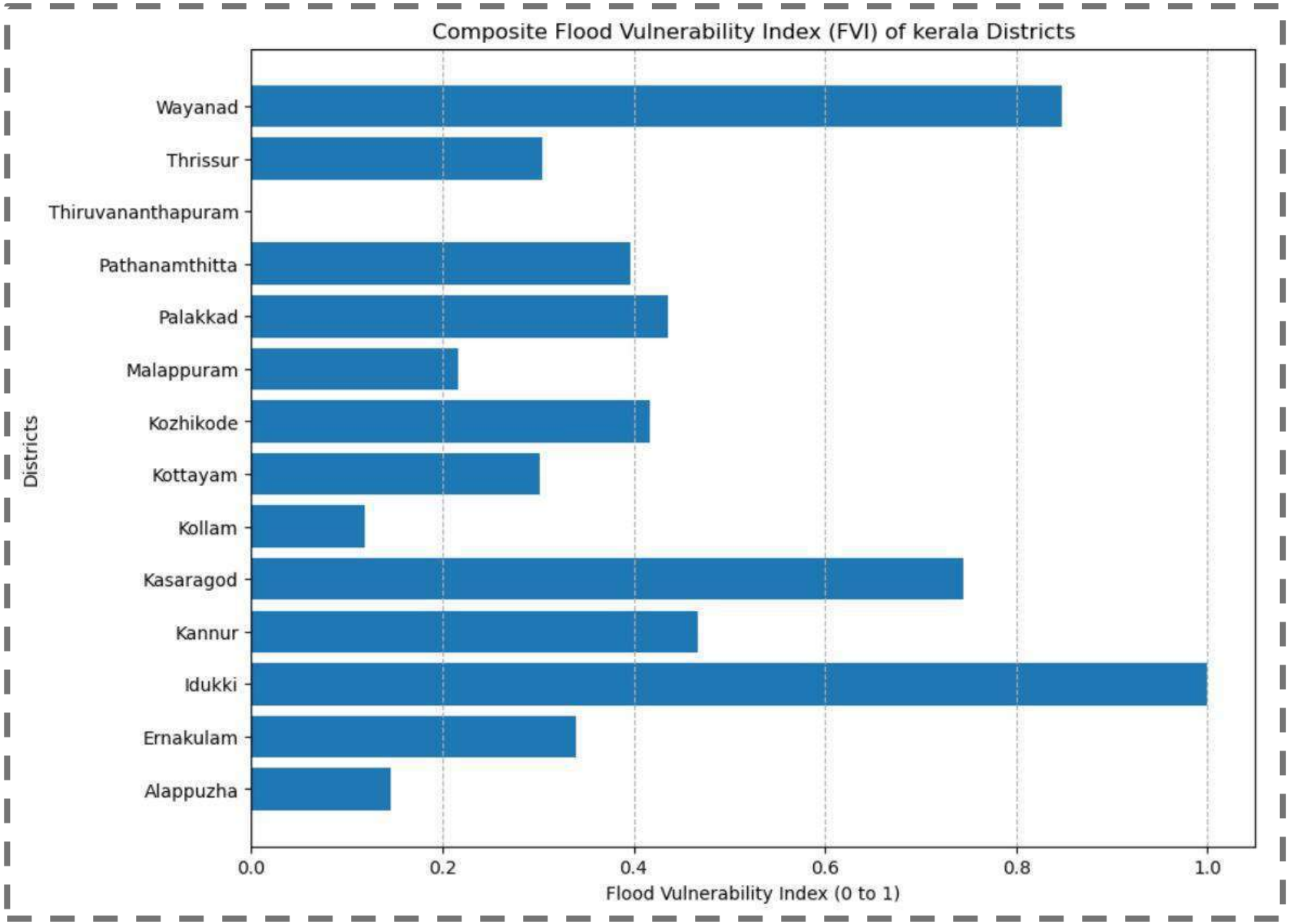
WITH COMPOSITE INDICATORS AND VARIANCE AS WEIGHT

RESULTS

Final Normalized FVI Ranks

	district	fvi	FVI_Normalized
0	Alappuzha	-15.765831	0.145885
1	Ernakulam	-4.207401	0.339218
2	Idukki	35.297558	1.000000
3	Kannur	3.453215	0.467354
4	Kasaragod	20.055195	0.745048
5	Kollam	-17.371476	0.119028
6	Kottayam	-6.490934	0.301022
7	Kozhikode	0.398037	0.416251
8	Malappuram	-11.533594	0.216676
9	Palakkad	1.557789	0.435650
10	Pathanamthitta	-0.810719	0.396033
11	Thiruvananthapuram	-24.487594	0.000000
12	Thrissur	-6.300771	0.304203
13	Wayanad	26.206528	0.847938

PLOT


Composite Flood Vulnerability Index (FVI) of kerala Districts
Districts
Wayanad
Thrissur
Thiruvananthapuram
Pathanamthitta
Palakkad
Malappuram
Kozhikode
Kottayam
Kollam
Kasaragod
Kannur
Idukki
Ernakulam
Alappuzha
0.0
0.2
0.4
0.6
0.8
1.0
Flood Vulnerability Index (0 to 1)

ORIGINAL INDICATORS WEIGHTS

USING VARIANCE WEIGHTED EIGENVECTOR AVERAGE OF ORIGINAL INDICATORS

RESULT: ORIGNAL INDICATORS WEIGHT	
population density	14.463037
mean slope (degrees)	15.235441
Normal rainfall SW Monsoon (mm)	14.079056
percentage katcha houses	14.380464
percentage agriculture dependent population	14.205148
number of primary healthcare centres	15.088172
per capita GDDP	12.548682

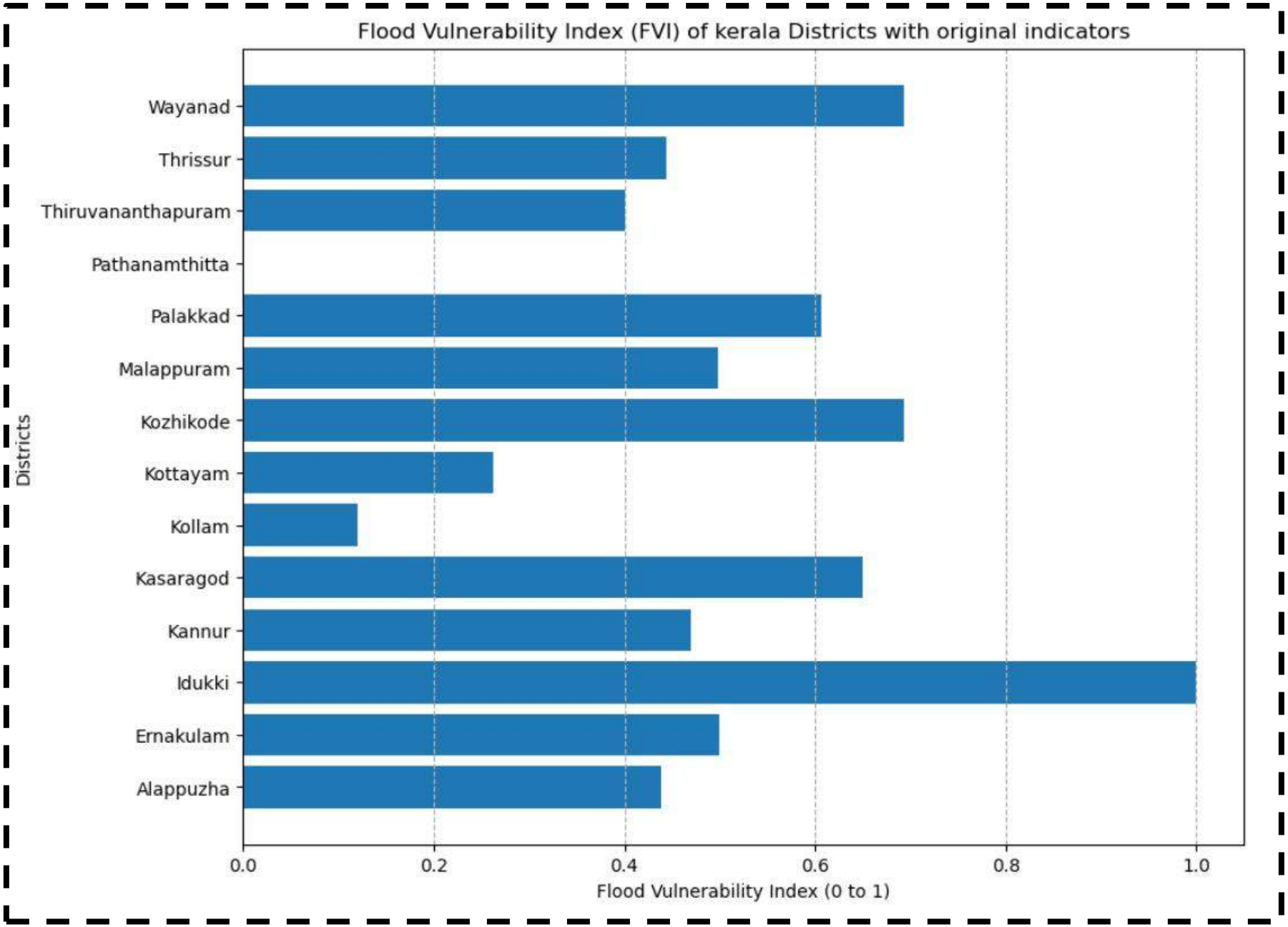
FVI

WITH ORIGINAL INDICATORS AND ITS DETRMINED WEIGHT

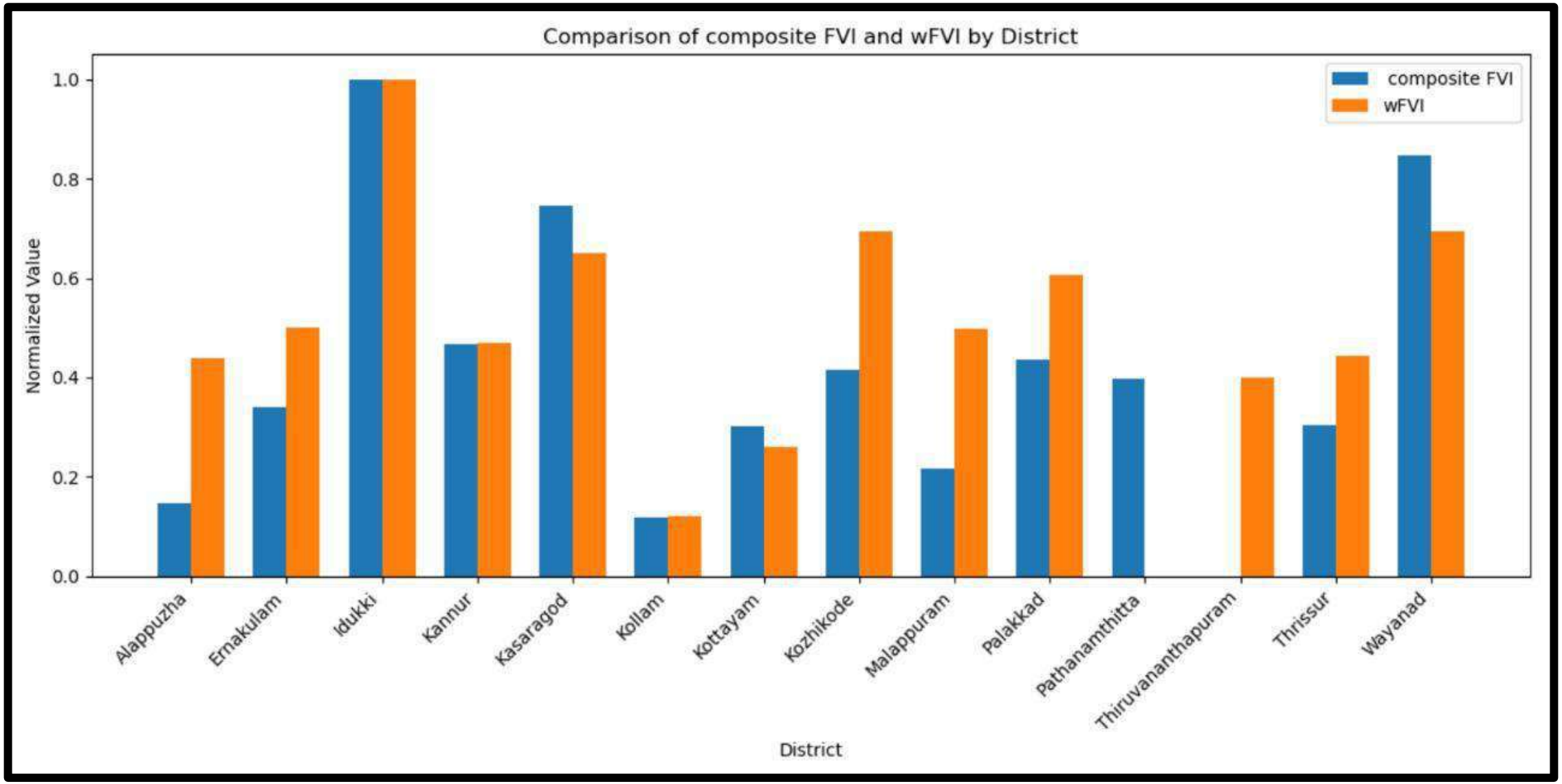
Final Normalized wFVI Ranks

	district	wfvi	wFVI_Normalized
0	Alappuzha	-5.885206	0.438729
1	Ernakulam	2.015781	0.499419
2	Idukki	67.184345	1.000000
3	Kannur	-1.812548	0.470013
4	Kasaragod	21.582561	0.649718
5	Kollam	-47.391995	0.119902
6	Kottayam	-28.938444	0.261650
7	Kozhikode	27.229261	0.693092
8	Malappuram	1.834549	0.498027
9	Palakkad	15.953202	0.606477
10	Pathanamthitta	-63.001573	0.000000
11	Thiruvananthapuram	-10.798884	0.400986
12	Thrissur	-5.286400	0.443329
13	Wayanad	27.315352	0.693753

PLOT


Flood Vulnerability Index (FVI) of kerala Districts with original indicators
Districts
Wayanad
Thrissur
Thiruvananthapuram
Pathanamthitta
Palakkad
Malappuram
Kozhikode
Kottayam
Kollam
Kasaragod
Kannur
Idukki
Ernakulam
Alappuzha
0.0
0.2
0.4
0.6
0.8
1.0
Flood Vulnerability Index (0 to 1)

COMPARING FVI



REFRENCES

1. Pathak, S. D., Kulshrestha, M., & Kulshreshtha, M. (2021). *Flood vulnerability assessment using data envelopment analysis – The case of Narmada river basin districts in central India. Natural Hazards*, 108(3), 2753–2776. https://doi.org/10.1007/s11069-021-04780-5

2. Intergovernmental Panel on Climate Change (IPCC). (2007). *Glossary of terms used in the IPCC Fourth Assessment Report*. In: *Climate Change 2007: Synthesis Report*. Cambridge University Press. https://www.ipcc.ch/site/assets/uploads/2018/02/ar4_syr_full_report.pdf

3. Intergovernmental Panel on Climate Change (IPCC). (2021). *Glossary*. In: *Climate Change 2021: The Physical Science Basis*. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (edited by V. Masson-Delmotte et al.). Cambridge University Press. https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Glossary.pdf

4. Liew, D. Y. C., Ros, F. C., & Harun, A. N. (2019). Developing composite indicators for flood vulnerability assessment: Effect of weight and aggregation techniques. *International Journal of Advanced Trends in Computer Science and Engineering*, *8*(3), 383–392 . https://doi.org/10.30534/ijatcse/2019/08832019

5. Iyengar, N. S., & Sudarshan, P. (1982). A method of classifying regions from multivariate data. *Economic and Political Weekly*, *17*(51), 2047–2052 .

6. Nardo, M., Saisana, M., Saltelli, A., & Tarantola, S. (2005). *Tools for composite indicators building* (Report No. EUR 21682 EN). European Commission, Joint Research Centre . http://farmweb.jrc.cec.eu.int/ci/bibliography.htm

7. Smits, J., & Huisman, J. (2023). *The GDL Climate Change Vulnerability Index (GVI)* (GDL Working Paper Series No. 23-101). Global Data Lab, Institute for Management Research, Radboud University. https://globaldatalab.org

8. Doyle, C., Sullivan, J., Mahtta, R., & Pandey, B. (2017). *Assessing biophysical and social vulnerability to natural hazards in Uttarakhand, India*. The World Bank Group. http://www.worldbank.org/water

(CC5900: LCA for Corporate Sustainability and Climate Change)

Scope 2 Emissions – Purchased Grid Electricity at IIT Hyderabad

PRESENTED BY:

NITESHKUMAR S K - CC25MTECH11002

UNDER SUPERVISION OF:

VENKATA KRISHNA KUMAR UPADHYAYULA

ADJUNCT FACULTY:

DEPT. OF CLIMATE CHANGE,

IIT HYDERABAD

INTRODUCTION

Scope 2 emissions refer to indirect greenhouse gas (GHG) emissions from the generation of purchased electricity, steam, heating, or cooling consumed by an organization.

- For IIT Hyderabad, this mainly includes electricity purchased from the state electricity grid for campus operations.

➢ Organizational Boundary:

The assessment follows the operational control approach, including electricity consumption across all campus infrastructure and facilities under IIT Hyderabad operations.

➢ Campus Activities Covered:

Academic buildings and laboratories

Administrative offices

Hostels and residential buildings

Lighting, cooling systems, and other campus infrastructure

➢ Reason for Selection:

- Electricity consumption is a major energy use in campus operations.
- Grid electricity in India is still partly generated using fossil fuels, leading to GHG emissions.
- Scope 2 missions can be quantified using electricity consumption data and emission factors, making it suitable for this analysis.

ELECTRICITY SUPPLIER

Electricity for IIT Hyderabad campus operations is supplied by:
Telangana State Southern Power Distribution Company Limited (TGSPDCL) (Telangana Electricity Distribution System) Formerly known as TSSPDCL, this state-owned utility is the sole provider for the Sangareddy district where the campus is located.

BASELINE YEAR

The baseline year selected is FY 2024–25 (based on collected data)

It represents a complete annual cycle (Jan –Dec) of electricity consumption data.

The baseline will be used to measure future emission reductions.

METHODOLOGY

Conversion of electricity consumption into emission

Total Emissions = Electricity Consumption (kWh) × Emission Factor

DATA

Source: Construction and Maintenance Division, IITH

IITH Campus Power Consumption details for the FY; 2024-25

Sl. No.	Year	Month	Total Power Cosumption in kWh	Total Power Cosumption in kVAh	Power factor	CMD (Contract Maximum Demand)	RMD (Recorded Maximum Demand)	Total paid Amount
1	2024	April	34,64,944.00	35,57,252.00	0.9741	7000 kVA	7564 kVA	₹ 3,03,32,551.00
2	2024	May	35,84,452.00	35,79,568.00	1.0014	7000 kVA	7596 kVA	₹ 3,34,71,913.00
3	2024	June	32,21,040.00	33,00,188.00	0.9760	7000 kVA	6776 kVA	₹ 2,97,93,168.00
4	2024	July	28,33,876.00	28,83,752.00	0.9827	7000 kVA	5600 kVA	₹ 2,59,37,724.00
5	2024	August	33,07,780.00	33,14,868.00	0.9979	7000 kVA	5960 kVA	₹ 2,95,86,368.00
6	2024	September	25,28,444.00	26,36,652.00	0.9590	7000 kVA	6048 kVA	₹ 2,41,89,492.00
7	2024	October	30,40,232.00	31,15,220.00	0.9759	7000 kVA	6212 kVA	₹ 2,81,35,636.00
8	2024	November	27,49,644.00	28,15,364.00	0.9767	7000 kVA	5600 kVA	₹ 2,51,29,412.00
9	2024	December	26,43,428.00	27,05,804.00	0.9769	7000 kVA	5600 kVA	₹ 2,42,62,442.00
10	2025	January	26,64,500.00	27,16,472.00	0.9809	7000 kVA	5600 kVA	₹ 2,42,62,443.00
11	2025	February	26,43,660.00	27,02,524.00	0.9782	7000 kVA	6940 kVA	₹ 2,42,62,444.00
12	2025	March	36,05,420.00	36,79,356.00	0.9799	7000 kVA	7164 kVA	₹ 2,42,62,445.00

EMISSION FACTOR

Emission Factors (tCO2/MWh) (incl. Imports)	2020-21	2021-22	2022-23	2023-24	2024-25
Weighted Average Emission Rate (2)	0.786	0.810	0.823	0.841	0.831
Weighted Average Grid Emission Rate (Incl. RES,Captive) (2)	**0.698**	**0.711**	**0.716**	**0.727**	**0.710**
Simple Operating Margin (1) (2)	0.940	0.960	0.971	0.961	0.961
Build Margin (not adjusted for imports)	0.865	0.869	0.867	0.552	0.512
Combined Margin (1) (2)	0.903	0.915	0.919	0.757	0.736

(1) Operating margin is based on the data for the same year. This corresponds to the *ex post option* given in "Tool to Calculate the Emission Factor for an Electricity System", Ver. 7.0 (p.16)

(2) Adjustments for imports from other Indian grids are based on operating margin of exporting grid. For imports from other countries, an emission factor of zero is used. See "Tool to Calculate the Emission Factor for an Electricity System", Ver. 7.0 (p.10 & 11), options a+b

Source: Central Electricity Authority (CEA), Government of India. CO_2 Baseline Database for the Indian Power Sector, Version 21.0, 2025.

0.710 t CO_2/MWh = 0.710 kg CO_2/kWh

https://cea.nic.in/cdm-co2-baseline-database/

IIT HYDERABAD SCOPE 2 EMISSION (PURCHASED ELECTRICITY)

EMISSIONS (0.710) KG CO2

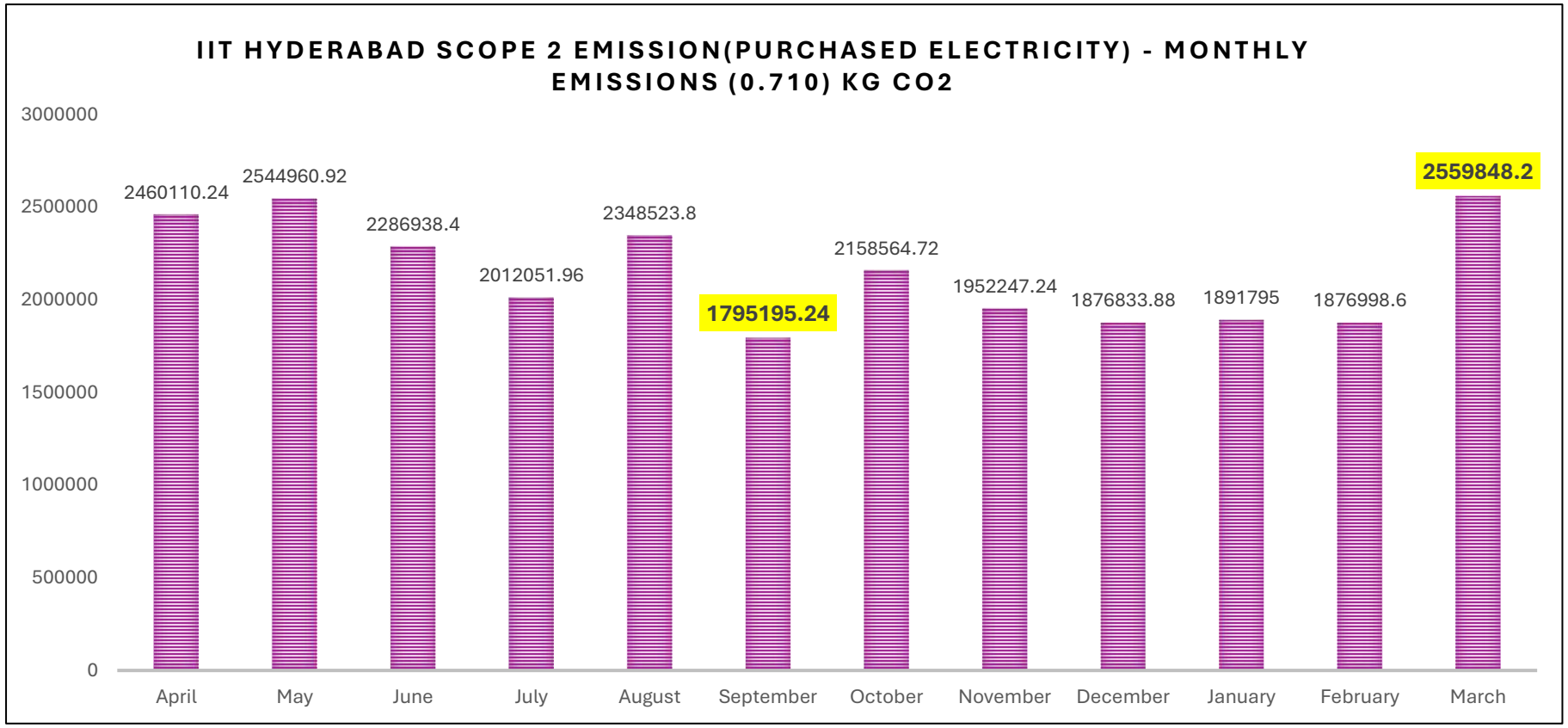
Year	Month	Consumption kWh	Emissions (0.710) kg CO2
2024	April	3464944	2460110.24
2024	May	3584452	2544960.92
2024	June	3221040	2286938.4
2024	July	2833876	2012051.96
2024	August	3307780	2348523.8
2024	September	2528444	1795195.24
2024	October	3040232	2158564.72
2024	November	2749644	1952247.24
2024	December	2643428	1876833.88
2025	January	2664500	1891795
2025	February	2643660	1876998.6
2025	March	3605420	2559848.2
2024-25	**Annual**	**36287420**	**25764068.2**

TOTAL ANNUAL CONSUMPTION
36287420 kWh

TOTAL ANNUAL EMISSIONS
25764068 kg CO2 = 25764 tonnes CO2

MONTHLY EMISSION AND OBSERVATIONS

2024-25



WE CAN SEE A SEASONAL PATTERN

High emission months

March–May (SUMMER) – due to use of AC utilities

Low emission months

July, September–December (MONSOON & WINTER)

Highest consumption & emission is at **March** (2025)

Lowest consumption & emission is at **September** (2024)

Strategies and Institutional Policy

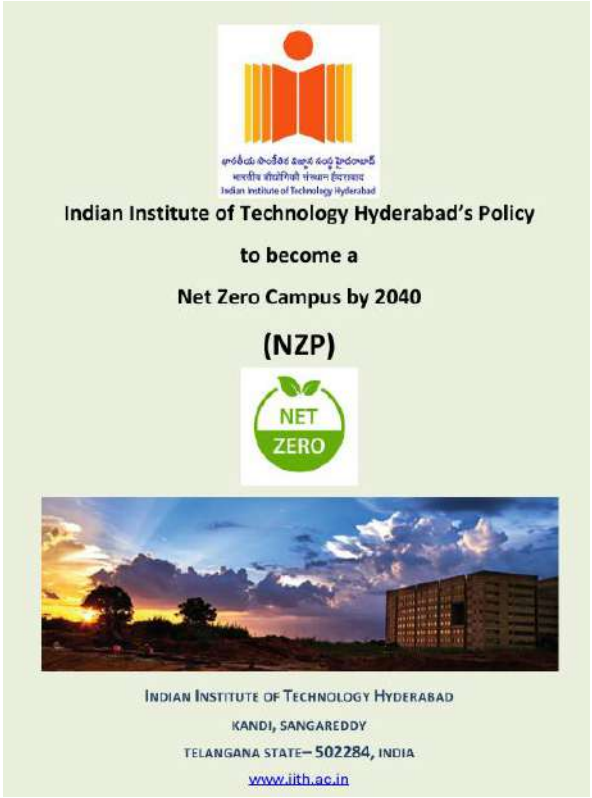
Under the **Net Zero Campus Policy 2040**, the institute plans to:

1. **Transition to a renewable energy–powered campus, aiming for over 95% energy from solar and other clean sources.**

2. **Expand renewable energy capacity to more than 18 MW of solar generation with battery storage.**

3. **Improve energy efficiency using LED lighting, BLDC fans, and high-efficiency HVAC systems.**

4. **Use smart energy monitoring systems and building management systems for real-time energy optimization.**



Impact

These initiatives will reduce dependence on fossil-fuel-based grid electricity and lower Scope 2 emissions, supporting IIT Hyderabad's Net-Zero Campus target by 2040.

REDUCTION AND FUTURE SCENARIO

Targeted solar generation capacity 2030 = 18 MW

with Capacity Utilization Factor (CUF) = 18–22% ~ 20%

Targeted energy from solar 2030 = 18 MW x 8760×0.20 = 31536000 kWh/year

ANNUAL CAMPUS CONSUMPTION
36287420 kWh

EMISSION REDUCTION = 31536000×0.710=22390560 kg CO2 =22365 tonnes CO2/year **REDUCED**

NET EMISSION (2030)
25764 tonnes - 22365 tonnes = 3399 tonnes CO_2

87% emission reduction

PROGRESS IN 2026

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

PV SYSTEM INSTALLATION – through end of 2025
OPERATION - from 2026

We may see the changes in emission in Fy 2025-26, 2026-27

Elite

NPTEL ONLINE CERTIFICATION

(Funded by the MoE, Govt. of India)

Skill India
कौशल भारत - कुशल भारत

This certificate is awarded to

NITESHKUMAR S K

for successfully completing the course

Environmental Impact Assessment

with a consolidated score of **96** %

Online Assignments	25/25	Proctored Exam	70.5/75

TOP TOPPER 1%

Total number of candidates certified in this course: **1266**

Prof. Kaushik Ghosh,
Professor(Chemistry)
Coordinator CEC

Jan-Apr 2025

(12 week course)

Prof. Ranjana Pathania,
Professor (BSBE)
Coordinator (NPTEL)

Indian Institute of Technology Roorkee

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

Roll No: NPTEL25AR03S658700486

To verify the certificate

No. of credits recommended: 3 or 4

Elite

NPTEL ONLINE CERTIFICATION

(Funded by the MoE, Govt. of India)

Skill India
कौशल भारत - कुशल भारत

This certificate is awarded to

NITESHKUMAR S K

for successfully completing the course

Research Methodology for Planning and Architectural Studies

with a consolidated score of **96** %

Online Assignments	25/25	Proctored Exam	71.25/75

TOP TOPPER 1%

Total number of candidates certified in this course: **262**

Prof. Kaushik Ghosh,
Professor(Chemistry)
Coordinator CEC

Jan-Apr 2025

(12 week course)

Prof. Ranjana Pathania,
Professor (BSBE)
Coordinator (NPTEL)

Indian Institute of Technology Roorkee

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

Roll No: NPTEL25AR05S358700008

To verify the certificate

No. of credits recommended: 3 or 4